ETSI TR 133 929-4 V19.0.0 (2026-10)



TECHNICAL REPORT

LTE;
5G;
Lawful Interception (LI) implementation guidance;
LI for data in 5G core
(3GPP TR 33.929-4 version 19.0.0 Release 19)



Reference
DTR/TSGS-0333929-4vj00

Keywords
5G,LTE

ETSI

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Contents

Intellectual Property Rights 2
Legal Notice 2
Modal verbs terminology 2
Foreword 6
Introduction 6
1 Scope 7
2 References 7
3 Definitions of terms, symbols and abbreviations 8
3.1 Terms 8
3.2 Symbols 8
3.3 Abbreviations 8
4 Data interception in 5G Core 9
4.1 General 9
4.2 Scope of illustration 9
5 LI in SMF/UPF 9
5.1 General 9
5.2 SMF/UPF flow diagrams 10
5.2.1 General 10
5.2.2 High level flow diagrams 10
5.2.2.1 General 10
5.2.2.2 Non-roaming scenario 10
5.2.2.2.1 PDU session establishment 10
5.2.2.2.2 PDU session modification 11
5.2.2.2.2.1 General 11
5.2.2.2.2.2 UE-initiated PDU session modification 11
5.2.2.2.2.3 Network-initiated PDU session modification 12
5.2.2.2.2.4 PDU session modification for change of access type 12
5.2.2.2.3 PDU session release 13
5.2.2.2.3.1 General 13
5.2.2.2.3.2 UE initiated PDU session release 13
5.2.2.2.3.3 Network initiated PDU session release 14
5.2.2.3 Home-routed roaming scenario (VPLMN) 14
5.2.2.3.1 PDU session establishment 14
5.2.2.3.2 PDU session modification 15
5.2.2.3.2.1 General 15
5.2.2.3.2.2 UE initiated PDU session modification 15
5.2.2.3.2.3 Network (VPLMN) initiated PDU session modification 16
5.2.2.3.2.4 Network (HPLMN) PDU session modification 17
5.2.2.3.2.5 PDU session modification for change of access type 17
5.2.2.3.3 PDU session release 18
5.2.2.3.3.1 General 18
5.2.2.3.3.2 UE initiated PDU session release 18
5.2.2.3.3.3 Network (VPLMN) initiated PDU session release 18
5.2.2.3.3.4 Network (HPLMN) initiated PDU session release 19
5.2.2.4 Home-routed roaming scenario (HPLMN) 20
5.2.2.4.1 PDU session establishment 20
5.2.2.4.2 PDU session modification 20
5.2.2.4.2.1 General 20
5.2.2.4.2.2 UE initiated PDU session modification 20
5.2.2.4.2.3 Network (VPLMN) initiated PDU session modification 21
5.2.2.4.2.4 Network (HPLMN) initiated PDU session modification 22
5.2.2.4.2.5 PDU session modification for change of access type – example 1 22

5.2.2.4.2.6 PDU session modification for change of access type – example 2 23
5.2.2.4.3 PDU session release 23
5.2.2.4.3.1 General 23
5.2.2.4.3.2 UE initiated PDU session release 23
5.2.2.4.3.3 Network (VPLMN) initiated PDU session release 24
5.2.2.4.3.4 Network (HPLMN) initiated PDU session release 25
5.2.3 Detailed flow diagrams 25
5.2.3.1 General 25
5.2.3.2 PDU session establishment 25
5.2.3.2.1 Non-roaming scenario 25
5.2.3.2.2 Home-routed roaming scenario (VPLMN) 26
5.2.3.2.3 Home-routed roaming scenario (HPLMN) 27
5.2.3.3 PDU session modification 28
5.2.3.3.1 Non-roaming scenario 28
5.2.3.3.1.1 UE-initiated PDU session modification 28
5.2.3.3.1.2 Network-initiated PDU session modification 29
5.2.3.3.1.3 PDU session modification due to change in access type 30
5.2.3.3.2 Home-routed roaming scenario (VPLMN) 32
5.2.3.3.2.1 Network-initiated PDU session modification 32
5.2.3.3.2.2 PDU session modification due to change in access type 33
5.2.3.3.3 Home-routed roaming scenario (HPLMN) 34
5.2.3.3.3.1 Network-initiated PDU session modification 34
5.2.3.3.3.2 PDU session modification due to change in access type (N3IWF in the same VPLMN) 35
5.2.3.3.3.3 PDU session modification due to change in access type (N3IWF in the HPLMN) 36
5.2.3.4 PDU session release 37
5.2.3.4.1 General 37
5.2.3.4.2 Non-roaming scenario 37
5.2.3.4.2.1 UE initiated PDU session release 37
5.2.3.4.2.2 Network initiated PDU session release 38
5.2.3.4.2.3 Network initiated PDU session release without any N1 or N2 message transfer 39
5.2.3.4.3 Home-routed roaming scenario (VPLMN) 40
5.2.3.4.3.1 UE initiated PDU session release 40
5.2.3.4.3.2 Network (HPLMN) initiated PDU session release 41
5.2.3.4.3.3 Network initiated PDU session release without any N1 or N2 message transfer 42
5.2.3.4.4 Home-routed roaming scenario (HPLMN) 43
5.2.3.4.4.1 UE-initiated PDU session release 43
5.2.3.4.4.2 Network (HPLMN) initiated PDU session release 44
5.2.3.4.4.3 Network initiated PDU session release without any N1 or N2 message transfer 45
5.3 Handover related flow diagrams 46
5.3.1 Introduction 46
5.3.1.1 Overview 46
5.3.1.2 Intra-system handover 47
5.3.1.3 Inter-system handover 47
5.3.1.4 DL data forwarding 47
5.3.2 Xn-based handover flows 47
5.3.2.1 Overview 47
5.3.2.1.1 General 47
5.3.2.1.2 Without UPF relocation in a non-roaming case 48
5.3.2.1.3 With UPF relocation in a non-roaming case 49
5.3.2.1.4 Without UPF relocation in roaming case 53
5.3.2.1.5 With UPF relocation in roaming case 55
5.3.2.2 Detailed flows - non-roaming, no UPF relocation 58
5.3.2.2.1 General 58
5.3.2.2.2 The flows 59
5.3.2.3 Detailed flows - non-roaming, UPF relocation 61
5.3.2.3.1 General 61
5.3.2.3.2 The flows 61
5.3.2.4 Detailed flows - home-routed roaming, no UPF relocation 64
5.3.2.4.1 General 64
5.3.2.4.2 LI in VPLMN 65
5.3.2.4.3 LI in HPLMN 66
5.3.2.5 Detailed flows - home-routed roaming, UPF relocation 68

5.3.2.5.1 General 68
5.3.2.5.2 LI in VPLMN 69
5.3.2.5.3 LI in HPLMN 71
5.3.3 N2-based handover flows 74
5.3.3.1 Overview 74
5.3.3.1.1 General 74
5.3.3.1.2 Direct DL data forwarding (non-roaming) 74
5.3.3.1.3 Direct DL data forwarding (home-routed roaming, LI in VPLMN) 78
5.3.3.1.4 Direct DL data forwarding (home-routed roaming, LI in HPLMN) 80
5.3.3.1.5 Indirect DL data forwarding (non-roaming) 82
5.3.3.1.6 Indirect DL data forwarding (home-routed roaming, LI in VPLMN) 89
5.3.3.1.7 Indirect DL data forwarding (home-routed roaming, LI in HPLMN) 93
5.3.3.2 Detailed flows - non-roaming, direct DL data forwarding 97
5.3.3.2.1 General 97
5.3.3.2.2 The flows 98
5.3.3.3 Detailed flows - non-roaming, direct DL data forwarding (with UPF relocation) 101
5.3.3.3.1 General 101
5.3.3.3.2 The flows 101
5.3.3.4 Detailed flows - non-roaming, indirect DL data forwarding (with UPF relocation) 105
5.3.3.4.1 General 105
5.3.3.4.2 The flows 106
5.3.3.5 Detailed flows - home-routed roaming, indirect DL data forwarding (with UPF relocation) 110
5.3.3.5.1 General 110
5.3.3.5.2 LI in VPLMN 111
5.3.3.5.3 LI in HPLMN 115
5.3.4 5G to 4G handover flows 119
5.3.4.1 Overview 119
5.3.4.1.1 General 119
5.3.4.1.2 Direct forwarding 120
5.3.4.1.3 Indirect forwarding in a non-roaming case 122
5.3.4.1.4 Indirect forwarding in a roaming case 124
5.3.4.2 Detailed flows - non-roaming, direct DL data forwarding 127
5.3.4.2.1 General 127
5.3.4.2.2 The flows 128
5.3.4.3 Detailed flows - non-roaming, indirect DL data forwarding 132
5.3.4.3.1 General 132
5.3.4.3.2 The flows 133
5.3.4.4 Detailed flows – home-routed roaming, indirect DL data forwarding 137
5.3.4.4.1 General 137
5.3.4.4.2 LI in VPLMN 138
5.3.4.4.3 LI in HPLMN 142
5.3.5 4G to 5G handover flows 146
5.3.5.1 Overview 146
5.3.5.1.1 General 146
5.3.5.1.2 Direct forwarding 147
5.3.5.1.3 Indirect forwarding in a non-roaming case 149
5.3.5.1.4 Indirect forwarding in a roaming case 151
5.3.5.2 Detailed flows - non-roaming, direct DL data forwarding 154
5.3.5.2.1 General 154
5.3.5.2.2 The flows 155
5.3.5.3 Detailed flows - non-roaming, indirect DL data forwarding 159
5.3.5.3.1 General 159
5.3.5.3.2 The flows 160
5.3.5.4 Detailed flows – home-routed roaming, indirect DL data forwarding 163
5.3.5.4.1 General 163
5.3.5.4.2 LI in VPLMN 164
5.3.5.4.3 LI in HPLMN 168

Annex A: Change history 173

History 174

Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services. See TR 33.929-1 [13].
- Part 2: LI for IMS based STIR/SHAKEN. See TR 33.929-2 [14].
- Part 3: LI for messaging services. See TR 33.929-3 [15].
- Part 4: LI for data in 5G core (present document).
- Part 5: LI for ID association caching. See TR 33.929-5 [16].
- Part 6: LI for IMS based RCS. See TR 33.929-6 [17].
- Part 7: LI location acquisition capabilities. See TR 33.929-7 [18].
- Part 8: LI for MMS. See TR 33.929-8 [19].
- Part 9: LI for PTC. See TR 33.929-9 [20].

In Part 1, the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2, the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3, the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4 (present document), the illustrations of LI for data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5, the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6, the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7, the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8, the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9, the illustrations of LI for PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3], TS 33.128 [4] and TR 33.928 [5].

The present document that covers the illustrations for LI for Data in 5G Core include the LI aspects of data interception focussing on the PDU session related events including the handover cases.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TR 33.928: "ADMF logic for provisioning Lawful Interception (LI)".

[6] ETSI TS 103 221-1: "Lawful Interception; Internal Network Interfaces; Part 1: X1".

[7] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[8] 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS)".

[9] 3GPP TS 29.244: "Interface between the Control Plane and the User Plane nodes".

[10] 3GPP TS 29.502: "5G System; Session Management Services; Stage 3".

[11] 3GPP TS 38.413: "NG Application Protocol (NGAP)".

[12] 3GPP TS 38.331: "Radio Resource Control (RRC) protocol specification".

[13] 3GPP TR 33.929-1: "Lawful Interception (LI) implementation guidance; LI for IMS-based services".

[14] 3GPP TR 33.929-2: "Lawful Interception (LI) implementation guidance; LI for IMS based STIR/SHAKEN".

[15] 3GPP TR 33.929-3: "Lawful Interception (LI) implementation guidance; LI for messaging services".

[16] 3GPP TR 33.929-5: "Lawful Interception (LI) implementation guidance; LI for ID Association Caching ".

[17] 3GPP TR 33.929-6: "Lawful Interception (LI) implementation guidance; LI for IMS based RCS".

[18] 3GPP TR 33.929-7: "Lawful Interception (LI) implementation guidance; Part 7: Location Services".

[19] 3GPP TR 33.929-8: "Lawful Interception (LI) implementation guidance; LI for MMS".

[20] 3GPP TR 33.929-9: "Lawful Interception (LI) implementation guidance; LI for PTC".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5GMS	5G Media Streaming
AAnF	AKMA Anchor Function
ADMF	LI Administration Function
AF	Application Function
AKMA	Authentication and Key Management for Applications
AMF	Access and Mobility Management Function
CC	Content of Communication
CSP	Communication Service Provider
HPLMN	Home PLMN
IRI	Intercept Related Information
LALS	Lawful Access Location Services
LEMF	Law Enforcement Monitoring Facility
LI	Lawful Interception
LI_X2	Lawful Interception Internal Interface 2
LI_X3	Lawful Interception Internal Interface 3
MCPTT	Mission Critical Push To Talk
MDF	Mediation and Delivery Function
MDF2	Mediation and Delivery Function 2
MDF3	Mediation and Delivery Function 3
NF	Network Function
PDU	Packet Data Unit
PFCP	Packet Forwarding Control Protocol
PLMN	Public Land Mobile Network
PoC	Push to talk over Cellular
POI	Point Of Interception
PTC	Push to Talk over Cellular
SMF	Session Management Function
TF	Triggering Function
UE	User Equipment
UPF	User Plane Function
V-SMF	Visted SMF
V-UPF	Visited UPF
VPLMN	Visited PLMN

xCC LI_X3 Communications Content
xIRI LI_X2 Intercept Related Information

4 Data interception in 5G Core

4.1 General

As defined in TS 33.127 [3], TS 33.128 [4] and illustrated in TR 33.928 [5], the following NFs participate in providing the interception in the 5G core when the service type provisioned to the respective POIs is "Data" (see ETSI TS 103 221-1 [6] and TR 33.928 [5]):

- AMF.
- SMF.
- UPF.
- UDM.
- NEF.
- AAnF.
- 5GMS AF.
- NWDAF.

In many cases, the scope of interception provided by the LI functions present in the above NFs depends on the services offered by the CSP.

4.2 Scope of illustration

The present document illustrates cases for LI for 5G core network only when it is found that such illustrations provide additional clarity. In doing so, the present document focusses on the LI at the SMF/UPF from the NFs listed in clause 4.1.

In illustrating the LI for SMF/UPF, the focus is on the PDU session related events that result in LI reporting. Such PDU session related events include various handover scenarios as well.

5 LI in SMF/UPF

5.1 General

The subsequent clauses provide a few examples of flow-diagrams that illustrate the LI in SMF/UPF. There are two sets of flow-diagrams:

- High-level flow diagrams.
- Detailed flow-diagrams.

The high-level flow diagrams focus on the target UE and the NF that provides the LI functions. These flow-diagrams illustrate the various trigger conditions defined in TS 33.128 [4] for the generation of xIRIs.

The detailed flow diagrams expand the scope of involved NFs in the 5G core network and these flows help to understand the LI in the scheme of overall services provided within the 5G core network. The detailed flow diagrams are provided only when it is found that such flow diagrams may provide additional illustrations. In other words, not all the high-level flow-diagrams have the equivalent detailed flow-diagrams.

5.2 SMF/UPF flow diagrams

5.2.1 General

The subsequent clauses provide a series of flow diagrams that illustrate PDU session related xIRIs as defined in TS 33.128 [4]. The flow diagrams include the following cases:

- PDU session establishment.
- PDU session modification.
- PDU session release.

MA PDU session related flow-diagrams high-light any key differences they may have.

The high-level flow diagrams focus on the xIRI generation. Few of those high-level flow-diagrams are expanded to the detailed flows to illustrate the generation of LI_T3 triggers and the xCC generation. Detailed flow diagrams are provided only when it is found that such flows may provide additional illustrations.

5.2.2 High level flow diagrams

5.2.2.1 General

The following scenarios are illustrated:

- SMF, non-roaming.
- V-SMF, home-routed roaming.
- H-SMF, home-routed roaming.

In the home-routed roaming case, LI provided in the VPLMN is independent of the LI provided in the HPLMN.

The NAS messages and the 5GSM state changes are defined in TS 24.501 [8].

5.2.2.2 Non-roaming scenario

5.2.2.2.1 PDU session establishment

The flow-diagram in figure 5.2.2.2.1-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the SMF for a non-roaming scenario.

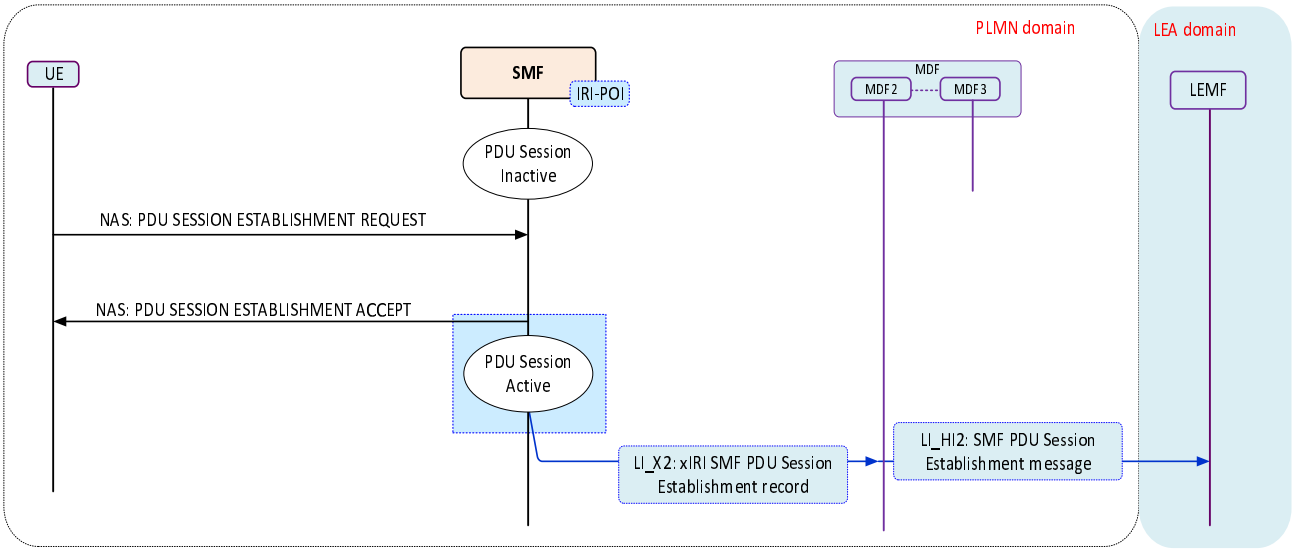


Figure 5.2.2.2.1-1: High-level flow diagram for PDU session establishment (non-roaming)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI is SMF sending a NAS message PDU SESSION ESTABLISHMENT ACCEPT to the target UE and the 5GSM state changing into PDU Session Active.

While drawing this flow, the TS 23.502 [7], figure 5.3.2.2.1-1 is referred to.

5.2.2.2.2 PDU session modification

5.2.2.2.2.1 General

PDU session modification related scenarios include the following cases:

- UE-initiated PDU session modification.
- Network initiated PDU session modification.
- PDU session modification due to the change of access type (e.g. non-3GPP access to 3GPP access).

The NAS messages and the 5GSM state changes are defined in TS 24.501 [8].

5.2.2.2.2.2 UE-initiated PDU session modification

The flow-diagram in figure 5.2.2.2.2.2-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the SMF for a UE-initiated PDU session modification.

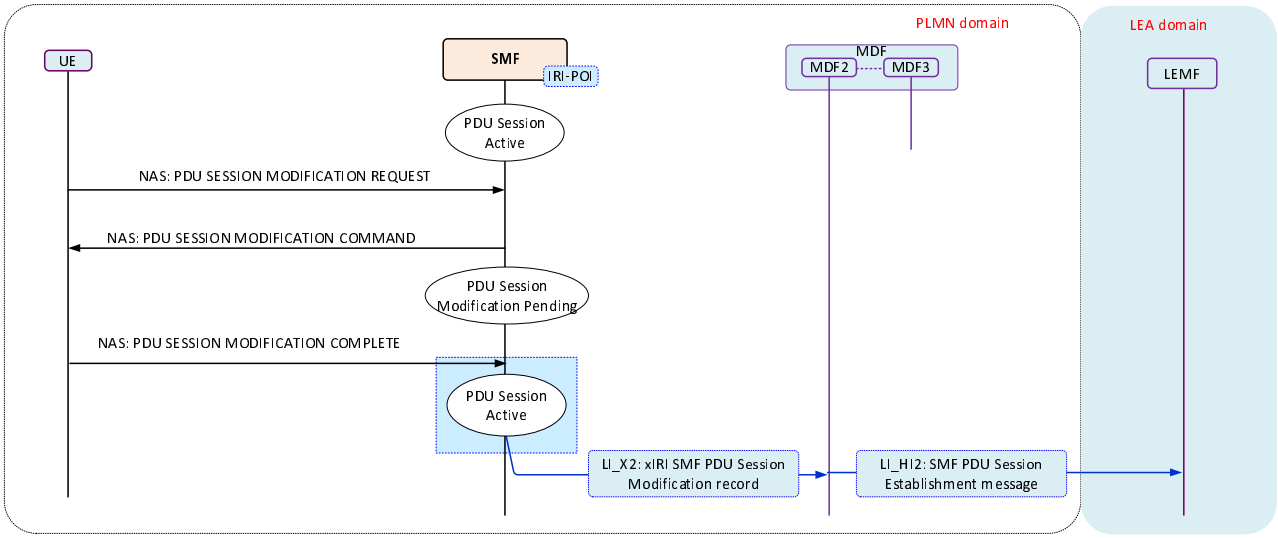


Figure 5.2.2.2.2.2-1: High-level flow diagram for PDU session modification (UE-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the reception of NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state returns to PDU Session Active.

While drawing this flow, the TS 23.502 [7], figure 5.3.2.2.2-1 is referred to.

5.2.2.2.2.3 Network-initiated PDU session modification

The flow-diagram in figure 5.2.2.2.2.3-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the SMF for network-initiated PDU session modification.

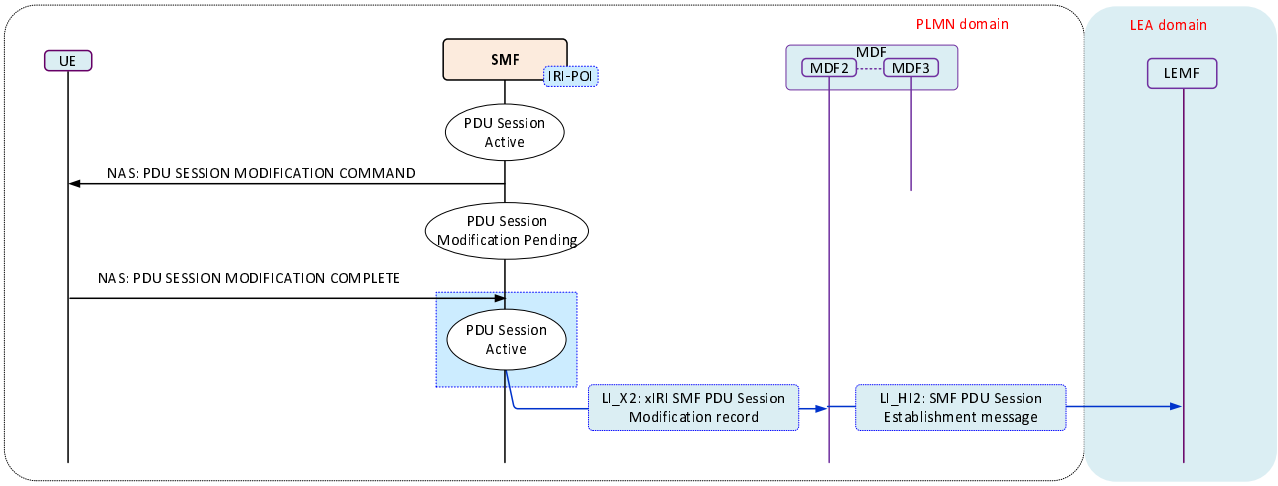


Figure 5.2.2.2.2.3-1: High-level flow diagram for PDU session modification (Network-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the reception of NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state returns to PDU Session Active.

While drawing these flows, the TS 23.502 [7], figure 5.3.2.2.2-1 is referred to.

5.2.2.2.2.4 PDU session modification for change of access type

The flow-diagram in figure 5.2.2.2.2.4-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the SMF when the target UE changes access type (e.g. non-3GPP access to 3GPP access).

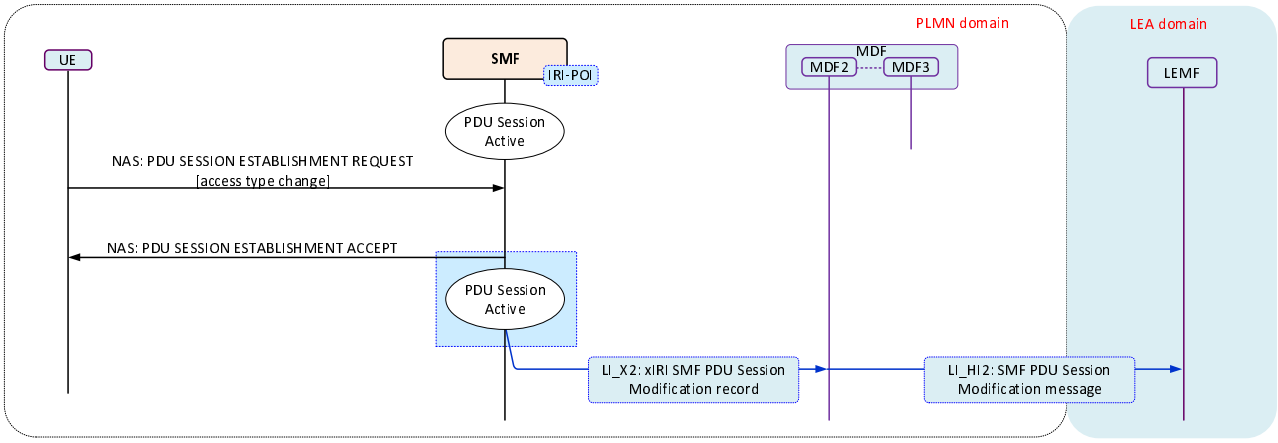


Figure 5.2.2.2.2.4-1: High-level flow diagram for PDU session modification (access type change)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the sending of the NAS message PDU SESSION ESTABLISHMENT ACCEPT to the target UE and the 5GSM state remains in PDU Session Active.

While drawing these flows, the TS 23.502 [7], figure 4.9.2.1-1 is referred to.

5.2.2.2.3 PDU session release

5.2.2.2.3.1 General

PDU session release related scenarios include the following cases:

- UE-initiated PDU session release.
- Network initiated PDU session release.

The NAS messages and the 5GSM state changes are defined in TS 24.501 [8].

5.2.2.2.3.2 UE initiated PDU session release

The flow-diagram in figure 5.2.2.2.3.2-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the SMF for a UE-initiated PDU session release.

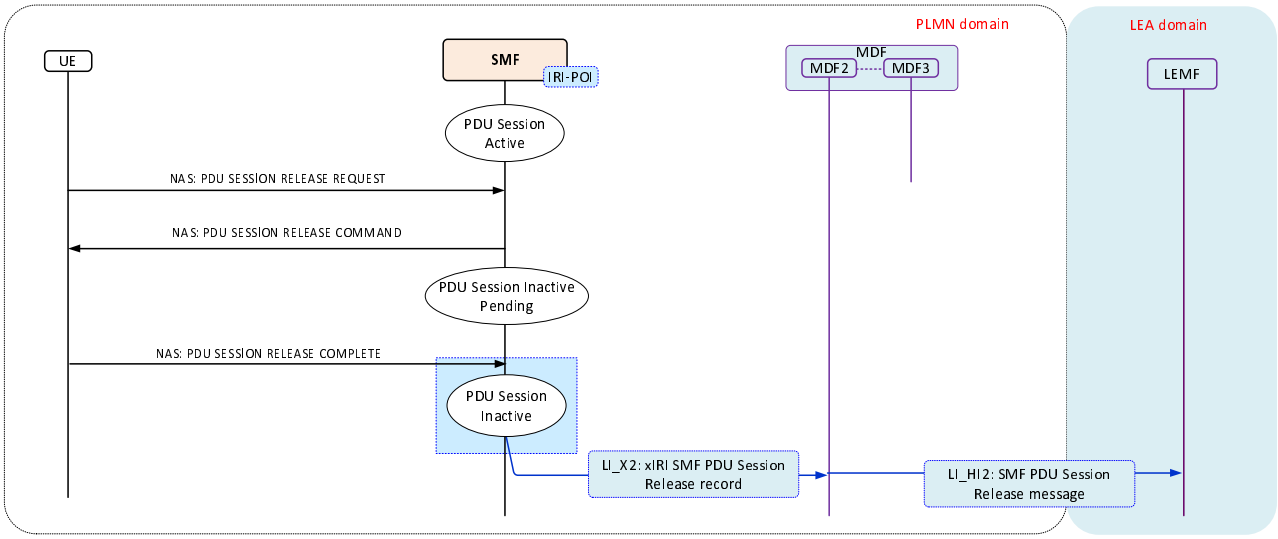


Figure 5.2.2.2.3.2-1: High-level flow diagram for PDU session release (UE-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the reception of NAS message PDU SESSION RELEASE COMPLETE from the target UE and the 5GSM state becomes PDU Session Inactive.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.2-1 is referred to.

5.2.2.2.3.3 Network initiated PDU session release

The flow-diagram in figure 5.2.2.2.3.3-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the SMF for network-initiated PDU session release.

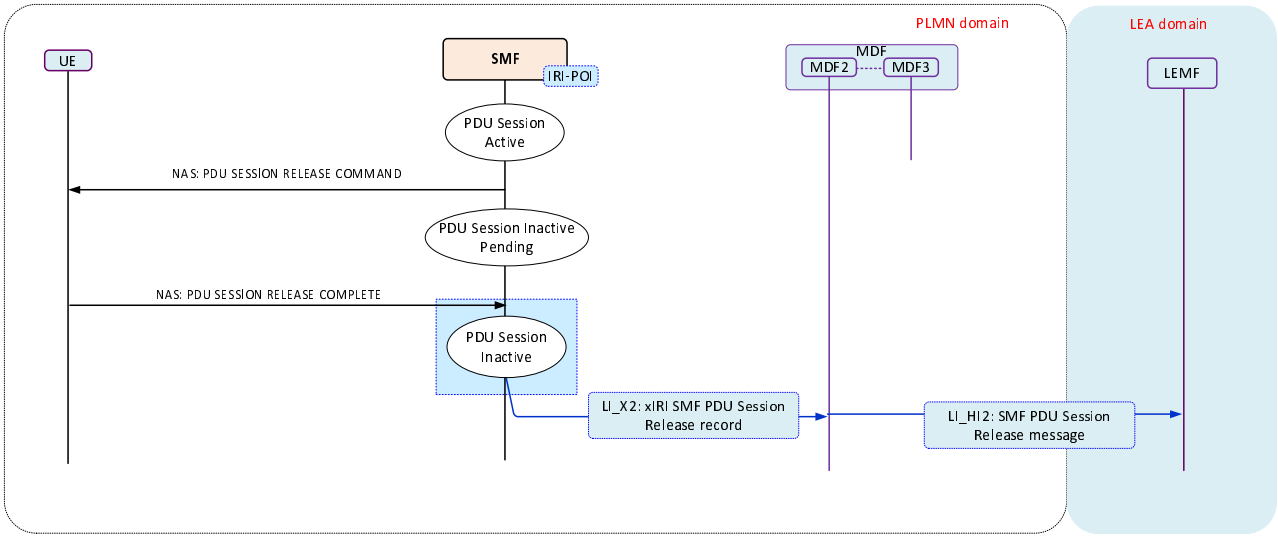


Figure 5.2.2.2.3.3-1: High-level flow diagram for PDU session release (network-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the reception of NAS message PDU SESSION RELEASE COMPLETE from the target UE and the 5GSM state becomes to PDU Session Inactive.

While drawing these flows, the TS 23.502 [7], figure 4.3.4.2-1 is referred to.

5.2.2.3 Home-routed roaming scenario (VPLMN)

5.2.2.3.1 PDU session establishment

The flow-diagram in figure 5.2.2.3.1-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the V-SMF for a home-routed roaming scenario. In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

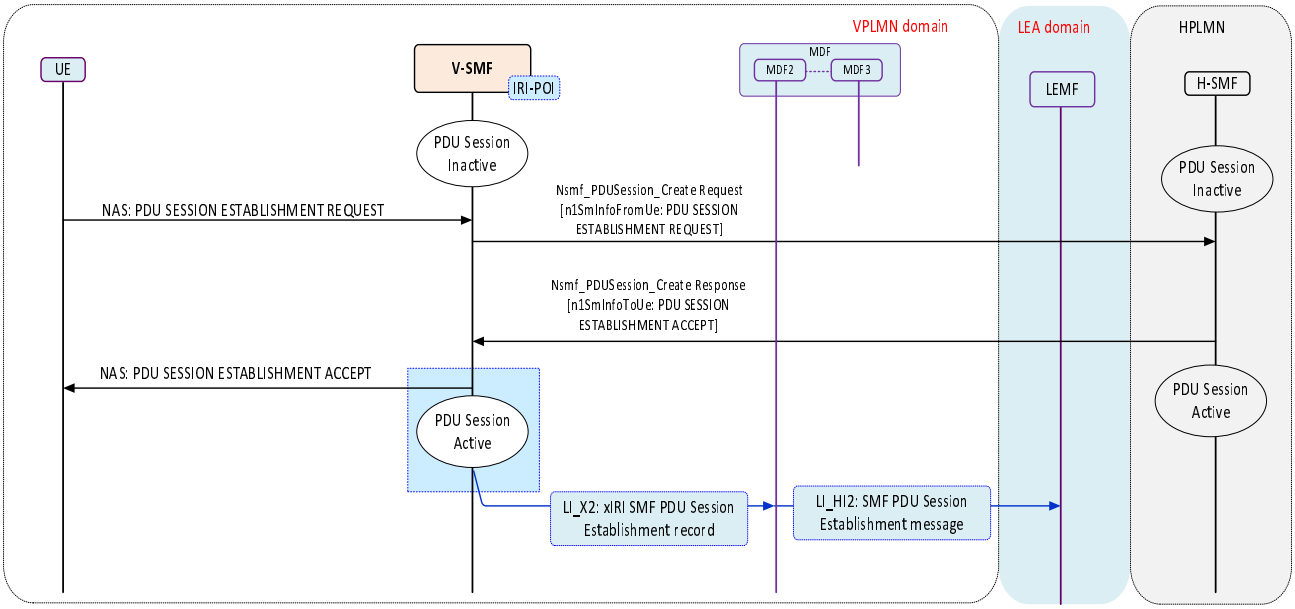


Figure 5.2.2.3.1-1: High-level flow diagram for PDU session establishment (VPLMN)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the V-SMF is same as the trigger condition used to generate in a non-roaming case (see clause 5.2.2.2.1).

While drawing this flow, the TS 23.502 [7], figure 5.3.2.2.2-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.3.2 PDU session modification

5.2.2.3.2.1 General

PDU session modification related scenarios include the following cases:

- UE-initiated PDU session modification.
- Network (VPLMN) initiated PDU session modification.
- Network (HPLMN) initiated PDU session modification.
- PDU session modification due to the change of access type (e.g. non-3GPP access to 3GPP access).

The NAS messages and the 5GSM state changes are defined in TS 24.501 [8].

5.2.2.3.2.2 UE initiated PDU session modification

The flow-diagram in figure 5.2.2.3.2.2-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the V-SMF for a UE-initiated PDU session modification. In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

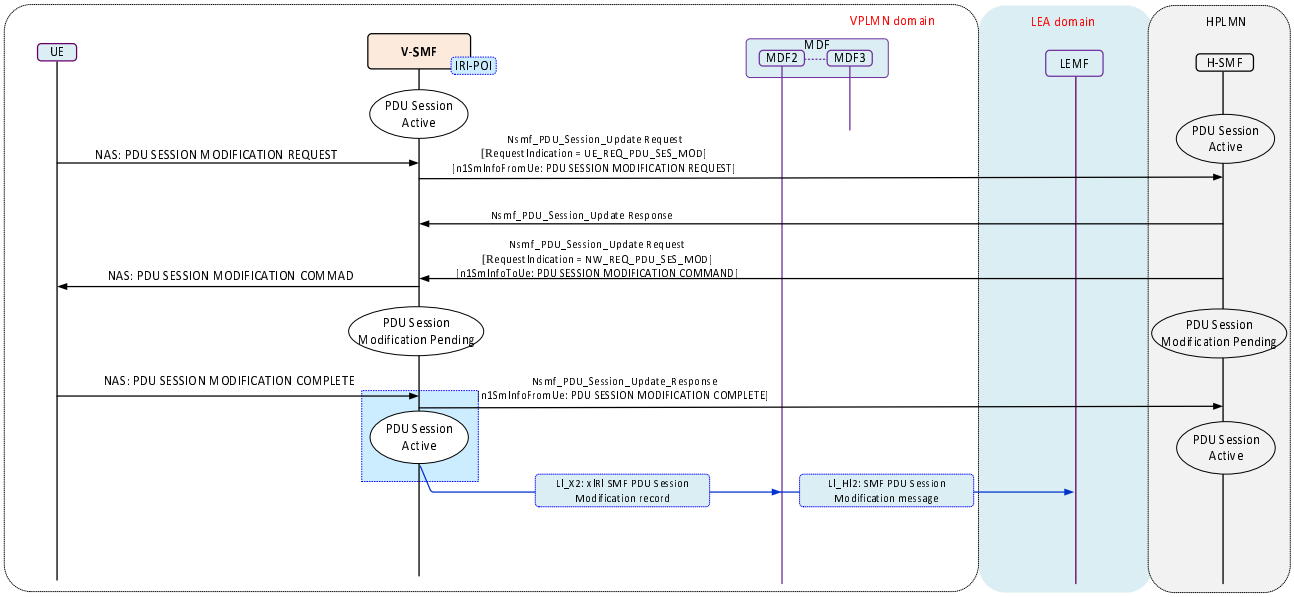


Figure 5.2.2.3.2.2-1: High-level flow diagram for PDU session modification (UE-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the V-SMF is on the reception of NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state returns to PDU Session Active. Note that the trigger conditions for the generation of xIRI in the V-SMF is same as the trigger condition used to generate in a non-roaming case (see clause 5.2.2.2.2.2).

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.3.2.3 Network (VPLMN) initiated PDU session modification

The flow-diagram in figure 5.2.2.3.2.3-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the V-SMF for a VPLMN-initiated PDU session modification. In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

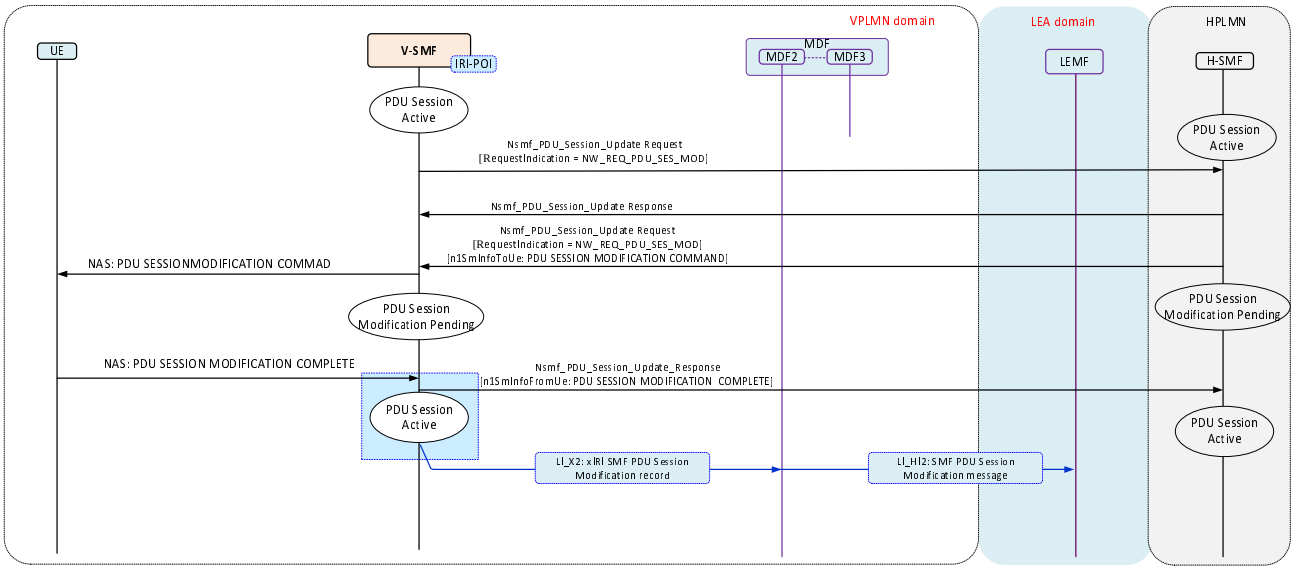


Figure 5.2.2.3.2.3-1: High-level flow diagram for PDU session modification (Network-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the reception of NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state returns to PDU Session Active. Note that the trigger conditions for the generation of xIRI in the V-SMF is same as the trigger condition used to generate in a non-roaming case (see clause 5.2.2.2.2.3).

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.3.2.4 Network (HPLMN) PDU session modification

The flow-diagram in figure 5.2.2.3.2.4-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the V-SMF for a HPLMN-initiated PDU session modification. In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

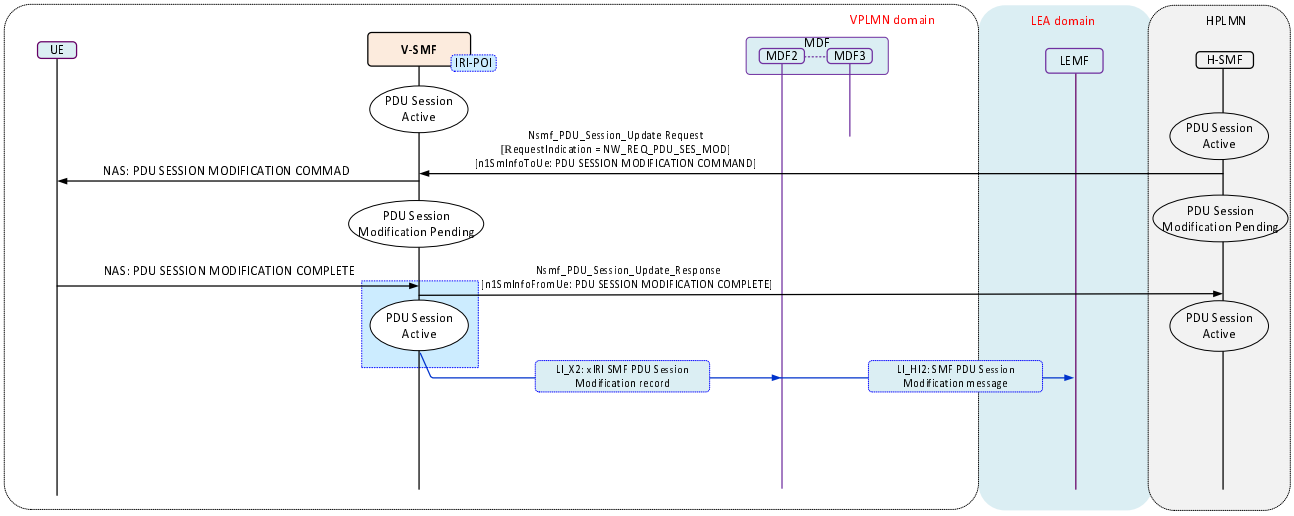


Figure 5.2.2.3.2.4-1: High-level flow diagram for PDU session modification (Network-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the reception of NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state returns to PDU Session Active. Note that the trigger conditions for the generation of xIRI in the V-SMF is same as the trigger condition used to generate in a non-roaming case (see clause 5.2.2.2.2.3).

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.3.2.5 PDU session modification for change of access type

The flow-diagram in figure 5.2.2.3.2.5-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the V-SMF when the target UE changes access type (e.g. non-3GPP access to 3GPP access).

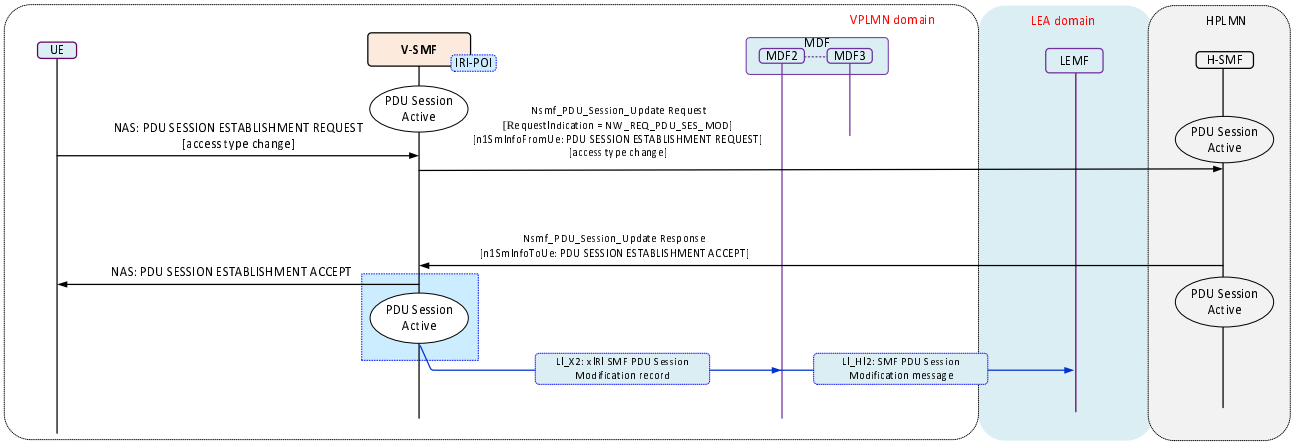


Figure 5.2.2.3.2.5-1: High-level flow diagram for PDU session modification (access type change)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the V-SMF is on the sending of the NAS message PDU SESSION ESTABLISHMENT ACCEPT to the target UE and the 5GSM state remains in PDU

Session Active. Note that the trigger conditions for the generation of xIRI in the V-SMF is same as the trigger condition used to generate in a non-roaming case (see clause 5.2.2.2.2.4).

While drawing this flow, the TS 23.502 [7], figure 4.9.2.3.1-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.3.3 PDU session release

5.2.2.3.3.1 General

PDU session release related scenarios illustrated in this clause include the following cases:

- UE-initiated PDU session release.
- Network (VPLMN) initiated PDU session release.
- Network (HPLMN) initiated PDU session release.

The NAS messages and the 5GSM state changes are defined in TS 24.501 [8].

5.2.2.3.3.2 UE initiated PDU session release

The flow-diagram in figure 5.2.2.3.3.2-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the V-SMF for a UE-initiated PDU session release. In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

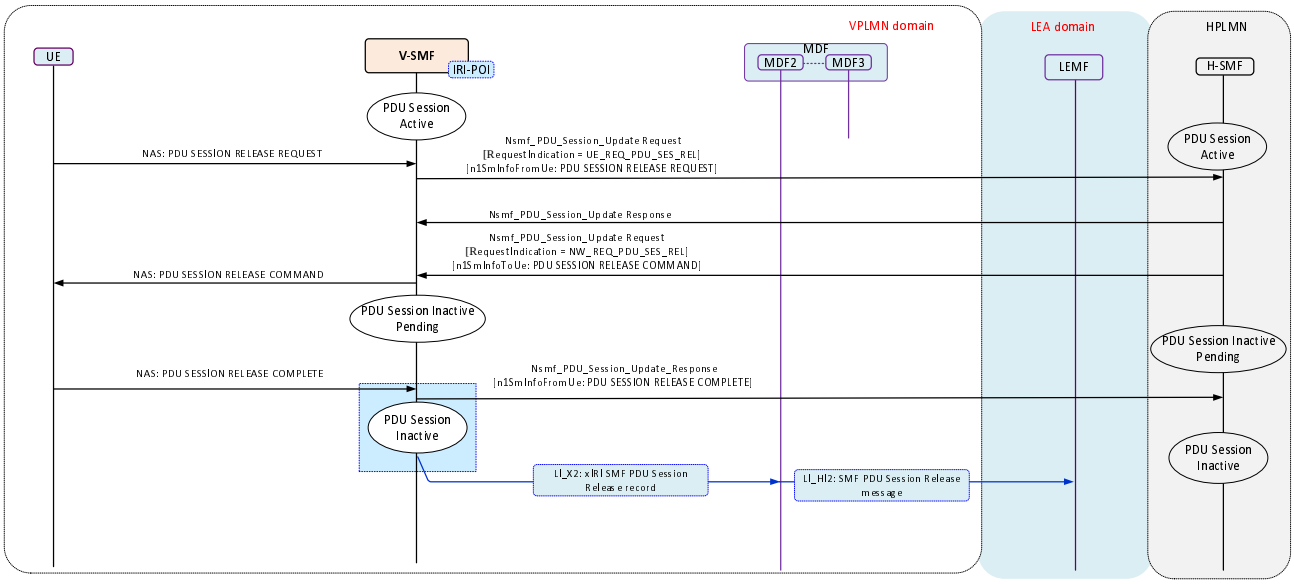


Figure 5.2.2.3.3.2-1: High-level flow diagram for PDU session release (UE-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the V-SMF is on the reception of NAS message PDU SESSION RELEASE COMPLETE from the target UE and the 5GSM state and the 5GSM state becomes PDU Session Inactive. Note that the trigger conditions for the generation of xIRI in the V-SMF is same as the trigger condition used to generate in a non-roaming case (see clause 5.2.2.2.3.2).

While drawing this flow, the TS 23.502 [7], figure 4.3.4.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.3.3.3 Network (VPLMN) initiated PDU session release

The flow-diagram in figure 5.2.2.3.3.3-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the V-SMF for a VPLMN-initiated PDU session release. In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

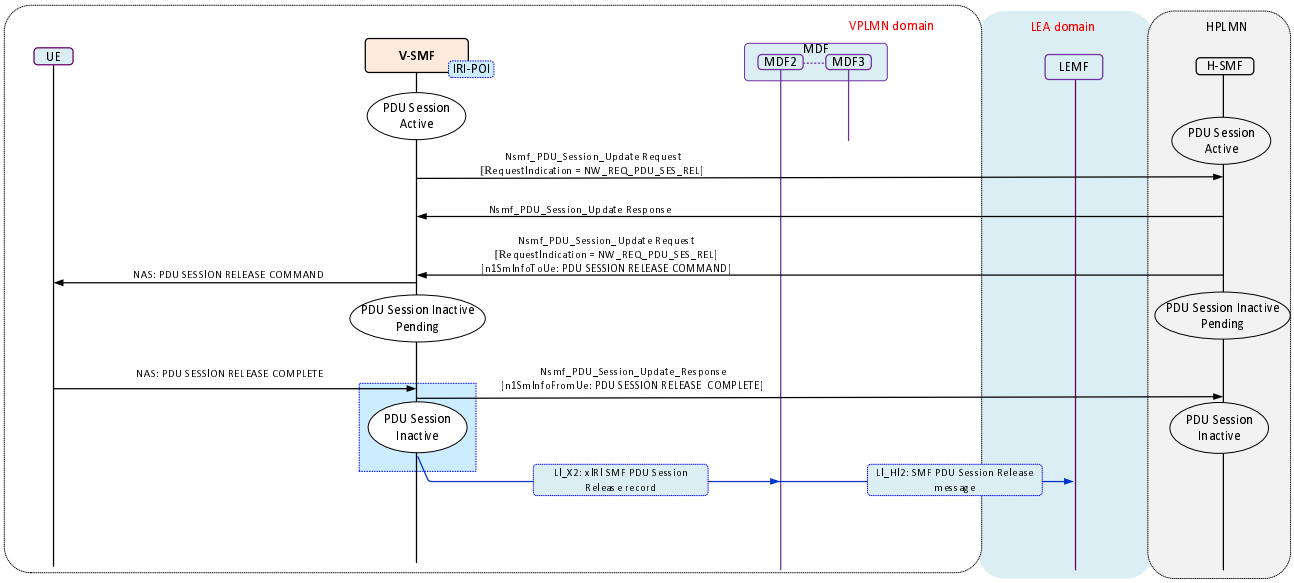


Figure 5.2.2.3.3.3-1: High-level flow diagram for PDU session release (network-initiated - VPLMN)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the reception of NAS message PDU SESSION RELEASE COMPLETE from the target UE and the 5GSM state becomes to PDU Session Inactive. Note that the trigger conditions for the generation of xIRI in the V-SMF is same as the trigger condition used to generate in a non-roaming case (see clause 5.2.2.2.3.3).

While drawing this flow, the TS 23.502 [7], figure 4.3.4.3-1 is referred to. The session management related messages and parameters (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.3.3.4 Network (HPLMN) initiated PDU session release

The flow-diagram in figure 5.2.2.3.3.4-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the V-SMF for a HPLMN-initiated PDU session release. In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

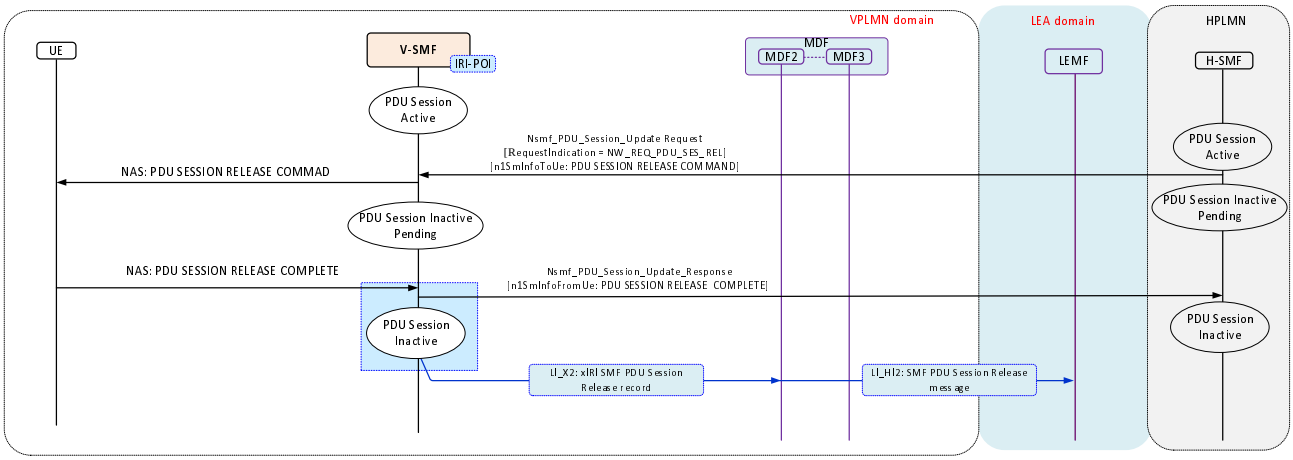


Figure 5.2.2.3.3.4-1: High-level flow diagram for PDU session release (network-initiated - HPLMN)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the SMF is on the reception of NAS message PDU SESSION RELEASE COMPLETE from the target UE and the 5GSM state becomes to PDU Session Inactive. Note that the trigger conditions for the generation of xIRI in the V-SMF is same as the trigger condition used to generate in a non-roaming case (see clause 5.2.2.2.3.3).

While drawing this flow, the TS 23.502 [7], figure 4.3.4.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4 Home-routed roaming scenario (HPLMN)

5.2.2.4.1 PDU session establishment

The flow-diagram in figure 5.2.2.4.1-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the H-SMF for a home-routed roaming scenario. In this illustration, the LI does not apply to the VPLMN. In the illustration, the LI is on the outbound roaming UE-A.

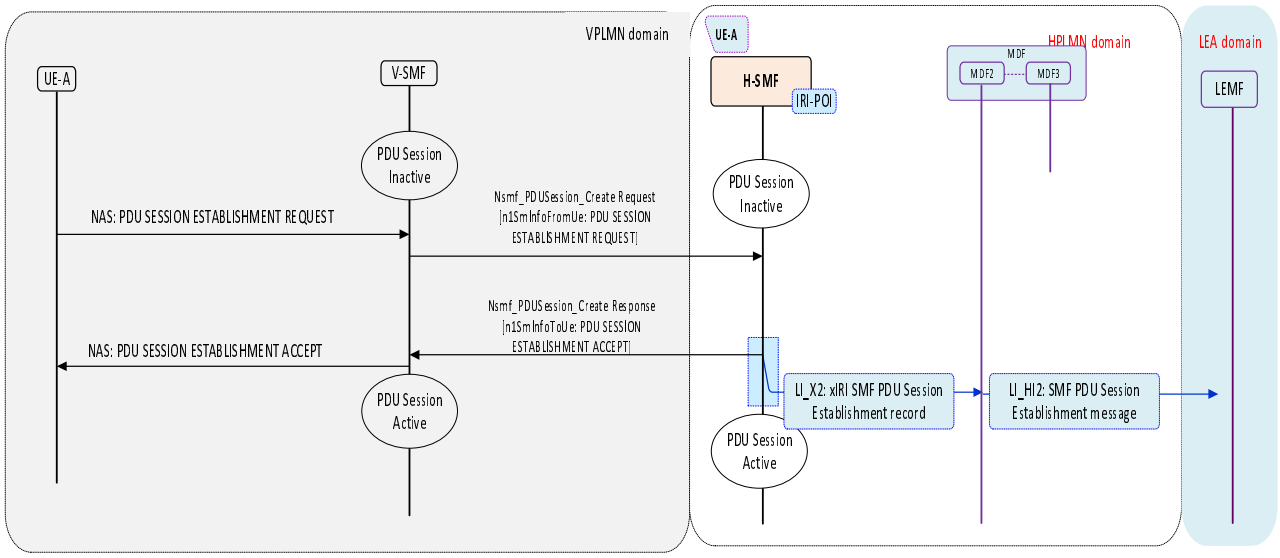


Figure 5.2.2.4.1-1: High-level flow diagram for PDU session establishment (HPLMN)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on the sending Hsmf_ PDUSession_Create Response message that includes the NAS message PDU SESSION ESTABLISHMENT ACCEPT in the n1smInfoToUe field and even though the 5GSM state in the H-SMF is also changed to PDU Session Active, the trigger is not dependent on that.

While drawing this flow, the TS 23.502 [7], figure 5.3.2.2.2-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4.2 PDU session modification

5.2.2.4.2.1 General

PDU session modification related scenarios include the following cases:

- UE-initiated PDU session modification.
- Network (VPLMN) initiated PDU session modification.
- Network (HPLMN) initiated PDU session modification.
- PDU session modification due to the change of access type (e.g. non-3GPP access to 3GPP access).

The two examples are shown for the access type change case.

The NAS messages and the 5GSM state changes are defined in TS 24.501 [8].

5.2.2.4.2.2 UE initiated PDU session modification

The flow-diagram in figure 5.2.2.4.2.2-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the H-SMF for a UE-initiated PDU session modification. In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

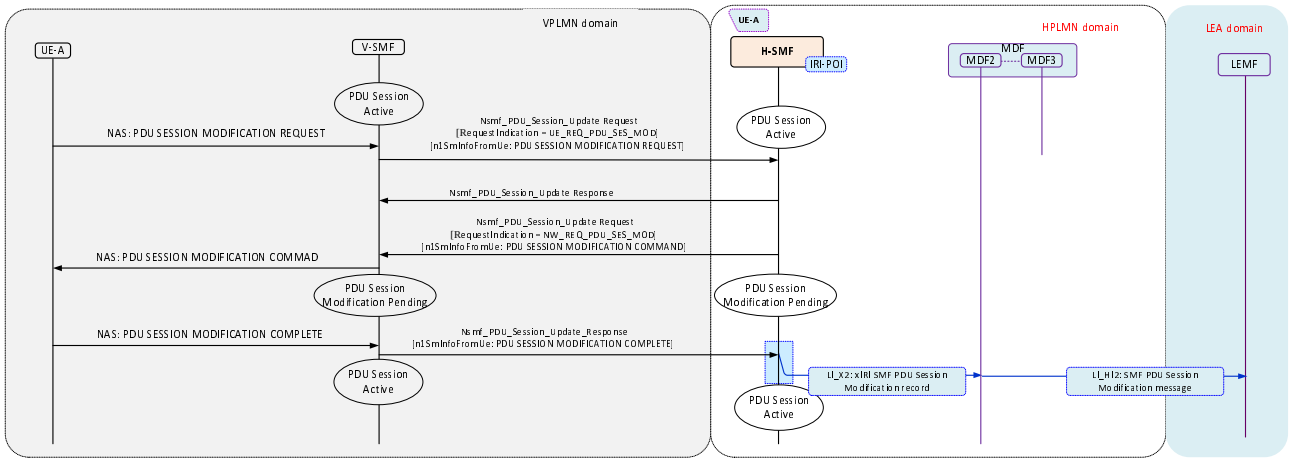


Figure 5.2.2.4.2.2-1: High-level flow diagram for PDU session modification (UE-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on receiving the Hsmf_PDUSession_Update Response message that includes the NAS message PDU SESSION MODIFICATION COMPLETE in the n1smInfoFromUe field and even though the 5GSM state in the H-SMF is also changed to PDU Session Active, the trigger is not dependent on that.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4.2.3 Network (VPLMN) initiated PDU session modification

The flow-diagram in figure 5.2.2.4.2.3-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the H-SMF for a VPLMN-initiated PDU session modification. In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

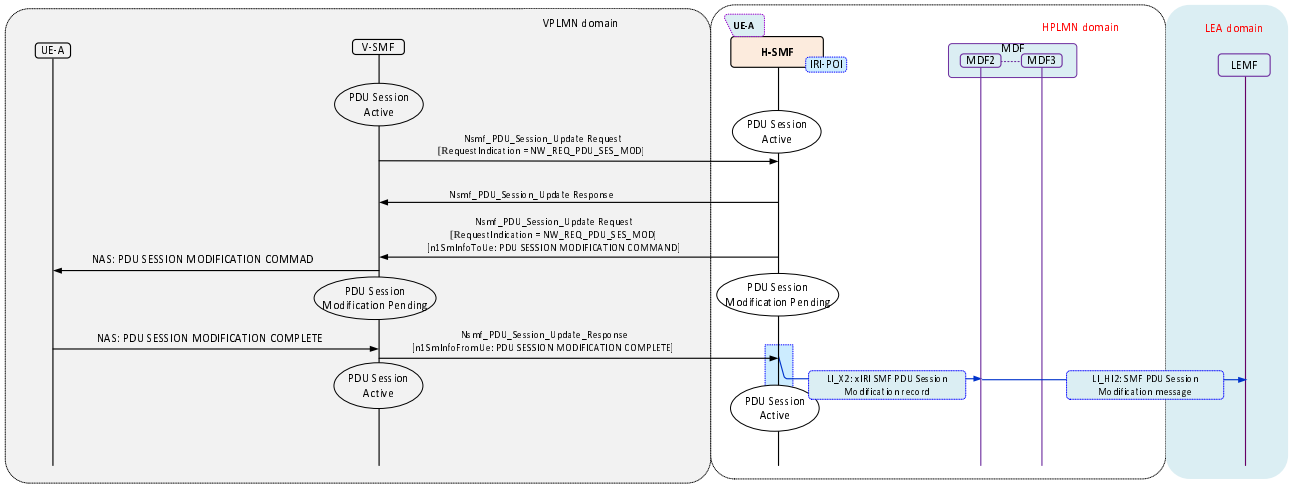


Figure 5.2.2.4.2.3-1: High-level flow diagram for PDU session modification (VPLMN-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on receiving the Hsmf_PDUSession_Update Response message that includes the NAS message PDU SESSION MODIFICATION COMPLETE in the n1smInfoFromUe field and even though the 5GSM state in the H-SMF is also changed to PDU Session Active, the trigger is not dependent on that.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4.2.4 Network (HPLMN) initiated PDU session modification

The flow-diagram in figure 5.2.2.4.2-3 illustrates a high-level view of xIRI generation from the IRI-POI present in the H-SMF for a HPLMN-initiated PDU session modification. In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

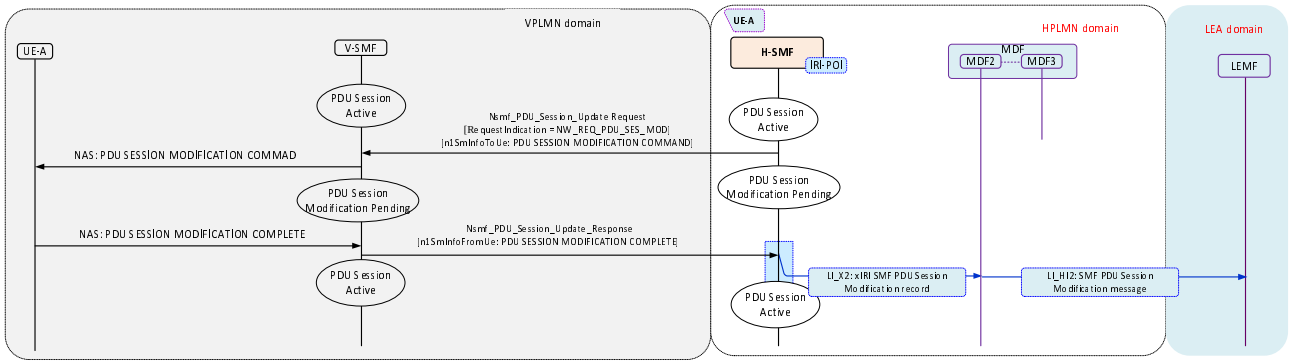


Figure 5.2.2.4.2.4-1: High-level flow diagram for PDU session modification (Network-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on receiving the Hsmf_ PDUSession_Update Response message that includes the NAS message PDU SESSION MODIFICATION COMPLETE in the n1smInfoFromUe field and even though the 5GSM state in the H-SMF is also changed to PDU Session Active, the trigger is not dependent on that.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4.2.5 PDU session modification for change of access type – example 1

The flow-diagram in figure 5.2.2.4.2.5-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the H-SMF when the target UE changes access type (e.g. non-3GPP access to 3GPP access) and the VPLMN is aware of that (i.e. the same V-SMF is handling the session before and after the handover).

In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

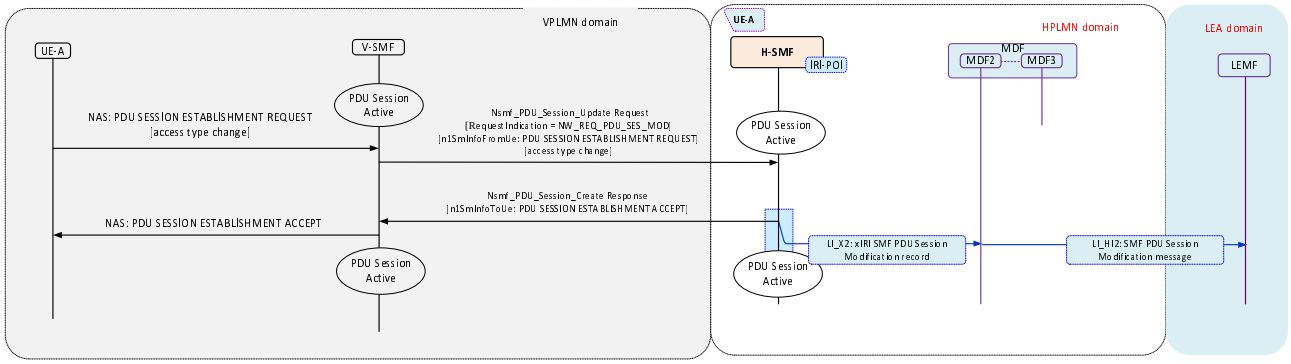


Figure 5.2.2.4.2.5-1: High-level flow diagram for PDU session modification (access type change) – example 1

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on sending the Hsmf_ PDUSession_Update Response message that includes the NAS message PDU SESSION ESTABLISHMENT ACCEPT in the n1smInfoToUe field and even though the 5GSM state in the H-SMF is also changed to PDU Session Active, the trigger is not dependent on that.

While drawing this flow, the TS 23.502 [7], figure 4.9.2.3.1-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4.2.6 PDU session modification for change of access type – example 2

The flow-diagram in figure 5.2.2.4.2.5-1 illustrates another high-level view of xIRI generation from the IRI-POI present in the H-SMF when the target UE changes access type (e.g. non-3GPP access to 3GPP access) and the VPLMN is not aware of that. For example, the UE may be connected via non-3GPP access where the N3IWF is present in the HPLMN. In the illustration, V-SMF is not aware of the PDU session before the handover.

In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

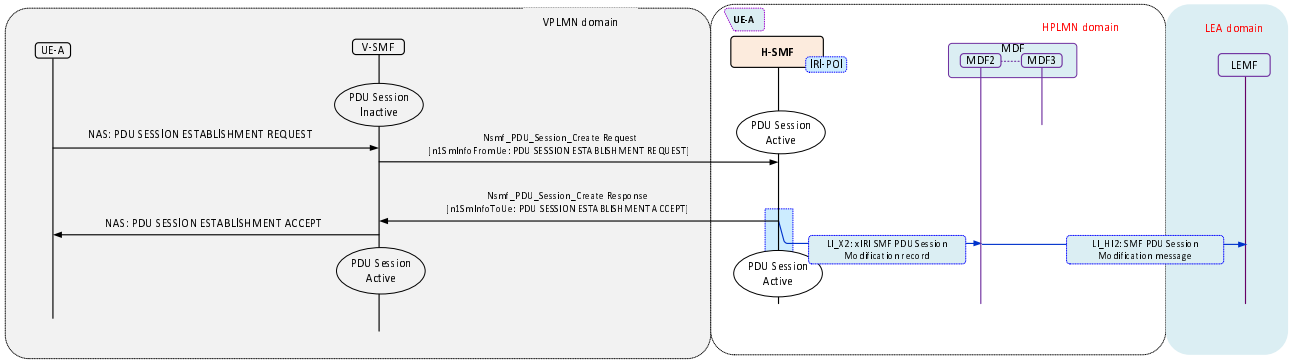


Figure 5.2.2.4.2.6-1: High-level flow diagram for PDU session modification (access type change) – example 2

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on sending the Hsmf_ PDUSession_Create Response message that includes the NAS message PDU SESSION ESTABLISHMENT ACCEPT in the n1smInfoToUe field.

While drawing this flow, the TS 23.502 [7], figure 4.9.2.4.2-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4.3 PDU session release

5.2.2.4.3.1 General

PDU session release related scenarios include the following cases:

- UE-initiated PDU session release.
- Network (VPLMN) initiated PDU session release.
- Network (HPLMN) initiated PDU session release.

The NAS messages and the 5GSM state changes are defined in TS 24.501 [8].

5.2.2.4.3.2 UE initiated PDU session release

The flow-diagram in figure 5.2.2.4.3.2-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the H-SMF for a UE-initiated PDU session release. In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

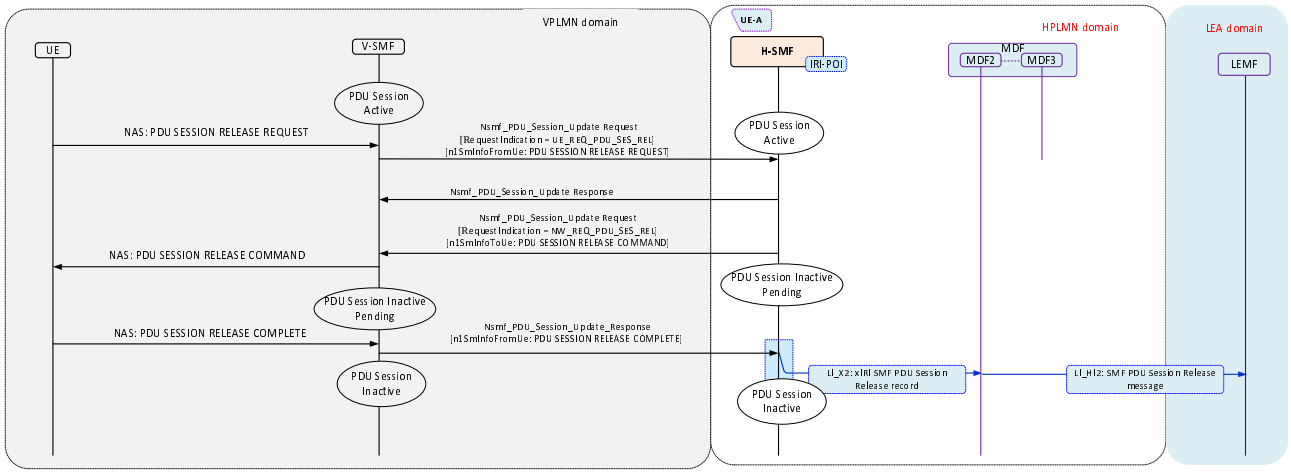


Figure 5.2.2.4.3.2-1: High-level flow diagram for PDU session release (UE-initiated)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on receiving the Hsmf_PDUSession_Update Response message that includes the NAS message PDU SESSION RELEASE COMPLETE in the n1smInfoFromUe field and even though the 5GSM state in the H-SMF is also changed to PDU Session InActive, the trigger is not dependent on that.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4.3.3 Network (VPLMN) initiated PDU session release

The flow-diagram in figure 5.2.2.4.3.3-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the H-SMF for a VPLMN-initiated PDU session release. In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

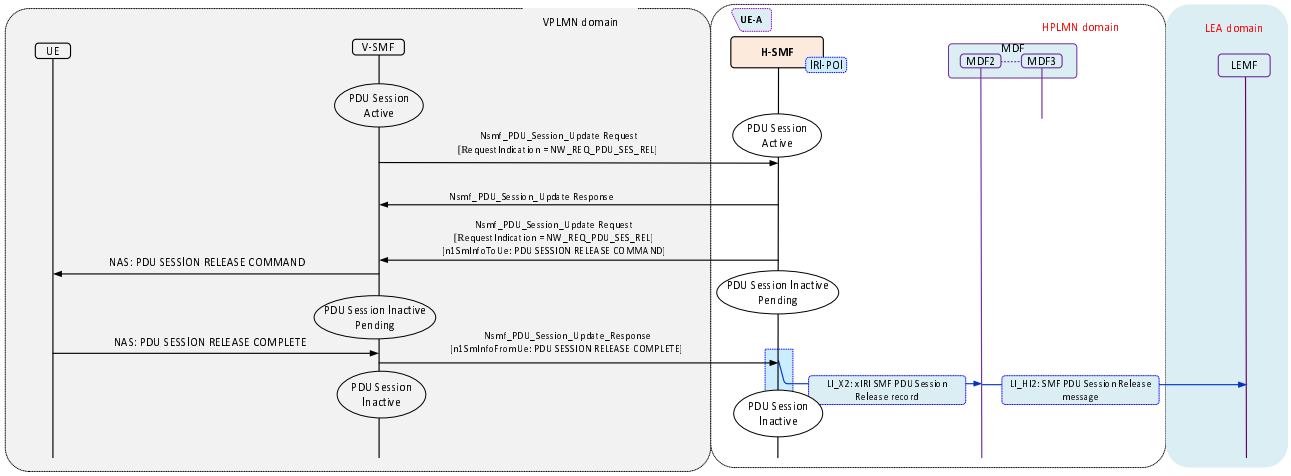


Figure 5.2.2.4.3.3-1: High-level flow diagram for PDU session release (network-initiated - VPLMN)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on receiving the Hsmf_PDUSession_Update Response message that includes the NAS message PDU SESSION RELEASE COMPLETE in the n1smInfoFromUe field and even though the 5GSM state in the H-SMF is also changed to PDU Session Active, the trigger is not dependent on that.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

5.2.2.4.3.4 Network (HPLMN) initiated PDU session release

The flow-diagram in figure 5.2.2.4.3.4-1 illustrates a high-level view of xIRI generation from the IRI-POI present in the H-SMF for a VPLMN-initiated PDU session release. In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

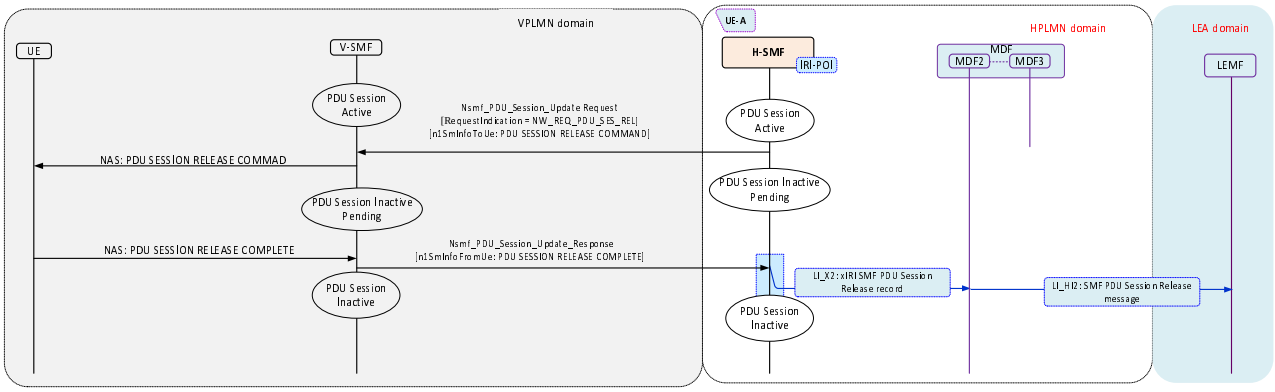


Figure 5.2.2.4.3.4-1: High-level flow diagram for PDU session release (network-initiated - HPLMN)

As defined in TS 33.128 [4], the trigger conditions for the generation of xIRI in the H-SMF is on receiving the Hsmf_ PDUSession_Update Response message that includes the NAS message PDU SESSION RELEASE COMPLETE in the n1smInfoFromUe field and even though the 5GSM state in the H-SMF is also changed to PDU Session Active, the trigger is not dependent on that.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.3-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10].

The flow-diagram in figure 5.2.3.3.1.3-1 illustrates a detailed view of PDU session modification for access type change for a non-roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the SMF, LI_T3 triggering from the CC-TF present in the SMF and the xCC generation from the CC-POI present in the UPF.

5.2.3 Detailed flow diagrams

5.2.3.1 General

The following LI scenarios are illustrated:

- SMF, non-roaming.
- V-SMF, home-routed roaming.
- H-SMF, home-routed roaming.

In the home-routed roaming case, LI provided in the VPLMN is independent of the LI provided in the HPLMN.

The NAS messages and the 5GSM state changes are defined in TS 24.501 [8].

5.2.3.2 PDU session establishment

5.2.3.2.1 Non-roaming scenario

The flow-diagram in figure 5.2.3.2.1-1 illustrates a detailed view of PDU session establishment steps for a non-roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the SMF, LI_T3 triggering from the CC-TF present in the SMF and the xCC generation from the CC-POI present in the UPF.

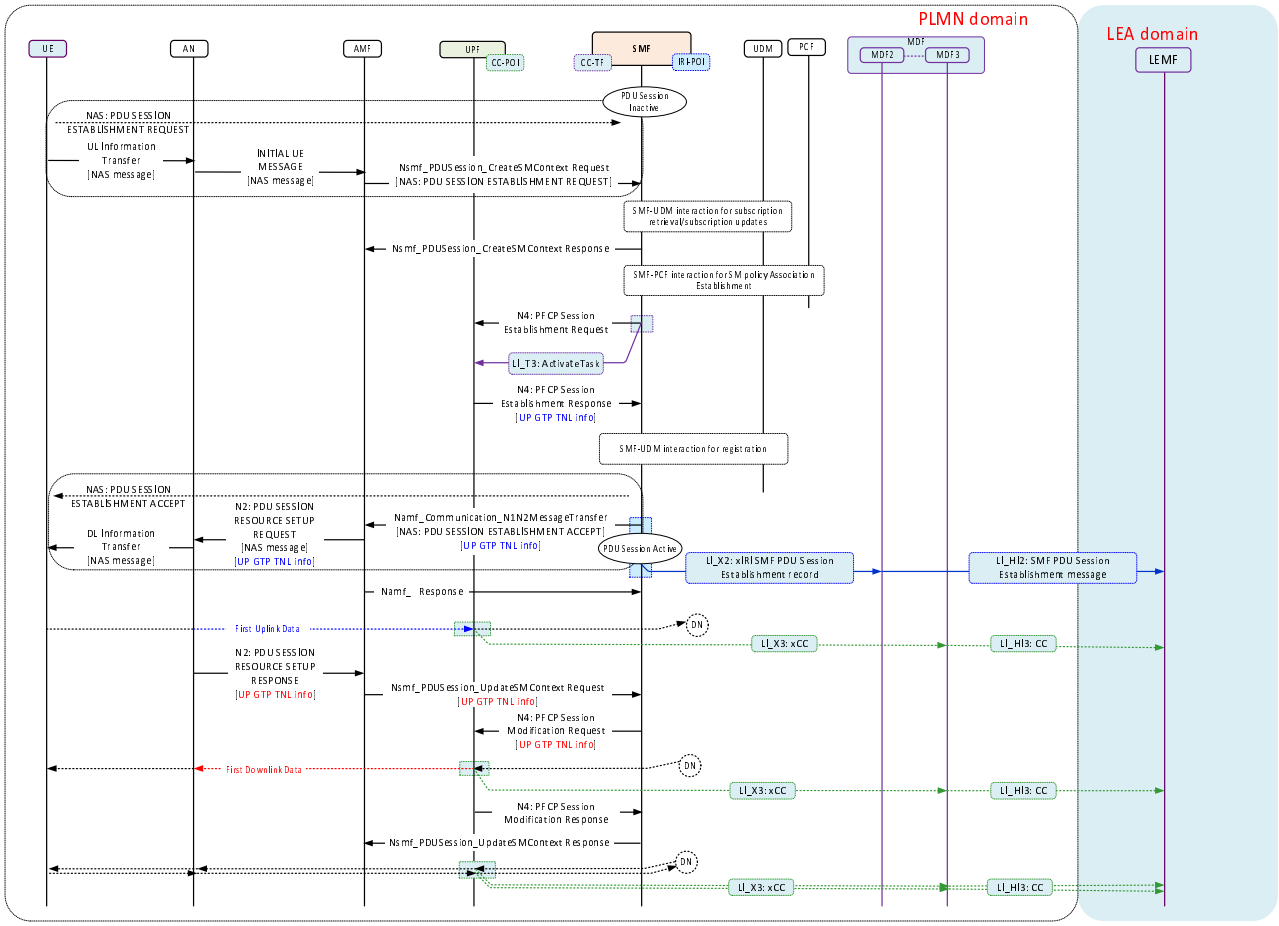


Figure 5.2.3.2.1-1: Detailed flow diagram for PDU session establishment (non-roaming)

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is SMF sending a NAS message PDU SESSION ESTABLISHMENT ACCEPT to the target UE and the 5GSM state changing into PDU Session Active.
- The LI_T3 Activate Task trigger is sent from CC-TF present in the SMF to the CC-POI present in the UPF when the SMF sends the N4: PFCP Session Establishment Request to the UPF.
- The UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 5.3.2.2.1-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.2.2 Home-routed roaming scenario (VPLMN)

The flow-diagram in figure 5.2.3.2.2-1 illustrates a detailed view of PDU session establishment steps for a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the V- SMF, LI_T3 triggering from the CC-TF present in the V-SMF and the xCC generation from the CC-POI present in the V-UPF.

In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

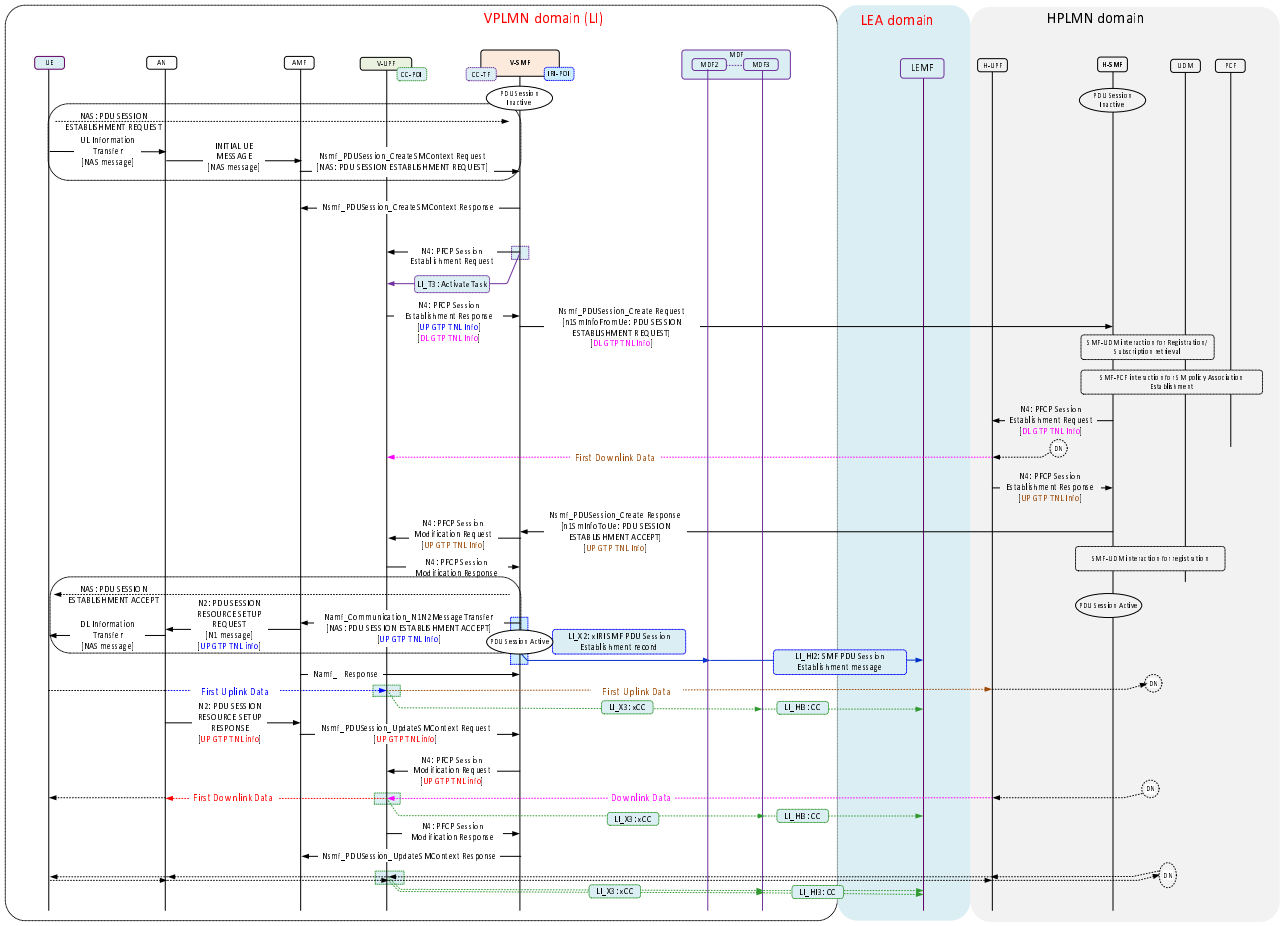


Figure 5.2.3.2.2-1: Detailed flow diagram for PDU session establishment (VPLMN)

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is V-SMF sending a NAS message PDU SESSION ESTABLISHMENT ACCEPT to the target UE and the 5GSM state changing into PDU Session Active.
- The LI_T3 Activate Task trigger is sent from CC-TF present in the V-SMF to the CC-POI present in the V-UPF when the V-SMF sends the N4: PFCP Session Establishment Request to the V-UPF.
- The V-UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the V-UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 5.3.2.2.2-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10]. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.2.3 Home-routed roaming scenario (HPLMN)

The flow-diagram in figure 5.2.3.2.3-1 illustrates a detailed view of PDU session establishment steps for a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the H- SMF, LI_T3 triggering from the CC-TF present in the H-SMF and the xCC generation from the CC-POI present in the H-UPF.

In this illustration, the LI does not apply to the VPLMN. In the illustration, the LI is on the outbound roaming UE-A.

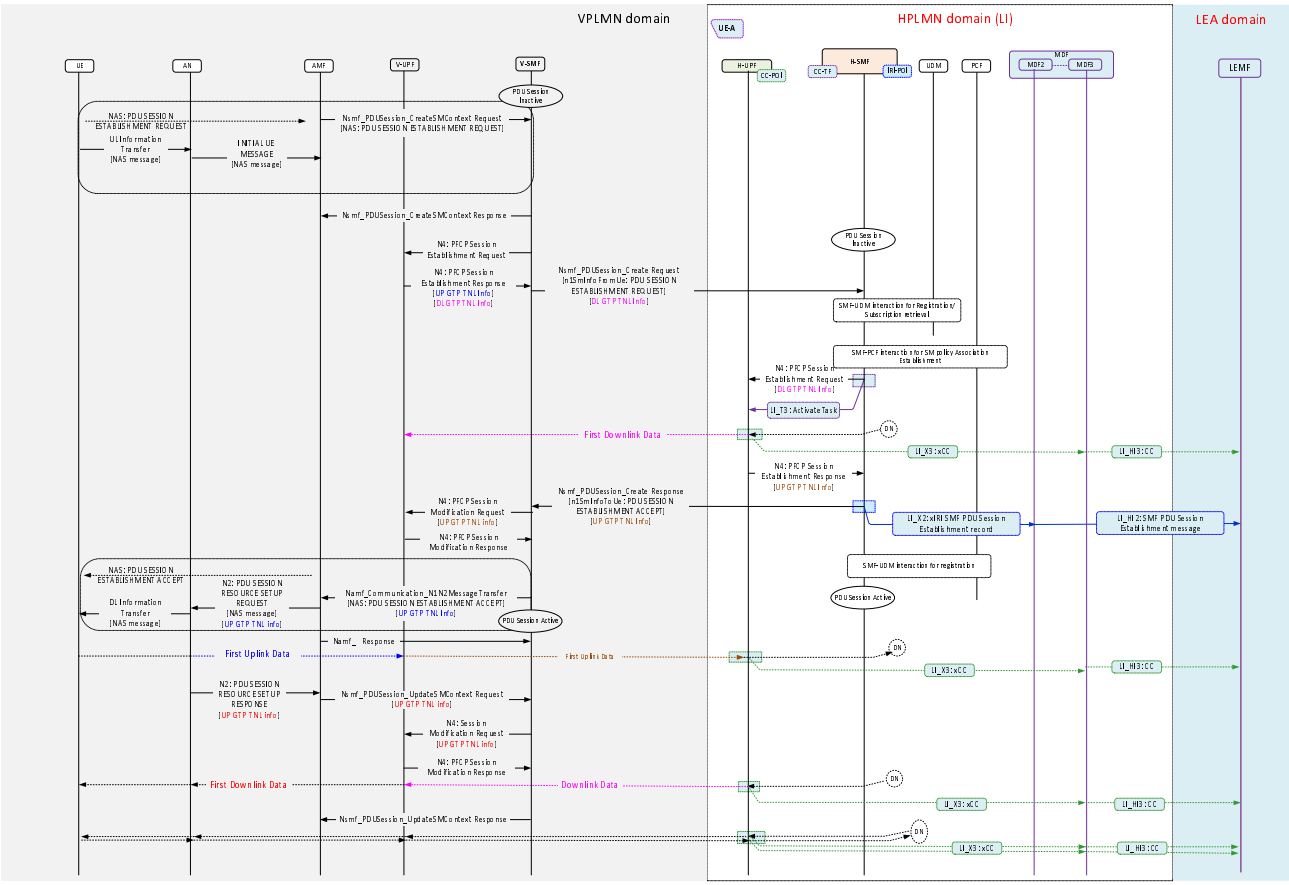


Figure 5.2.3.2.3-1: Detailed flow diagram for PDU session establishment (HPLMN)

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is H-SMF sending Hsmf_ PDUSession_Create Response message that includes the NAS message PDU SESSION ESTABLISHMENT ACCEPT in the n1smInfoToUe field. Even though the 5GSM state in the H-SMF is also changed to PDU Session Active, the trigger is not dependent on that.

- The LI_T3 Activate Task trigger is sent from CC-TF present in the H-SMF to the CC-POI present in the H-UPF when the H-SMF sends the N4: PFCP Session Establishment Request to the H-UPF.

- The H-UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the H-UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 5.3.2.2.2-1 is referred to. The session management related messages and parameter (i.e. between the V-SMF and H-SMF) are defined in TS 29.502 [10]. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.3 PDU session modification

5.2.3.3.1 Non-roaming scenario

5.2.3.3.1.1 UE-initiated PDU session modification

The flow-diagram in figure 5.2.3.3.1.1-1 illustrates a detailed view of PDU session modification steps for a non-roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the SMF, LI_T3 triggering from the CC-TF present in the SMF and the xCC generation from the CC-POI present in the UPF.

In this illustration, the case of UE-initiated session modification is considered.

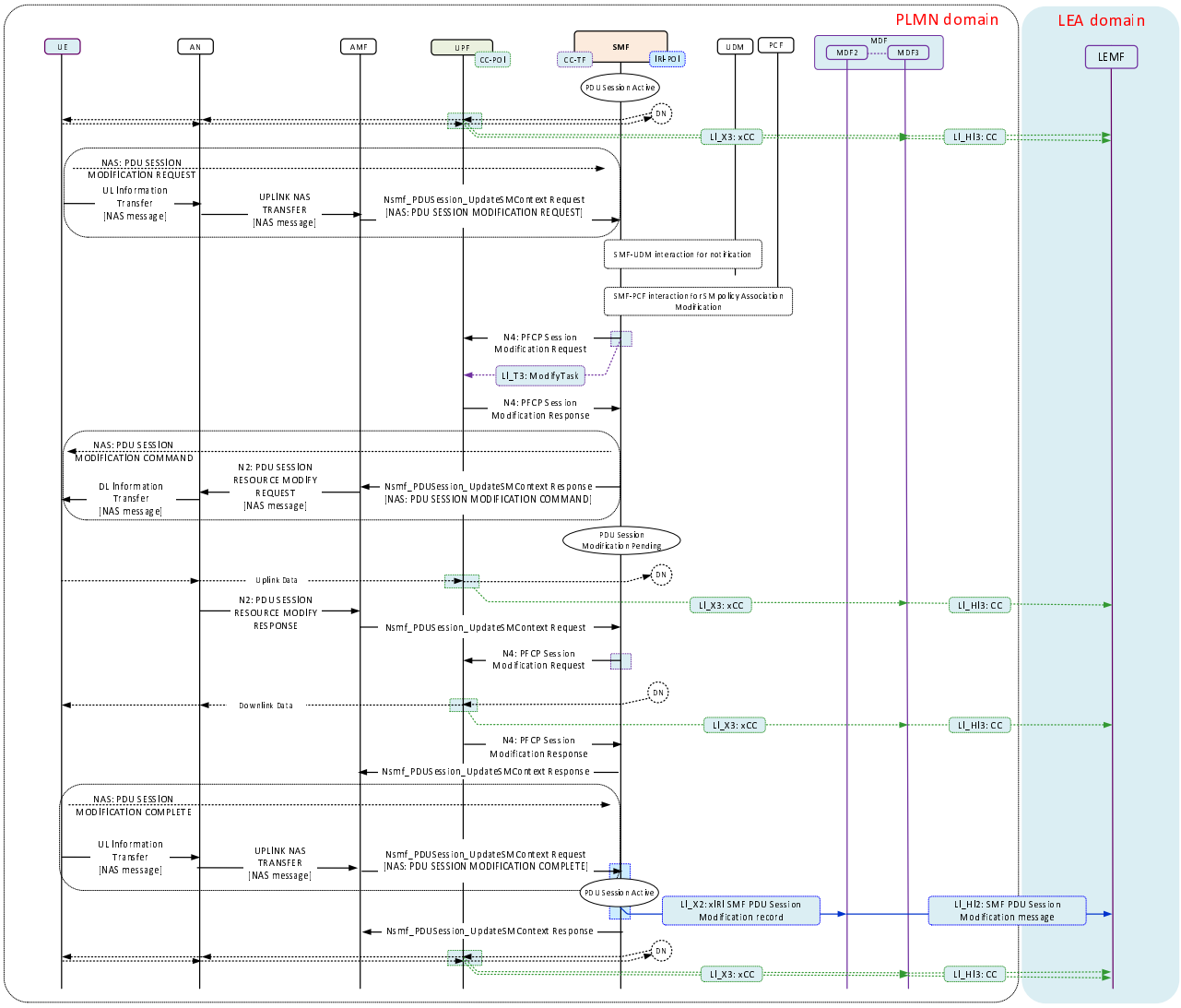


Figure 5.2.3.3.1.1-1: Detailed flow diagram for PDU session modification (non-roaming) – UE-initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is SMF receiving the NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state changing into PDU Session Active.
- The LI_T3 Modify Task trigger may be sent from CC-TF present in the SMF to the CC-POI present in the UPF when the SMF sends the N4: PFCP Session Modification Request to the UPF. If the PDU session modification does not alter the CC-interception being done at the CC-POI in the UPF, the LI_T3 Modify Task may not be sent.
- The CC-POI in UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.2-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.3.1.2 Network-initiated PDU session modification

The flow-diagram in figure 5.2.3.3.1.2-1 illustrates a detailed view of PDU session modification steps for a non-roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the SMF, LI_T3 triggering from the CC-TF present in the SMF and the xCC generation from the CC-POI present in the UPF.

In this illustration, the case of SMF-initiated session modification is considered.

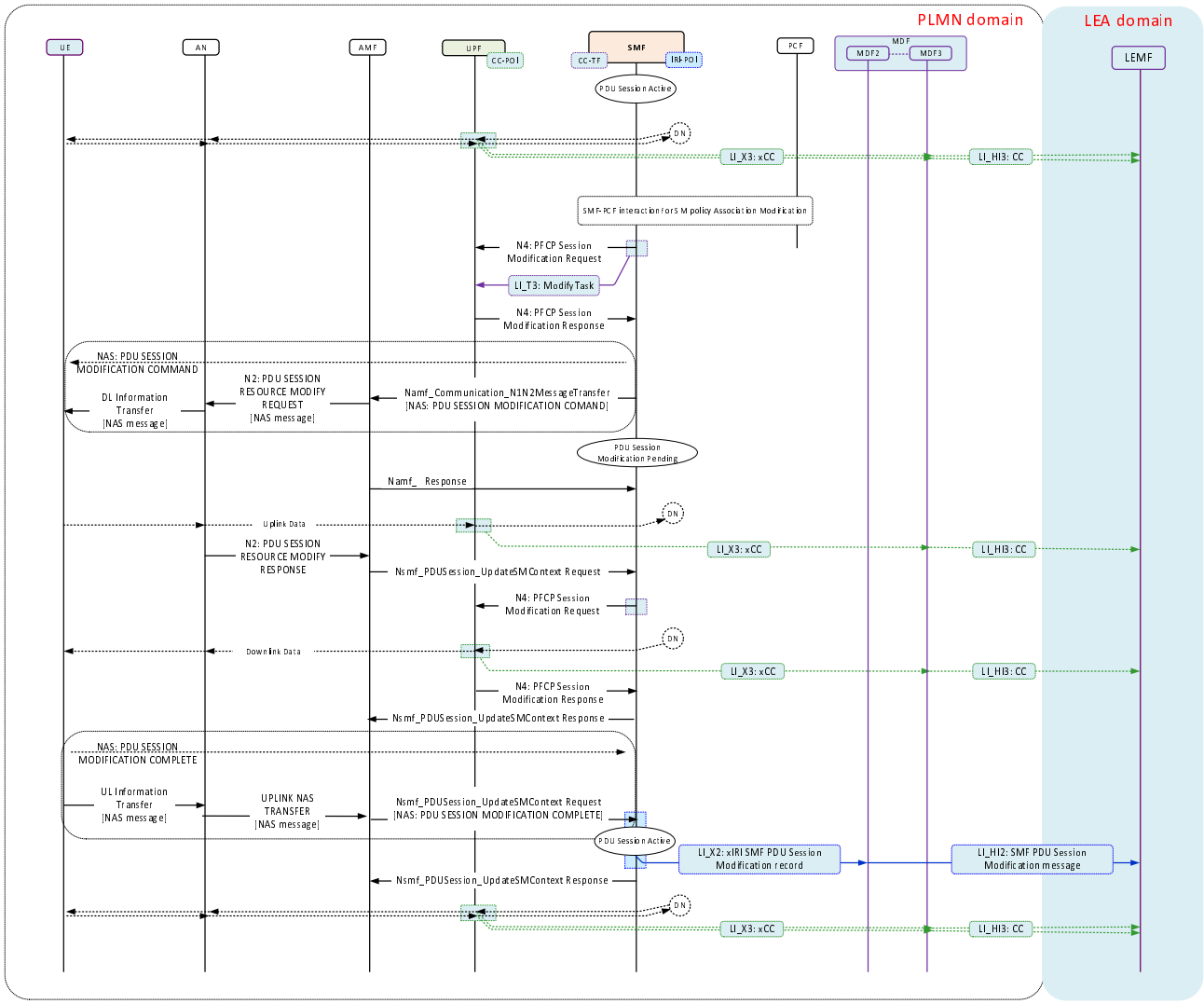


Figure 5.2.3.3.1.2-1: Detailed flow diagram for PDU session modification (non-roaming) – network-initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is SMF receiving the NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state changing into PDU Session Active.
- The LI_T3 Modify Task trigger is sent from CC-TF present in the SMF to the CC-POI present in the UPF when the SMF sends the N4: PFCP Session Modification Request to the UPF. For network-initiated modification, it is quite possible that the CC-interception is impacted (e.g. new QoS flow) and therefore, this is a good example where LI_T3: Modify Task is sent.
- The CC-POI in UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.2-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.3.1.3 PDU session modification due to change in access type

The flow-diagram in figure 5.2.3.3.1.3-1 illustrates a detailed view of PDU session modification for access type change for a non-roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the SMF, LI_T3 triggering from the CC-TF present in the SMF and the xCC generation from the CC-POI present in the UPF.

In this illustration, the case of access type change from non-3GPP access to 3GPP access is considered.

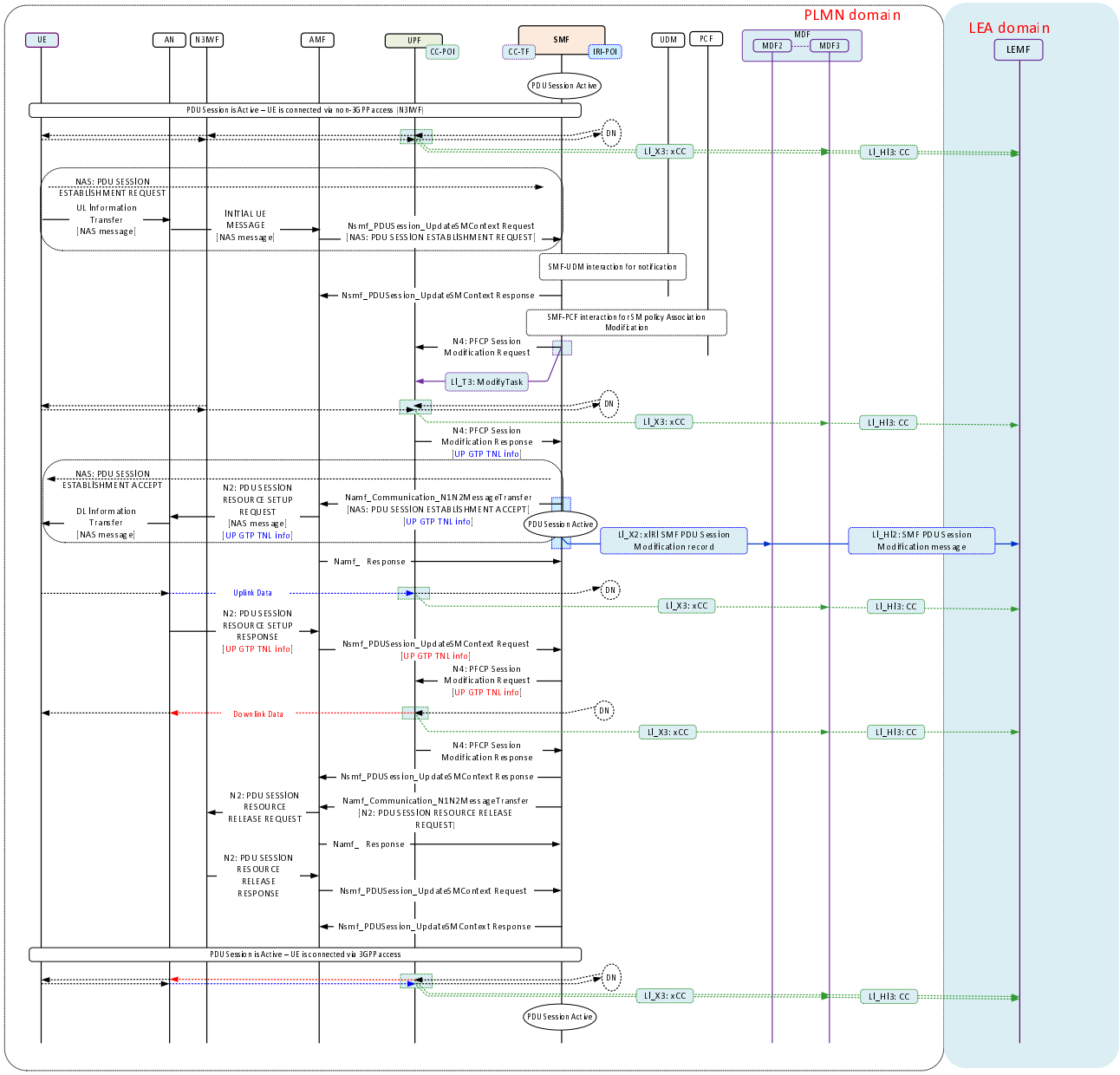


Figure 5.2.3.3.1.3-1: Detailed flow diagram for PDU session modification (non-roaming) – access type change

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is SMF sending the NAS message PDU SESSION ESTABLISHMENT ACCEPT to the target UE and the 5GSM state changing into PDU Session Active.
- The LI_T3 Modify Task trigger is sent from CC-TF present in the SMF to the CC-POI present in the UPF when the SMF sends the N4: PFCP Session Modification Request to the UPF.
- The UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 4.9.2.1-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.3.2 Home-routed roaming scenario (VPLMN)

5.2.3.3.2.1 Network-initiated PDU session modification

The flow-diagram in figure 5.2.3.3.2.1-1 illustrates a detailed view of PDU session modification steps for a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the V-SMF, LI_T3 triggering from the CC-TF present in the V-SMF and the xCC generation from the CC-POI present in the V-UPF.

In this illustration, the case of V-SMF-initiated session modification is considered.

In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

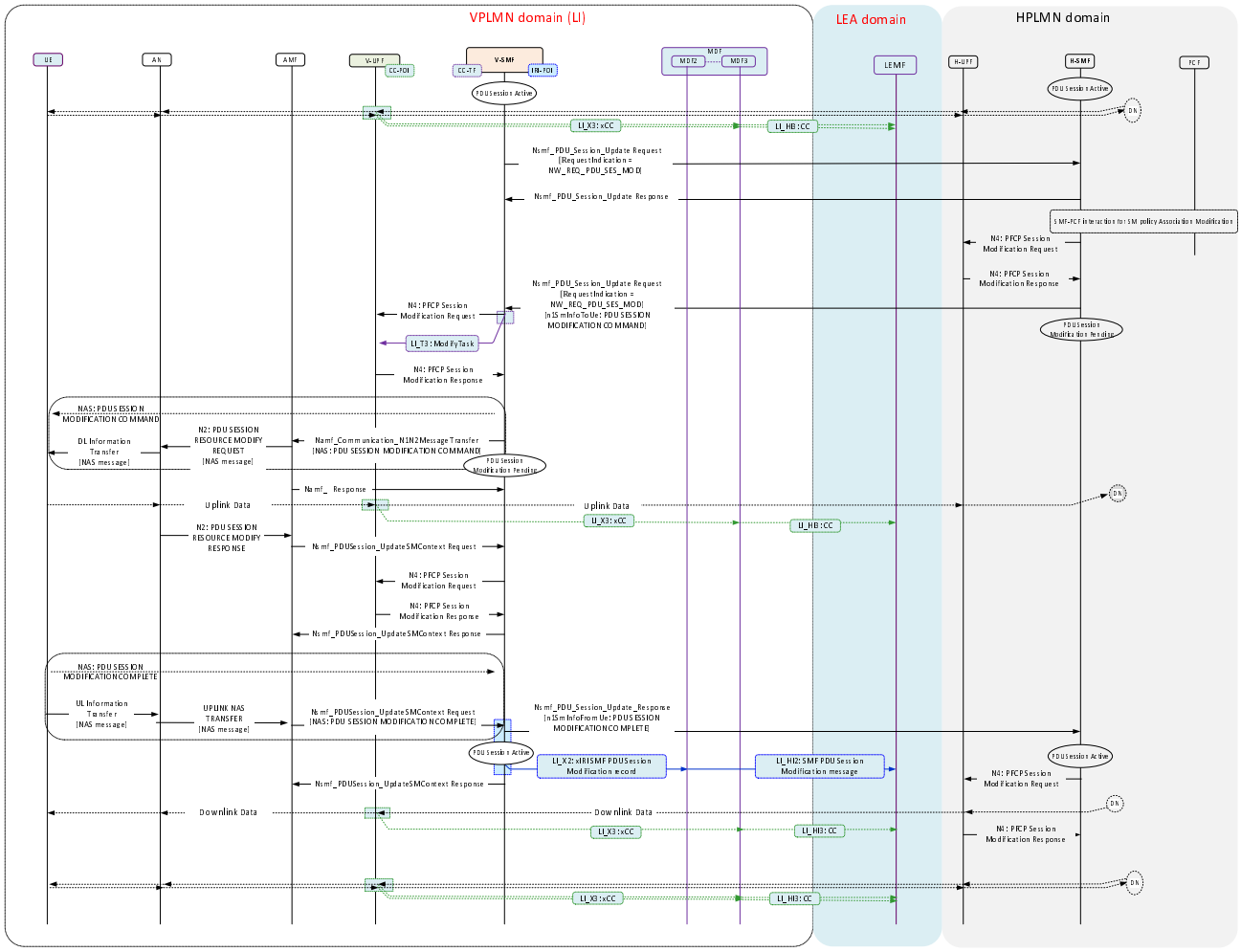


Figure 5.2.3.3.2.1-1: Detailed flow diagram for PDU session modification (VPLMN) – network initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is V-SMF receiving the NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state changing into PDU Session Active.
- The LI_T3 Modify Task trigger may be sent from CC-TF present in the V-SMF to the CC-POI present in the V-UPF when the V-SMF sends the N4: PFCP Session Modification Request to the V-UPF. If the PDU session modification does not alter the CC-interception being done at the CC-POI in the UPF, the LI_T3 Modify Task may not be sent.
- The V-UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the V-UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.3.2.2 PDU session modification due to change in access type

The flow-diagram in figure 5.2.3.3.2.2-1 illustrates a detailed view of PDU session modification for access type change in a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the V-SMF, LI_T3 triggering from the CC-TF present in the V-SMF and the xCC generation from the CC-POI present in the V-UPF.

In this illustration, the case of access type change from non-3GPP access to 3GPP access is considered.

In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

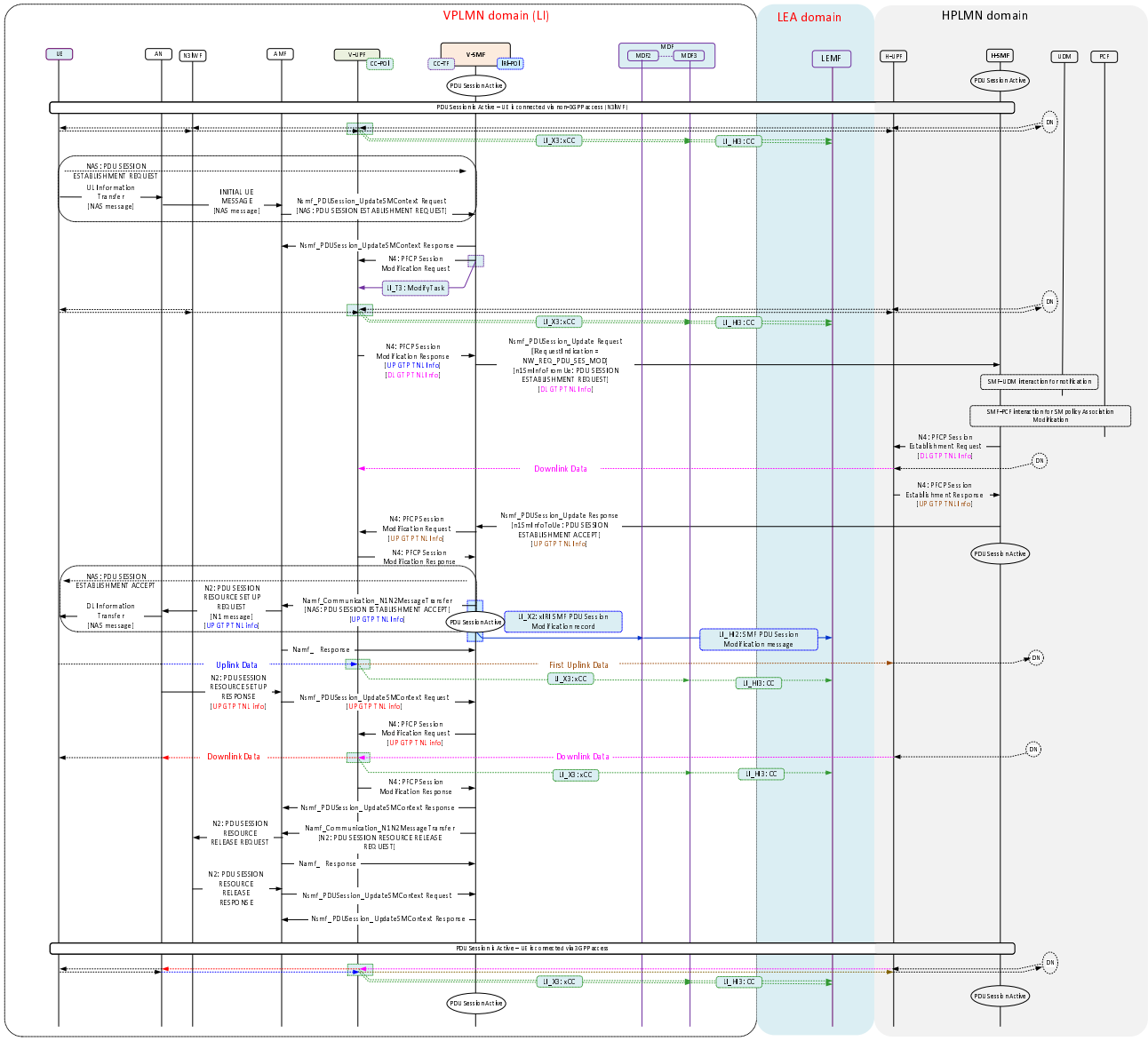


Figure 5.2.3.3.2.2-1: Detailed flow diagram for PDU session modification (VPLMN) – access type change

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is V-SMF sending the NAS message PDU SESSION ESTABLISHMENT ACCEPT to the target UE and the 5GSM state changing into PDU Session Active.
- The LI_T3 Modify Task trigger is sent from CC-TF present in the V-SMF to the CC-POI present in the V-UPF when the V-SMF sends the N4: PFCP Session Modification Request to the V-UPF.

- The V-UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the V-UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 4.9.2.3.1-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.3.3 Home-routed roaming scenario (HPLMN)

5.2.3.3.3.1 Network-initiated PDU session modification

The flow-diagram in figure 5.2.3.3.3.1-1 illustrates a detailed view of PDU session modification steps for a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the H-SMF, LI_T3 triggering from the CC-TF present in the H-SMF and the xCC generation from the CC-POI present in the H-UPF.

In this illustration, the case of H-SMF-initiated session modification is considered.

In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

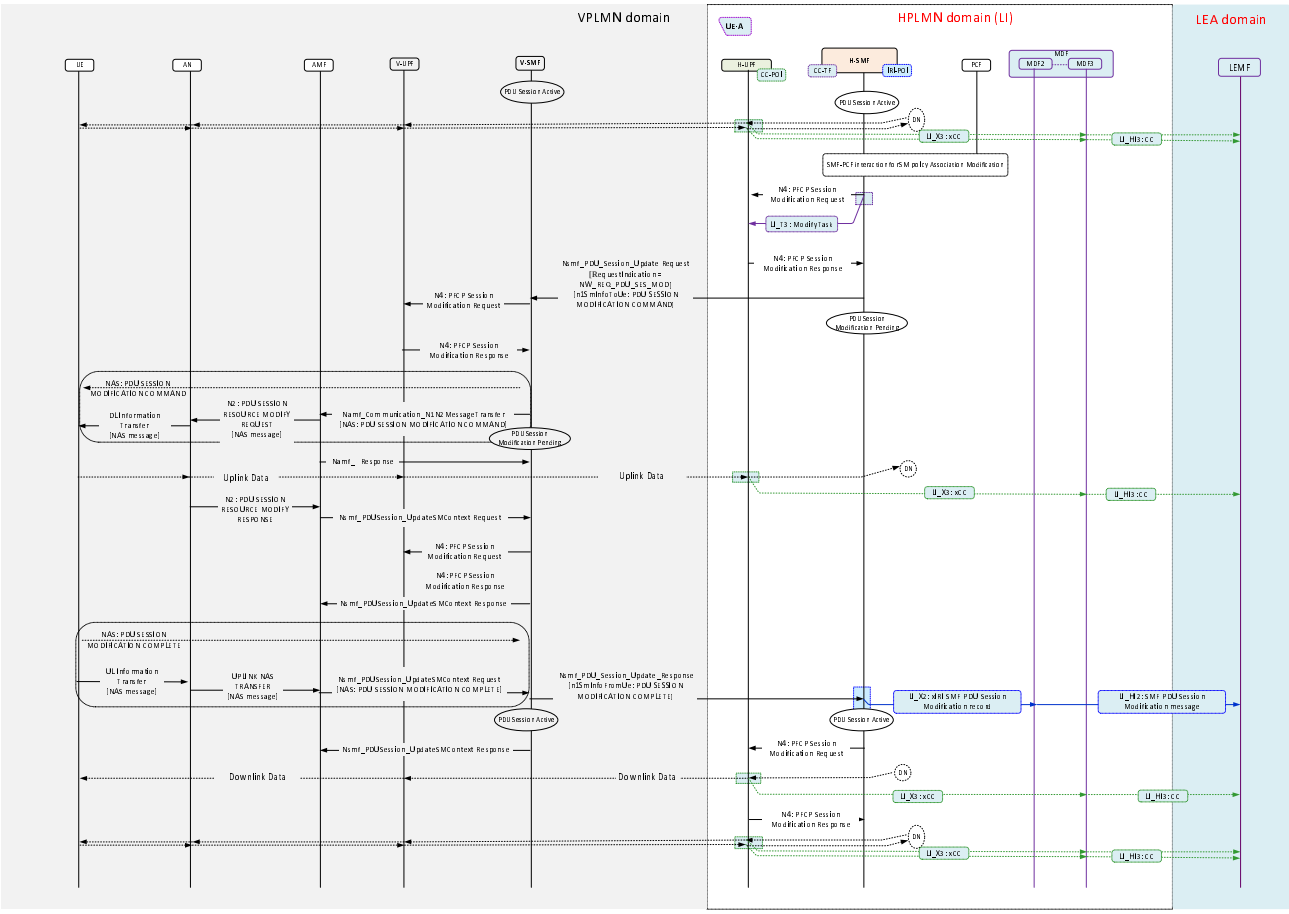


Figure 5.2.3.3.3.1-1: Detailed flow diagram for PDU session modification (HPLMN) – network initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is H-SMF receiving the Nsmf_PDU_Session_Update Response from the V-SMF with N1SmInfoFromUe containing the message PDU SESSION MODIFICATION COMPLETE.

- The LI_T3 Modify Task trigger is sent from CC-TF present in the H-SMF to the CC-POI present in the H-UPF when the H-SMF sends the N4: PFCP Session Modification Request to the H-UPF. For network-initiated modification, it is quite possible that the CC-interception is impacted (e.g. new QoS flow) and therefore, this is a good example where LI_T3: Modify Task is sent.

- The H-UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the H-UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.3.3.2 PDU session modification due to change in access type (N3IWF in the same VPLMN)

The flow-diagram in figure 5.2.3.3.3.2-1 illustrates a detailed view of PDU session modification for access type change in a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the H-SMF, LI_T3 triggering from the CC-TF present in the H-SMF and the xCC generation from the CC-POI present in the H-UPF.

In this illustration, the case of access type change from non-3GPP access to 3GPP access is considered. And the N3IWF is in the same VPLMN. In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

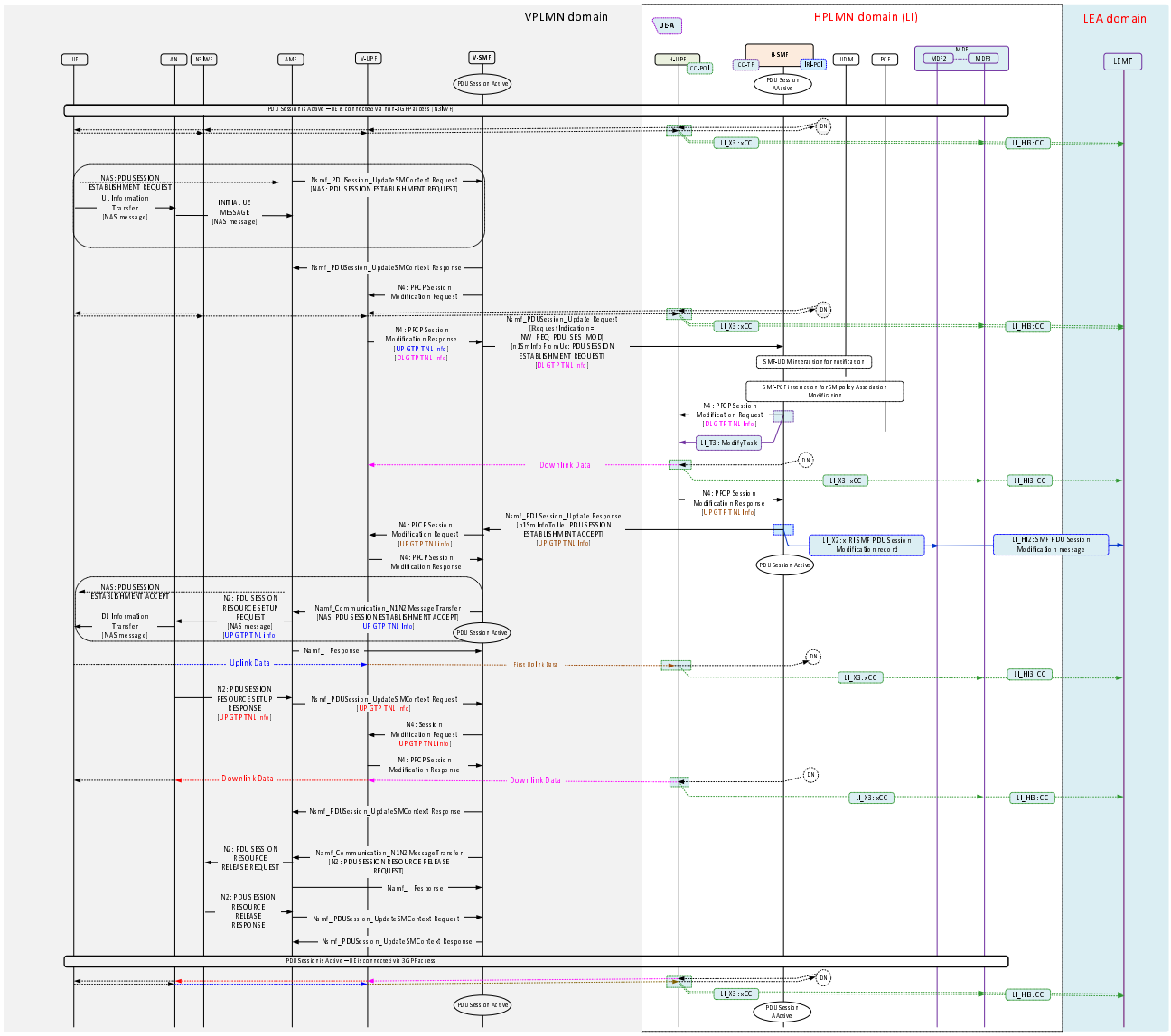


Figure 5.2.3.3.3.2-1: Detailed flow diagram for PDU session modification (HPLMN) – access type change (N3IWF in the same VPLMN)

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is H-SMF sending the Nsmf_PDU_Session_Update Response to the V-SMF with N1SmInfoToUe containing the message PDU SESSION ESTABLISHMENT ACCEPT.

- The LI_T3 Modify Task trigger is sent from CC-TF present in the H-SMF to the CC-POI present in the H-UPF when the H-SMF sends the N4: PFCP Session Modification Request to the H-UPF.

- The H-UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the H-UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 4.9.2.3.1-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.3.3.3 PDU session modification due to change in access type (N3IWF in the HPLMN)

The flow-diagram in figure 5.2.3.3.3.3-1 illustrates a detailed view of PDU session modification for access type change in a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the H-SMF, LI_T3 triggering from the CC-TF present in the H-SMF and the xCC generation from the CC-POI present in the H-UPF.

In this illustration, the case of access type change from non-3GPP access to 3GPP access is considered. And the N3IWF is in the HPLMN. In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

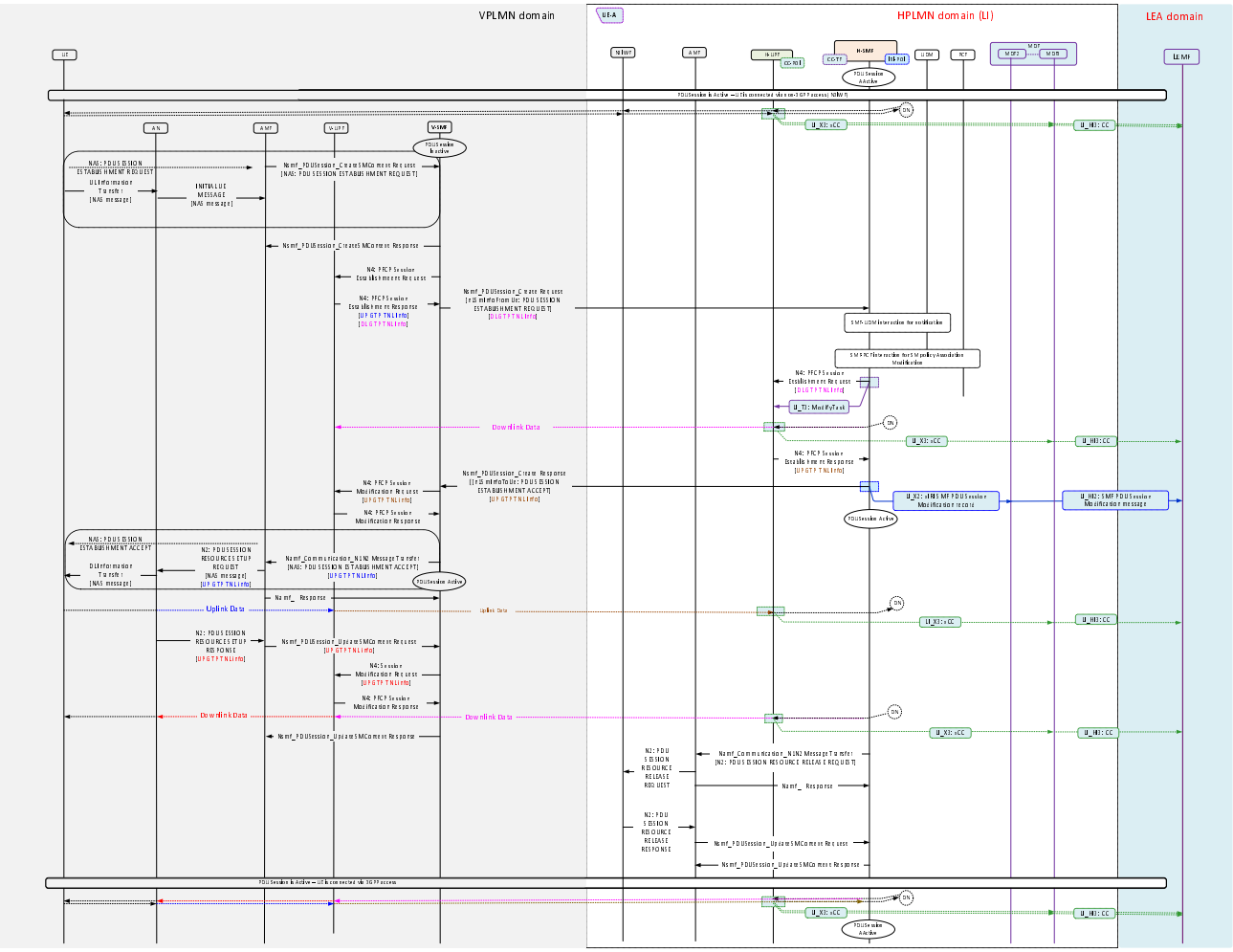


Figure 5.2.3.3.3.3-1: Detailed flow diagram for PDU session modification (HPLMN) – access type change (N3IWF in the HPLMN)

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is H-SMF sending the Nsmf_PDU_Session_Create Response to the V-SMF with N1SmInfoToUe containing the message PDU SESSION ESTABLISHMENT ACCEPT.
- The LI_T3 Modify Task trigger is sent from CC-TF present in the H-SMF to the CC-POI present in the H-UPF when the H-SMF sends the N4: PFCP Session Modification Request to the H-UPF.
- The H-UPF generates the xCC based on the LI_T3 trigger received from the CC-TF. The xCC is generated from the user-plane packets that traverse through the H-UPF on the PDU session.

While drawing this flow, the TS 23.502 [7], figure 4.9.2.3.2-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4 PDU session release

5.2.3.4.1 General

The PDU session release may be initiated by the UE or the network. In the latter case, PDU session release may be initiated by the PCF, SMF, AMF or the AN.

The PDU session may also be released without any NAS messages to the UE or also without any N2 message exchanges with the AN.

5.2.3.4.2 Non-roaming scenario

5.2.3.4.2.1 UE initiated PDU session release

The flow-diagram in figure 5.2.3.4.2.1-1 illustrates a detailed view of PDU session release steps for a non-roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the SMF, LI_T3 triggering from the CC-TF present in the SMF and the xCC generation from the CC-POI present in the UPF.

In this illustration, the case of UE initiated PDU session release is considered.

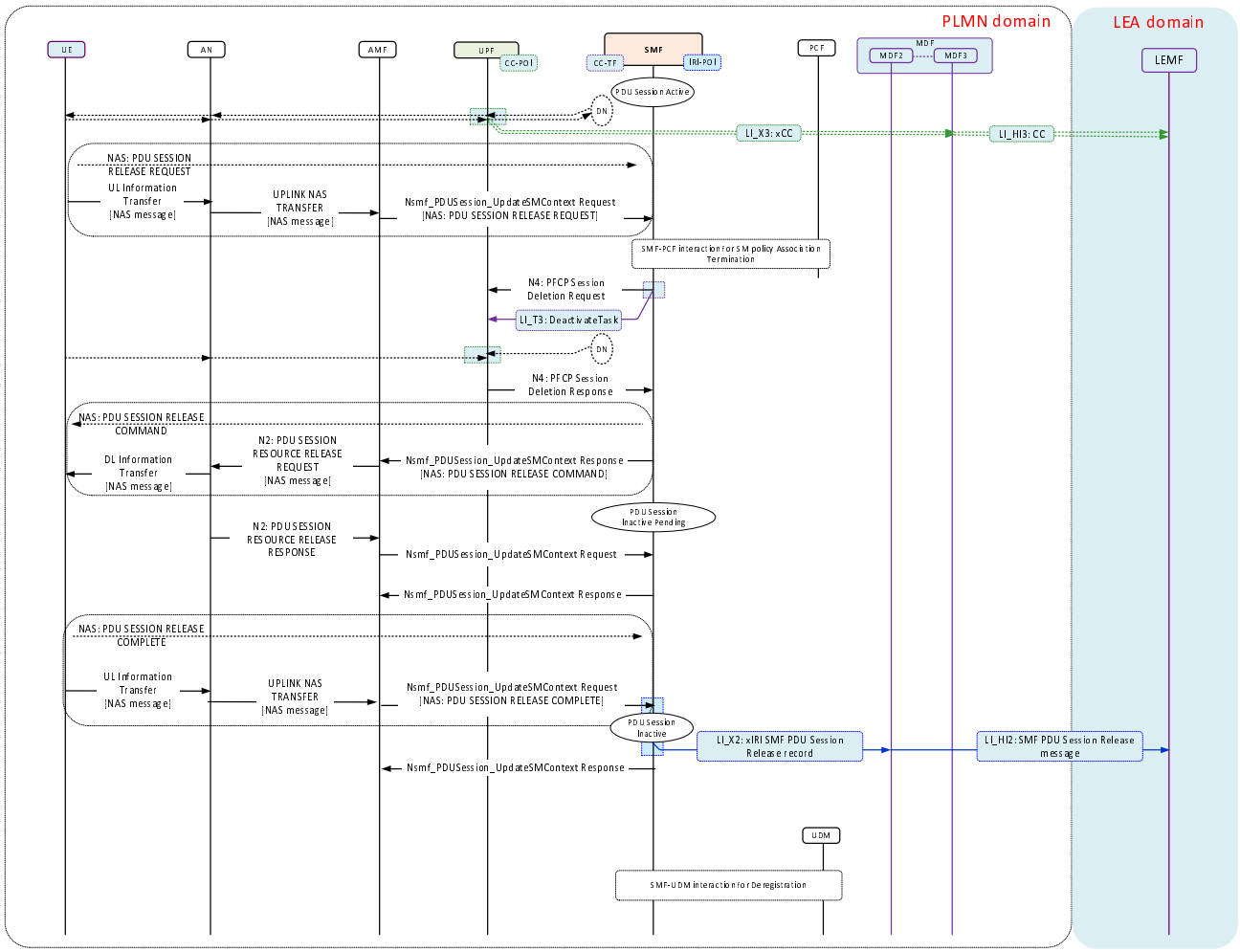


Figure 5.2.3.4.2.1-1: Detailed flow diagram for PDU session release (non-roaming) – UE-initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is SMF receiving the NAS message PDU SESSION RELEASE COMPLETE from the target UE and the 5GSM state changing into PDU Session Inactive.
- The LI_T3 Deactivate Task trigger may be sent from CC-TF present in the SMF to the CC-POI present in the UPF when the SMF sends the N4: PFCP Session Deletion Request to the UPF.
- The CC-POI in UPF stops generating the xCC upon receiving the LI_T3: Deactivate Task received from the CC-TF.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.2-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4.2.2 Network initiated PDU session release

The flow-diagram in figure 5.2.3.4.2.2-1 illustrates a detailed view of PDU session release steps for a non-roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the SMF, LI_T3 triggering from the CC-TF present in the SMF and the xCC generation from the CC-POI present in the UPF.

In this illustration, the case of SMF initiated PDU session release is considered.

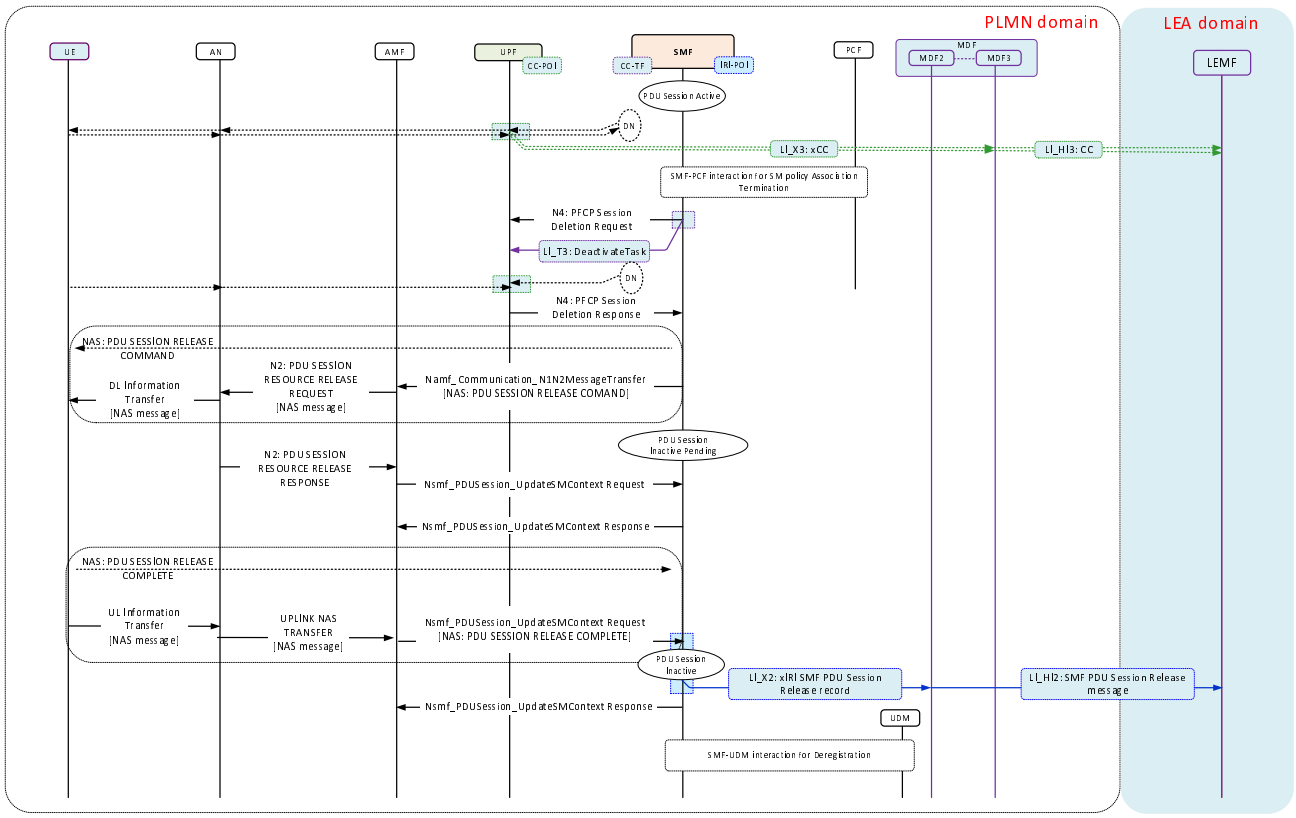


Figure 5.2.3.4.2.2-1: Detailed flow diagram for PDU session release (non-roaming) – network-initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is SMF receiving the NAS message PDU SESSION RELEASE COMPLETE from the target UE and the 5GSM state changing into PDU Session Inactive.
- The LI_T3 Deactivate Task trigger is sent from CC-TF present in the SMF to the CC-POI present in the UPF when the SMF sends the N4: PFCP Deletion Request to the UPF.
- The CC-POI in UPF stops generating the xCC upon receiving the LI_T3: Deactivate Task received from the CC-TF.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.2-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4.2.3 Network initiated PDU session release without any N1 or N2 message transfer

The flow-diagram in figure 5.2.3.4.2.3-1 illustrates a detailed view of PDU session release steps for a non-roaming scenario without the transfer of any N1 (i.e. NAS) or N2 messages. In the illustration, the release procedure is initiated by the AMF.

The flow illustrates the xIRI generation from the IRI-POI present in the SMF, LI_T3 triggering from the CC-TF present in the SMF and the xCC generation from the CC-POI present in the VPF.

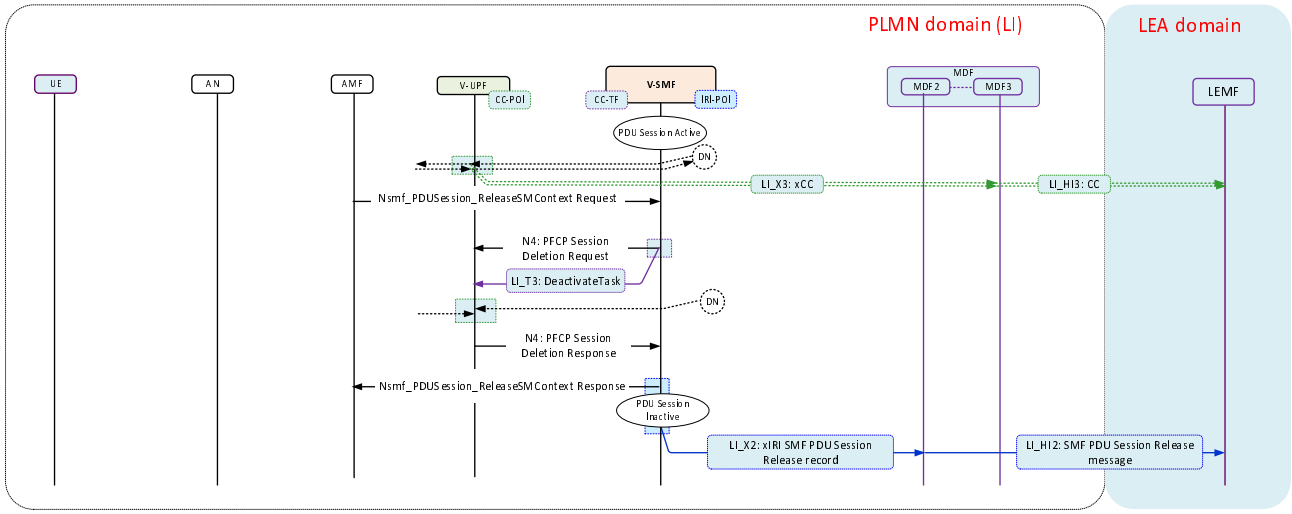


Figure 5.2.3.4.2.3-1: Detailed flow diagram for PDU session release without any N1 or N2 message transfer

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is SMF sending the Nsmf_PDUSession_ReleaseSMContext Response to the AMF and the 5GSM state changing into PDU Session Inactive.
- The LI_T3 DeactivateTask trigger is sent from CC-TF present in the SMF to the CC-POI present in the UPF when the SMF sends the N4: PFCP Session Delete Request to the UPF.
- The CC-POI in UPF stops generating the xCC upon receiving the LI_T3: Deactivate Task received from the CC-TF.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.2-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4.3 Home-routed roaming scenario (VPLMN)

5.2.3.4.3.1 UE initiated PDU session release

The flow-diagram in figure 5.2.3.4.3.1-1 illustrates a detailed view of PDU session release steps for a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the V-SMF, LI_T3 triggering from the CC-TF present in the V-SMF and the xCC generation from the CC-POI present in the V-UPF.

In this illustration, the case of UE initiated PDU session release is considered.

In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

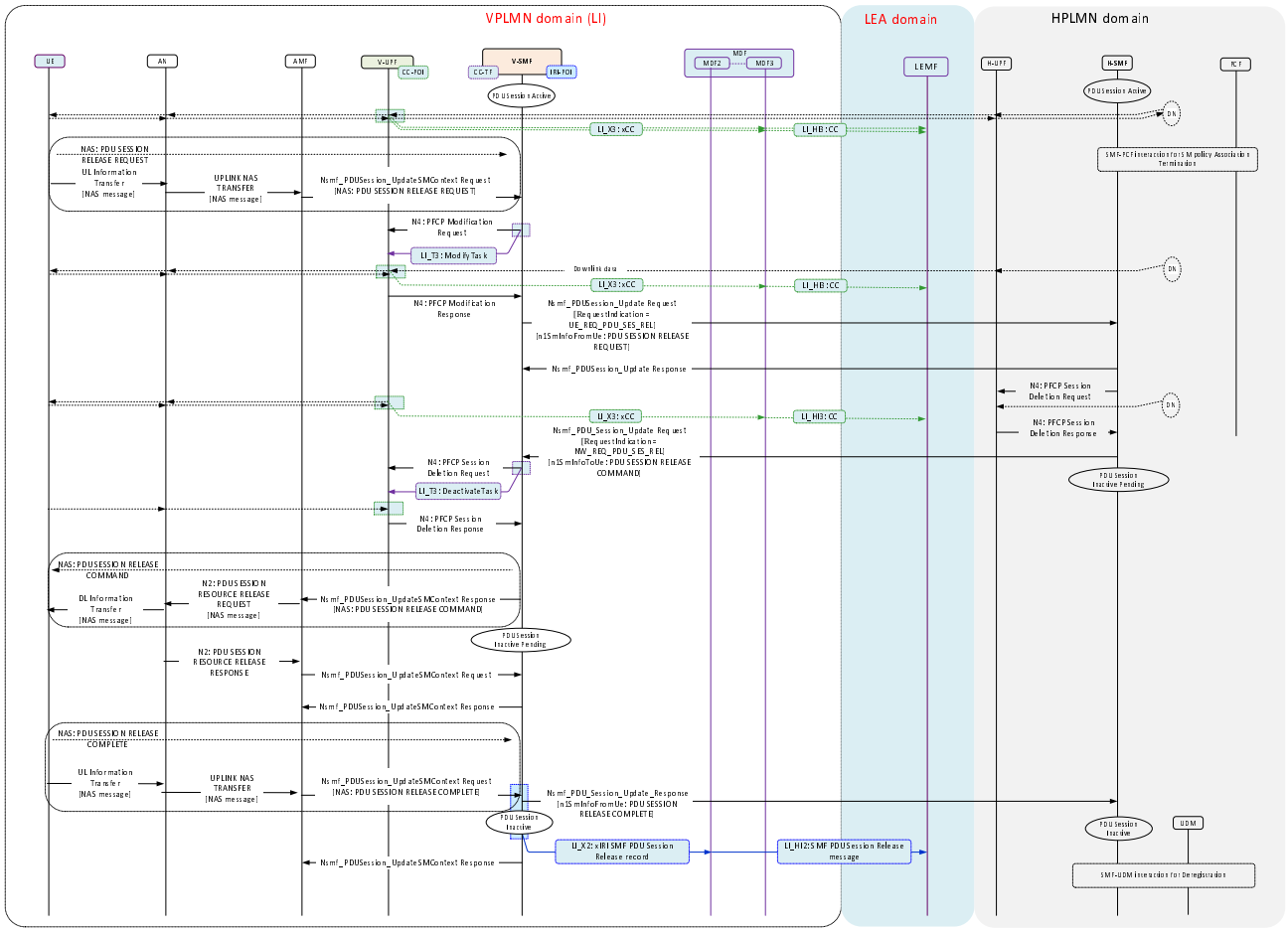


Figure 5.2.3.4.3.1-1: Detailed flow diagram for PDU session release (VPLMN) – UE initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is V-SMF receiving the NAS message PDU SESSION RELEASE COMPLETE from the target UE and the 5GSM state changing into PDU Session Inactive.

- The LI_T3 Modify Task trigger is sent from CC-TF present in the V-SMF to the CC-POI present in the V-UPF when the V-SMF sends the N4: PFCP Session Modification Request to the V-UPF. For network-initiated modification, it is quite possible that the CC-interception is impacted (e.g. new QoS flow) and therefore, this is a good example where LI_T3: Modify Task is sent.

- The CC-POI in V-UPF stops generating the xCC upon receiving the LI_T3: Deactivate Task received from the CC-TF.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4.3.2 Network (HPLMN) initiated PDU session release

The flow-diagram in figure 5.2.3.4.3.2-1 illustrates a detailed view of PDU session release steps for a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the V-SMF, LI_T3 triggering from the CC-TF present in the V-SMF and the xCC generation from the CC-POI present in the V-UPF.

In this illustration, the case of Network (HPLMN) initiated PDU session release is considered.

In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

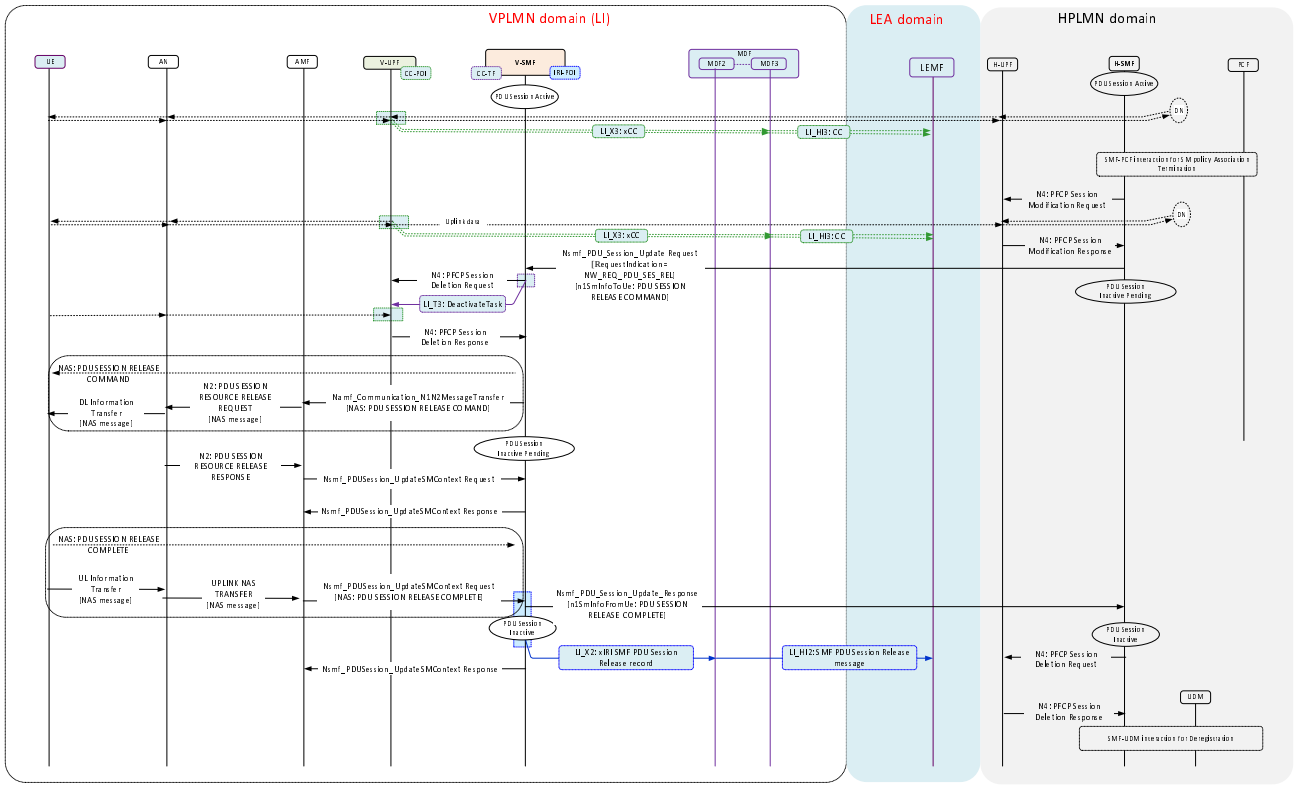


Figure 5.2.3.4.3.2-1: Detailed flow diagram for PDU session release (VPLMN) – Network (HPLMN) initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is V-SMF receiving the NAS message PDU SESSION MODIFICATION COMPLETE from the target UE and the 5GSM state changing into PDU Session Inactive.
- The LI_T3 Modify Task trigger is sent from CC-TF present in the V-SMF to the CC-POI present in the V-UPF when the V-SMF sends the N4: PFCP Session Modification Request to the V-UPF. For network-initiated modification, it is quite possible that the CC-interception is impacted (e.g. new QoS flow) and therefore, this is a good example where LI_T3: Modify Task is sent.
- The CC-POI in V-UPF stops generating the xCC upon receiving the LI_T3: Deactivate Task received from the CC-TF.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4.3.3 Network initiated PDU session release without any N1 or N2 message transfer

The flow-diagram in figure 5.2.3.4.3.3-1 illustrates a detailed view of PDU session release steps for a home-routed roaming scenario without the transfer of any N1 (i.e. NAS) or N2 messages. In the illustration, the release procedure is initiated by the AMF.

The flow illustrates the xIRI generation from the IRI-POI present in the V-SMF, LI_T3 triggering from the CC-TF present in the V-SMF and the xCC generation from the CC-POI present in the V-UPF.

In this illustration, the LI does not apply to the HPLMN. In this case, the LI is on an inbound roaming UE.

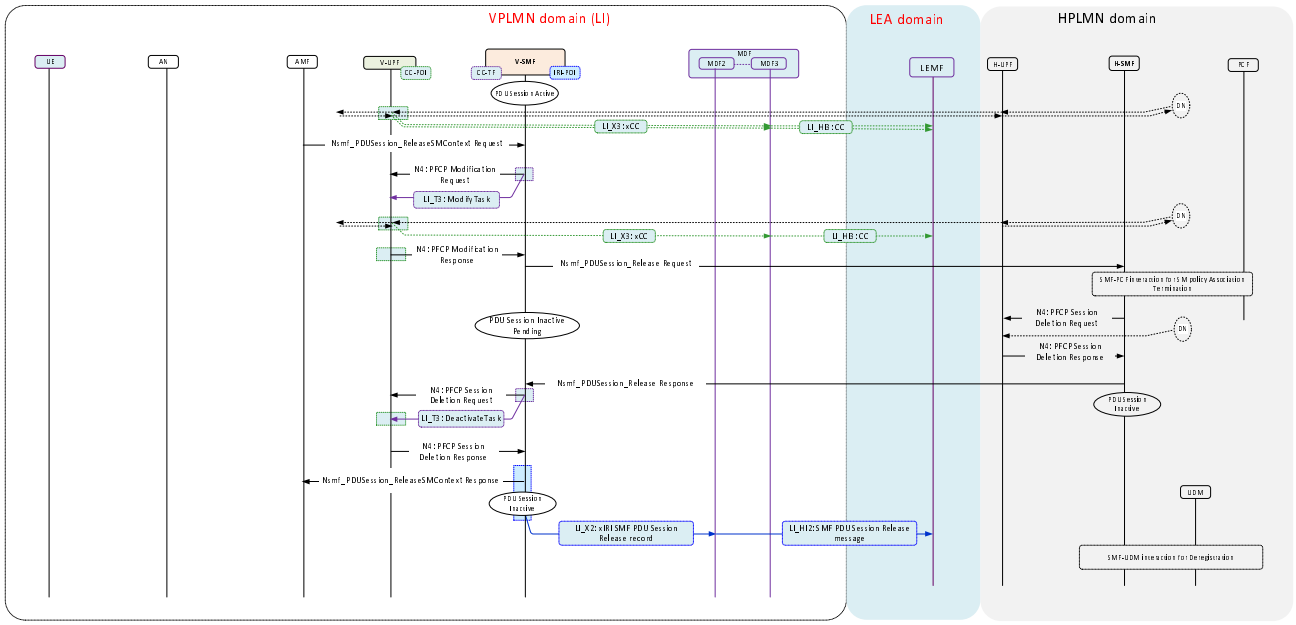


Figure 5.2.3.4.3.3-1: Detailed flow diagram for PDU session release without any N1 or N2 message transfer (VPLMN)

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is the V-SMF sending the Nsmf_PDUSession_ReleaseSMContext Response to the AMF and the 5GSM state changing into PDU Session Inactive.
- The LI_T3 DeactivateTask trigger is sent from CC-TF present in the V-SMF to the CC-POI present in the V-UPF when the V-SMF sends the N4: PFCP Session Delete Request to the V-UPF.
- The CC-POI in V-UPF stops generating the xCC upon receiving the LI_T3: Deactivate Task received from the CC-TF.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.3-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4.4 Home-routed roaming scenario (HPLMN)

5.2.3.4.4.1 UE-initiated PDU session release

The flow-diagram in figure 5.2.3.4.4.1-1 illustrates a detailed view of PDU session release steps for a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the H-SMF, LI_T3 triggering from the CC-TF present in the H-SMF and the xCC generation from the CC-POI present in the H-UPF.

In this illustration, the case of UE-initiated session release is considered.

In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

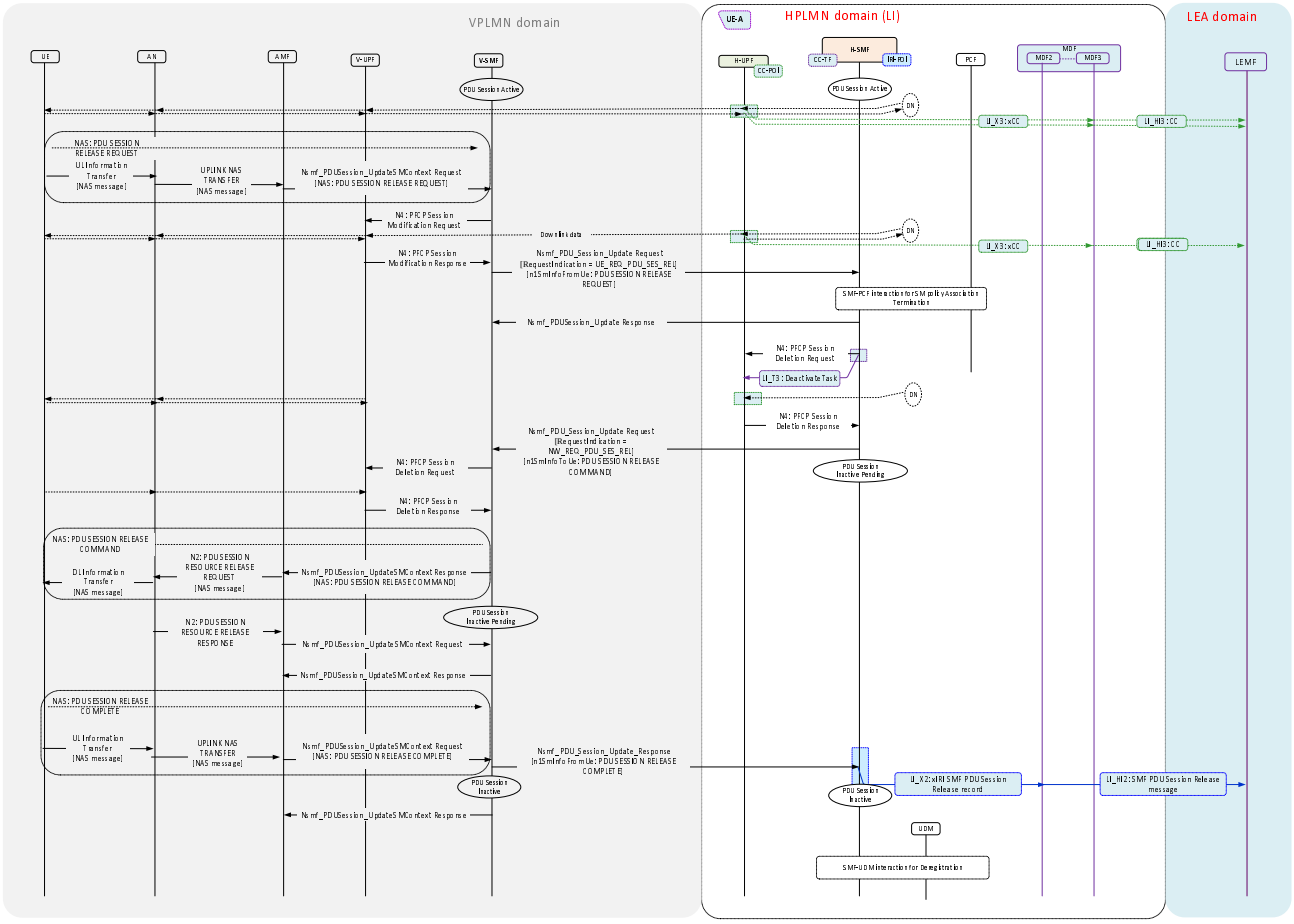


Figure 5.2.3.4.4.1-1: Detailed flow diagram for PDU session release (HPLMN) – UE initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is H-SMF receiving the Nsmf_PDU_Session_Update Response from the V-SMF with N1SmInfoFromUe containing the message PDU SESSION RELEASE COMPLETE.
- The LI_T3 Deactivate Task trigger is sent from CC-TF present in the H-SMF to the CC-POI present in the H-UPF when the H-SMF sends the N4: PFCP Deletion Request to the H-UPF.
- The CC-POI in H-UPF stops generating the xCC upon receiving the LI_T3: Deactivate Task received from the CC-TF

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4.4.2 Network (HPLMN) initiated PDU session release

The flow-diagram in figure 5.2.3.4.4.2-1 illustrates a detailed view of PDU session release steps for a home-routed roaming scenario. The flow illustrates the xIRI generation from the IRI-POI present in the H-SMF, LI_T3 triggering from the CC-TF present in the H-SMF and the xCC generation from the CC-POI present in the H-UPF.

In this illustration, the case of network (HPLMN) initiated session release is considered.

In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

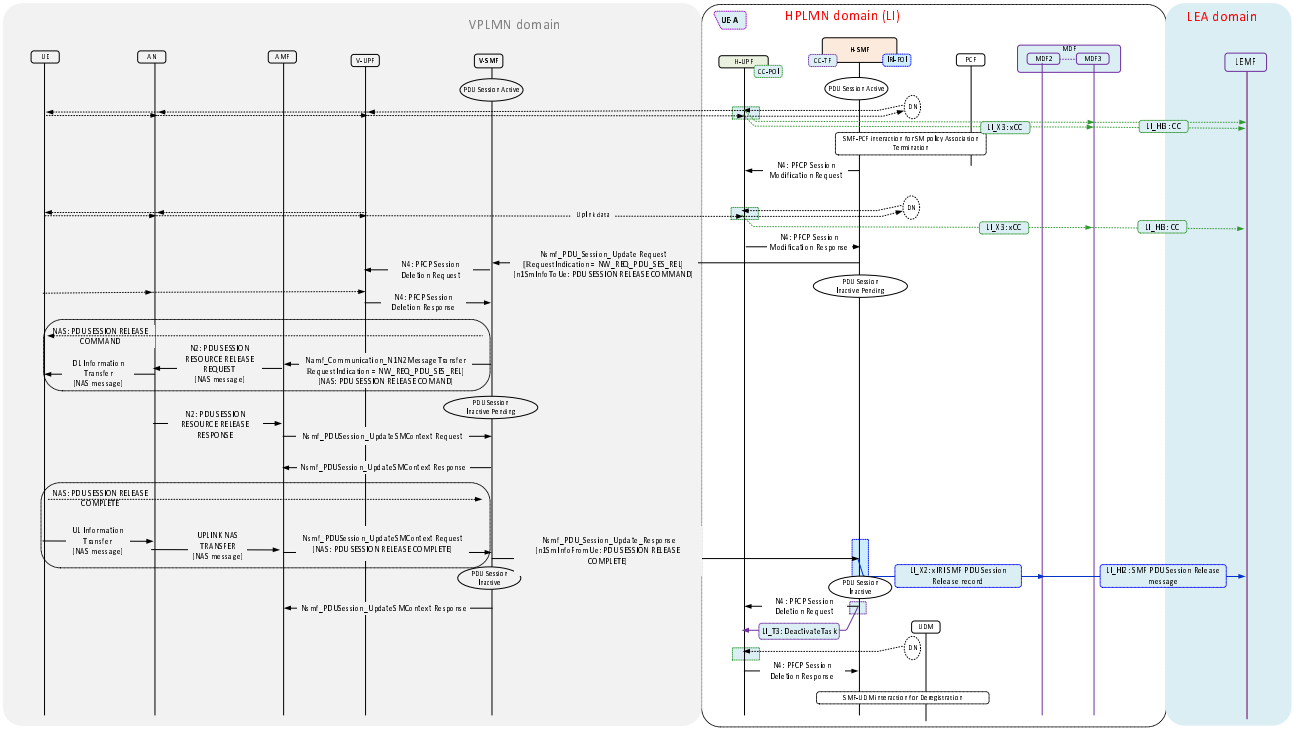


Figure 5.2.3.4.4.2-1: Detailed flow diagram for PDU session release (HPLMN) – network (HPLMN) initiated

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is H-SMF receiving the Nsmf_PDU_Session_Update Response from the V-SMF with N1SmInfoFromUe containing the message PDU SESSION RELEASE COMPLETE.
- The LI_T3 Deactivate Task trigger is sent from CC-TF present in the H-SMF to the CC-POI present in the H-UPF when the H-SMF sends the N4: PFCP Deletion Request to the H-UPF.
- The CC-POI in H-UPF stops generating the xCC upon receiving the based the LI_T3: Deactivate Task received from the CC-TF.

While drawing this flow, the TS 23.502 [7], figure 4.3.3.3-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.2.3.4.4.3 Network initiated PDU session release without any N1 or N2 message transfer

The flow-diagram in figure 5.2.3.4.4.3-1 illustrates a detailed view of PDU session release steps for a home-routed roaming scenario without the transfer of any N1 (i.e. NAS) or N2 messages. In the illustration, the release procedure is initiated by the AMF present in the VPLMN.

The flow illustrates the xIRI generation from the IRI-POI present in the H-SMF, LI_T3 triggering from the CC-TF present in the H-SMF and the xCC generation from the CC-POI present in the H-UPF.

In this illustration, the LI does not apply to the VPLMN. In this case, the LI is on an outbound roaming UE.

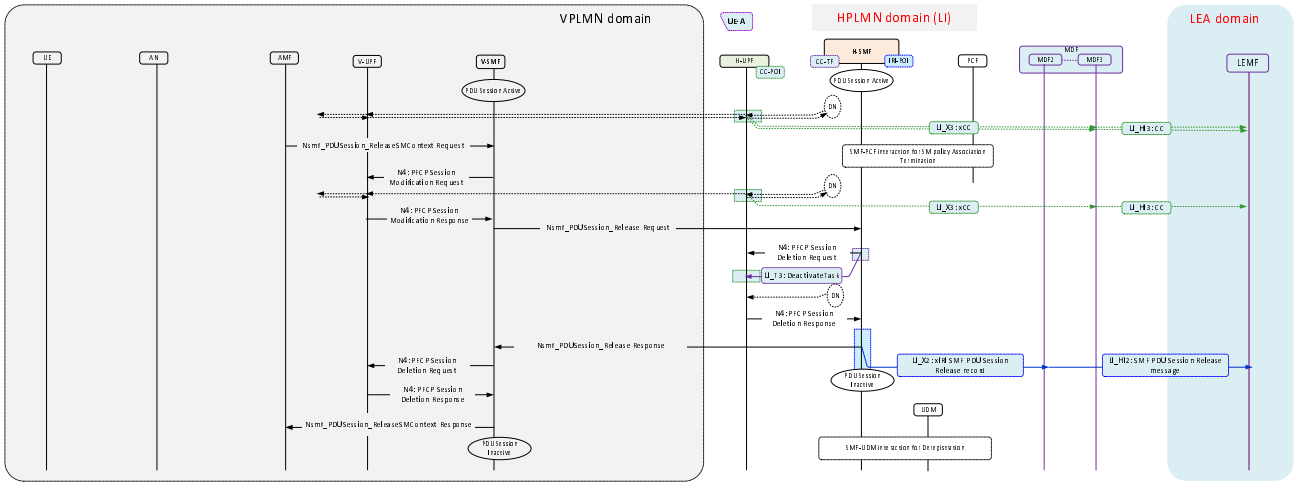


Figure 5.2.3.4.4.3-1: Detailed flow diagram for PDU session release without any N1 or N2 message transfer (HPLMN)

As defined in TS 33.128 [4]:

- The trigger conditions for the generation of xIRI is the H-SMF sending the Nsmf_PDUSession_Release Response to the V-SMF (in VPLMN).
- The LI_T3 DeactivateTask trigger is sent from CC-TF present in the H-SMF to the CC-POI present in the H-UPF when the H-SMF sends the N4: PFCP Session Delete Request to the H-UPF.
- The CC-POI in H-UPF stops generating the xCC upon receiving the LI_T3: Deactivate Task from the CC-TF.

While drawing this flow, the TS 23.502 [7], figure 4.3.4.3-1 is referred to. The PFCP session related messages are defined in TS 29.244 [9].

5.3 Handover related flow diagrams

5.3.1 Introduction

5.3.1.1 Overview

The subsequent clauses provide a series of flow-diagrams that illustrate LI aspects for various handover scenarios where the target mobile device moves from one radio network area to another radio network area. The radio network as mentioned here is referring to the Access Network (AN). The AN can be either an E-UTRAN (denoted as 4G) or an NG-RAN (denoted as 5G).

The flows cover only the inter-AN handover cases, In other words, the flows do not cover the case where the target mobile device moves within an AN space (e.g. from one cell to another cell). Depending on the scope of mobility that happens during a handover, the following two main categories are included in the illustrations:

- Intra-system handover scenarios.
- Inter-system handover scenarios.

A handover is considered to be intra-system handover when the mobility happens from AN to another AN of the same type (e.g. NG-RAN). A handover is considered to be inter-system handover when the mobility happens from one AN to another AN of different types (i.e. NG-RAN to E-UTRAN and vice-versa).

The scenarios gathered here are based on the stage-2 flow-diagrams given in TS 23.502 [7]. For some of the details the stage-3 specifications TS 29.244 [9], TS 38.331 [12], TS 38.413 [11] are also referenced.

5.3.1.2 Intra-system handover

The present document covers only the intra-systems handovers that happen between the NG-RANs. Intra-system handovers that happen between E-UTRANs are outside the scope of the present document. There are two cases of intra-system handovers illustrated:

- Xn-based handover scenarios.
- N2-based handover scenarios.

Xn-based handover is between the NG-RANs without a change in the AMF. Here, Xn refers to the interface between the two NG-RANs. N2-based handover is also between the NG-RANs, however, here the AMF may change. Here, N2 refers to the interface between the NG-RAN and the AMF. The N2-based handover procedure is used when an Xn2 based handover cannot be done (e.g. source NG-RAN does not have the Xn connectivity with the target NG-RAN).

5.3.1.3 Inter-system handover

There are two cases for inter-system handovers:

- 5G to 4G.
- 4G to 5G.

This is also referred to as N26-based handover. N26 refers to the interface between the AMF and the MME. When an inter-system handover happens, there is also a change in the mobility management functions (i.e. MME and AMF) and also the NF that handles the user plane packets (i.e. SGW and the UPF). The UPF+SGW-U provides the anchor point between the two systems.

In 5G to 4G case, the handover happens from NG-RAN to E-UTRAN. In the 4G to 5G case, the handover happens from E-UTRAN to NG-RAN.

5.3.1.4 DL data forwarding

The intra-system and inter-system handovers have the ability to have a continuity in the data flow. To accomplish that the Downlink (DL) data is forwarded, if necessary, buffered during the handover procedures. There are different methods that can be used for such DL data forwarding:

- Direct DL Data Forwarding.
- Indirect DL Data Forwarding.

Direct DL Data Forwarding happens between the ANs and the Indirect DL Data Forwarding happens via the intermediate UPFs. The selection of intermediate UPFs have different choices and hence, the steps taken can be different depending on how the intermediate UPFs are selected.

5.3.2 Xn-based handover flows

5.3.2.1 Overview

5.3.2.1.1 General

The Xn is an interface between the two NG-RAN nodes. As described in TS 23.502 [7] clause 4.9.1.2, the Xn-based handover is supported only for intra-AMF mobility with or without an UPF relocation. The source NG-RAN determines whether an Xn-based handover can be done.

Handover procedures apply to all QoS flows in the PDU session. The QoS flows that are not handed over are released. The CC interception happens in the anchor UPF (in a non-roaming case) even if there is an UPF relocation.

During handover to maintain the data flow continuity, the DL data may be forwarded from the old NG-RAN to the NG-RAN. Depending on whether or not UPF relocation happens, several scenarios are described in TS 23.502 [7].

5.3.2.1.2 Without UPF relocation in a non-roaming case

Scenario 1

Figure 5.3.2.1.2-1 shows the initial (i.e. before the handover), during the DL data forward and the final (after the successful handover) user-plane packet flow without an UPF relocation.

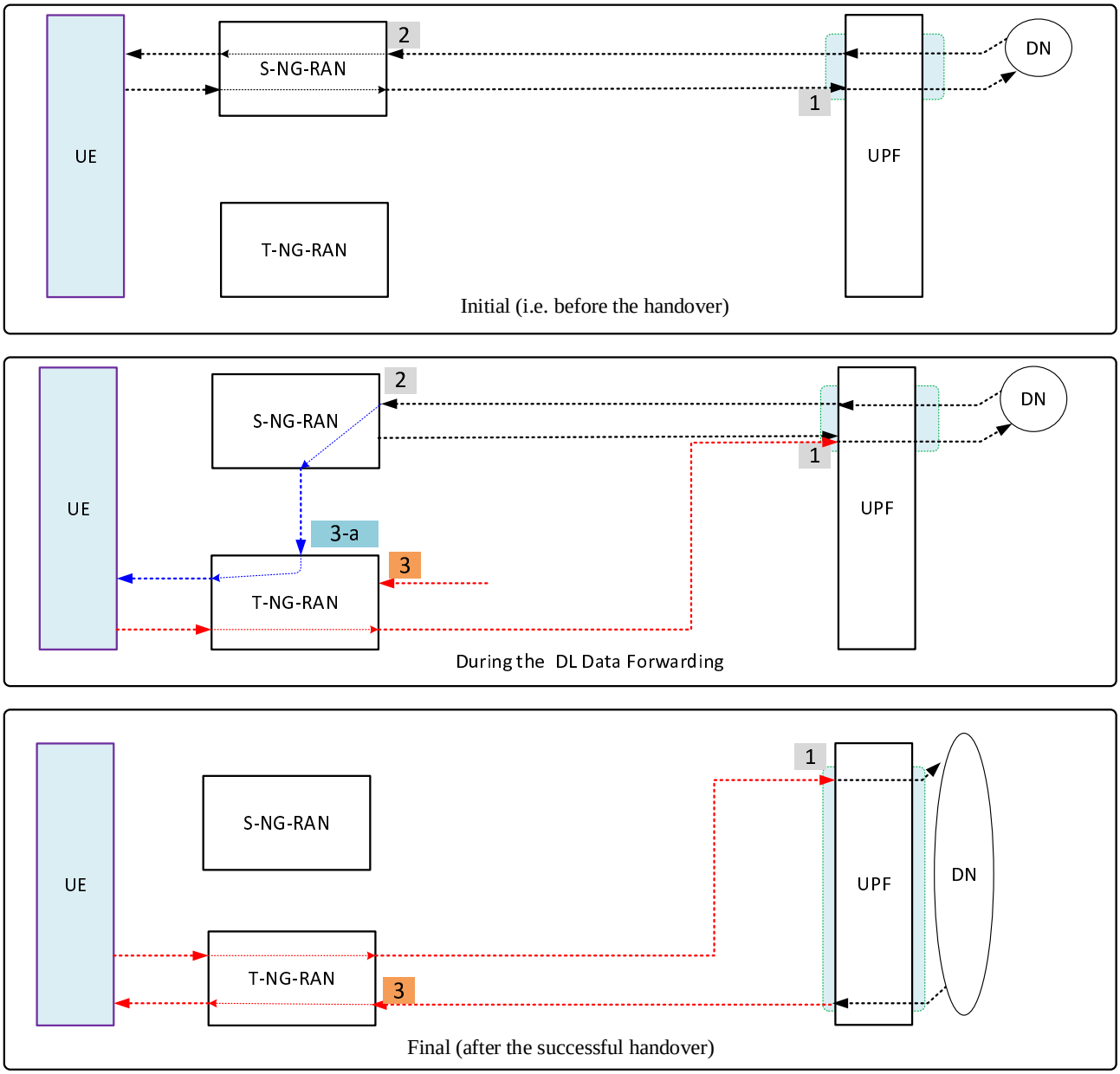


Figure 5.3.2.1.2-1: User plane packet flow during handover without UPF relocation (non-roaming)

The detailed flow diagram that depicts the above scenario is illustrated in clause 5.3.2.2.2.

Scenario 2

When an intermedia UPF is present, the initial (i.e. before the handover), during the DL data forward and the final (after the successful handover) user-plane packet flow when no UPF relocation happens is shown in figure 5.3.2.1.2-2. The intermediate UPF present before the handover is shown as S-UPF in figure 5.3.2.1.2-2.

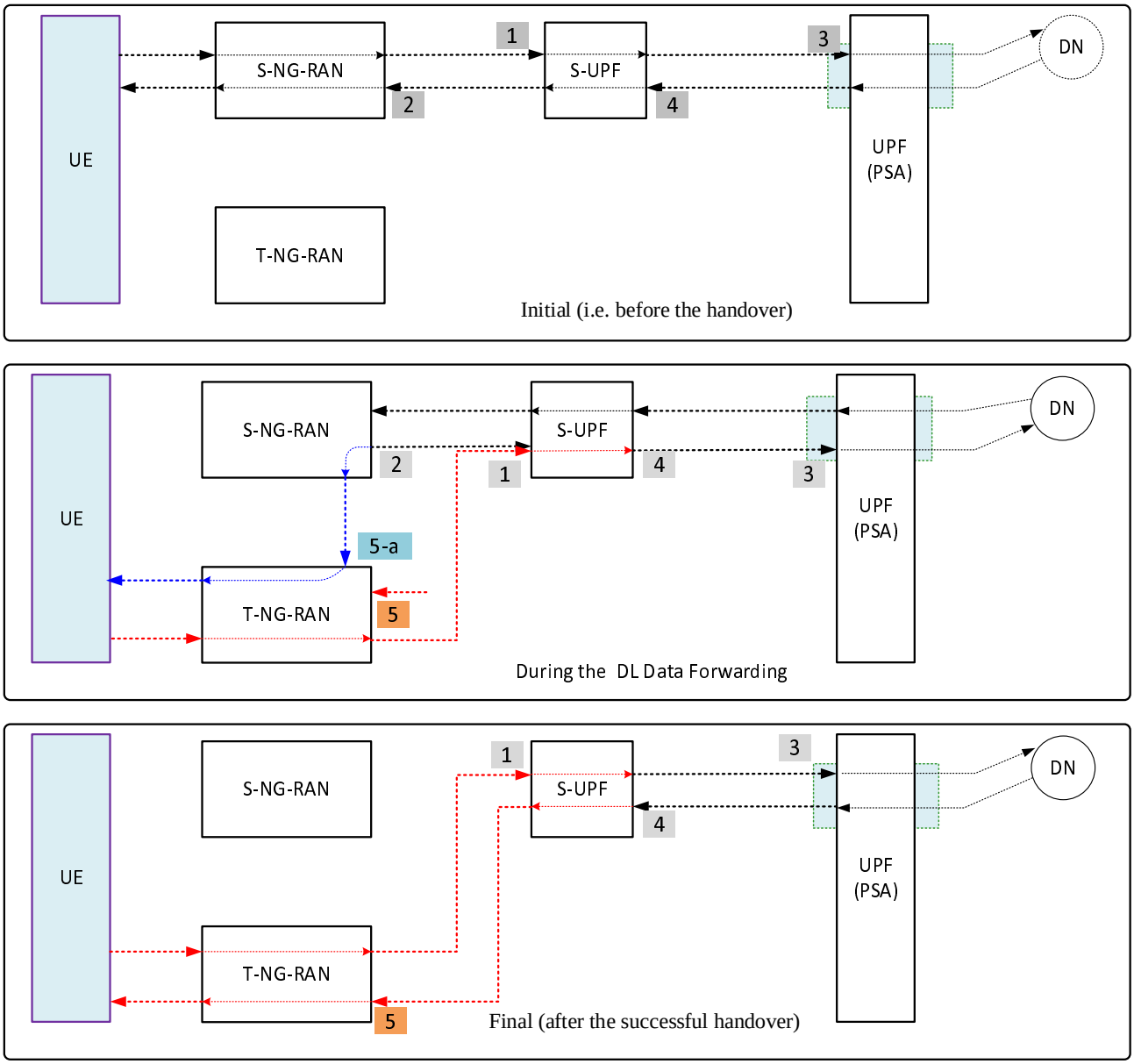


Figure 5.3.2.1.2-2: User plane packet flow during handover with an intermedia UPF without a relocation (non-roaming)

5.3.2.1.3 With UPF relocation in a non-roaming case

Scenario 1

Figure 5.3.2.1.3-1 shows the initial (i.e. before the handover), during the DL data forward and the final (after the successful handover) user-plane packet flow with an UPF relocation. The intermediate UPF (shown as T-UPF in figure 5.3.2.1.3-1) is inserted after the handover.

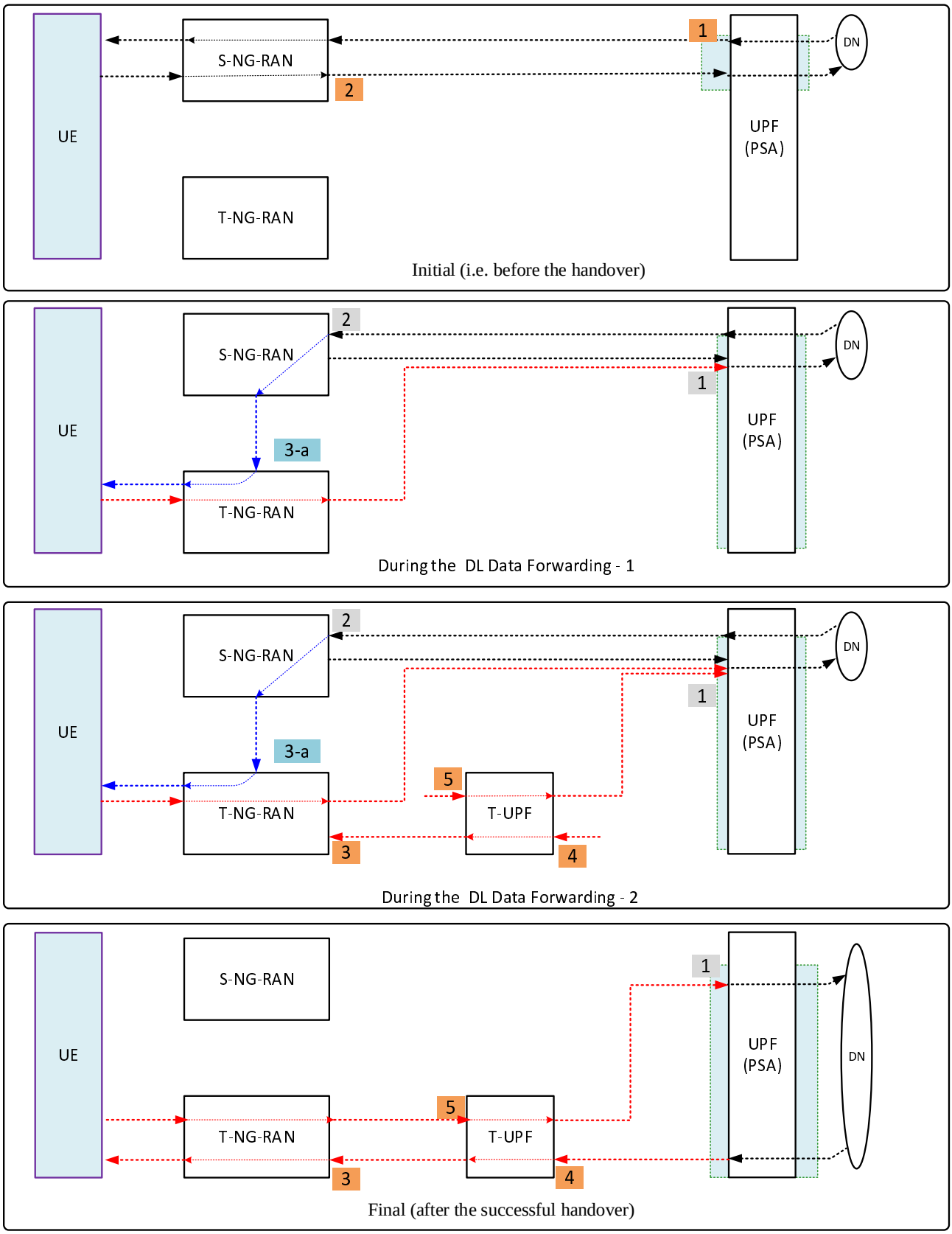


Figure 5.3.2.1.3-1: User plane packet flow during handover with UPF relocation (non-roaming, case 1)

Scenario 2

It is also possible that a different Tunnel End Point Identifier (shown as 6 in figure 4.3.1.3-2) can be used for the UL user plane packets at the UPF (PSA) after the handover.

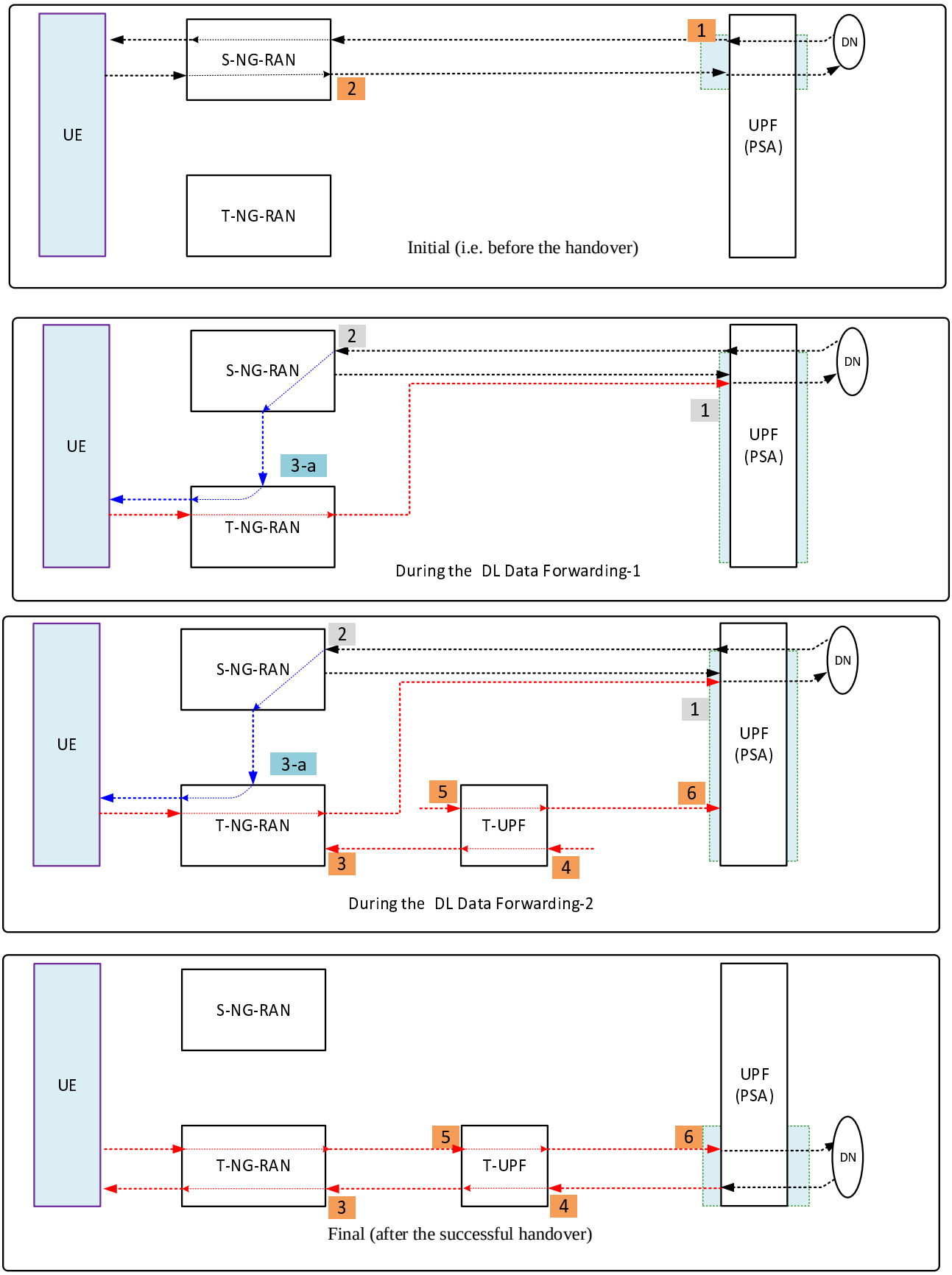


Figure 5.3.2.1.3-2: User plane packet flow during handover with UPF relocation (non-roaming, case 2)

Scenario 3

When an intermedia UPF is present, the initial (i.e. before the handover), during the DL data forward and the final (after the successful handover) user-plane packet flow with an UPF relocation is shown in figure 5.3.2.1.3-2. The two intermediate UPFs are shown in figure 5.3.2.1.3-3 as S-UPF and T-UPF.

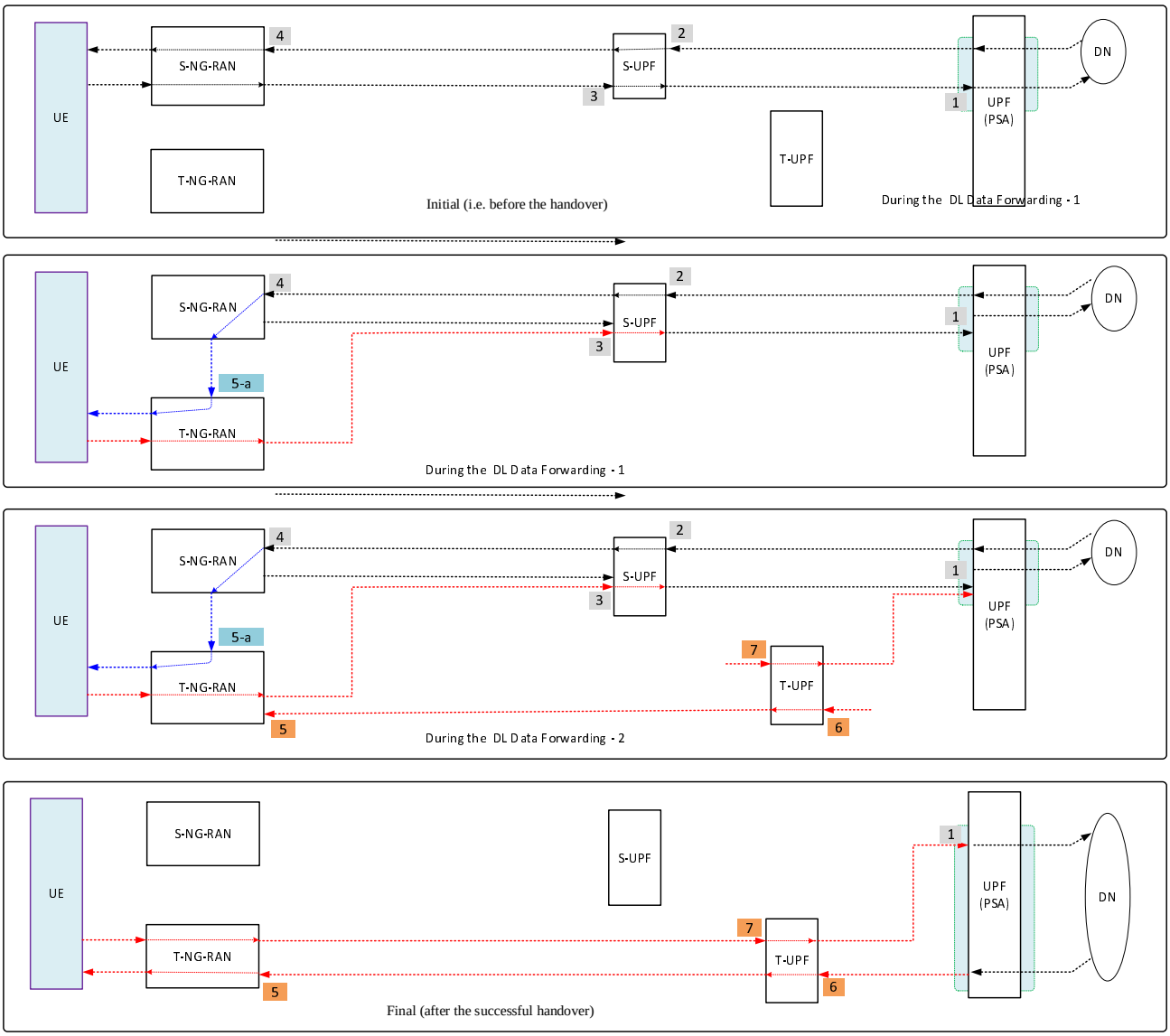


Figure 5.3.2.1.3-3: User plane packet flow during handover with intermediate UPFs (non-roaming, case 1)

Scenario 4

It is also possible that a different Tunnel End Point Identifier (similar to the way is shown as 6 in figure 5.3.2.1.3-4) can be used for the UL user plane packets at the UPF (PSA).

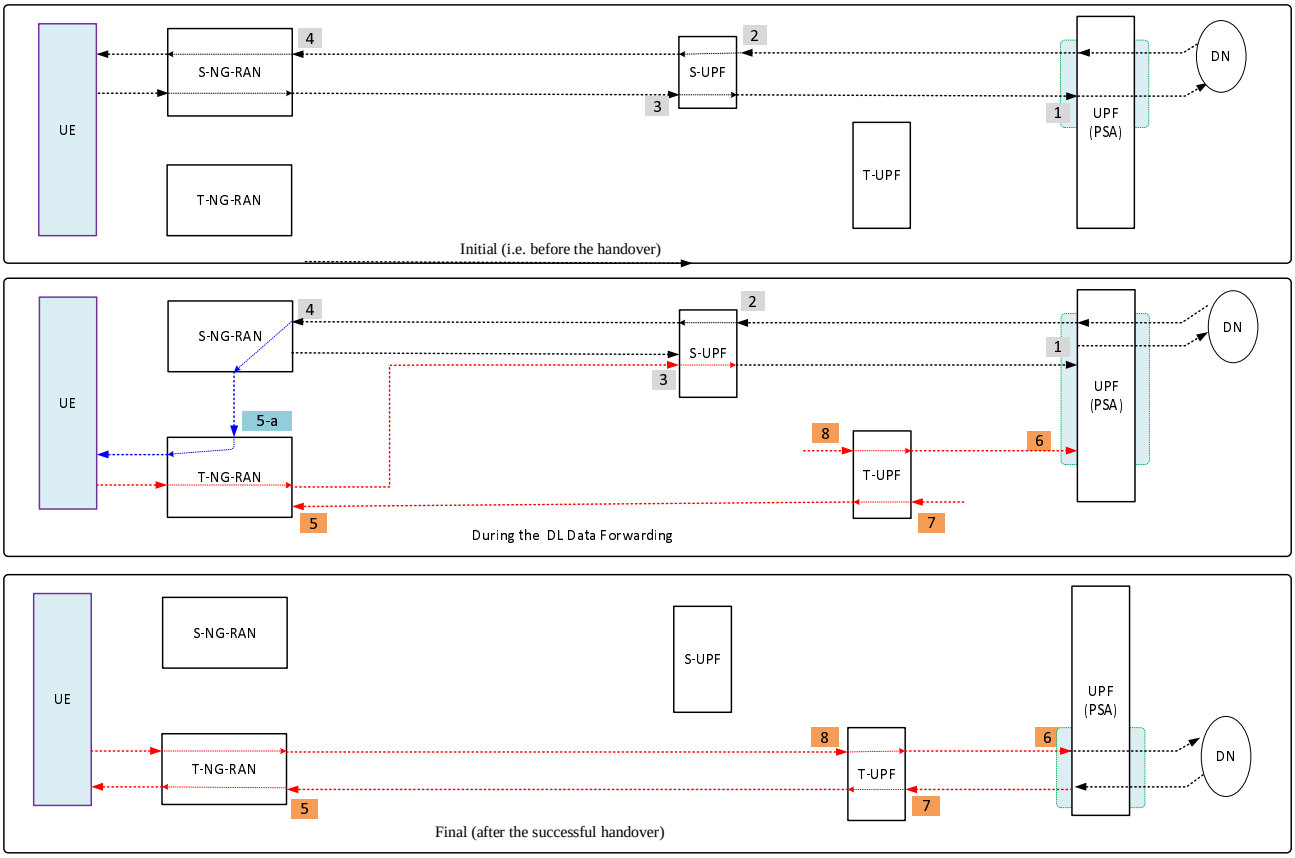


Figure 5.3.2.1.3-4: User plane packet flow during handover with intermediate UPFs (non-roaming, case 2)

The detailed flow diagrams that depict the above scenario is illustrated in clause 5.3.2.3.2.

5.3.2.1.4 Without UPF relocation in roaming case

Scenario 1: VPLMN

Figure 5.3.2.1.4-1 shows the initial (i.e. before the handover), during the DL data forward and the final (after the successful handover) user-plane packet flow without an UPF relocation in VPLMN with home-routed roaming scenario.

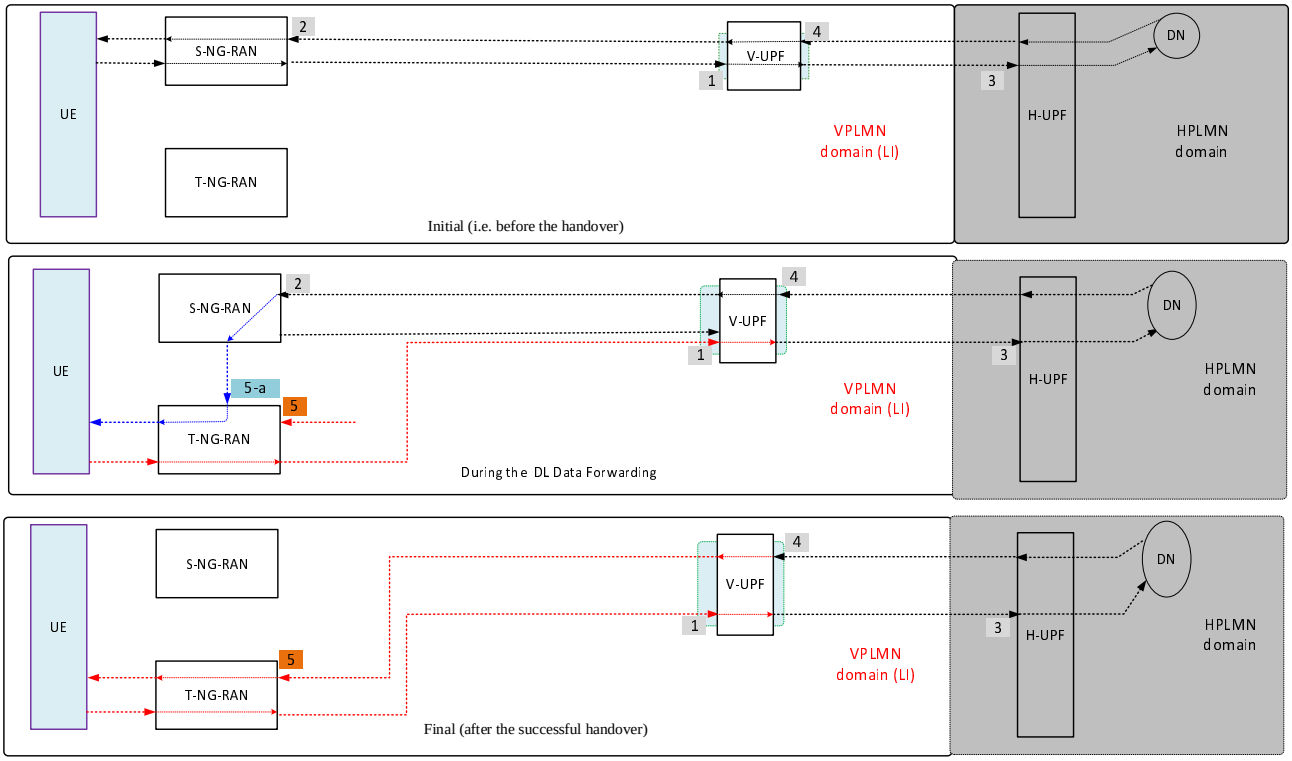


Figure 5.3.2.1.4-1: User plane packet flow during handover with roaming and without UPF relocation (LI in VPLMN)

The detailed flow diagram that depicts the above scenario is depicted in clause 5.3.2.4.2.

Scenario 2: HPLMN

Figure 5.3.2.1.4-2 shows the initial (i.e. before the handover), during the DL data forward and the final (after the successful handover) user-plane packet flow without an UPF relocation in VPLMN with home-routed roaming scenario when LI applies to HPLMN.

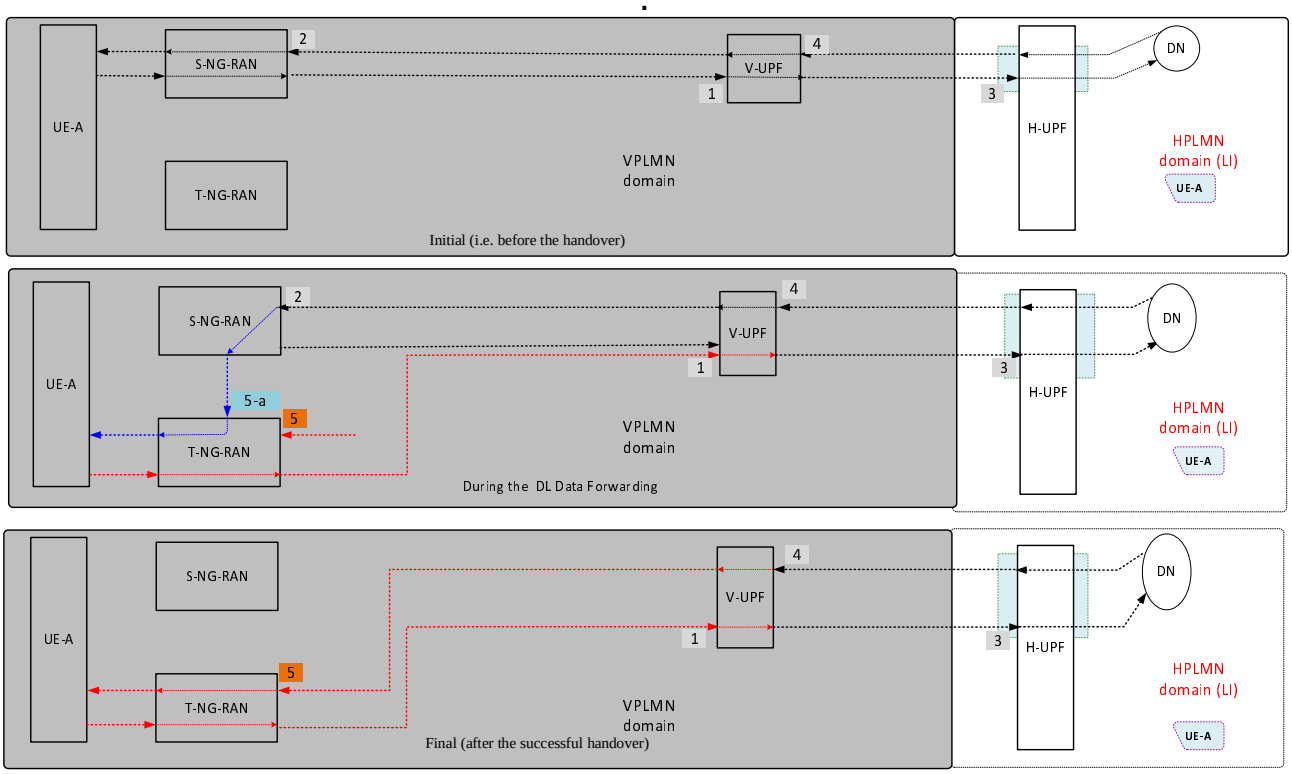


Figure 5.3.2.1.4-2: User plane packet flow during handover with roaming and without UPF relocation (LI in HPLMN)

In this case, if the same Tunnel End Points are used (as shown in figure 5.3.2.1.4-2), there may not be any reporting possible from the HPLMN as the SMF/UPF in the HPLMN may not be aware of the fact such a handover has occurred.

The detailed flow diagram that depicts the above scenario is depicted in clause 5.3.2.4.3.

5.3.2.1.5 With UPF relocation in roaming case

Scenario 1: VPLMN

Figure 5.3.2.1.5-1 shows the initial (i.e. before the handover), during the DL data forward and the final (after the successful handover) user-plane packet flow with V-UPF changing in VPLMN with home-routed roaming scenario.

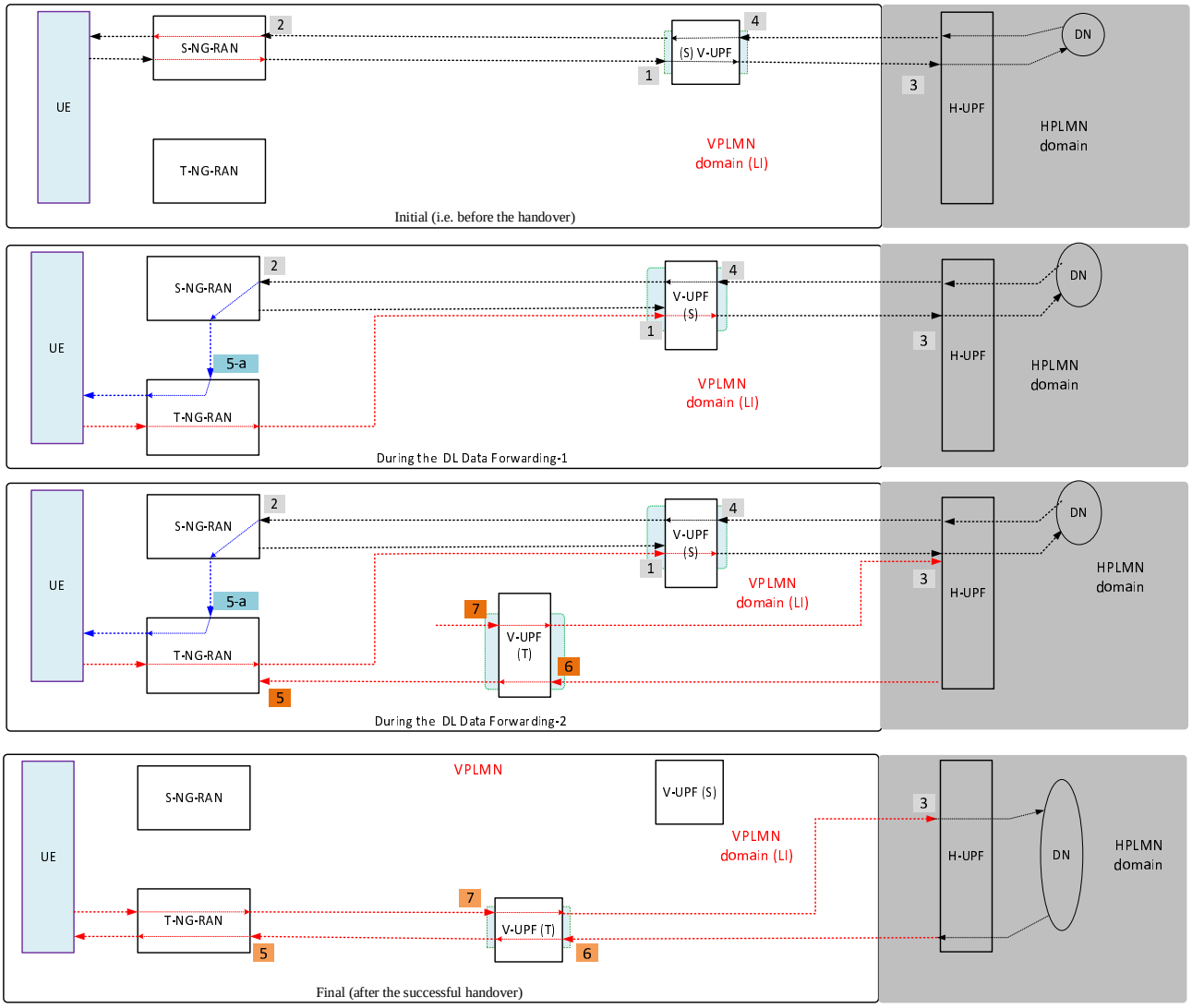


Figure 5.3.2.1.5-1: User plane packet flow during handover with roaming and UPF relocation (LI in VPLMN)

The detailed flow diagram that depicts the above scenario is depicted in clause 5.3.2.5.2.

It is also possible that a different Tunnel End Point Identifier can be used for the UL user plane packets at the H-UPF (as shown in figure 5.3.2.1.5-3).

Scenario 2: HPLMN

Figure 5.3.2.1.5-2 shows the initial (i.e. before the handover), during the DL data forward and the final (after the successful handover) user-plane packet flow with the UPF relocation in VPLMN with home-routed roaming scenario when LI applies to HPLMN.

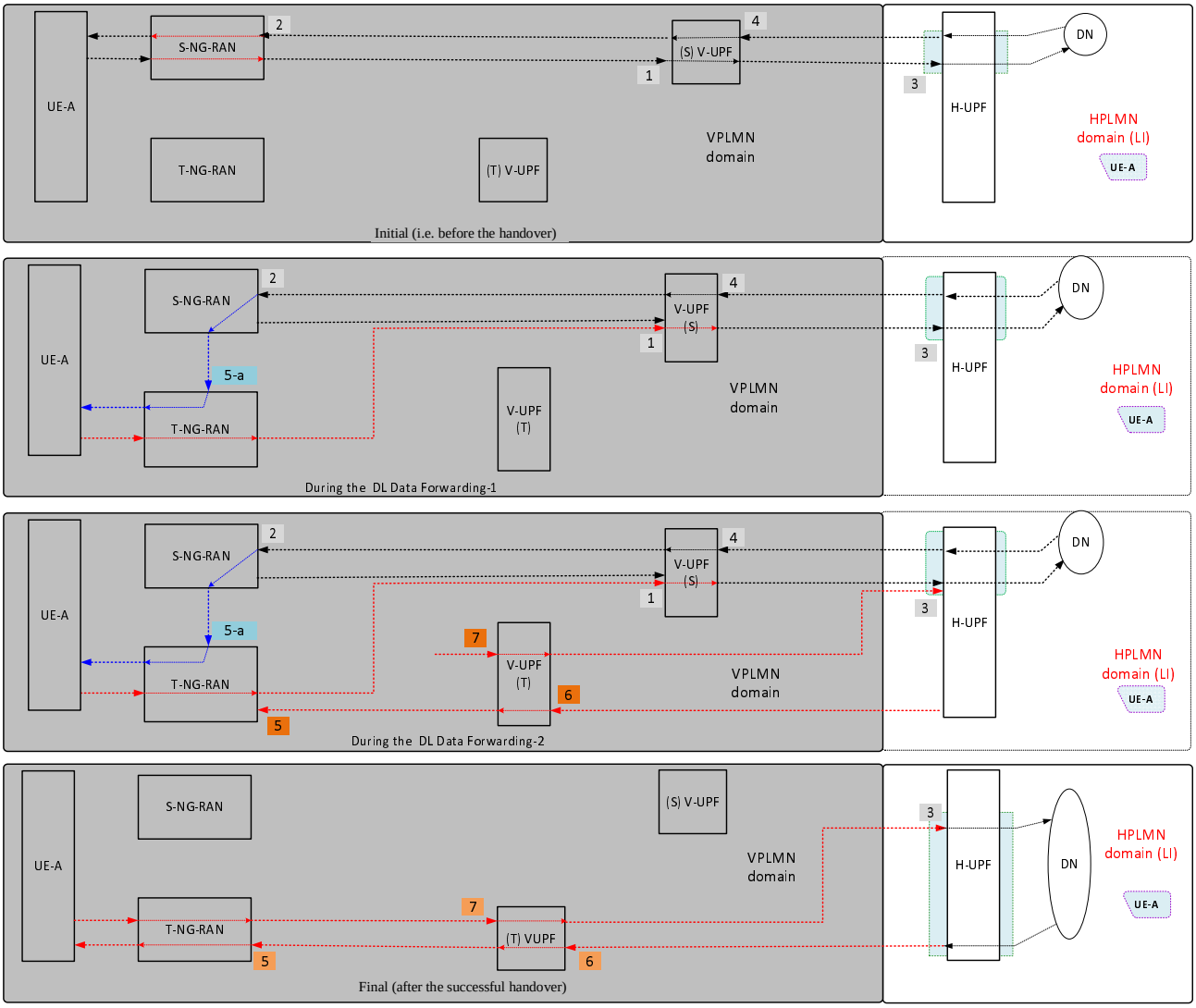


Figure 5.3.2.1.5-2: User plane packet flow during handover with roaming and UPF relocation (LI in HPLMN) – case 1

The detailed flow diagram that depicts the above scenario is depicted in clause 5.3.2.5.3.

Scenario 3: HPLMN

It is also possible that a different Tunnel End Point Identifier (shown as 8 on figure 5.3.2.1.5-3) can be used for the UL user plane packets at the H-UPF.

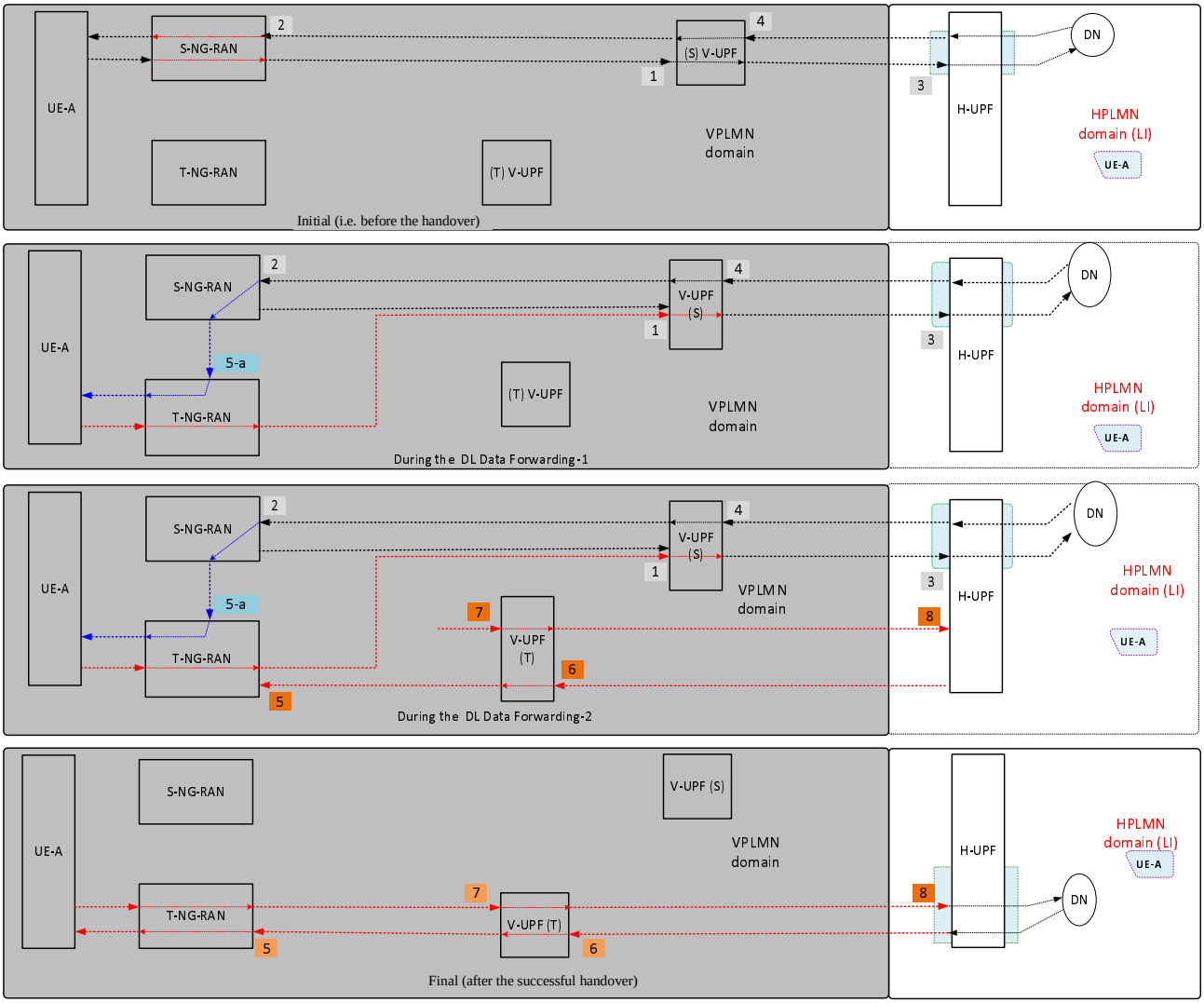


Figure 5.3.2.1.5-3: User plane packet flow during handover with roaming and with UPF relocation (LI in HPLMN) – case 2

5.3.2.2 Detailed flows - non-roaming, no UPF relocation

5.3.2.2.1 General

An Xn-based handover may happen without a change in the UPF (i.e. without an UPF relocation). This can be the case with or without an intermediate UPF prior to the handover. In this illustration, no intermediate UPF is inserted prior to the handover and no UPF relocation occurs during the handover. This flow is based on the Xn-based handover stage 2 flow in TS 23.502 [7], clause 4.9.1.2.2.

Within drawings, (source) NG-RAN is shown as S-NG-RAN and as S-5G. Likewise, the (target) NG-RAN is shown as T-NG-RAN and as T-5G. The numbers 1, 2, 3, 3-a correspond to the TEID (1), TEID (2), TEID (3) and TEID (3a) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier for UL UP packets at the UPF and the TEID (2) is the tunnel end point identifier for the DL UP packets at the (source) NG-RAN. The TEID (3a) is the tunnel end point identifier at the (target) NG-RAN for the DL data forwarded UP packets. The DL data forwarding happens from (source) NG-RAN to (target NG-RAN).

The TEID (3) is the tunnel end point identifier for the DL UP packets at the (target) NG-RAN. The same tunnel end point identifier (TEID (1)) is used for the UL UP packets at the UPF.

The generation of xCC from the UP packets done by the CC-POI present in the UPF.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.2.2.2 The flows

This flow is split into two parts, for easy reading purpose. The flow-diagram in figure 5.3.2.2.2-1 illustrates the first part and figure 5.3.2.2.2-2 illustrates the second part.

Flow 1 of 2

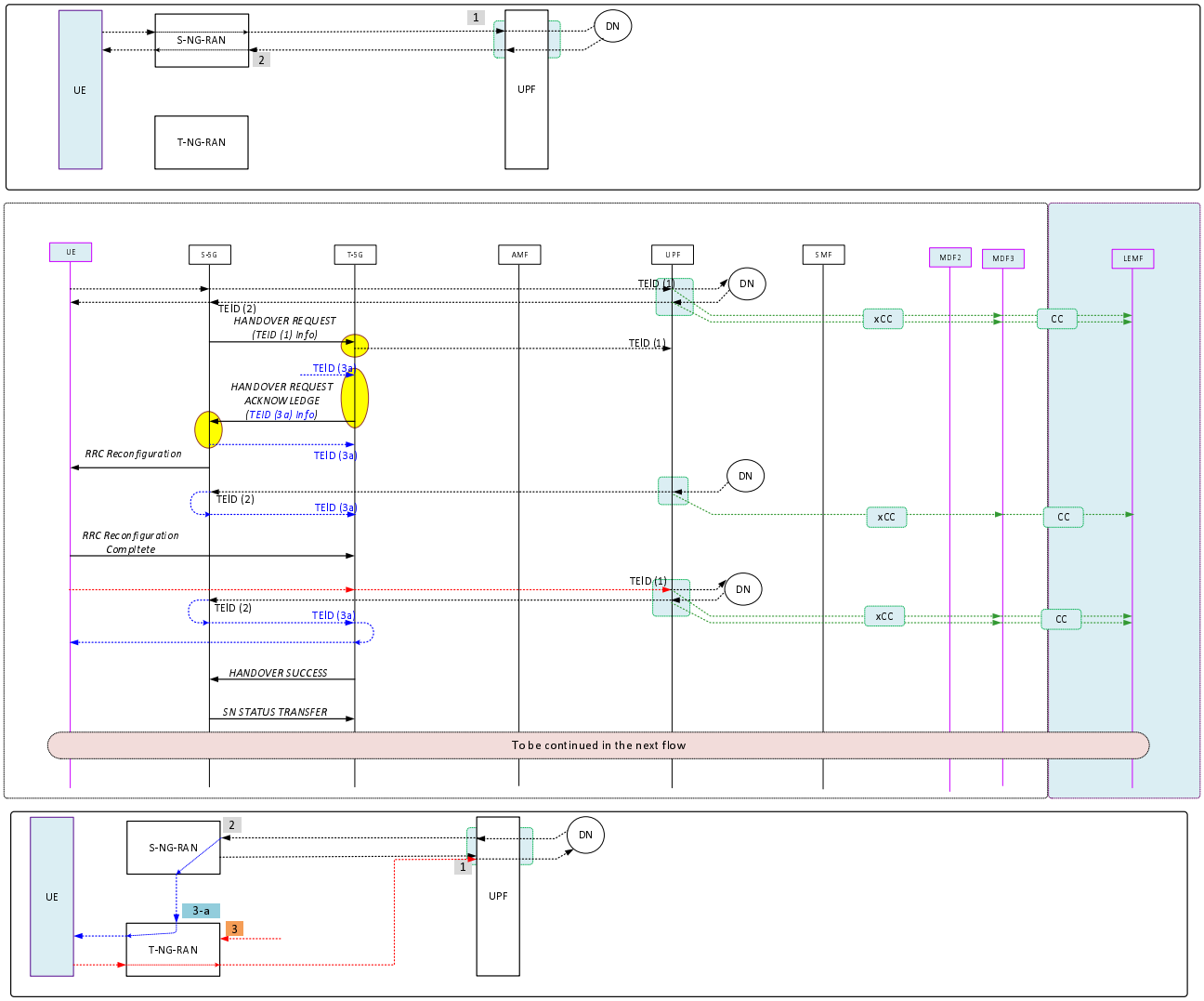


Figure 5.3.2.2.2-1: Xn-based handover with no UPF relocation (non-roaming) – flow 1 of 2

Flow 2 of 2

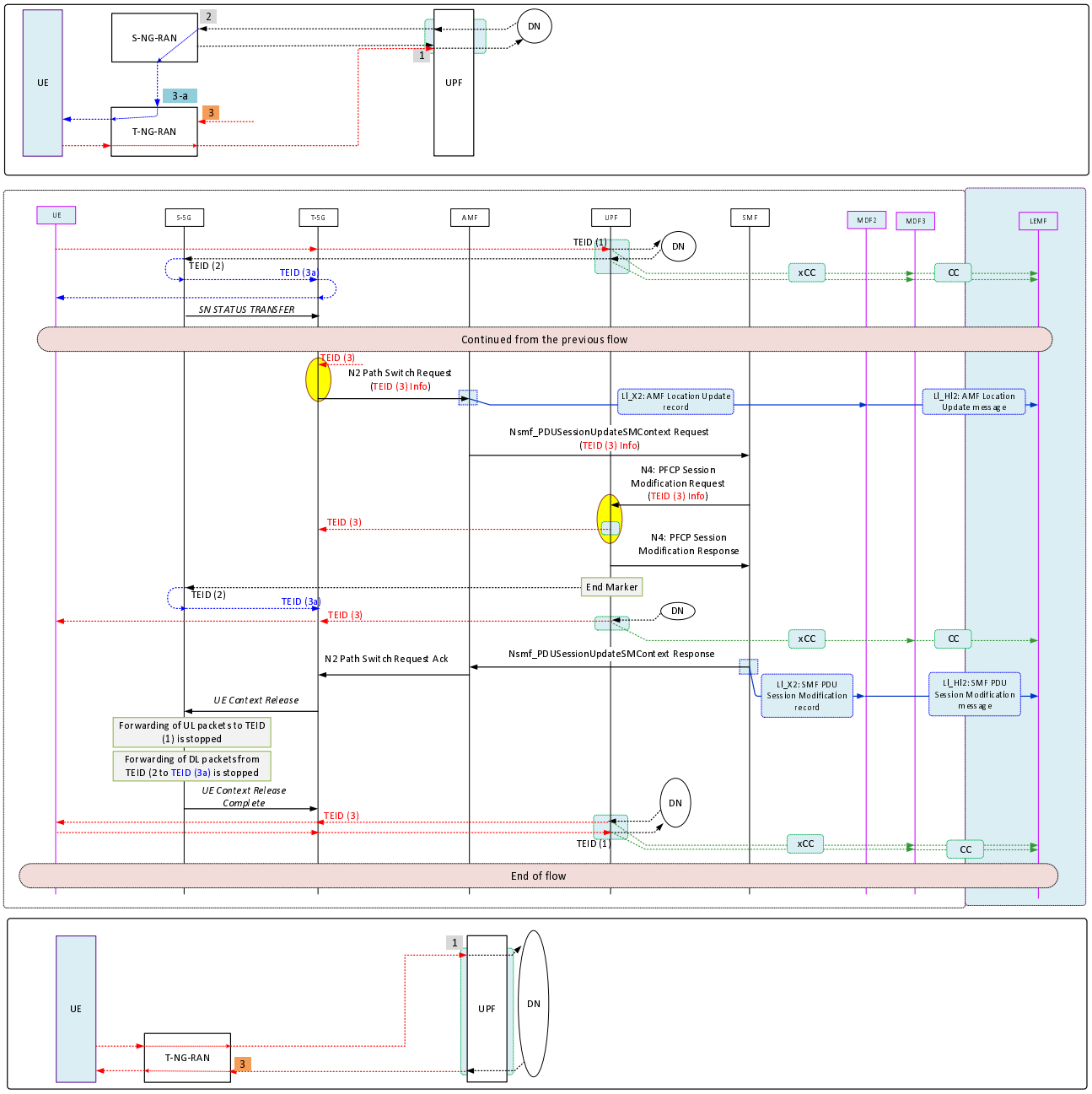


Figure 5.3.2.2.2-2: Xn-based handover with no UPF relocation (non-roaming) – flow 2 of 2

When the AMF receives a Path Switch Request from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF Location Update record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

As an alternative, SMF may request the UPF to allocate a different tunnel end point end point identifier for the UL UP packets.

5.3.2.3 Detailed flows - non-roaming, UPF relocation

5.3.2.3.1 General

With an Xn-based handover, a new intermediate UPF may be inserted when the SMF finds that a new UPF is needed with the (target) NG-RAN. An intermediate UPF may or may not be present prior to the handover. In this illustration, an intermediate UPF is present prior to the handover and is changed during the handover. This flow is based on the Xn-based handover stage 2 flow in TS 23.502 [7], clause 4.9.1.2.4.

Within drawings, two intermediate UPFs are shown as S-UPF (prior to the handover) and T-UPF (post-handover). The anchor UPF is shown as UPF (PSA). The (source) NG-RAN is shown as S-NG-RAN and as S-5G. Likewise, the (target) NG-RAN is shown as T-NG-RAN and as T-5G. The numbers 1, 2, 3, 4, 5, 5-a, 6, 7, 8 correspond to the TEID (1), TEID (2), TEID (3) and TEID (4), TEID (5), TEID (5a), TEID (6), TEID (7) and TEID (8) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at UPF (PSA) and the TEID (3) is the tunnel end point identifier at S-UPF for UL UP packets. TEID (2) is the tunnel end point identifier at S-UPF and TEID (4) is the tunnel end point identifier at the (source) NG-RAN for the DL UP packets. The TEID (5a) is the tunnel end point identifier at the (target) NG-RAN for the DL data forwarded UP packets. The DL data forwarding happens from (source) NG-RAN to (target NG-RAN).

The TEID (5) is the tunnel end point identifier for the DL UP packets at the (target) NG-RAN. The TEID (7) is the tunnel end point identifier for the DL UP packets at the T-UPF. The TEID (6) is the tunnel end point identifier for the UL traffic at the UPF (PSA). As an alternate (not shown), the SMF may choose to use the TEID (1) for the UL UP packets even after the handover. The TEID (8) is the tunnel end point identifier at the T-UPF for UL UP packets.

The generation of xCC from the UP packets done by the CC-POI present in the anchor UPF.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF (PSA) would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.2.3.2 The flows

This flow diagram is split into three parts, for easy reading purpose. The flow-diagram in figure 5.3.2.3.2-1 illustrates the first part. The flow-diagram in figure 5.3.2.3.2-2 illustrates the second part. The flow-diagram in figure 5.3.2.3.2-3 illustrates the third part.

Flow 1 of 3

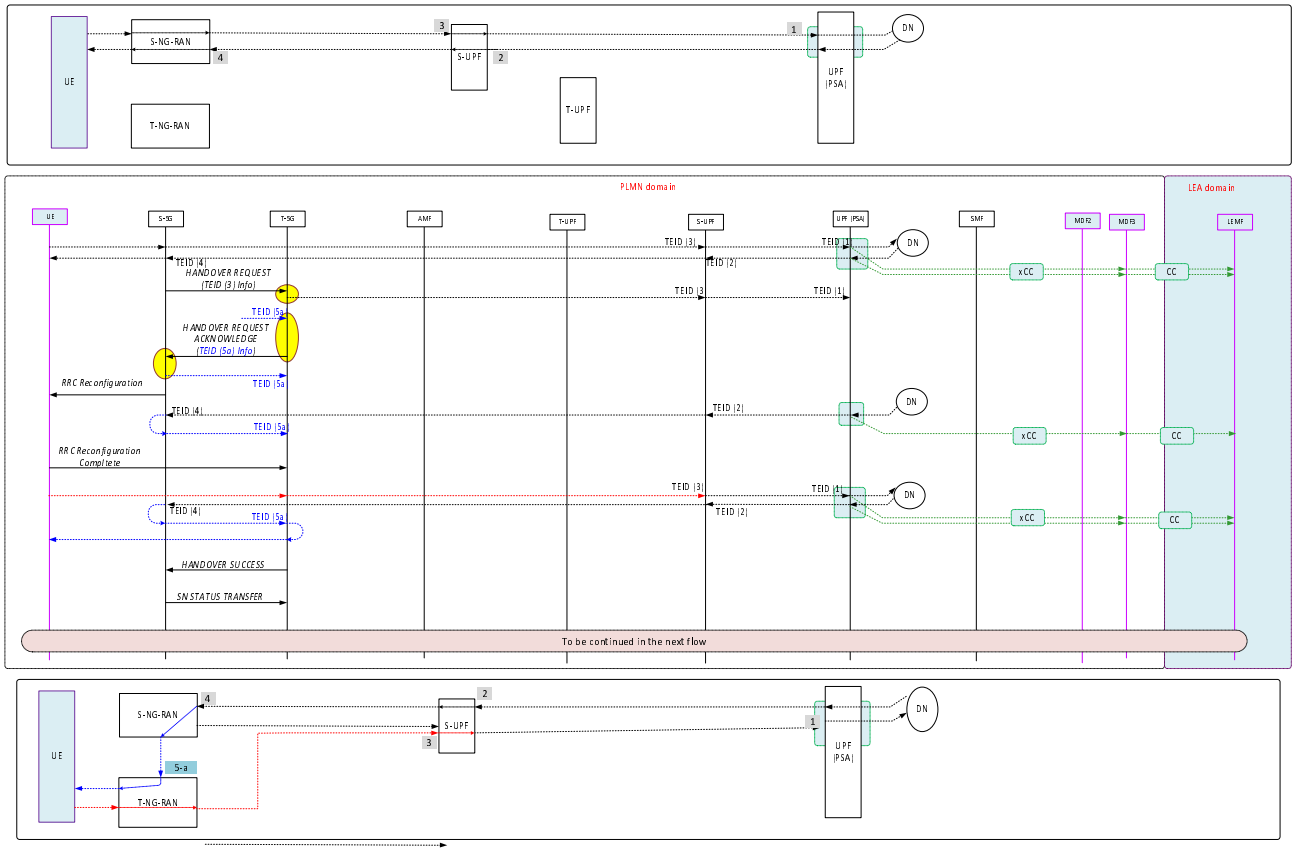


Figure 5.3.2.3.2-1: Xn-based handover with UPF relocation (non-roaming) – flow 1 of 3

Flow 2 of 3

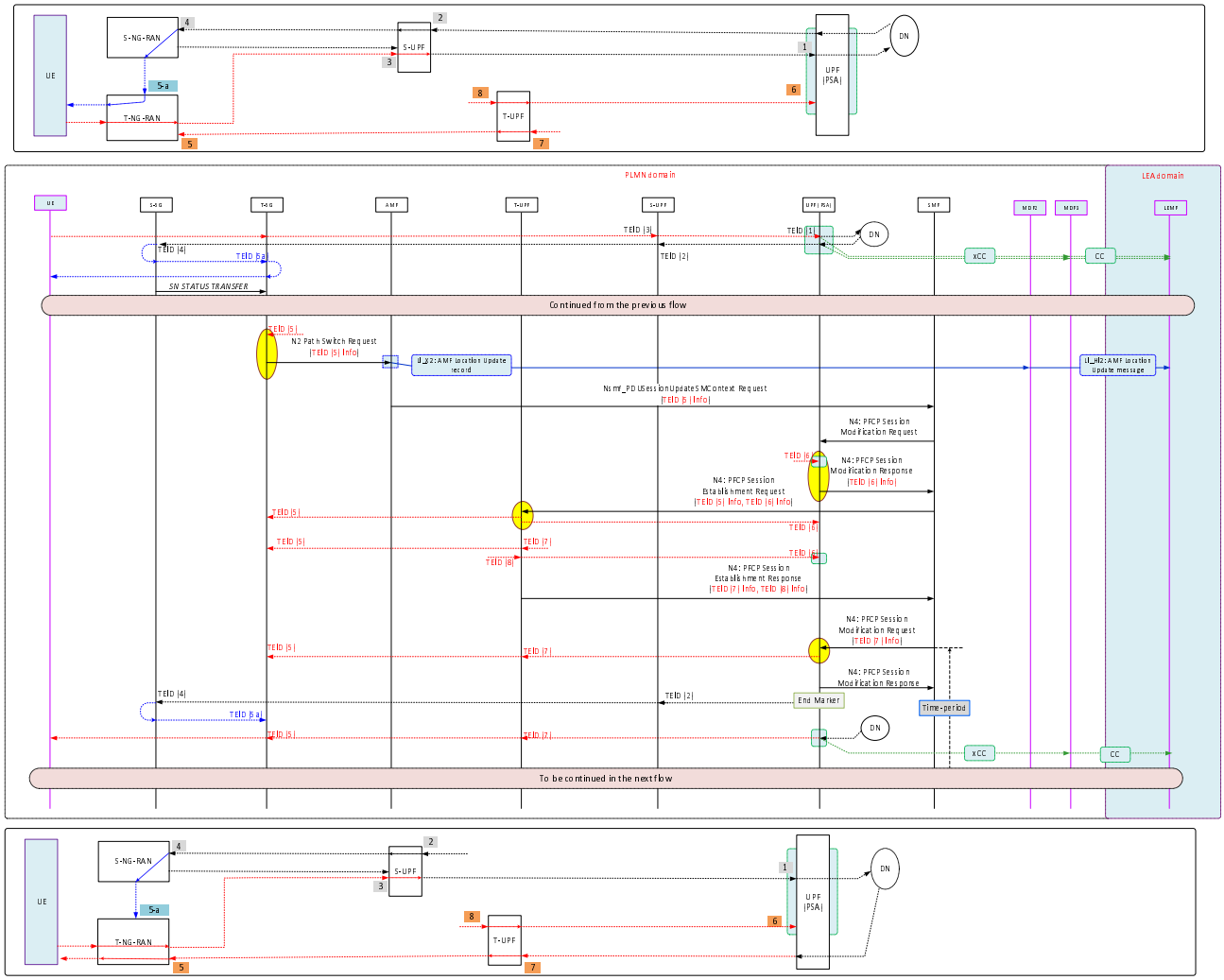


Figure 5.3.2.3.2-2: Xn-based handover with UPF relocation (non-roaming) – flow 2 of 3

When the AMF receives a Path Switch Request from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF Location Update record.

Flow 3 of 3

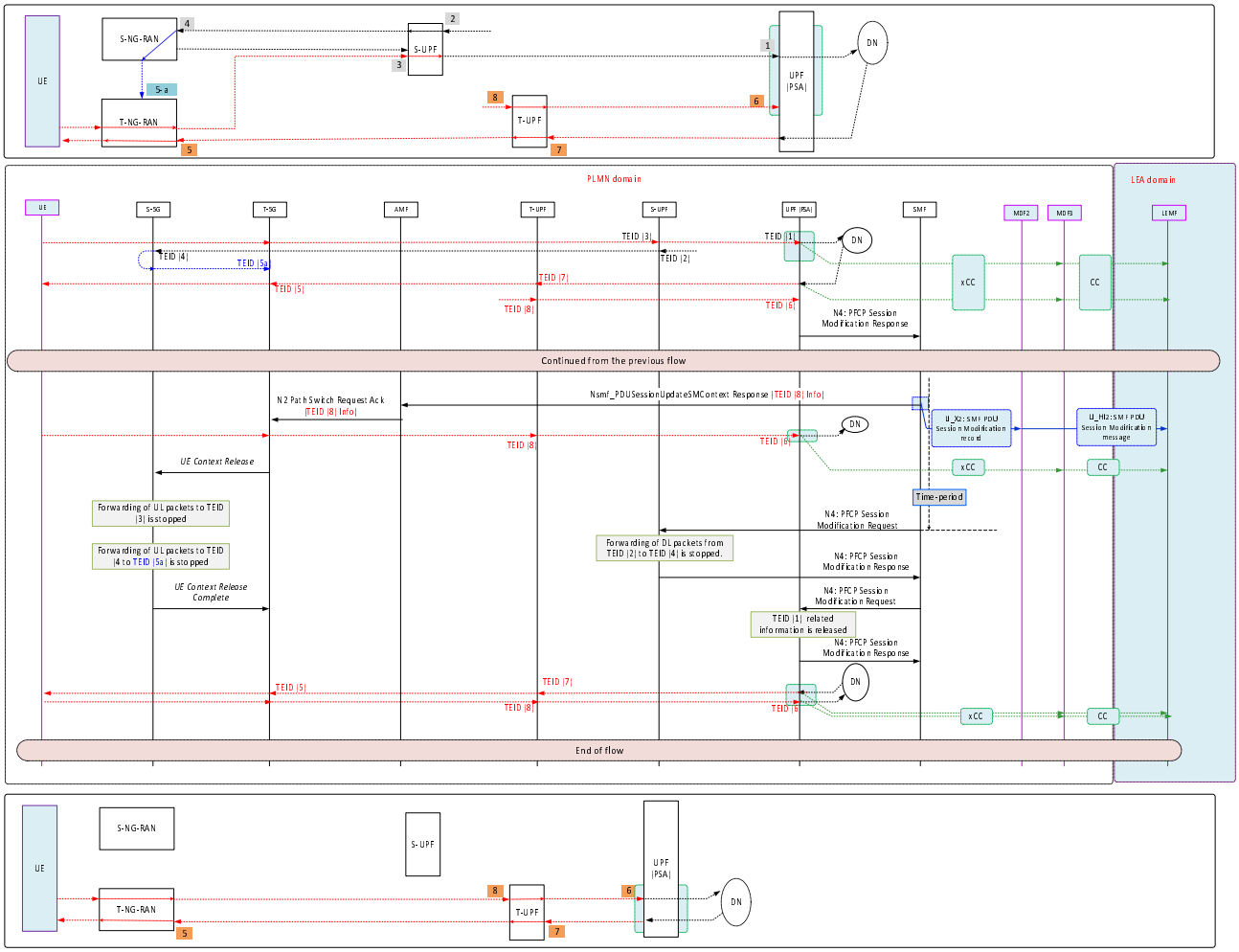


Figure 5.3.2.3.2-3: Xn-based handover with UPF relocation (non-roaming) – flow 3 of 3

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

5.3.2.4 Detailed flows - home-routed roaming, no UPF relocation

5.3.2.4.1 General

An Xn-based handover may happen without a change in the UPF (i.e. without an UPF relocation). In this illustration, LI applies to the VPLMN. No V-UPF relocation happens during the handover.

Within drawings, the (source) NG-RAN is shown as S-NG-RAN and as S-5G. Likewise, the (target) NG-RAN is shown as T-NG-RAN and as T-5G. The numbers 1, 2, 3, 4, 5, 5-a, 6, 7, 8 correspond to the TEID (1), TEID (2), TEID (3) and TEID (4), TEID (5) and TEID (5a) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at V-UPF and the TEID (3) is the tunnel end point identifier at H-UPF for UL UP packets. The TEID (4) is the tunnel end point identifier at V-UPF and TEID (2) is the tunnel end point identifier at the (source) NG-RAN for the DL UP packets. The TEID (5) is the tunnel end point identifier at the (target) NG-RAN for DL UP packets. The TEID (5a) is the tunnel end point identifier at the (target) NG-RAN for the DL data forwarded UP packets. The DL data forwarding happens from (source) NG-RAN to (target NG-RAN).

5.3.2.4.2 LI in VPLMN

The generation of xCC from the UP packets is done by the CC-POI present in the V-UPF. It is assumed that the PFCP session ID is used as the target identifier and therefore, the V-UPF would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

This flow is split into two parts, for easy reading purpose. The flow-diagram in figure 5.3.2.4.2-1 illustrates the first part and the flow diagram in figure 5.3.2.4.2-2 illustrates the second part.

Flow 1 of 2

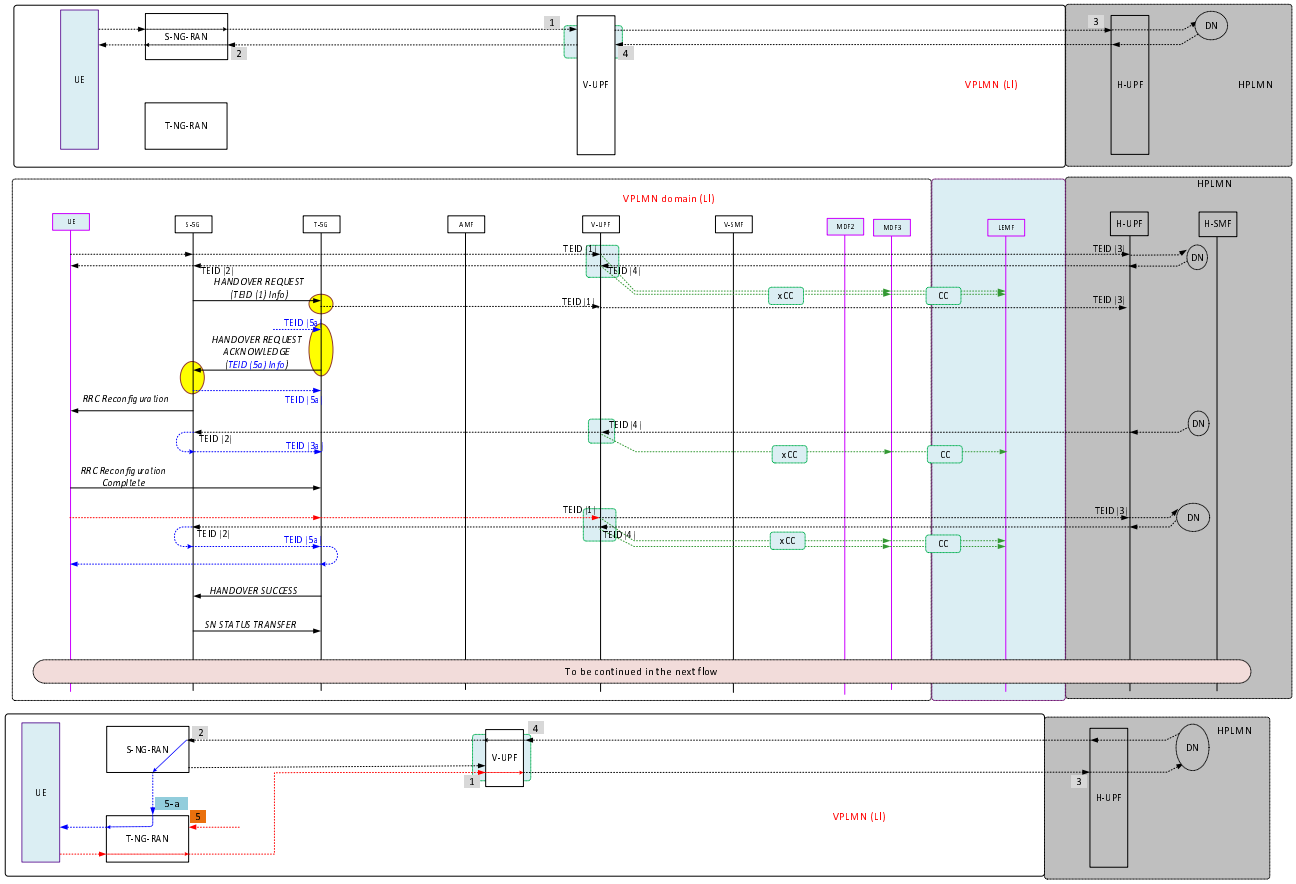


Figure 5.3.2.4.2-1: Xn-based handover with no UPF relocation (LI in VPLMN) – flow 1 of 2

Flow 2 of 2

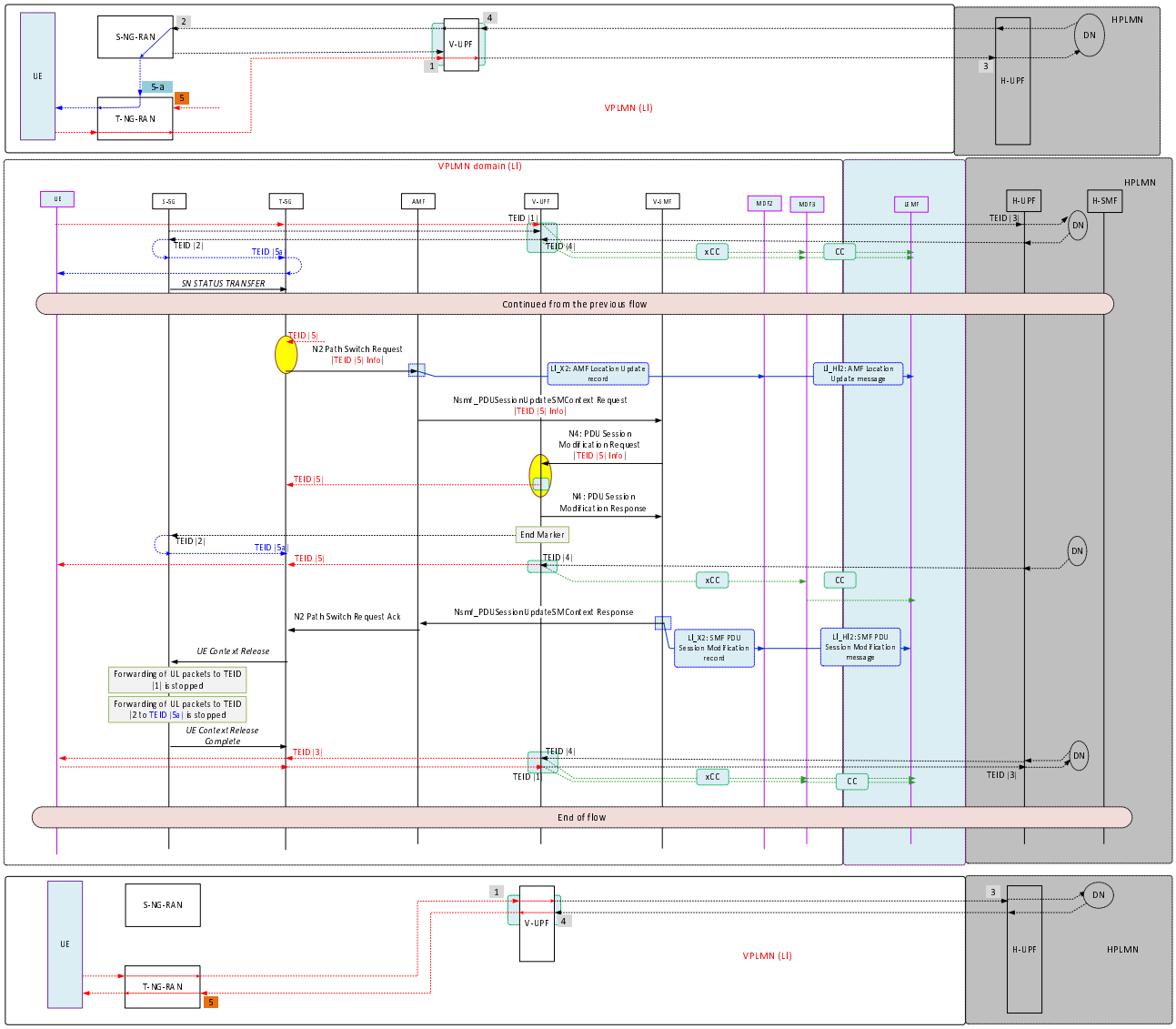


Figure 5.3.2.4.2-2: Xn-based handover with no UPF relocation (LI in VPLMN) – flow 2 of 2

When the AMF receives a Path Switch Request from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF Location Update record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

5.3.2.4.3 LI in HPLMN

The generation of xCC from the UP packets is done by the CC-POI present in the H-UPF.

This flow is split into two parts, for easy reading purpose. The flow-diagram in figure 5.3.2.4.3-1 illustrates the first part and the flow diagram in figure 5.3.2.4.3-2 illustrates the second part.

Flow 1 of 2

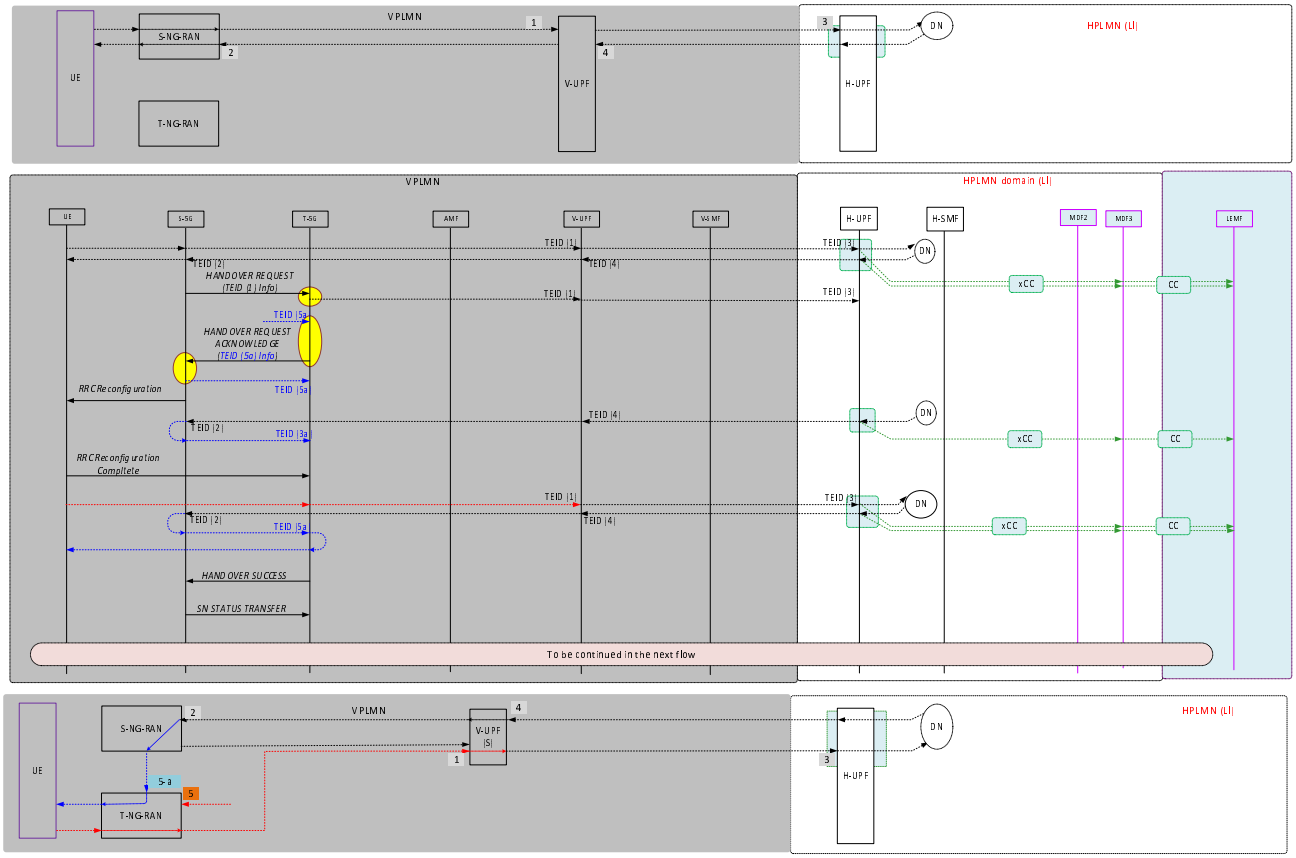


Figure 5.3.2.4.3-1: Xn-based handover with no UPF relocation (LI in HPLMN) – flow 1 of 2

Flow 2 of 2

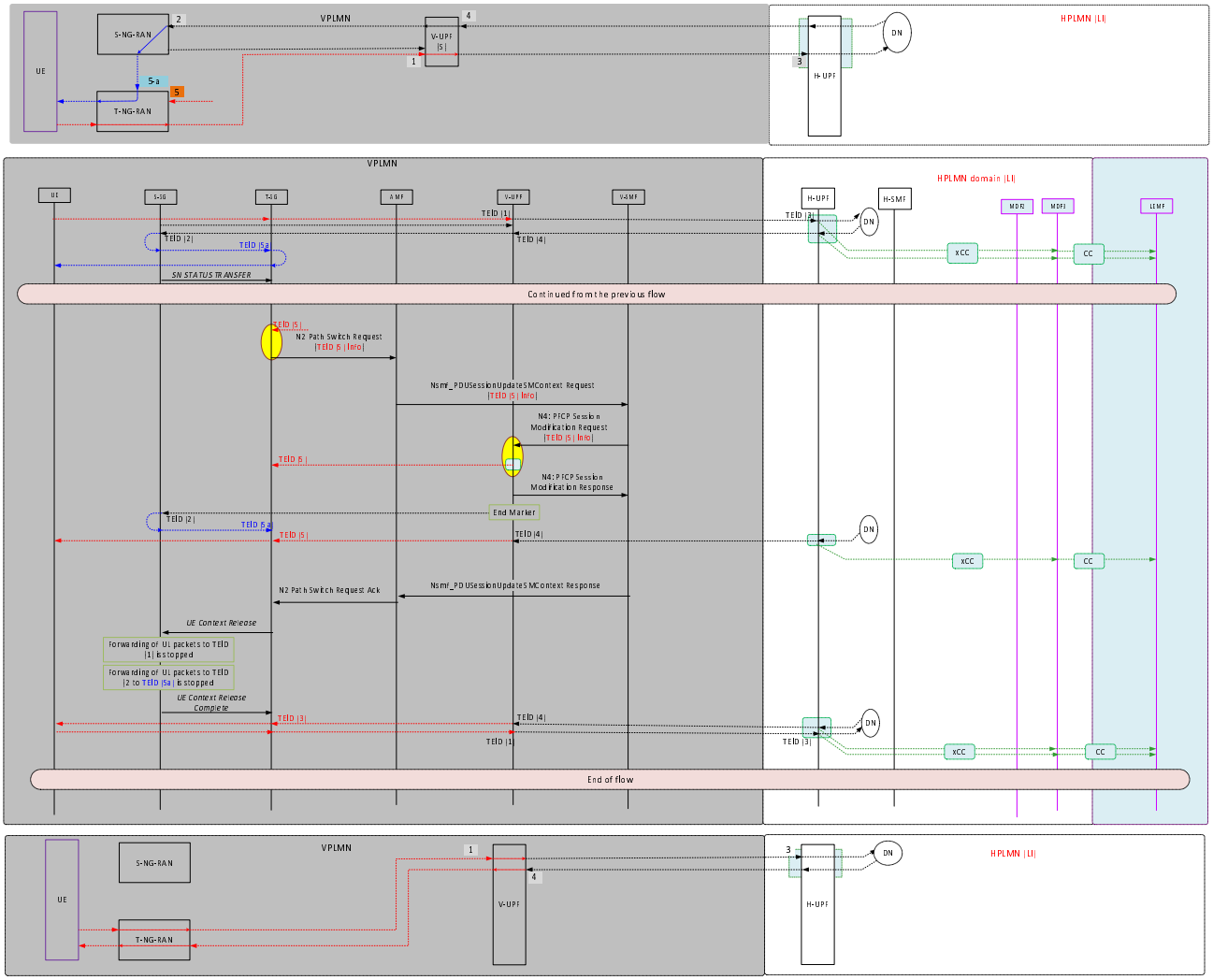


Figure 5.3.2.4.3-2: Xn-based handover with no UPF relocation (LI in HPLMN) – flow 2 of 2

As can be seen in the two flows, the fact that an Xn-based handover happened is not known at the HPLMN. Therefore, other than continuing the interception, there is nothing to report for an HPLMN in this scenario.

5.3.2.5 Detailed flows - home-routed roaming, UPF relocation

5.3.2.5.1 General

An Xn-based handover may happen with the change in the UPF (i.e. with an UPF relocation). In this illustration, the LI applies to the VPLMN and the V-UPF changes during the handover.

Within drawings, (source) NG-RAN is shown as S-NG-RAN and as S-5G and target NG-RAN is shown as T-NG-RAN and as T-5G. The V-UPF used before the handover is shown as (S) V-UPF and the V-UPF used after the handover is shown as (T) V-UPF. The numbers 1, 2, 3, 4, 5, 5-a, 6, 7 correspond to the TEID (1), TEID (2), TEID (3) and TEID (4), TEID (5), TEID (5-a), TEID (6) and TEID (7) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at (S) V-UPF and the TEID (3) is the tunnel end point identifier at H-UPF for UL UP packets. The TEID (4) is the tunnel end point identifier at (S) V-UPF and TEID (2) is the tunnel end point identifier at the (source) NG-RAN for the DL UP packets. The TEID (5a) is the tunnel end point identifier at the (target) NG-RAN for the DL data forwarded UP packets. The DL data forwarding happens from(source) NG-RAN to (target NG-RAN).

The TEID (6) is the tunnel end point identifier at (T) V-UPF and the TEID (5) is the tunnel end point identifier at the target NG-RAN for DL UP packets. The TEID (7) is the tunnel end point identifier at (T) V-UPF the UL UP packets. The TEID (3) the tunnel end point identifier for UL UP packets at the H-UPF is used as the same after the handover as well.

5.3.2.5.2 LI in VPLMN

The generation of xCC happens at (S) V-UPF before the handover and at (T) V-UPF after the handover. It is assumed that the PFCP session ID is used as the target identifier.

This flow is split into three parts, for easy reading purpose. The flow-diagram in figure 5.3.2.5.2-1 illustrates the first part, flow-diagram in figure 5.3.2.5.2-2 illustrates the second part and the flow-diagram in figure 5.3.2.5.2-3 illustrates the third part.

Flow 1 of 3

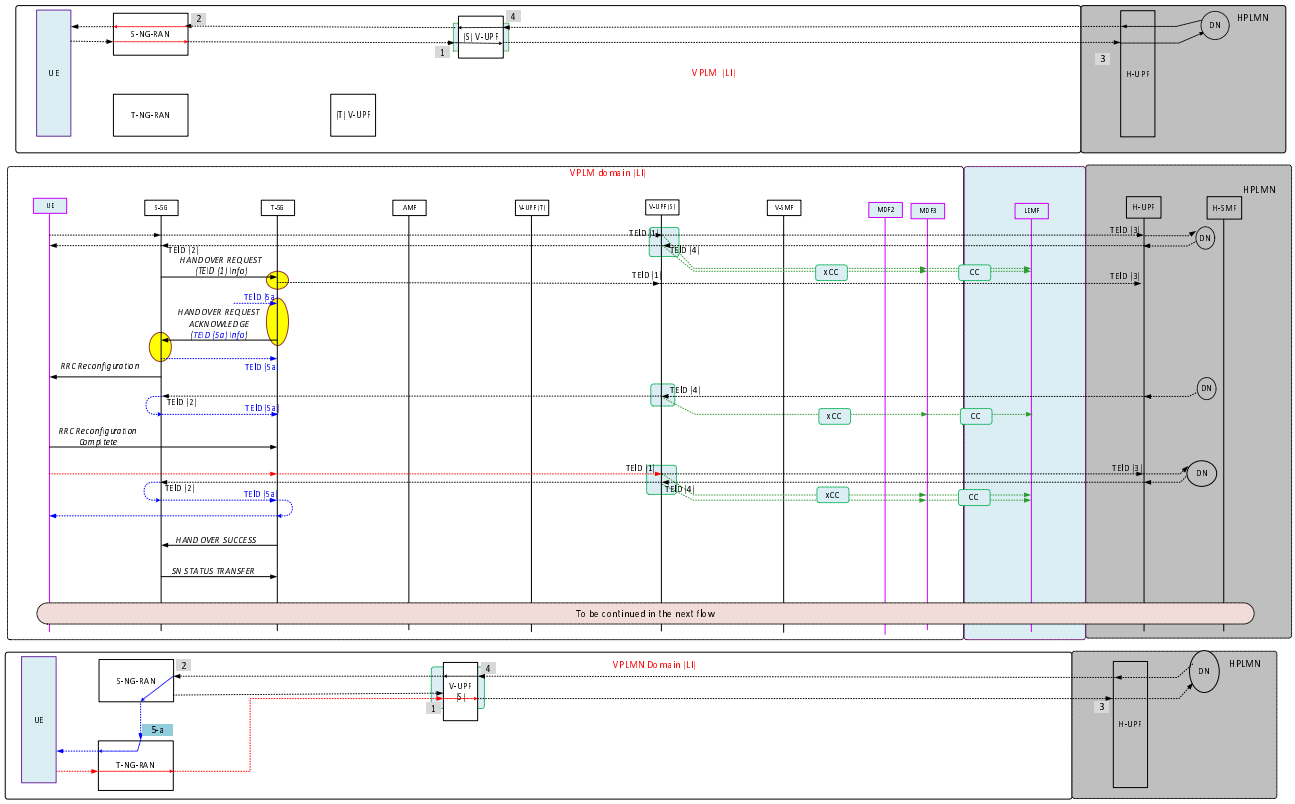


Figure 5.3.2.5.2-1: Xn-based handover with UPF relocation (LI in VPLMN) – flow 1 of 3

Flow 2 of 3

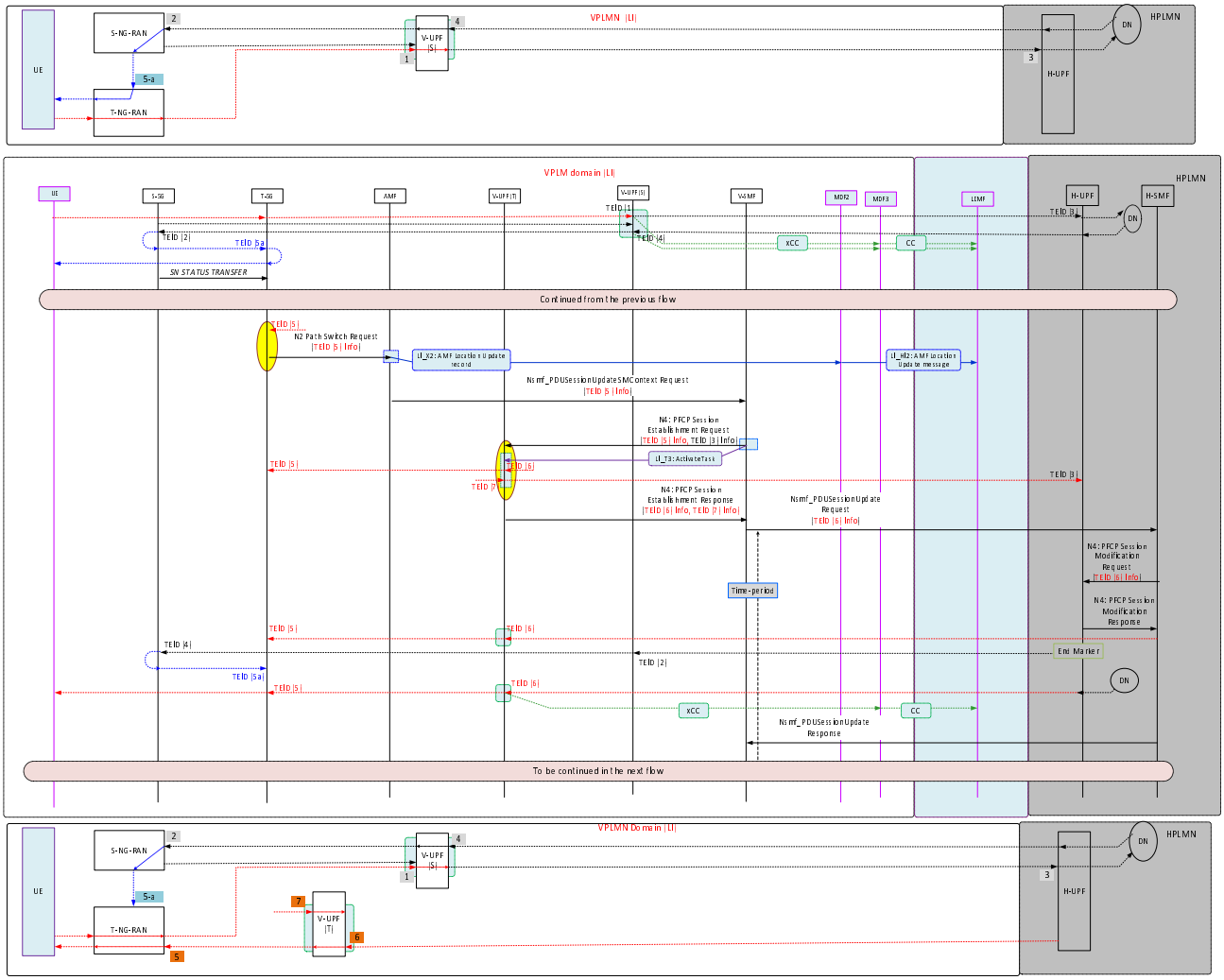


Figure 5.3.2.5.2-2: Xn-based handover with UPF relocation (LI in VPLMN) – flow 2 of 3

When the AMF receives a Path Switch Request from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF Location Update record.

When the PFCP Session Establishment is sent to the (T) V-UPF, the CC-TF in the SMF sends the LI_T3 ActivateTask to the CC-POI present in the (T) V-UPF to start the CC interception. The same correlation number is used for xCC before and after the handover.

Flow 3 of 3

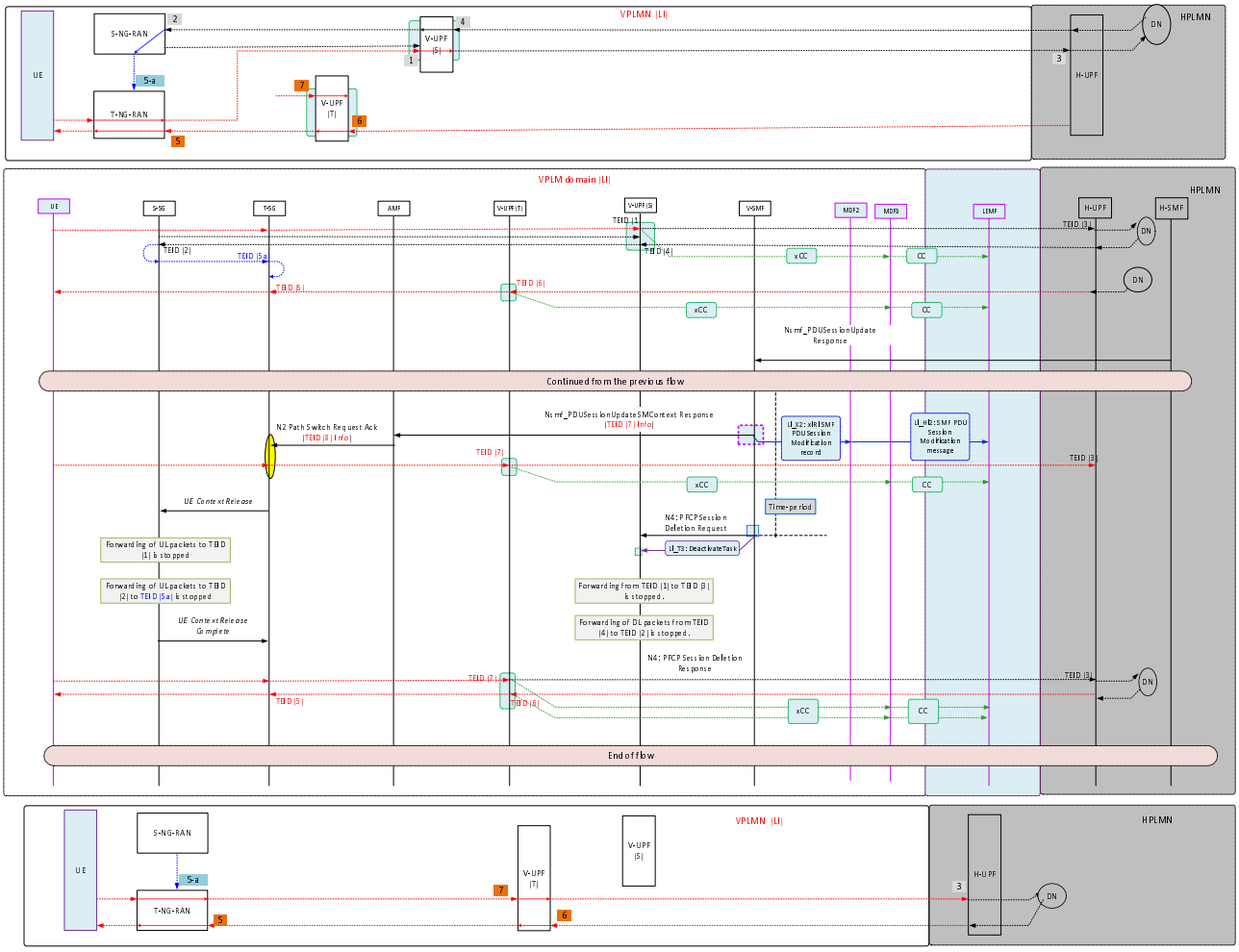


Figure 5.3.2.5.2-3: Xn-based handover with UPF relocation (LI in VPLMN) – flow 3 of 3

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDUSessionModification record.

When the SMF sends the N4: PFCP Session Deletion Request to the (S) V-UPF, the CC-TF present in the SMF sends the LI_T3: DeactivateTask to the CC-POI present in the (S) V-UPF to stop the CC interception since (S) V-UPF is no longer used after the handover.

5.3.2.5.3 LI in HPLMN

The generation of xCC happens at H-UPF. It is assumed that the PFCP session ID is used as the target identifier.

This flow is split into three parts, for easy reading purpose. The flow-diagram in figure 5.3.2.5.3-1 illustrates the first part, flow-diagram in figure 5.3.2.5.3-2 illustrates the second part and the flow-diagram in figure 5.3.2.5.3-3 illustrates the third part.

Flow 1 of 3

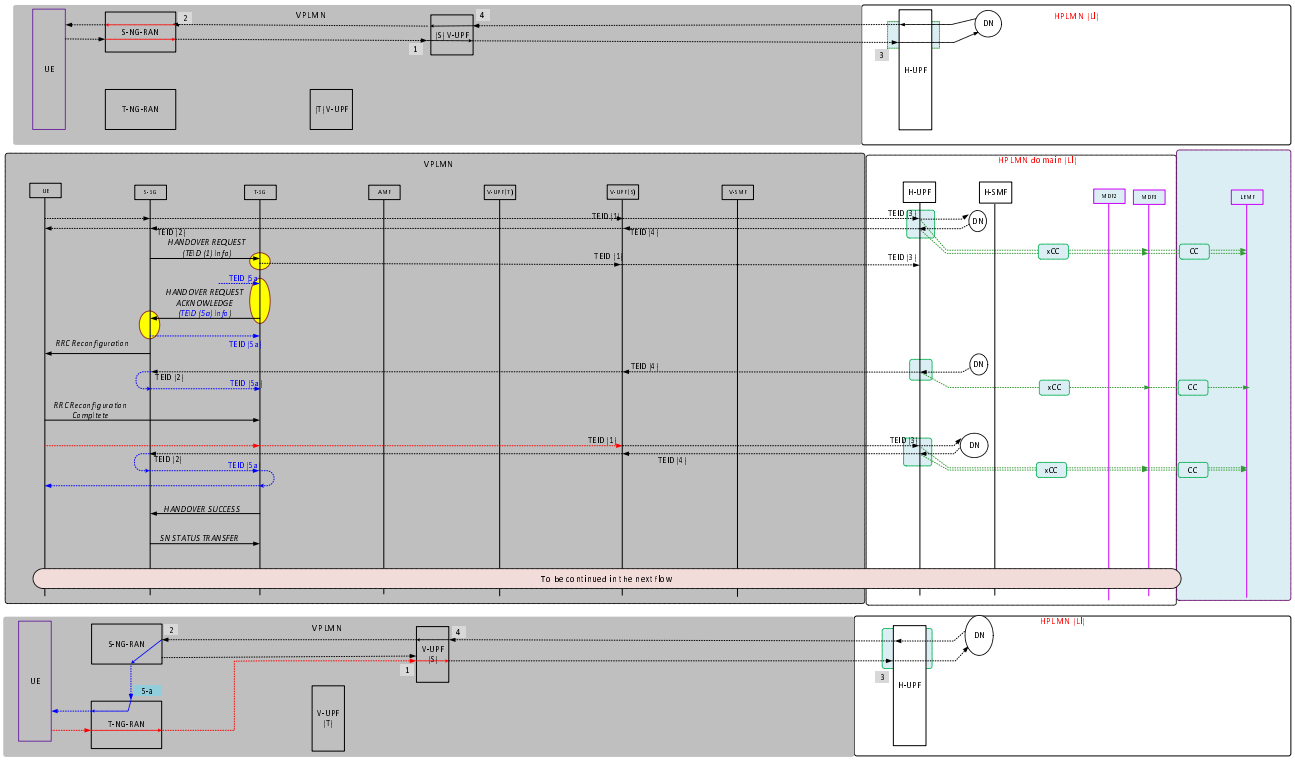


Figure 5.3.2.5.3-1: Xn-based handover with no UPF relocation (LI in HPLMN) – flow 1 of 3

Flow 2 of 3

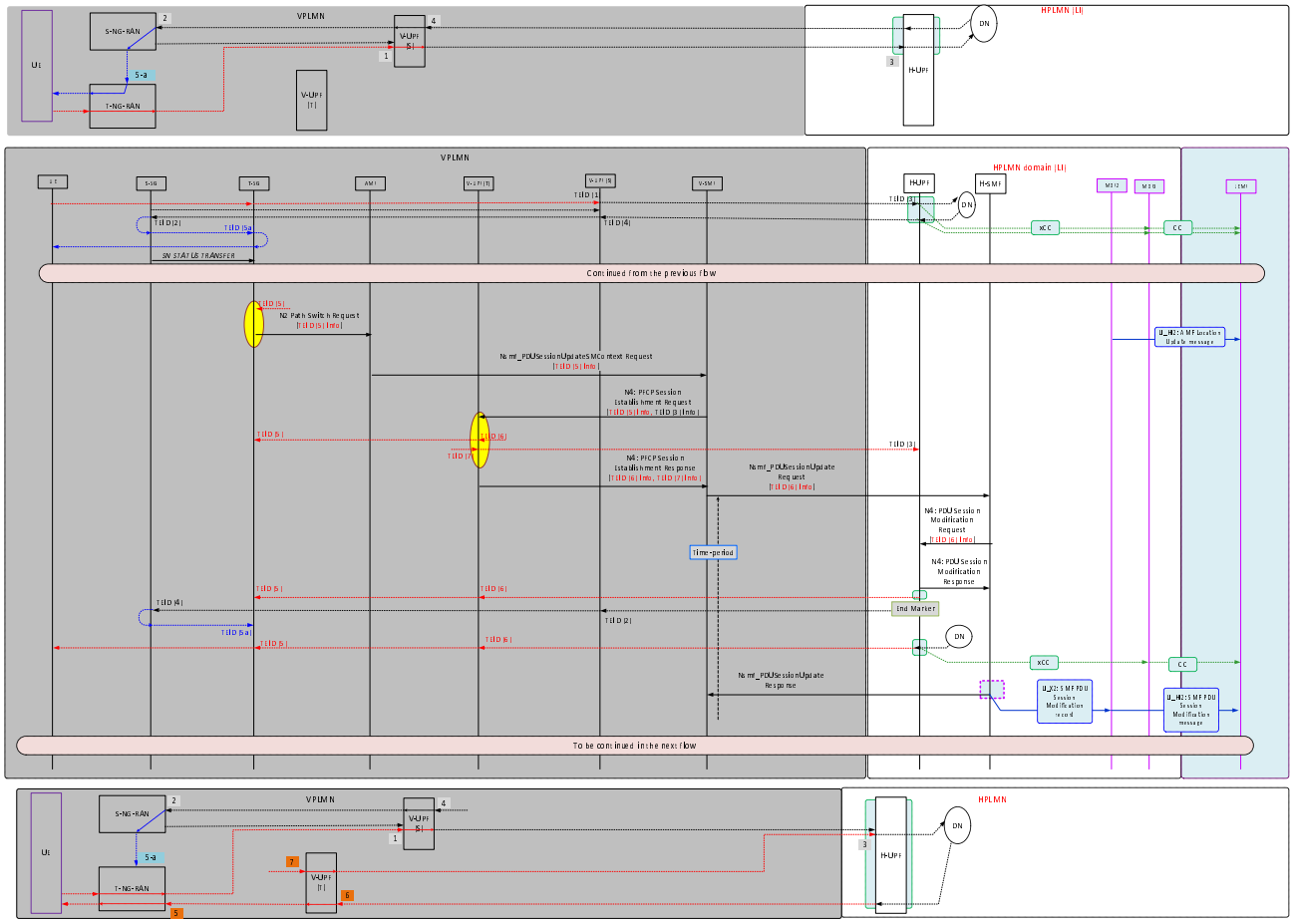


Figure 5.3.2.5.3-2: Xn-based handover with no UPF relocation (LI in HPLMN) – flow 2 of 3

When the H-SMF sends the Nsmf_PDUSessionUpdate Response to the V-SMF, the IRI-POI present in the H-SMF generates the xIRI SMF PDU Session Modification record.

Flow 3 of 3

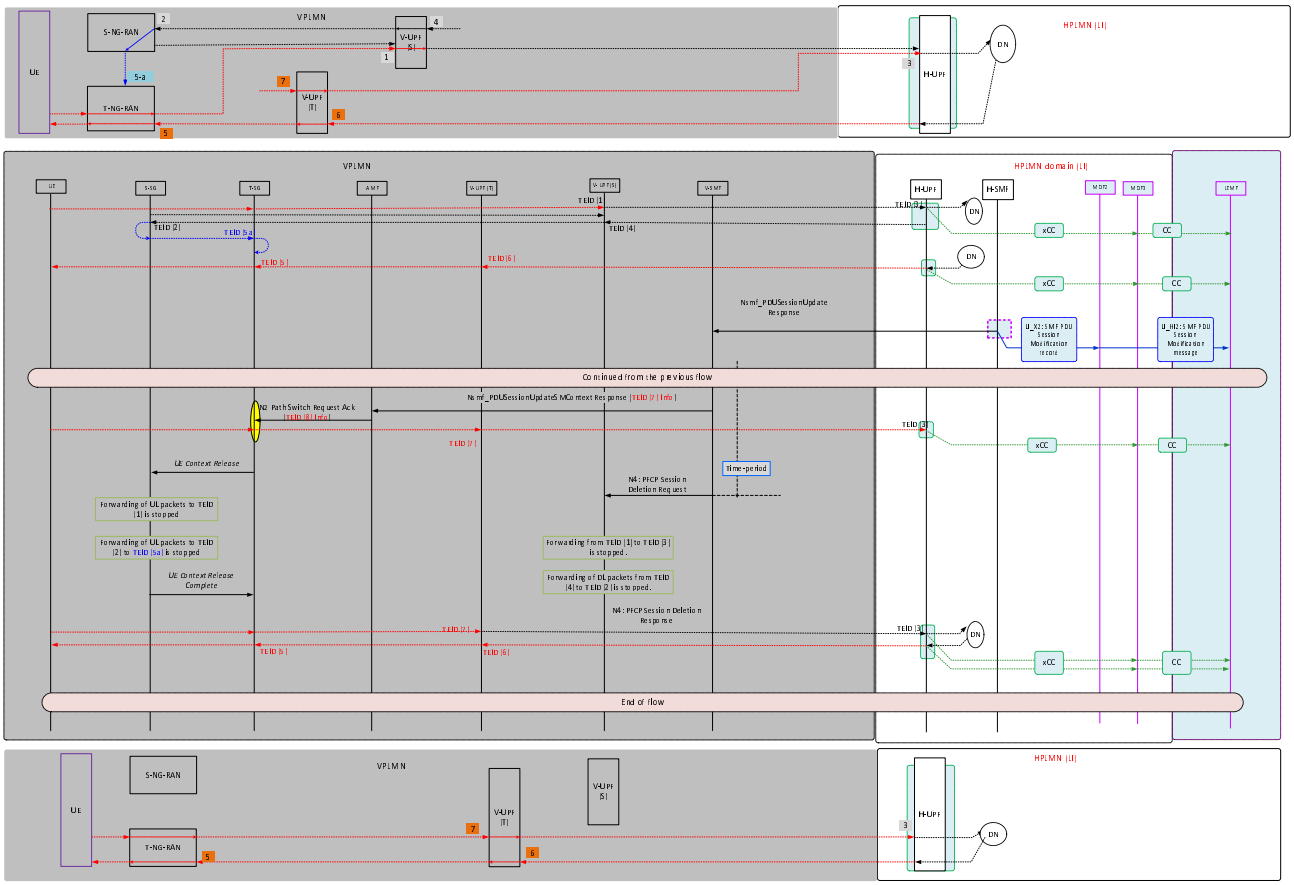


Figure 5.3.2.5.3-3: Xn-based handover with UPF relocation (LI in VPLMN) – flow 3 of 3

5.3.3 N2-based handover flows

5.3.3.1 Overview

5.3.3.1.1 General

The N2 is an interface between the NG-RAN and the AMF. As described in TS 23.502 [7] clause 4.9.1.3, the N2-based handover has to be used when there is no Xn connectivity between the NG-RANs involved in the handover. This procedure is also needed for inter-AMF mobility.

Handover procedures apply to all QoS flows in the PDU session. The QoS flows that are not handed over are released, The CC interception happens in the anchor UPF in the non-roaming case.

During handover to maintain the data flow continuity, the DL data may be forwarded from the old NG-RAN to the NG-RAN.

Depending on the UPF relocation needs, whether or not a separate intermediate UPF is used for DL data forwarding, several scenarios can happen.

5.3.3.1.2 Direct DL data forwarding (non-roaming)

In all the illustrations shown in this clause, the DL data forwarding happens directly from source NG-RAN to target NG-RAN. The CC interception in these cases always happen in the anchor UPF.

Scenario 1

Figure 5.3.3.1.2-1 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is for a non-roaming scenario. There are no intermediate UPFs in this case.

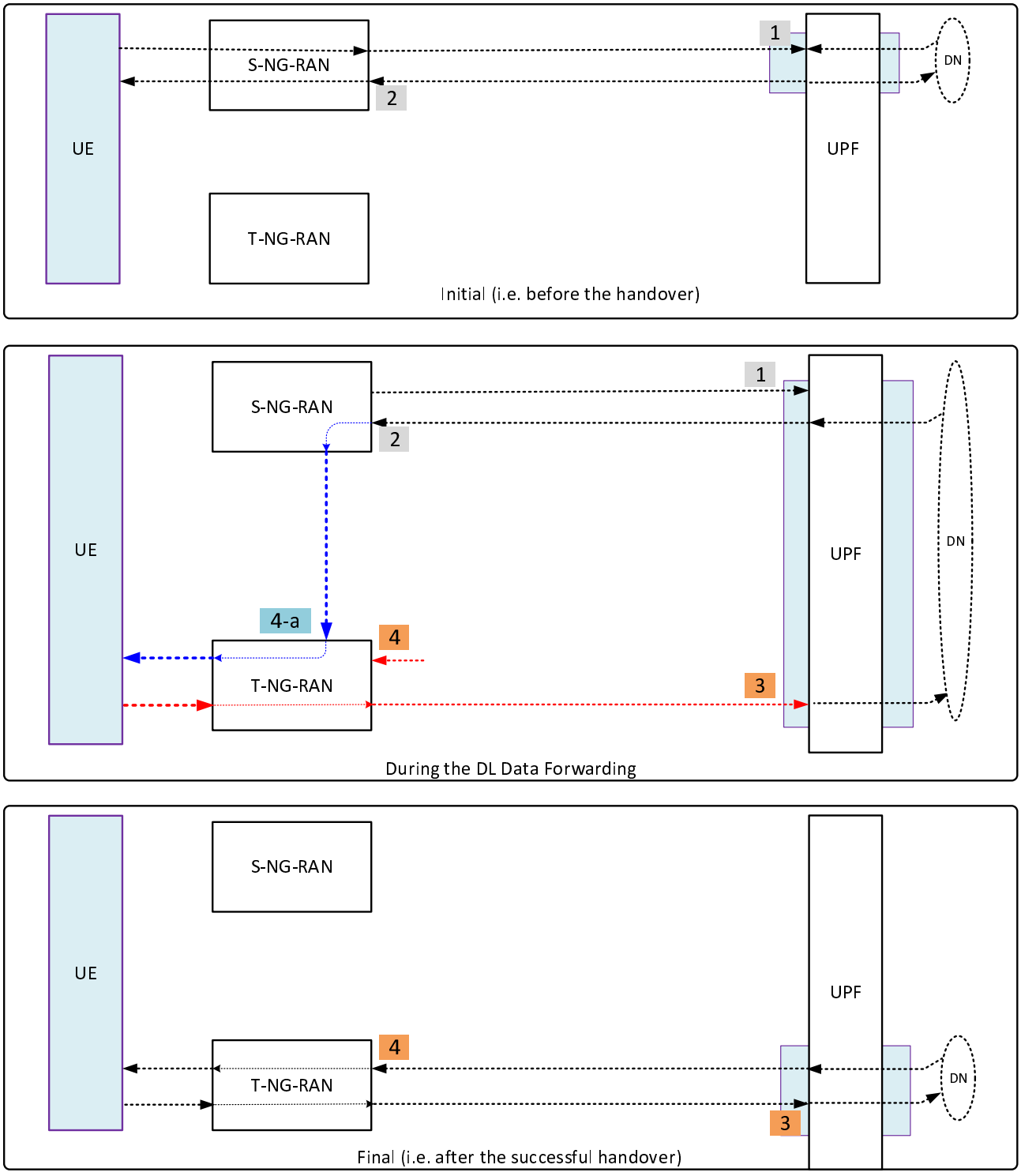


Figure 5.3.3.1.2-1: User plane packet flow during handover direct DL data forward (non-roaming) – case 1

The detailed flow diagrams that depict the above scenario are illustrated in clause 5.3.3.2.2.

Scenario 2

Figure 5.3.3.1.2-2 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an intermediate UPF

present, the relocation of the same does not happen during the handover. This intermediate UPF is shown as S-UPF in figure 5.3.3.1.2-2.

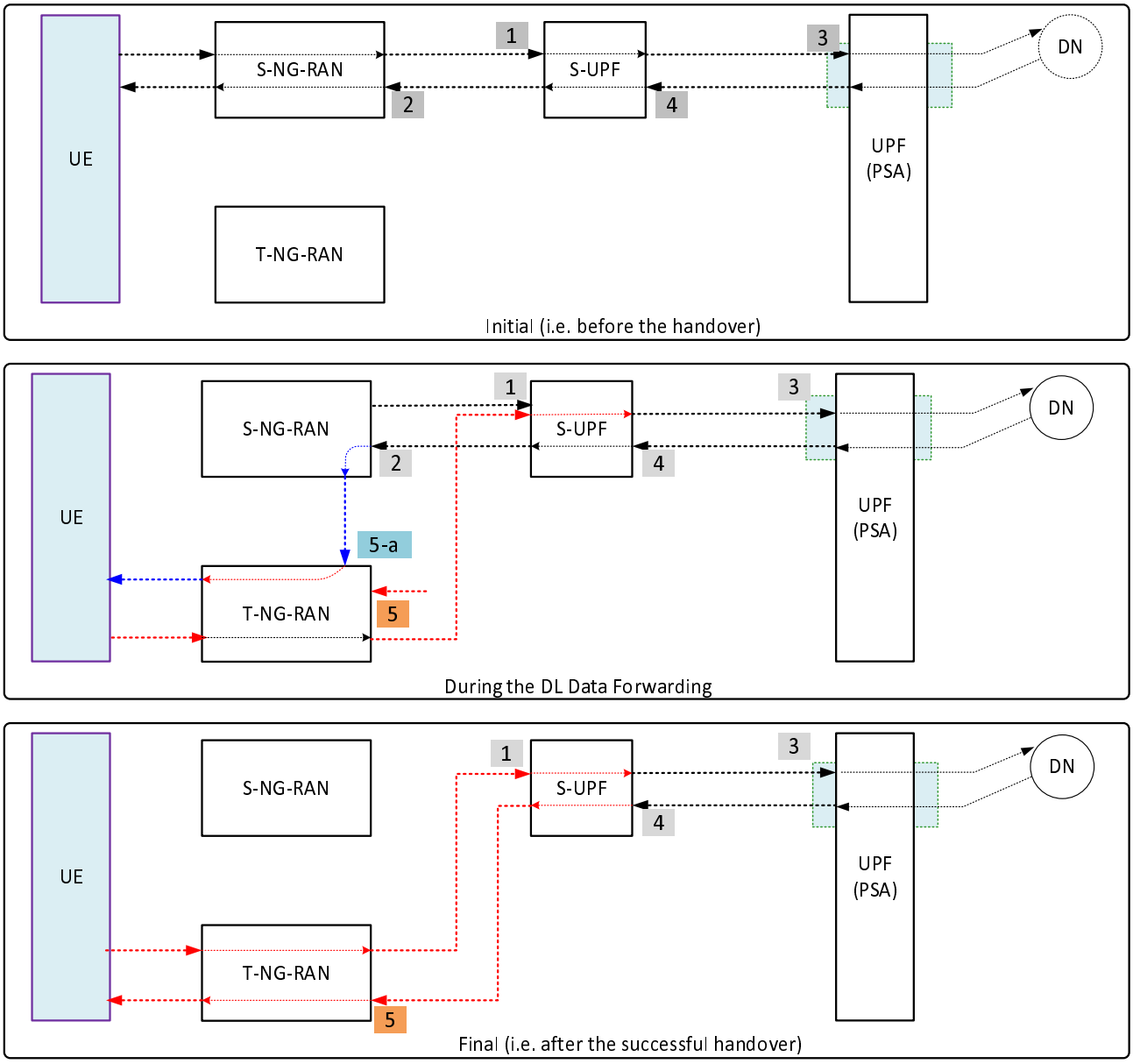


Figure 5.3.3.1.2-2: User plane packet flow during handover direct DL data forward (non-roaming) – case 2

Scenario 3

Figure 5.3.3.1.2-3 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an intermediate UPF present after the handover. This intermediate UPF is shown as T-UPF in figure 5.3.3.1.2-3 This is for a non-roaming scenario.

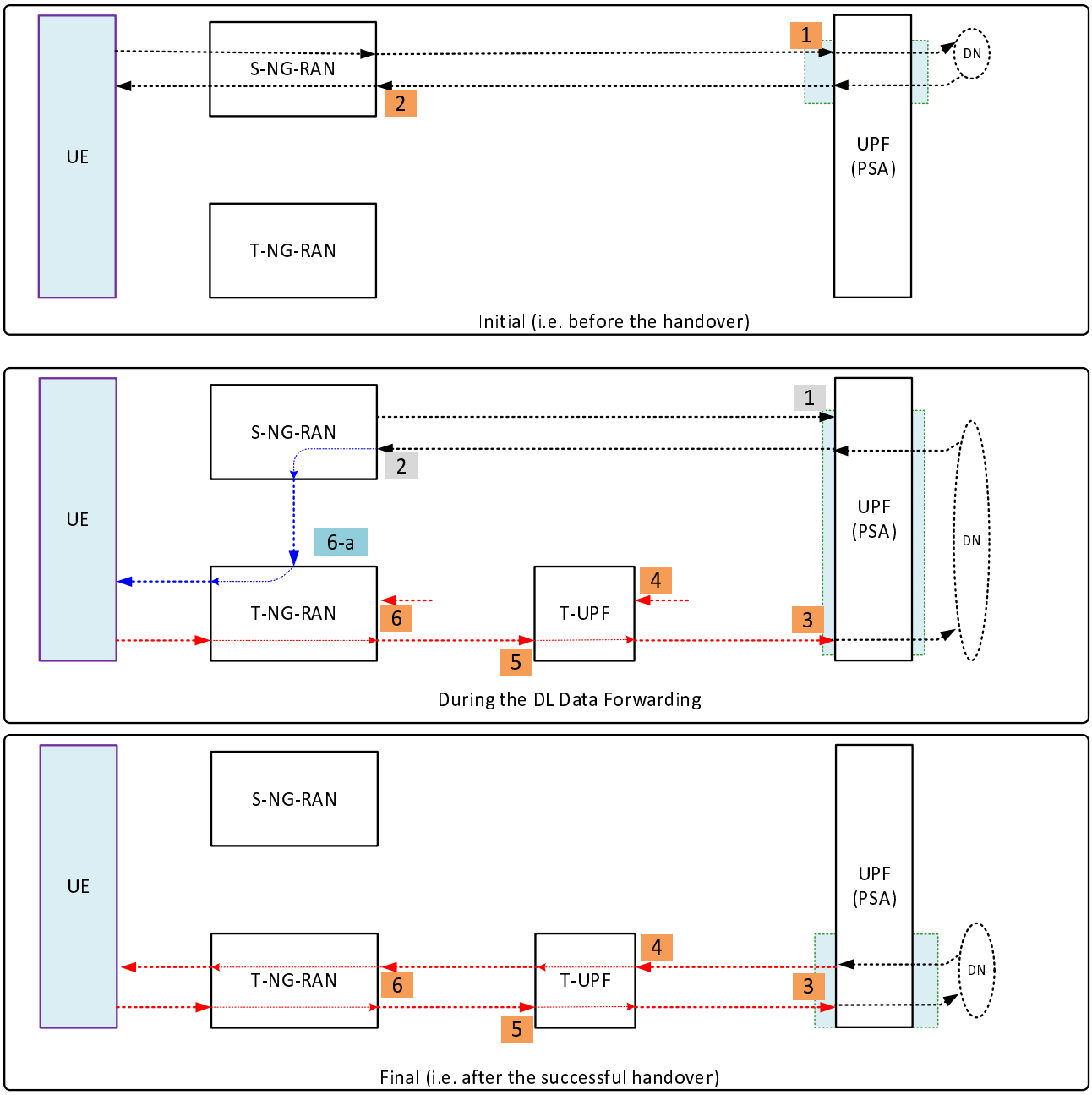


Figure 5.3.3.1.2-3: User plane packet flow during handover direct DL data forward (non-roaming) – case 3

Scenario 4

Figure 5.3.3.1.2-4 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an intermediate UPF which changes during the handover. The intermediate UPFs are shown as S-UPF (before the handover) and T-UPF (after the handover) in figure 5.3.3.1.2-4.

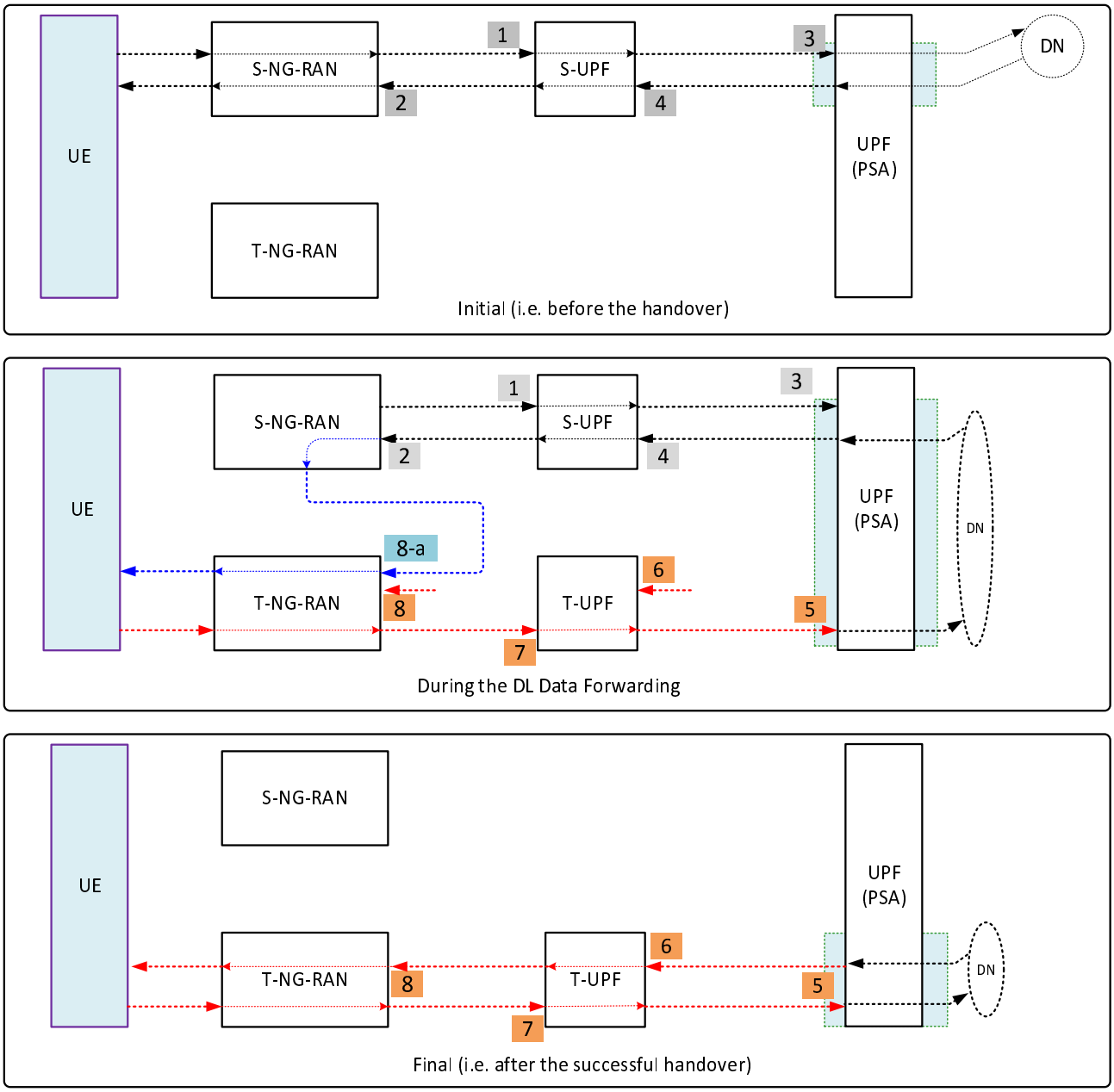


Figure 5.3.3.1.2-4: User plane packet flow during handover direct DL data forward (non-roaming) – case 4

The detailed flow diagrams that depict the above case are illustrated in clause 5.3.3.3.2.

5.3.3.1.3 Direct DL data forwarding (home-routed roaming, LI in VPLMN)

In all the illustrations shown in this clause, the DL data forwarding happens directly from source NG-RAN to target NG-RAN. The CC interception in these cases always happen in the V-UPF.

Scenario 1

Figure 5.3.3.1.3-1 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. The V-UPF remains the same after the handover.

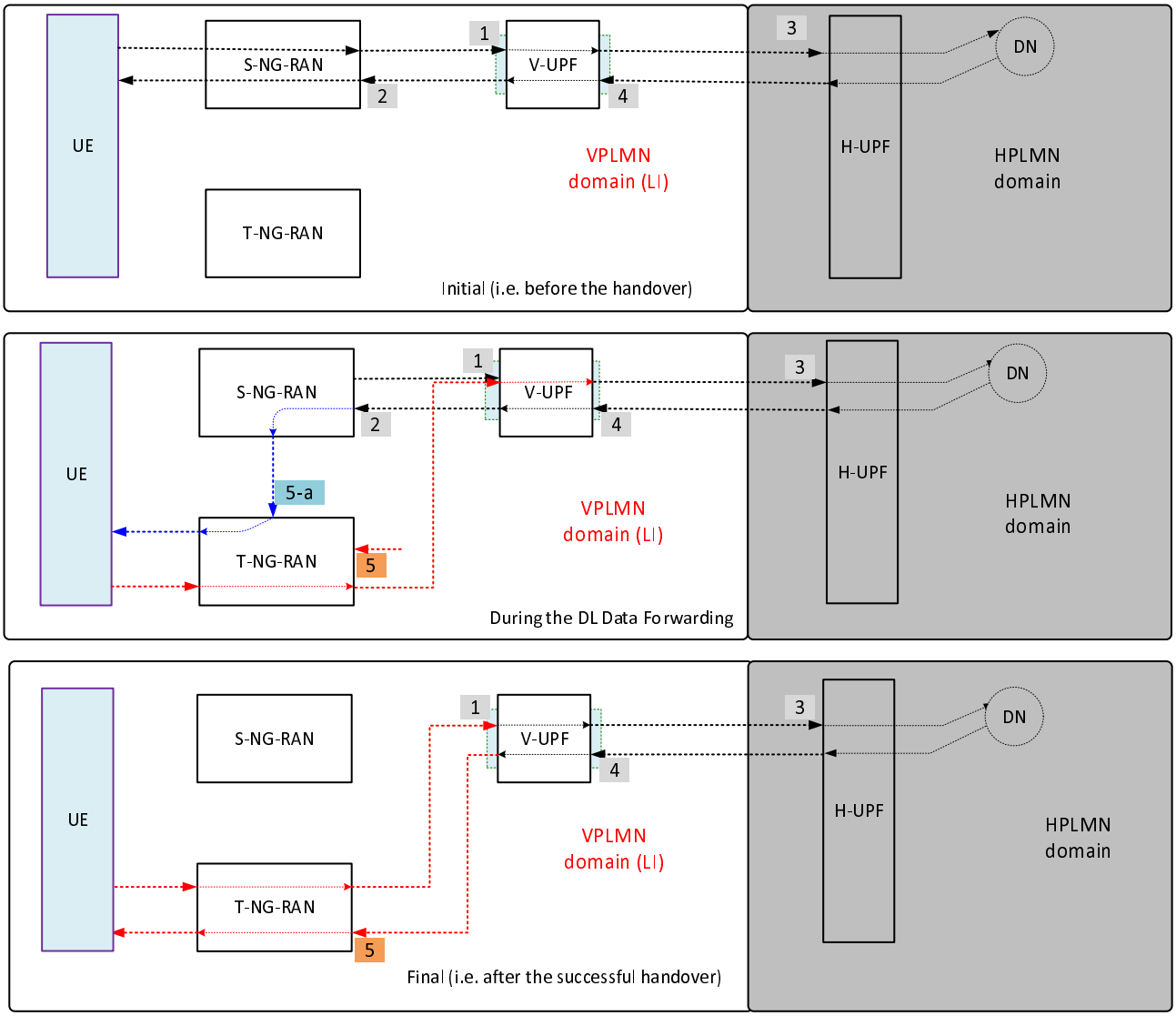


Figure 5.3.3.1.3-1: User plane packet flow during handover direct DL data forward (roaming, LI in VPLMN) – case 1

Scenario 2

Figure 5.3.3.1.3-2 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow when the V-UPF changes during the handover.

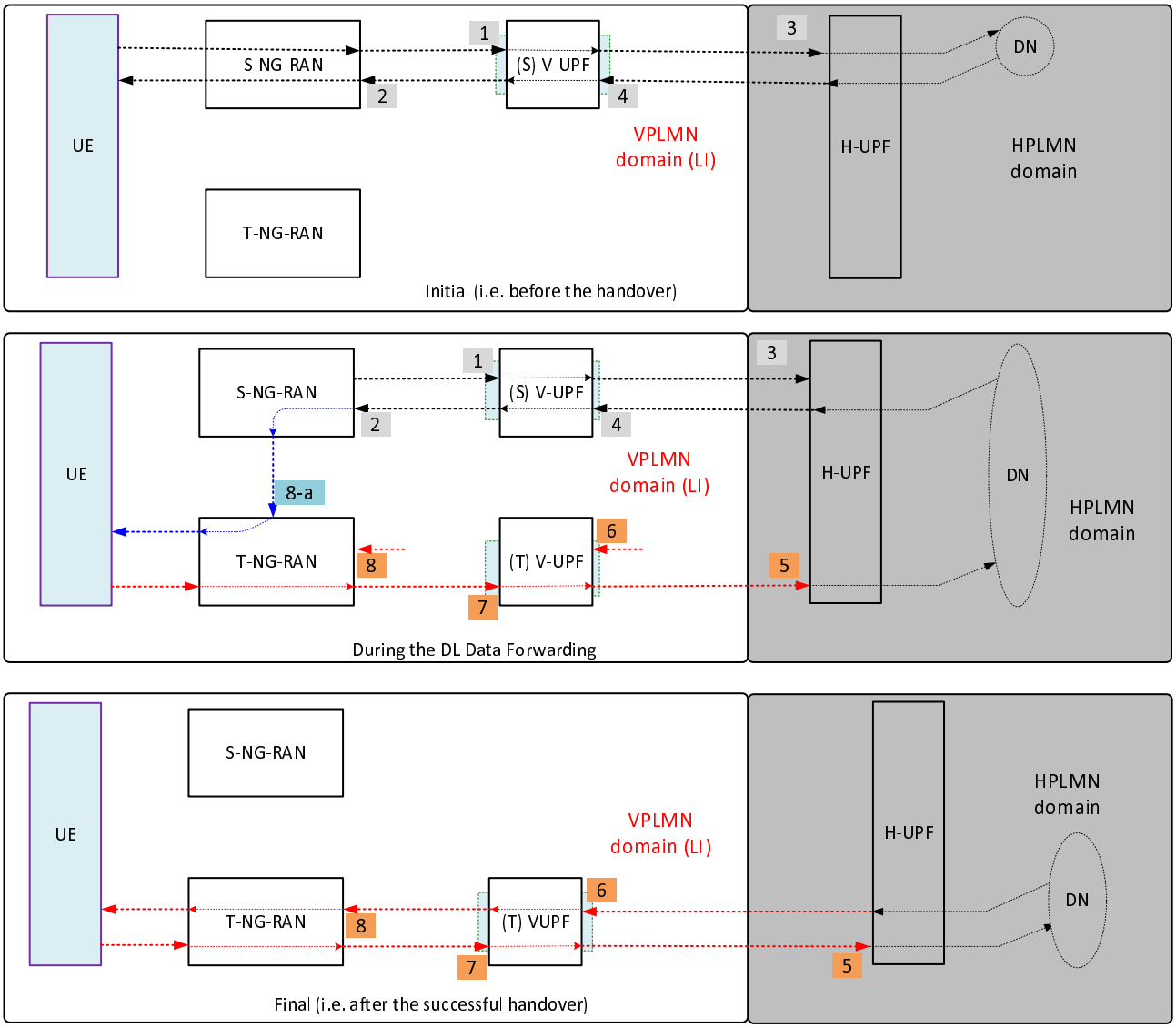


Figure 5.3.3.1.3-2: User plane packet flow during handover direct DL data forward (roaming, LI in VPLMN) – case 2

5.3.3.1.4 Direct DL data forwarding (home-routed roaming, LI in HPLMN)

In all the illustrations shown in this clause, the DL data forwarding happens directly from source NG-RAN to target NG-RAN. The CC interception in these cases always happen in the H-UPF.

Scenario 1

Figure 5.3.3.1.4-1 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. The V-UPF remains the same after the handover.

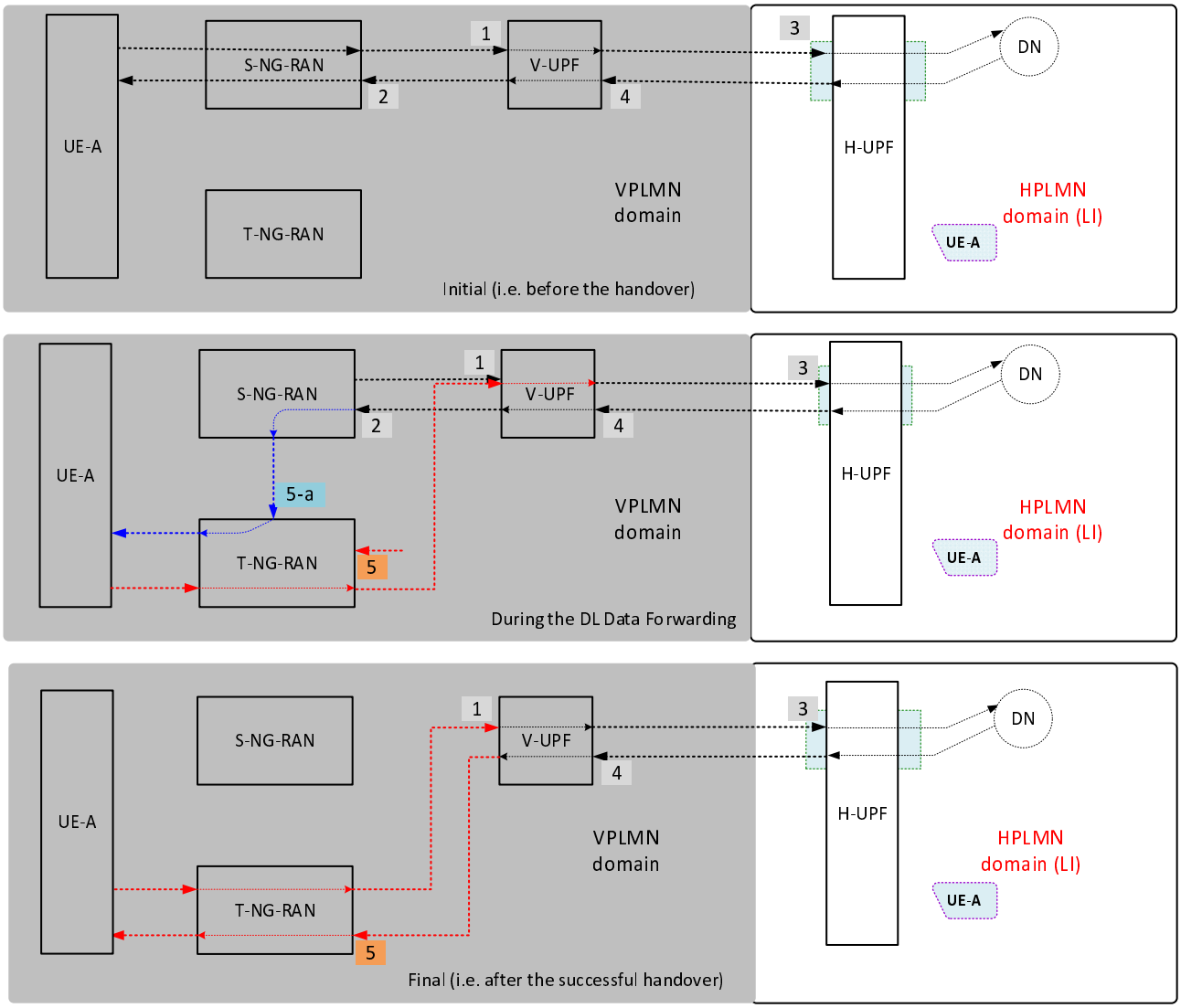


Figure 5.3.3.1.4-1: User plane packet flow during handover direct DL data forward (roaming, LI in HPLMN) – case 1

Scenario 2

Figure 5.3.3.1.4-2 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow when the V-UPF changes during the handover.

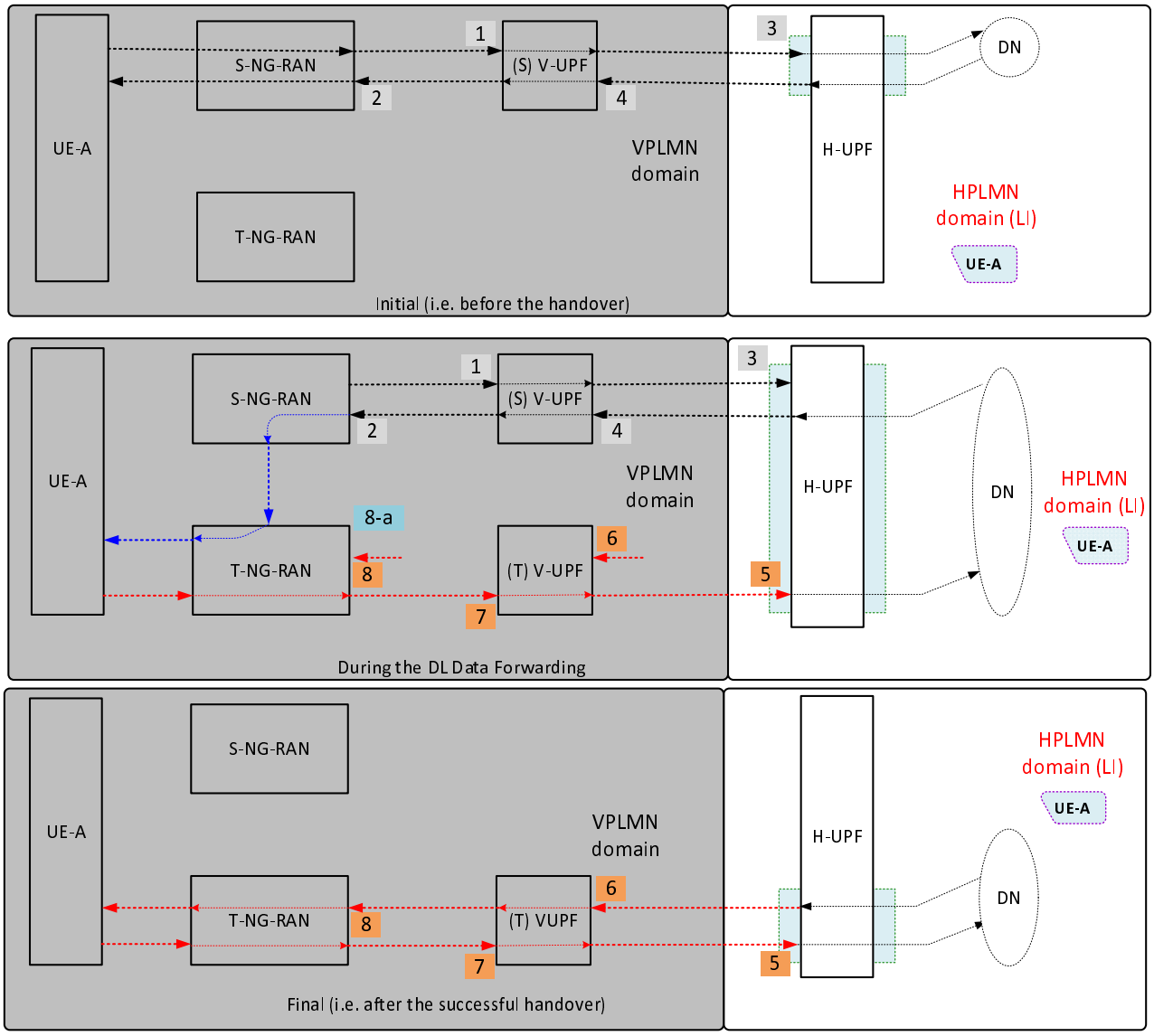


Figure 5.3.3.1.4-2: User plane packet flow during handover direct DL data forward (roaming, LI in HPLMN) – case 2

5.3.3.1.5 Indirect DL data forwarding (non-roaming)

In all the illustrations shown in this clause, indirect DL data forwarding happens. The CC interception is always done at the anchor UPF.

Scenario 1

Figure 5.3.3.1.5-1 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow without an UPF relocation. An intermediate UPF (shown as T-UPF (I)) is used for DL data forwarding.

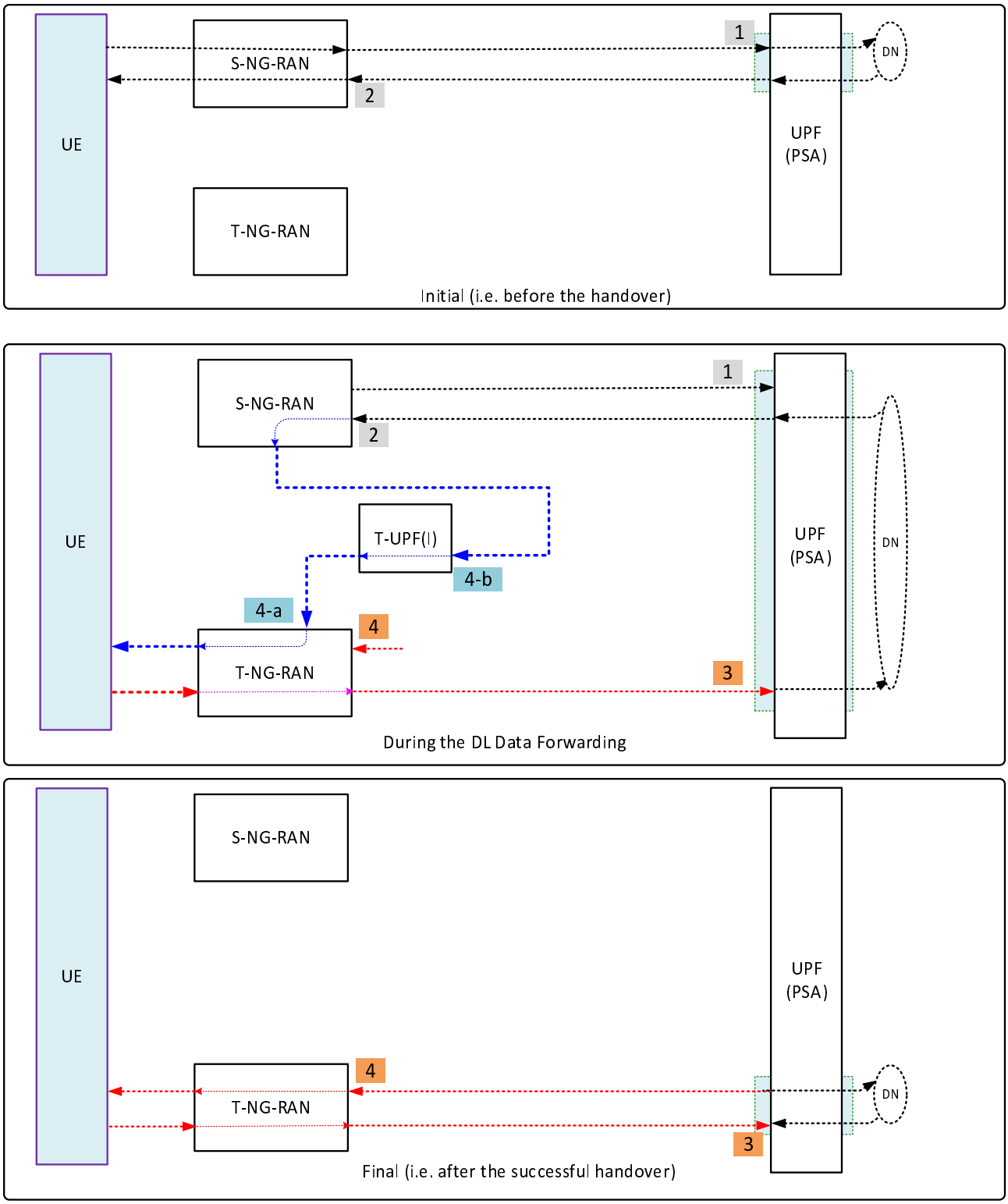


Figure 5.3.3.1.5-1: User plane packet flow during handover indirect DL data forward (non-roaming) – case 1

Scenario 2

Figure 5.3.3.1.5-2 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an intermediate UPF present, the relocation of the same does not happen during the handover. This intermediate UPF is shown as S-UPF in figure 5.3.3.1.5-2.

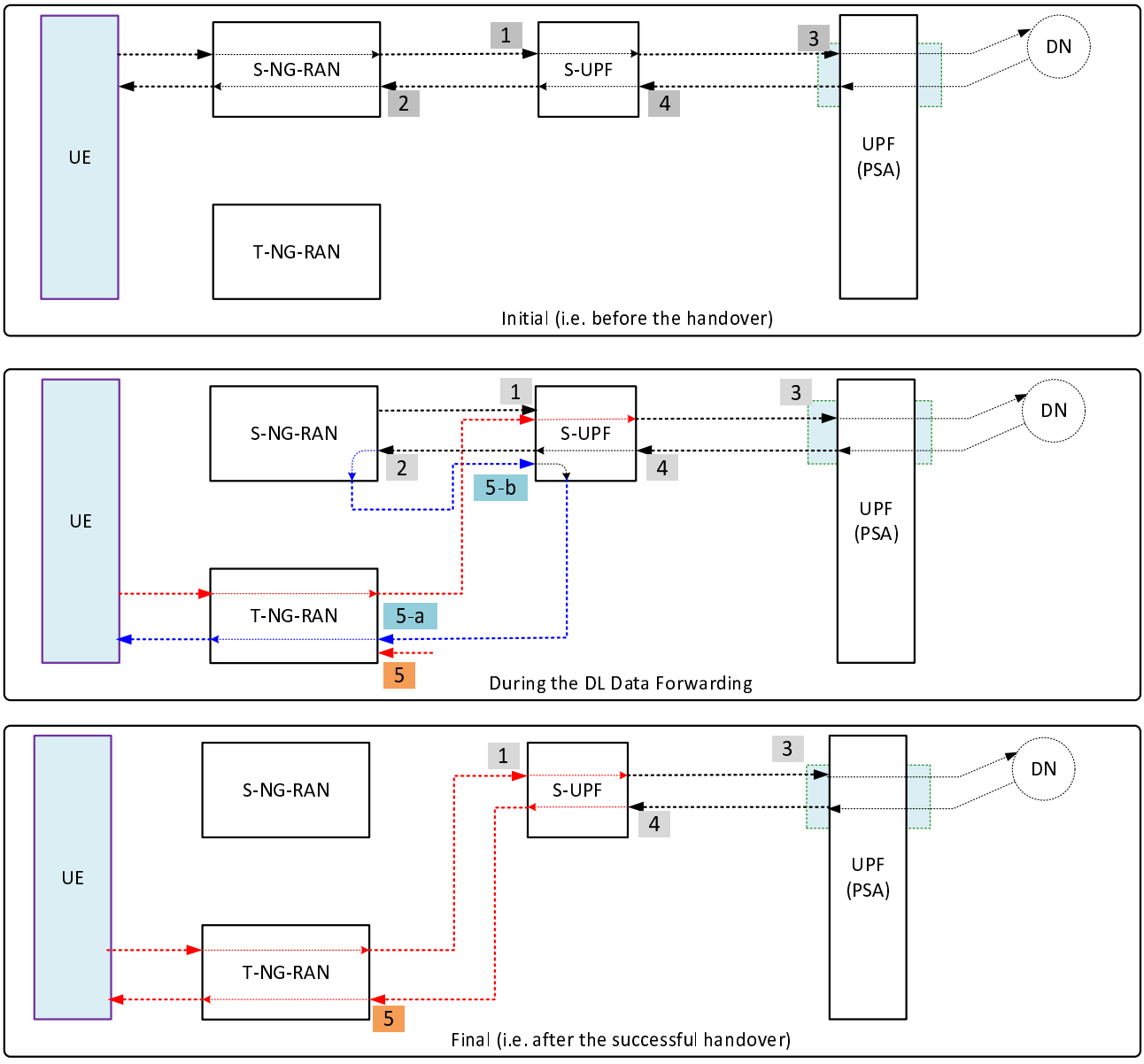


Figure 5.3.3.1.5-2: User plane packet flow during handover indirect DL data forward (non-roaming) – case 2

Scenario 3

Figure 5.3.3.1.5-3 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an intermediate UPF present, the relocation of the same does not happen during the handover. This intermediate UPF is shown as S-UPF in figure 5.3.3.1.5-3. A separate intermediate UPF (shown as S-UPF (I)) is used for DL data forwarding.

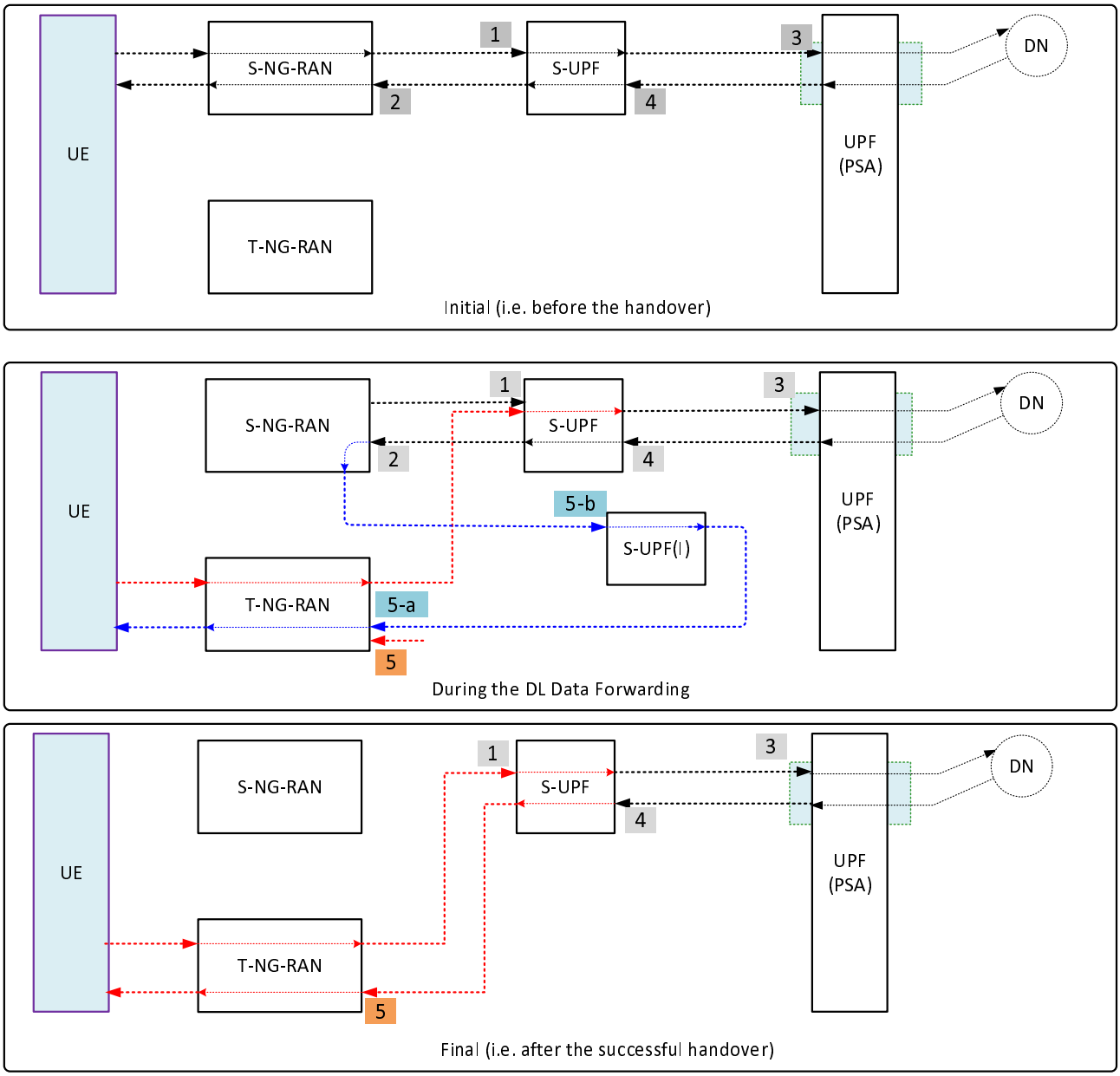


Figure 5.3.3.1.5-3: User plane packet flow during handover indirect DL data forward (non-roaming) – case 3

Scenario 4

Figure 5.3.3.1.5-4 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an UPF relocation in the sense an intermediate UPF is present after the handover. This intermediate UPF is shown as T-UPF in figure 5.3.3.1.5-4.

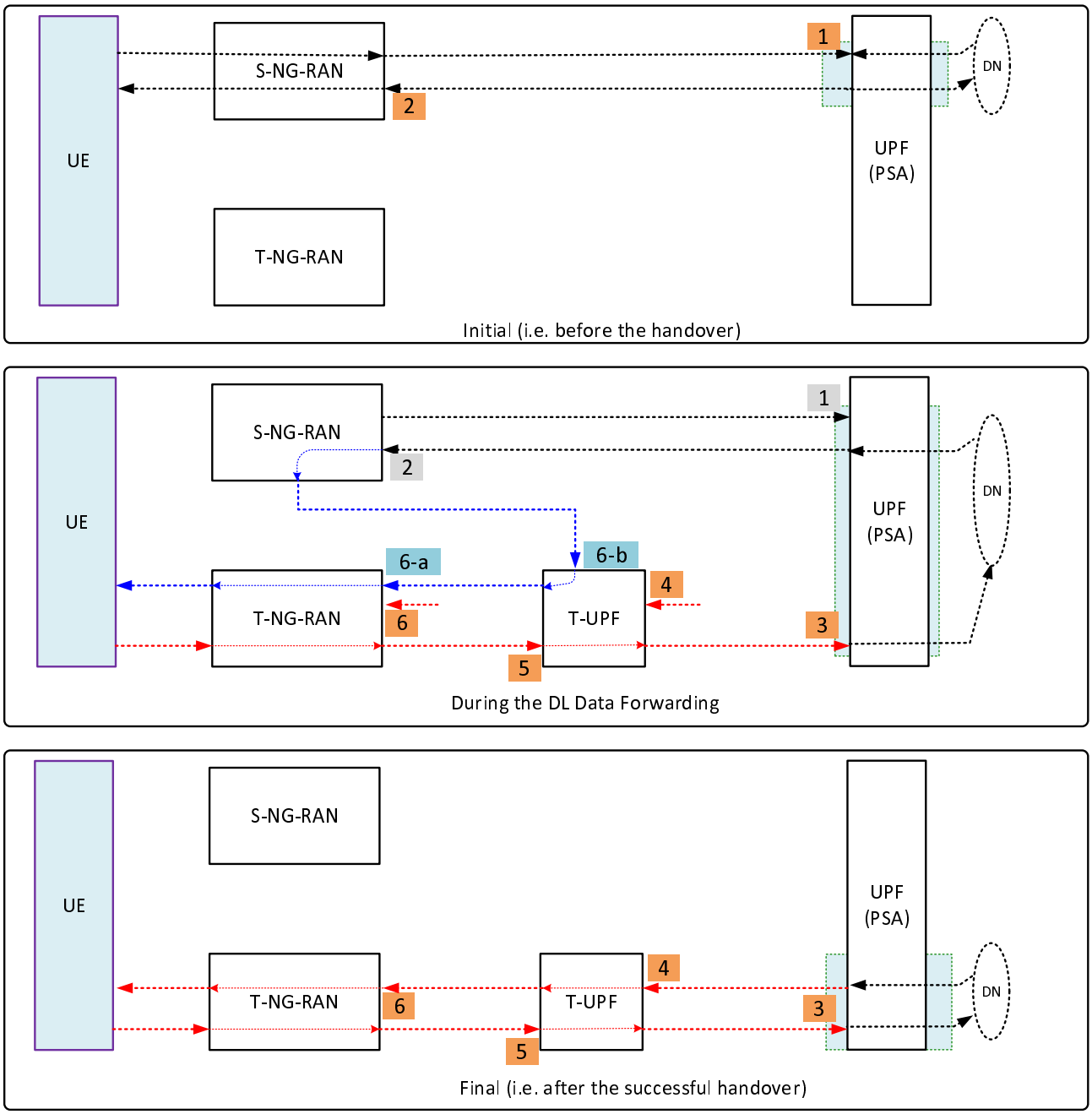


Figure 5.3.3.1.5-4: User plane packet flow during handover indirect DL data forward (non-roaming) – case 4

Scenario 5

Figure 5.3.3.1.5-5 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an UPF relocation in the sense an intermediate UPF is present after the handover. This intermediate UPF is shown as T-UPF in figure 5.3.3.1.5-5. A separate intermediate UPF (shown as T-UPF (I) is used for DL data forwarding.

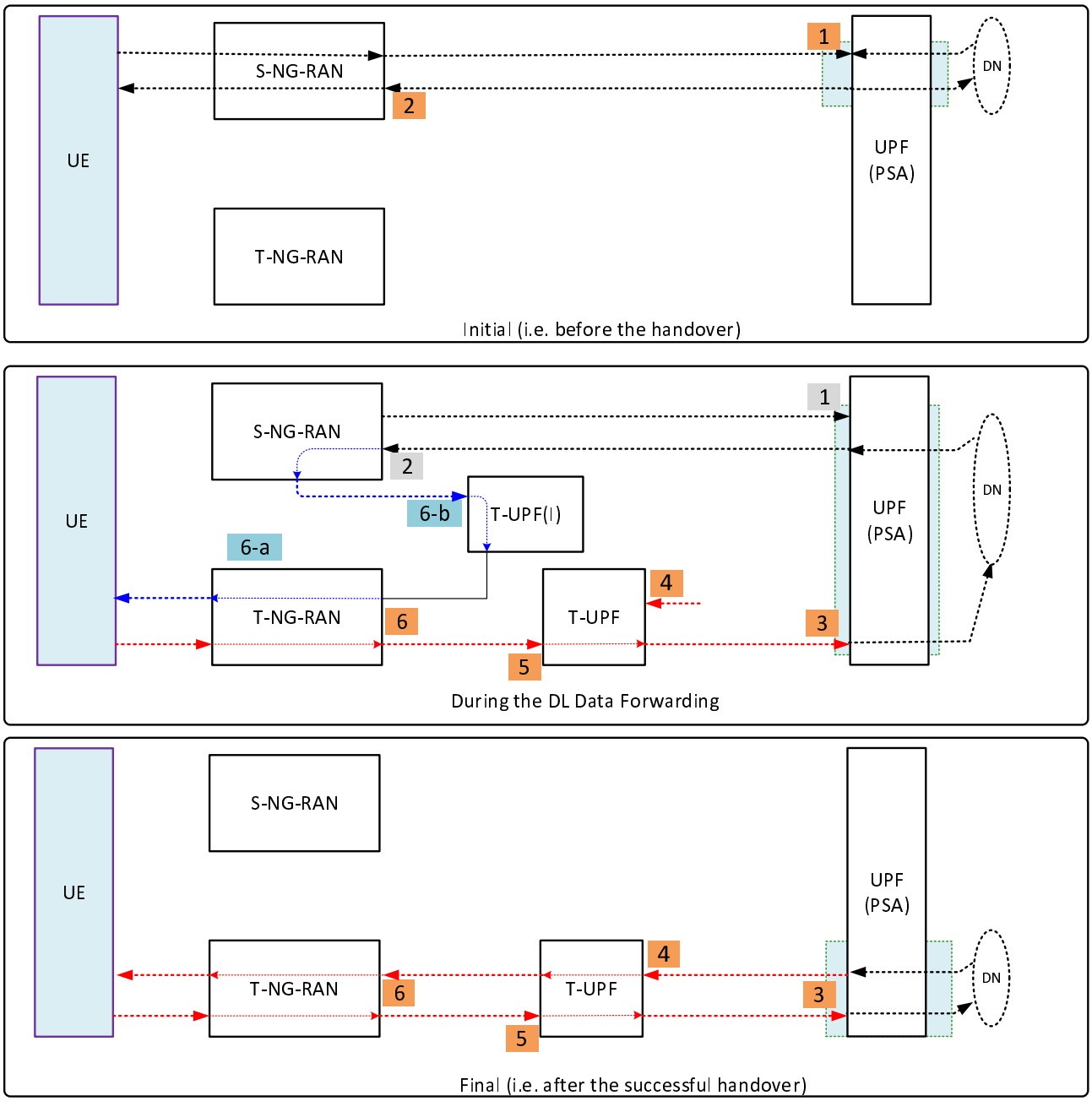


Figure 5.3.3.1.5-5: User plane packet flow during handover indirect DL data forward (non-roaming) – case 5

Scenario 6

Figure 5.3.3.1.5-6 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an intermediate UPF present which changes during the handover. The intermediate UPFs are shown as S-UPF (before the handover) and T-UPF (after the handover) in figure 5.3.3.1.5-6.

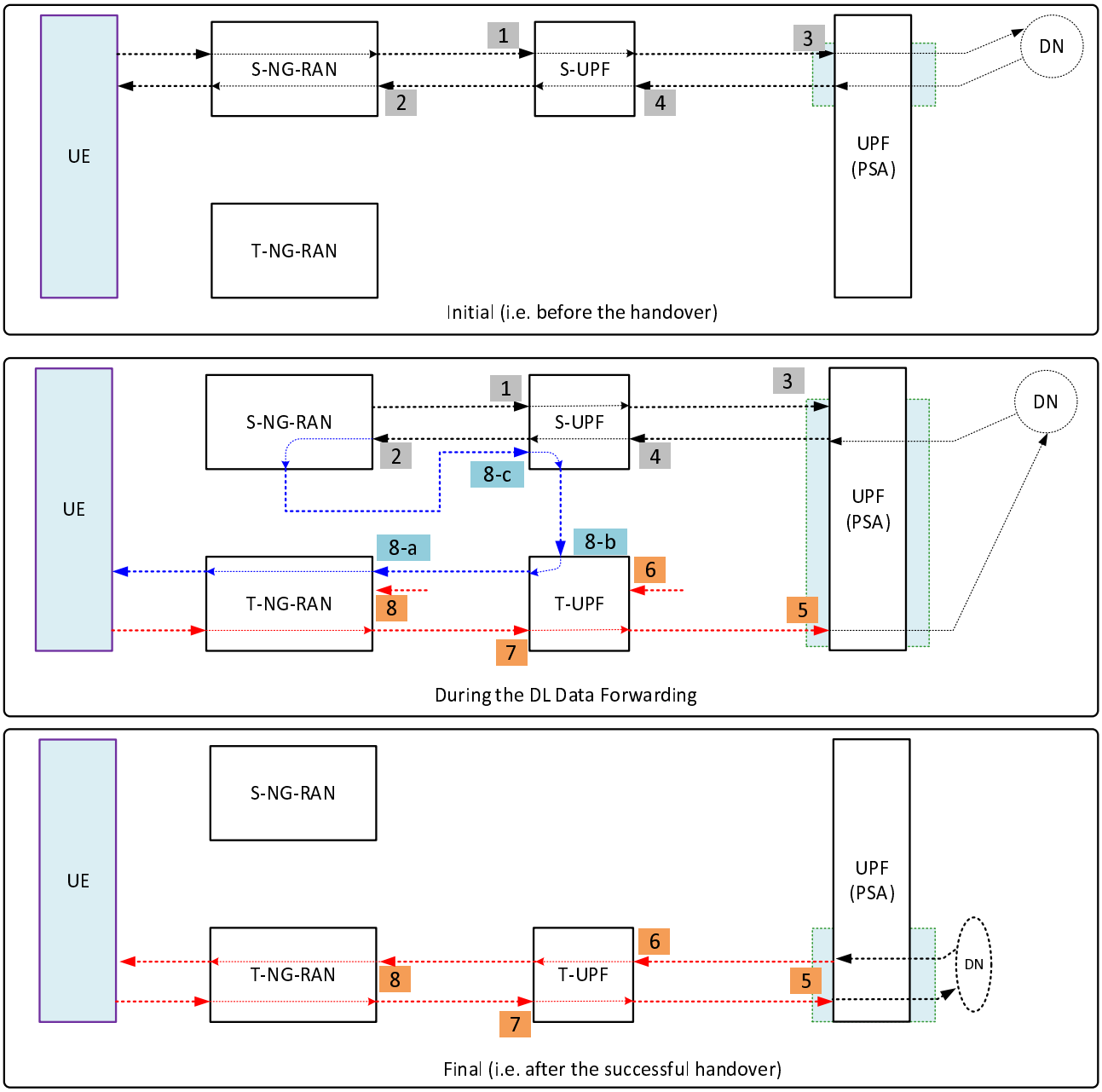


Figure 5.3.3.1.5-6: User plane packet flow during handover indirect DL data forward (non-roaming) – case 6

The detailed flow diagram that depicts the above scenario is illustrated in clause 5.3.3.4.2.

Scenario 7

Figure 5.3.3.1.5-7 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow when there is an intermediate UPF present which changes during the handover. The intermediate UPFs are shown as S-UPF (before the handover) and T-UPF (after the handover) in figure 5.3.3.1.5-7. In this illustration, two separate intermediate UPFs (shown as S-UPF (I) and T-UPF ((I)) are used for DL data forwarding.

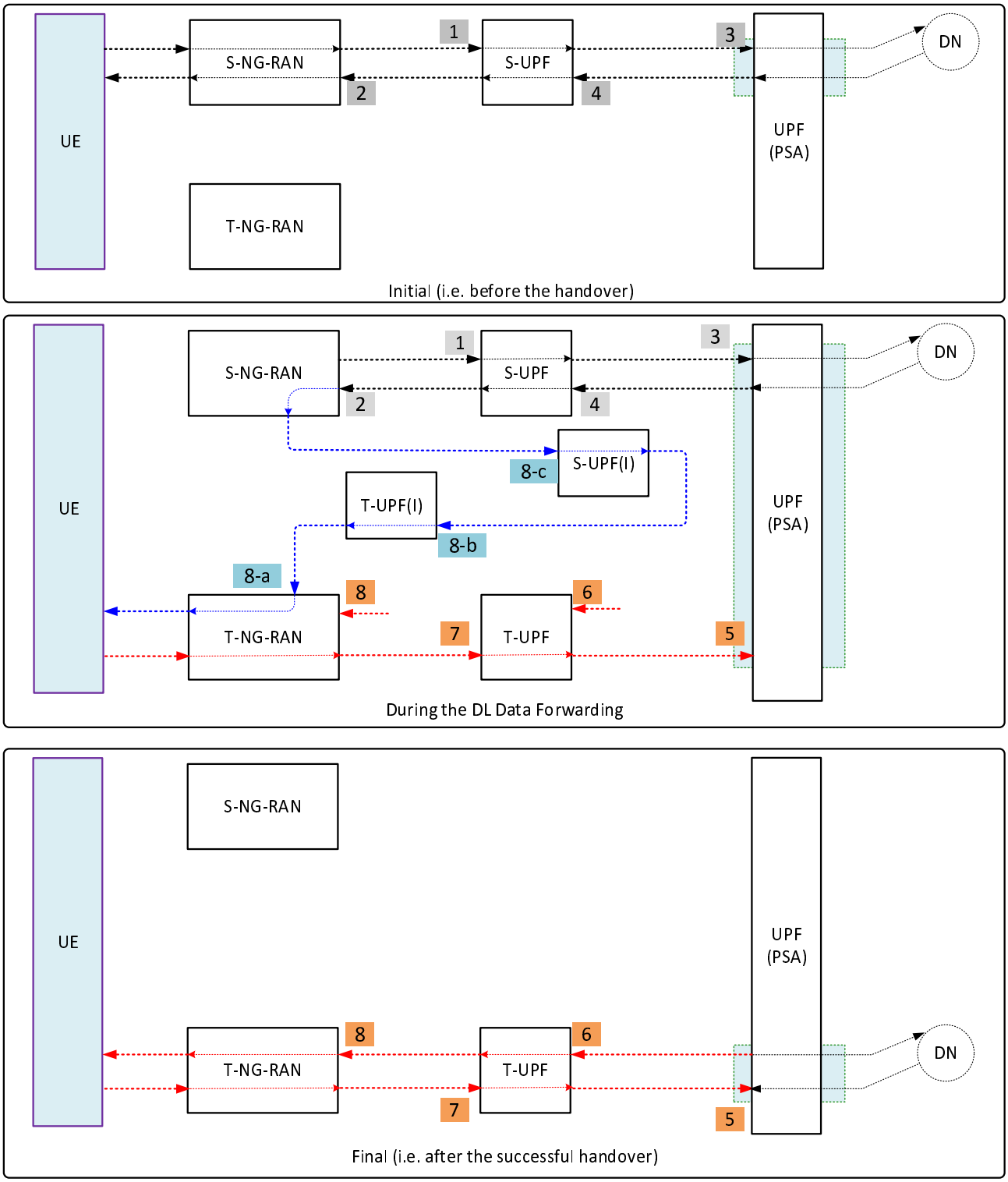


Figure 5.3.3.1.5-7: User plane packet flow during handover indirect DL data forward (non-roaming) – case 7

5.3.3.1.6 Indirect DL data forwarding (home-routed roaming, LI in VPLMN)

In all the illustrations shown in this clause, indirect DL data forwarding happens. The CC interception is done in V-UPF.

Scenario 1

Figure 5.3.3.1.6-1 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. Here, the same V-UPF is used before and after the handover.

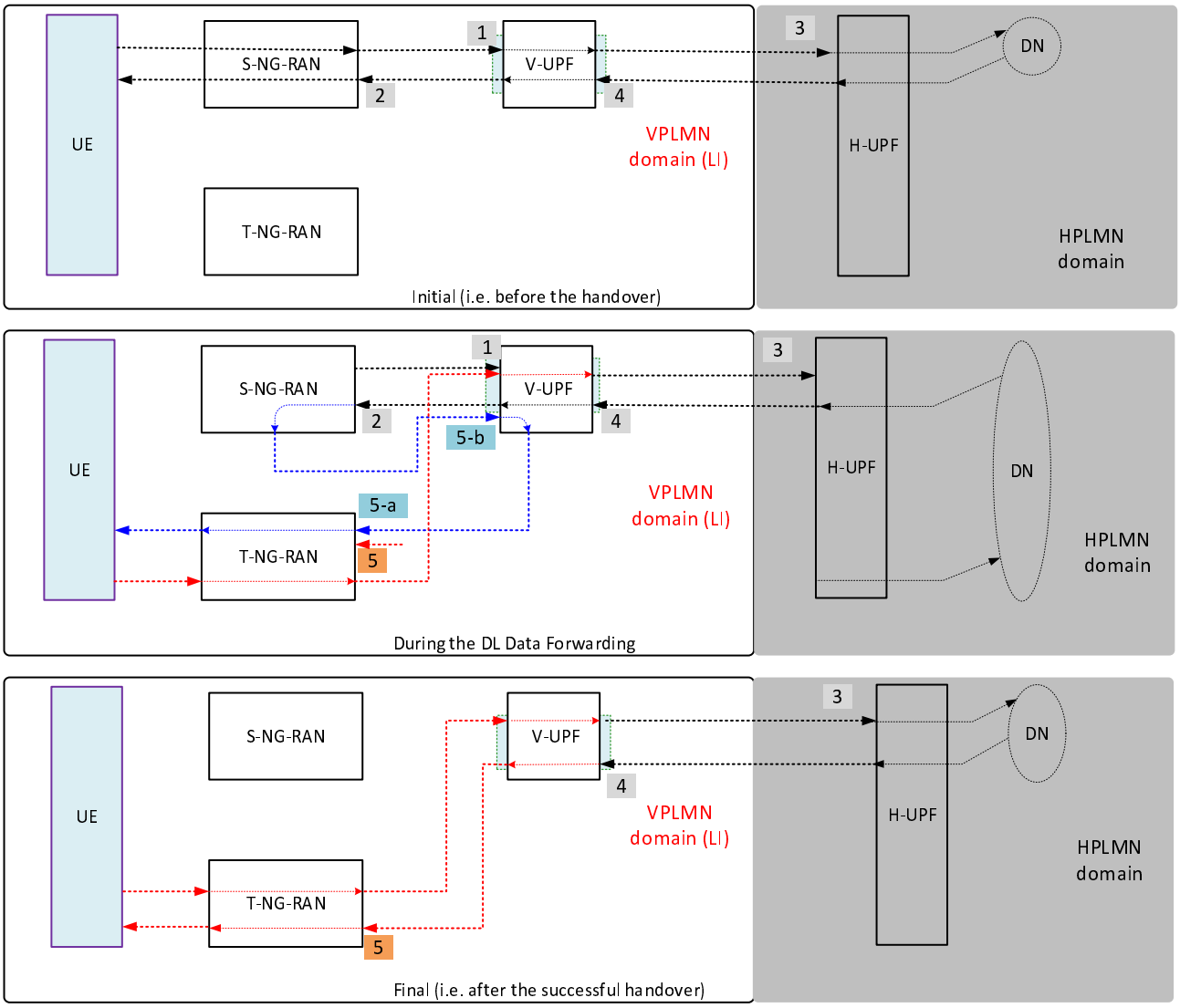


Figure 5.3.3.1.6-1: User plane packet flow during handover indirect DL data forward (roaming, LI in VPLMN) – case 1

Special care has to be taken in the V-UPF not to intercept the user plane packets that flow through the DL data forwarding tunnel.

Scenario 2

Figure 5.3.3.1.6-2 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. Here, the same V-UPF is used before and after the handover. A separate intermediate UPF (shown as S-UPF (I)) is inserted for DL data forwarding.

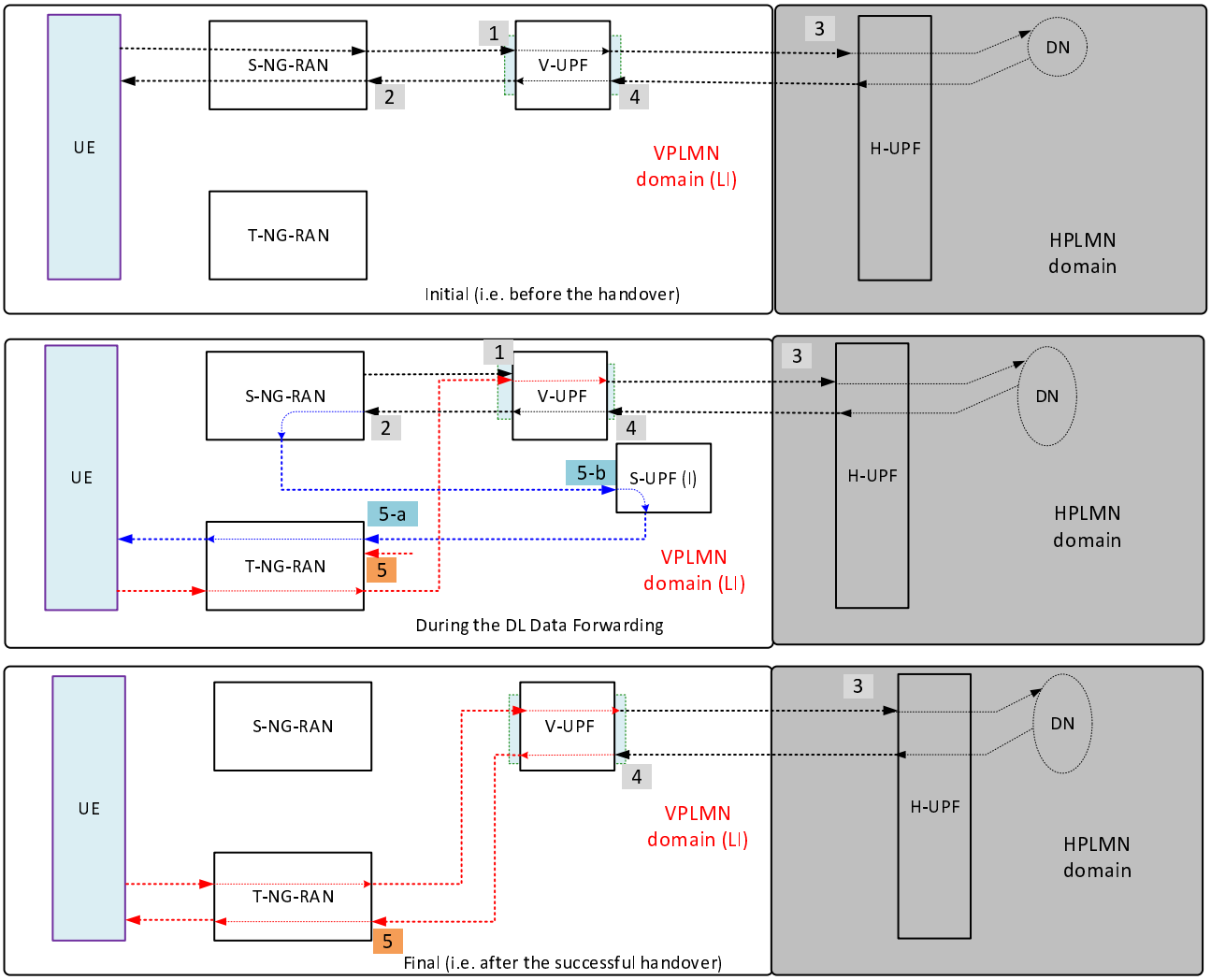


Figure 5.3.3.1.6-2: User plane packet flow during handover indirect DL data forward (roaming, LI in VPLMN) – case 2

Scenario 3

Figure 5.3.3.1.6-3 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. Here, the V-UPF relocation happens during the handover.

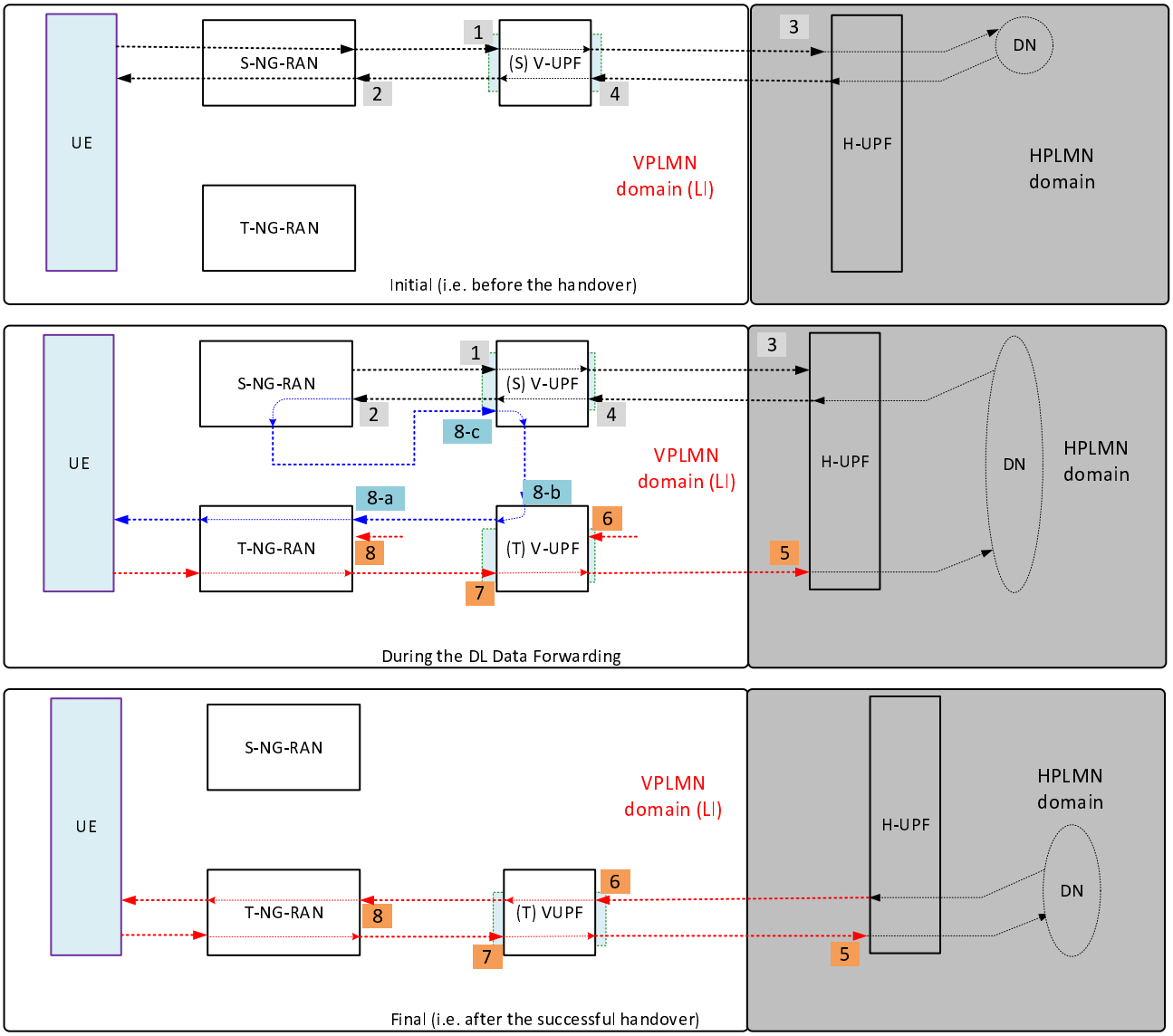


Figure 5.3.3.1.6-3: User plane packet flow during handover indirect DL data forward (roaming, LI in VPLMN) – case 3

Special care has to be taken in both V-UPFs not to intercept the user plane packets that flow through the DL data forwarding tunnel. The detailed flow diagrams depicting this scenario is illustrated in clause 5.3.3.5.2.

Scenario 4

Figure 5.3.3.1.6-4 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. Here, the V-UPF changes during the handover. In this illustration, two separate intermediate UPFs (shown as S-UPF (I) and T-UPF (I)) are used for DL data forwarding.

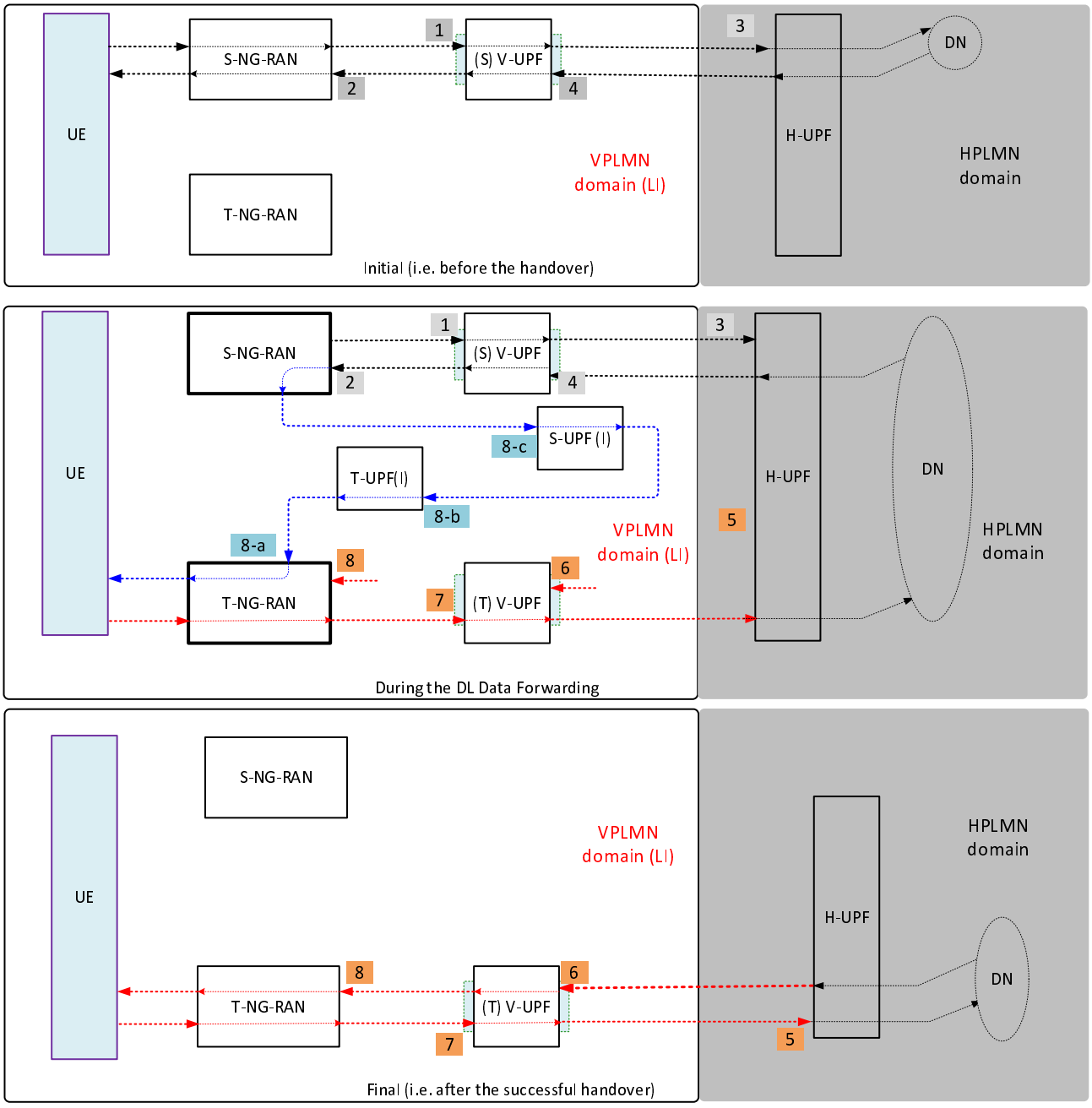


Figure 5.3.3.1.6-4: User plane packet flow during handover indirect DL data forward (roaming, LI in VPLMN) – case 4

5.3.3.1.7 Indirect DL data forwarding (home-routed roaming, LI in HPLMN)

In all the illustrations shown in this clause, indirect DL data forwarding happens. The CC interception is done in H-UPF.

Scenario 1

Figure 5.3.3.1.7-1 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. Here, the same V-UPF is used before and after the handover.

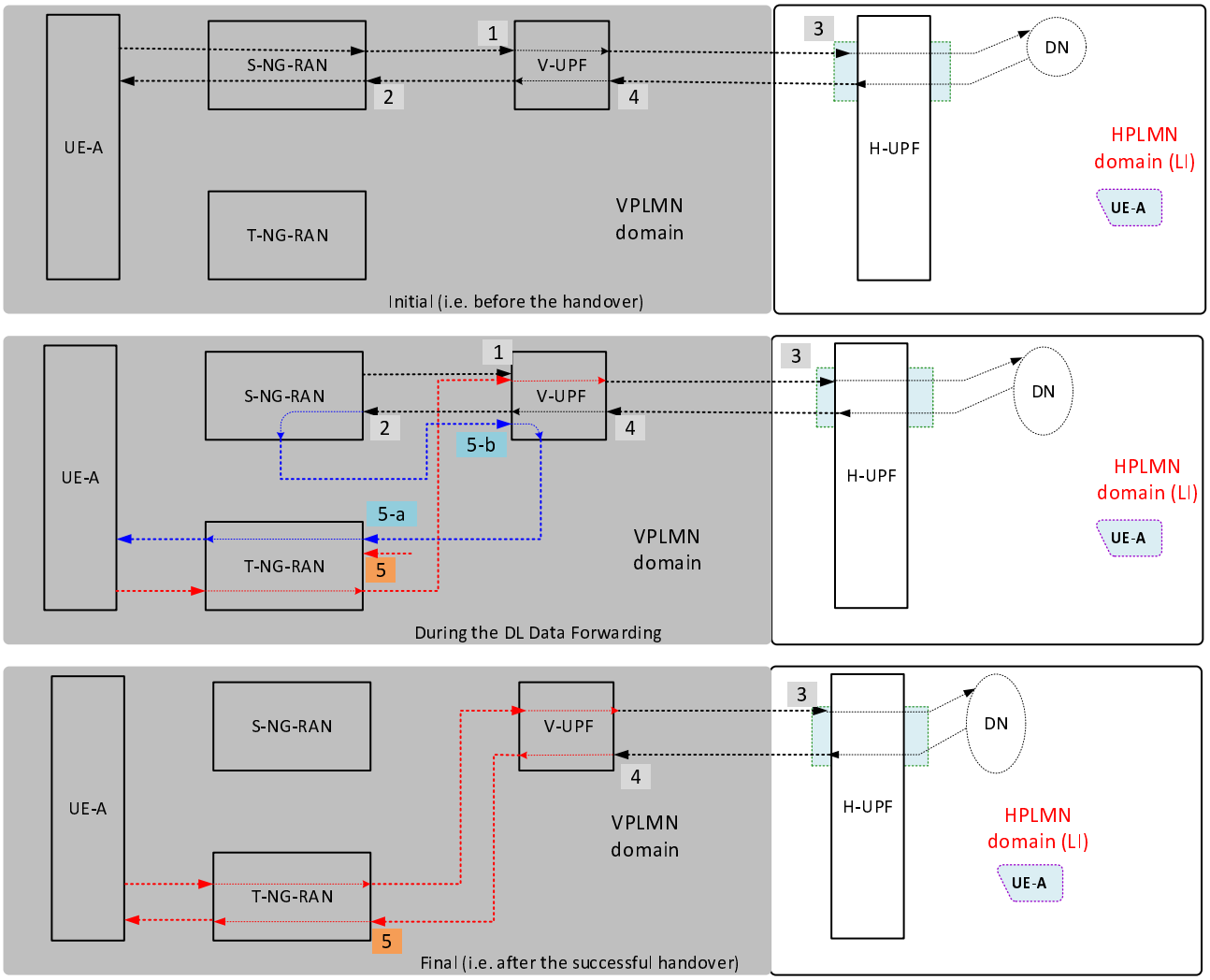


Figure 5.3.3.1.7-1: User plane packet flow during handover indirect DL data forward (roaming, LI in HPLMN) – case 1

Scenario 2

Figure 5.3.3.1.7-2 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. Here, the same V-UPF is used before and after the handover. A separate intermediate UPF (shown as S-UPF (I)) is inserted for DL data forwarding.

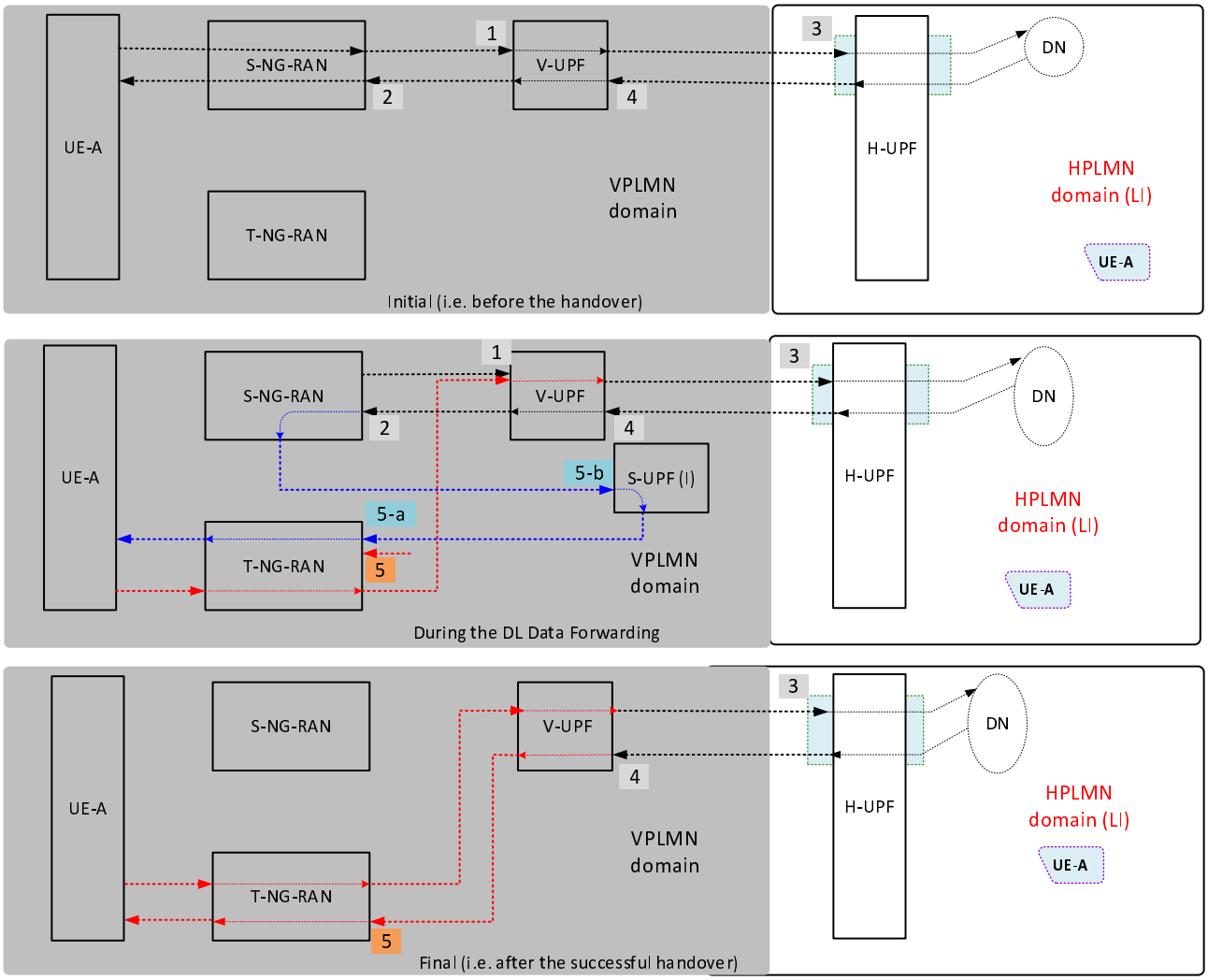


Figure 5.3.3.1.7-2: User plane packet flow during handover indirect DL data forward (roaming, , LI in HPLMN) – case 2

Scenario 3

Figure 5.3.3.1.7-3 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. Here, the V-UPF relocation happens during the handover.

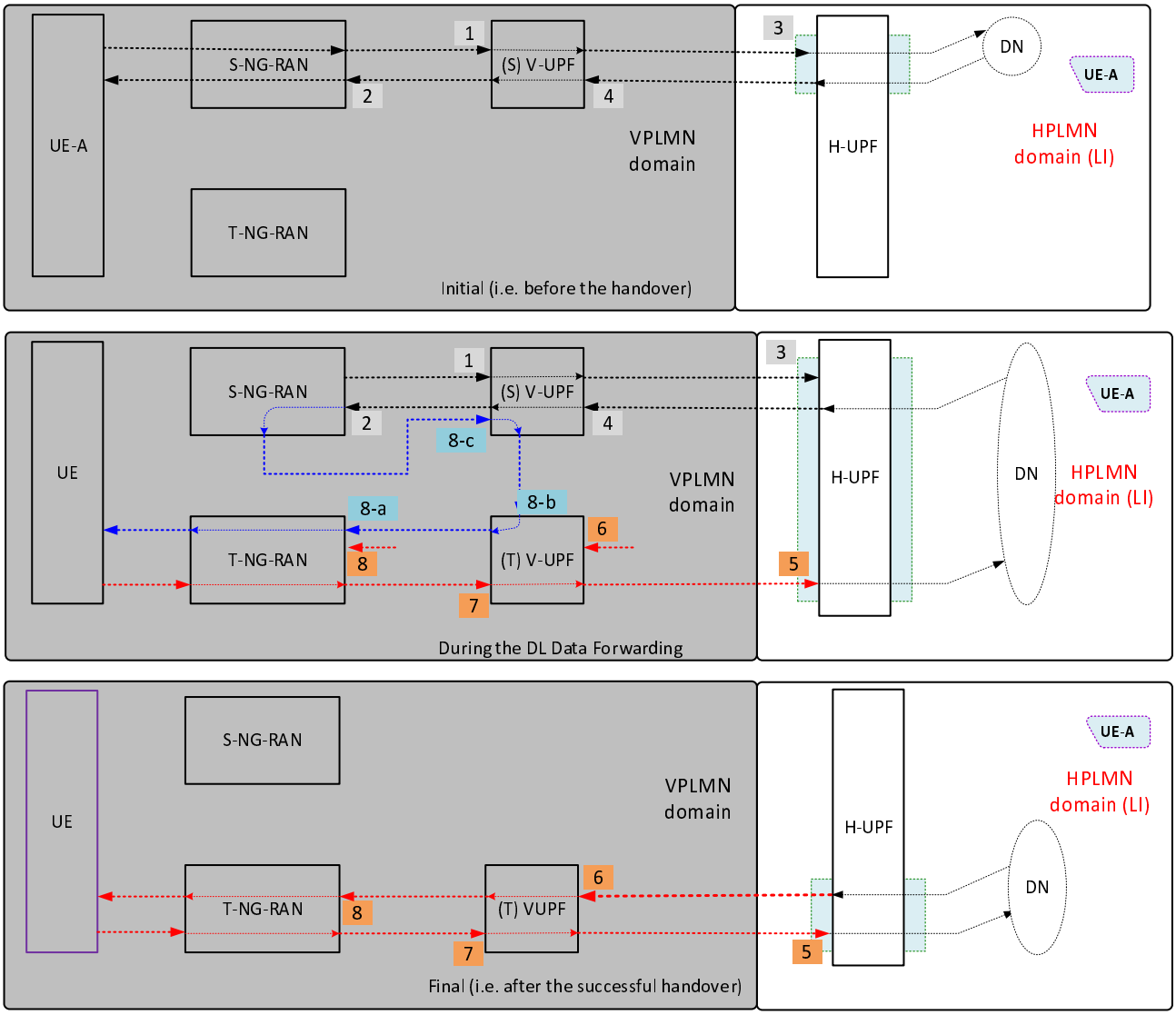


Figure 5.3.3.1.7-3: User plane packet flow during handover indirect DL data forward (roaming, , LI in HPLMN) – case 3

The detailed flow diagrams depicting this scenario is illustrated in clause 5.3.3.5.3.

Scenario 4

Figure 5.3.3.1.7-4 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. Here, the V-UPF changes during the handover. In this illustration, two separate intermediate UPFs (shown as S-UPF (I) and T-UPF (I)) are used for DL data forwarding.

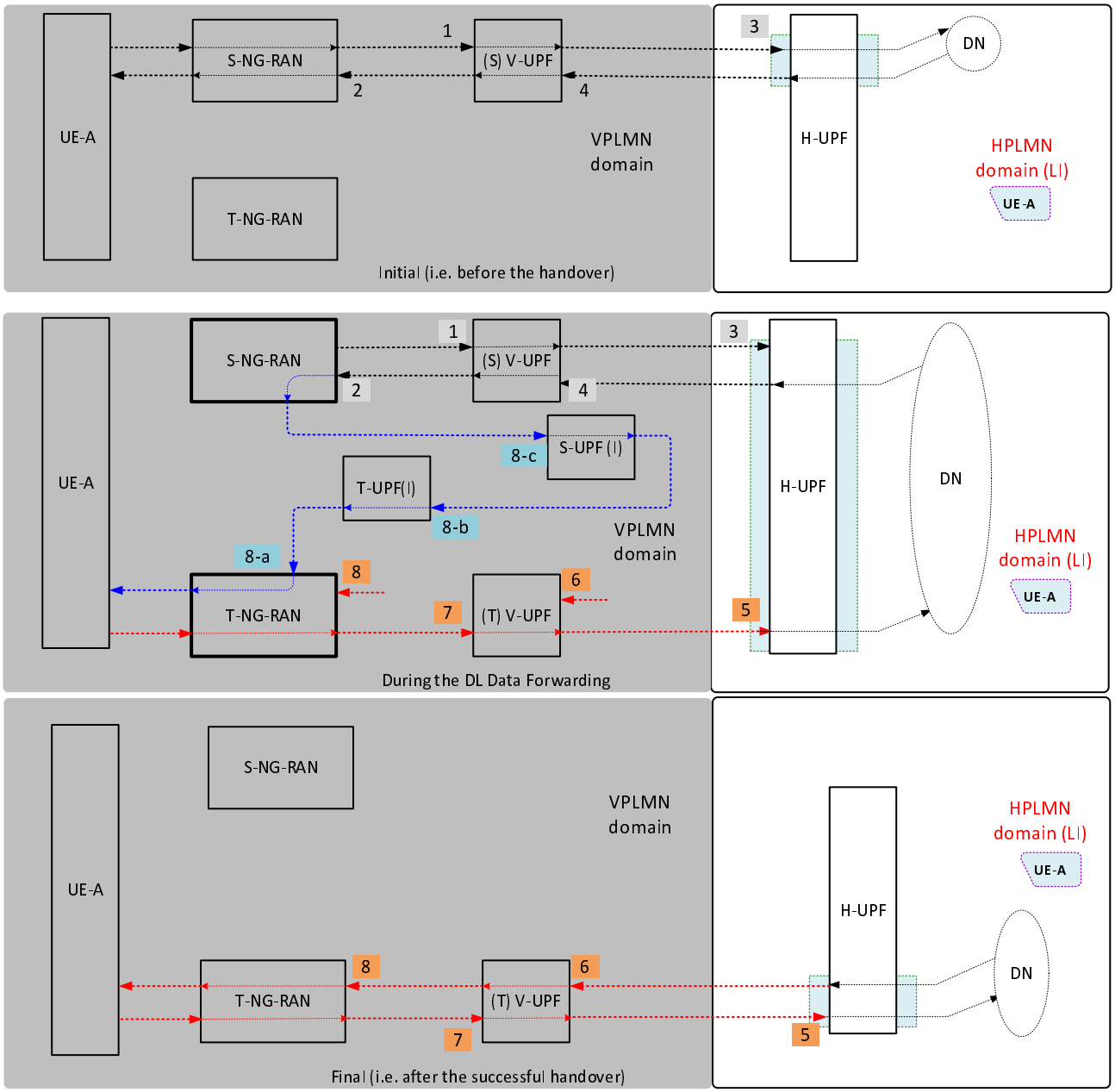


Figure 5.3.3.1.7-4: User plane packet flow during handover indirect DL data forward (roaming, LI in HPLMN) – case 4

5.3.3.2 Detailed flows - non-roaming, direct DL data forwarding

5.3.3.2.1 General

An N-2 based handover may happen when the source NG-RAN determines that an Xn-based handover cannot be done. The same criteria used with the Xn based handover is applicable here as well as far as the UPF relocation is concerned. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the former case. The illustrations given here assume that there is no UPF relocation and without any intermediate UPFs before or after the handover. This flow is based on the Xn-based handover stage 2 flow in TS 23.502 [7], clause 4.9.1.3.

Within drawings, (source) NG-RAN is shown as S-NG-RAN and as S-5G. Likewise, the (target) NG-RAN is shown as T-NG-RAN and as T-5G. The numbers 1, 2, 3, 4 and 4-a correspond to the TEID (1), TEID (2), TEID (3), TEID (4) and TEID (4a) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier for UL UP packets at the UPF and the TEID (2) is the tunnel end point identifier for the DL UP packets at the (source) NG-RAN. The TEID (4a) is the tunnel end point identifier at the (target) NG-RAN for the DL data forwarded UP packets. The DL data forwarding happens from (source) NG-RAN to (target) NG-RAN.

The TEID (4) is the tunnel end point identifier for the DL UP packets at the (target) NG-RAN. The TEID (3) is the tunnel end point identifier for UL UP packets at the UPF.

The generation of xCC from the UP packets done by the CC-POI present in the UPF.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.3.2.2 The flows

This flow is split into two parts, for easy reading purpose. The flow-diagram in figure 5.3.3.2.2-1 illustrates the first part and figure 5.3.3.2.2-2 illustrates the second part.

Flow 1 of 2

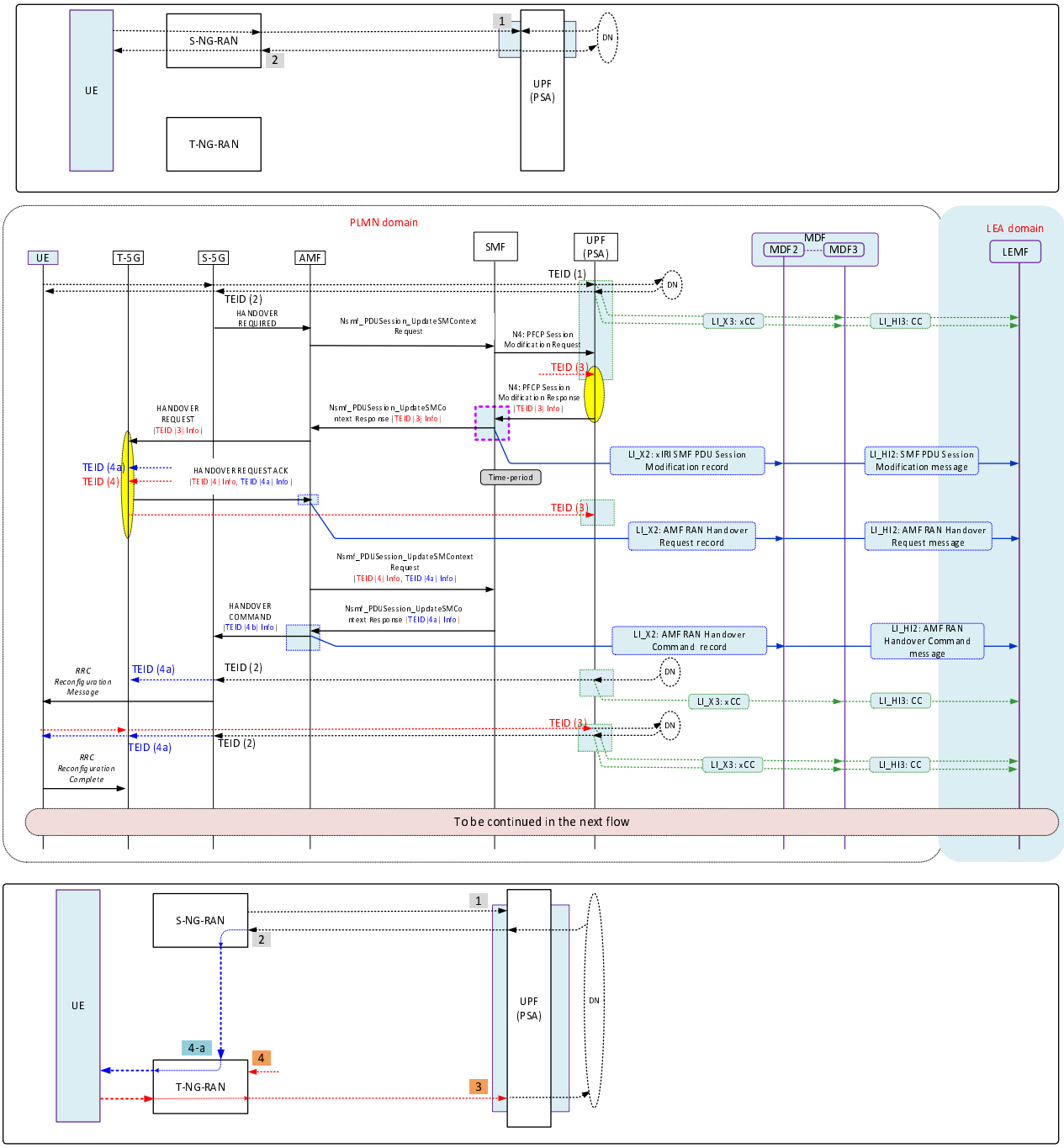


Figure 5.3.3.2.2-1: N2-based handover with no UPF relocation (non-roaming) – flow 1 of 2

When the AMF receives a HANDOVER REQUEST ACK from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Request record.

When the AMF sends the HANDOVER COMMAND to the (source) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Command record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDUSessionModification record.

Flow 2 of 2

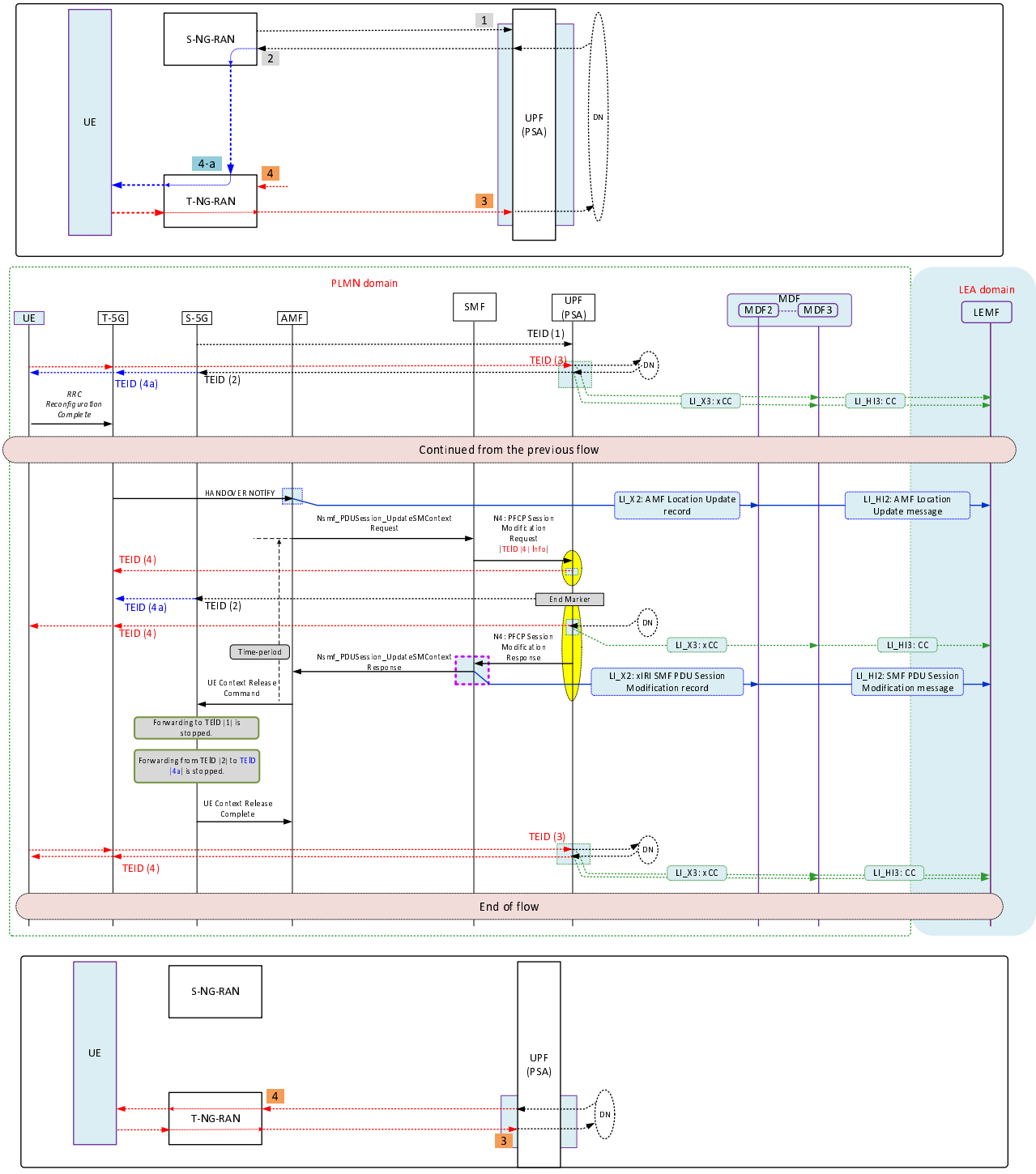


Figure 5.3.3.2.2-2: N2-based handover with no UPF relocation (non-roaming) – flow 2 of 2

When the AMF receives a HANDOVER NOTIFY from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI Update Location Request record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

5.3.3.3 Detailed flows - non-roaming, direct DL data forwarding (with UPF relocation)

5.3.3.3.1 General

An N-2 based handover may happen when the source NG-RAN determines that an Xn-based handover cannot be done. The same criteria used with the Xn based handover is applicable here as well as far as the UPF relocation is concerned. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the former case. The illustrations given here assume that an intermediate UPF is present before the handover and it is changed to another intermediate UPF after the handover. This flow is based on the N2-based handover stage 2 flow in TS 23.502 [7], clause 4.9.1.3.

Within drawings, two intermediate UPFs are shown as S-UPF (prior to the handover) and T-UPF (post-handover). The anchor UPF is shown as UPF (PSA). The (source) NG-RAN is shown as S-NG-RAN and as S-5G. Likewise, the (target) NG-RAN is shown as T-NG-RAN and as T-5G. The numbers 1, 2, 3, 4, 5, 6, 7. 8 and 8-a correspond to the TEID (1), TEID (2), TEID (3), TEID (4), TEID (5), TEID (6), TEID (7), TEID (8) and TEID (8a) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at S-UPF and the TEID (3) is the tunnel end point identifier at UPF (PSA) for UL UP packets. TEID (2) is the tunnel end point identifier at (source) NG-RAN and the TEID (4) is the tunnel end point identifier at the S-UPF for the DL UP packets. The TEID (8a) is the tunnel end point identifier at the (target) NG-RAN for the DL data forwarded UP packets. The direct DL data forwarding happens from (source) NG-RAN to (target) NG-RAN.

The TEID (8) is the tunnel end point identifier for the DL UP packets at the (target) NG-RAN. The TEID (6) is the tunnel end point identifier for the DL UP packets at the T-UPF. The TEID (5) is the tunnel end point identifier for the UL traffic at the UPF (PSA). The TEID (7) is the tunnel end point identifier at the T-UPF for UL UP packets.

The generation of xCC from the UP packets done by the CC-POI present in the anchor UPF.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.3.3.2 The flows

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.3.3.2-1 illustrates the first part. The flow-diagram in figure 5.3.3.3.2-2 illustrates the second part. The flow-diagram in figure 5.3.3.3.2-3 illustrates the third part. The flow-diagram in figure 5.3.3.3.2-4 illustrates the fourth part.

Flow 1 of 4

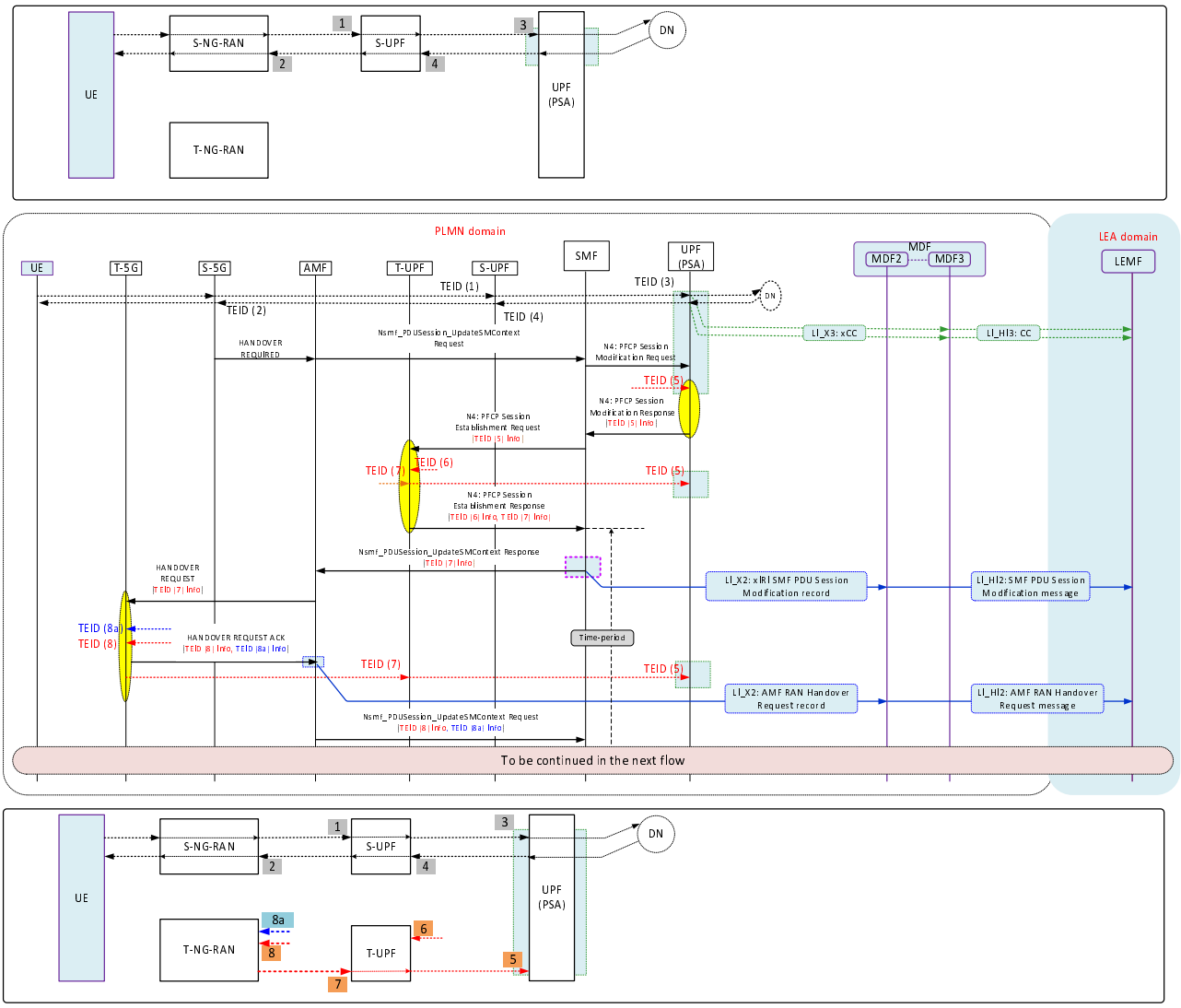


Figure 5.3.3.3.2-1: N2-based handover with UPF relocation (non-roaming) – flow 1 of 4

When the AMF receives a HANDOVER REQUEST ACK from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Request record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 2 of 4

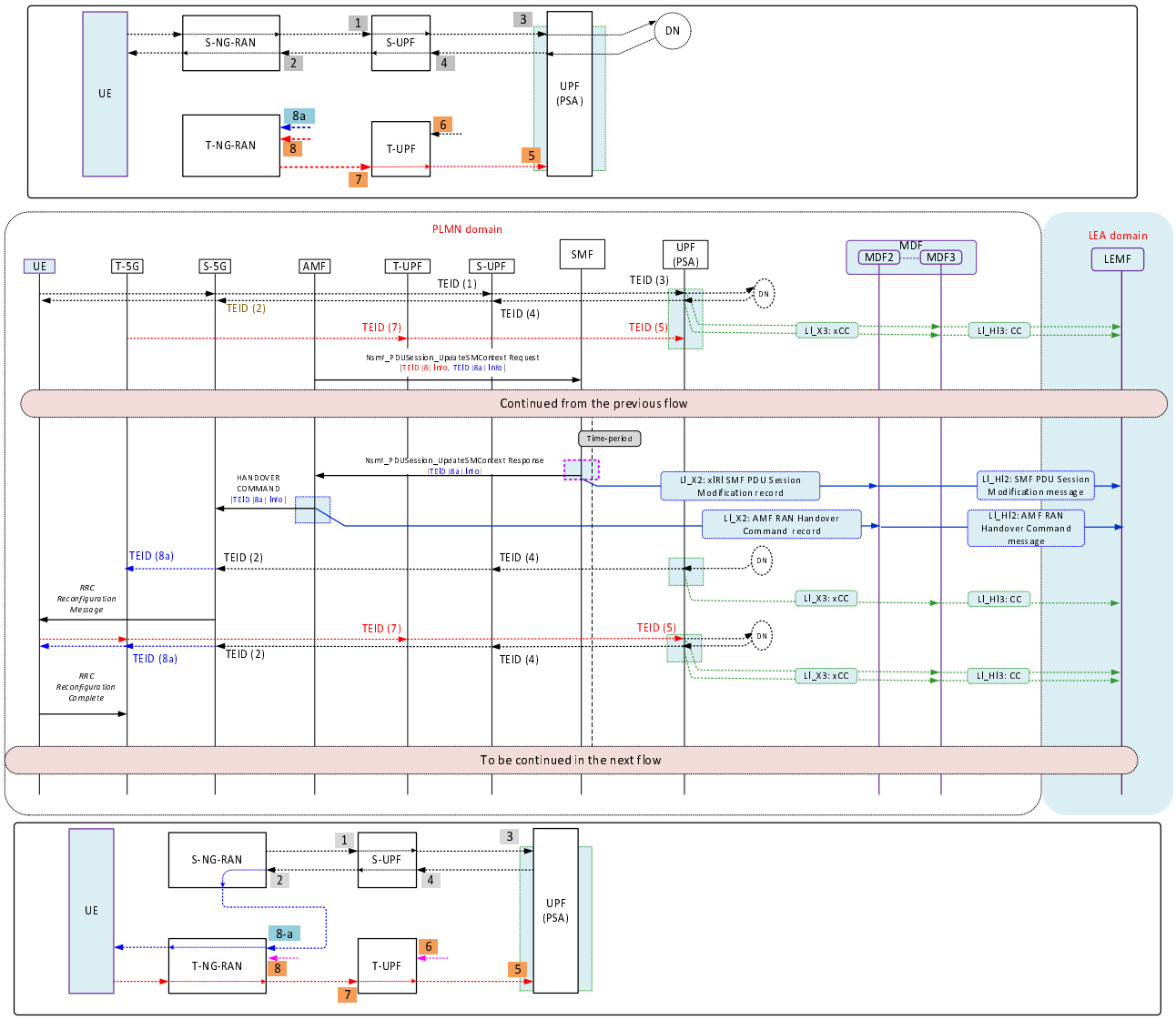


Figure 5.3.3.3.2-2: N2-based handover with UPF relocation (non-roaming) – flow 2 of 4

When the AMF sends the HANDOVER COMMAND to the (source) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Command record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 3 of 4

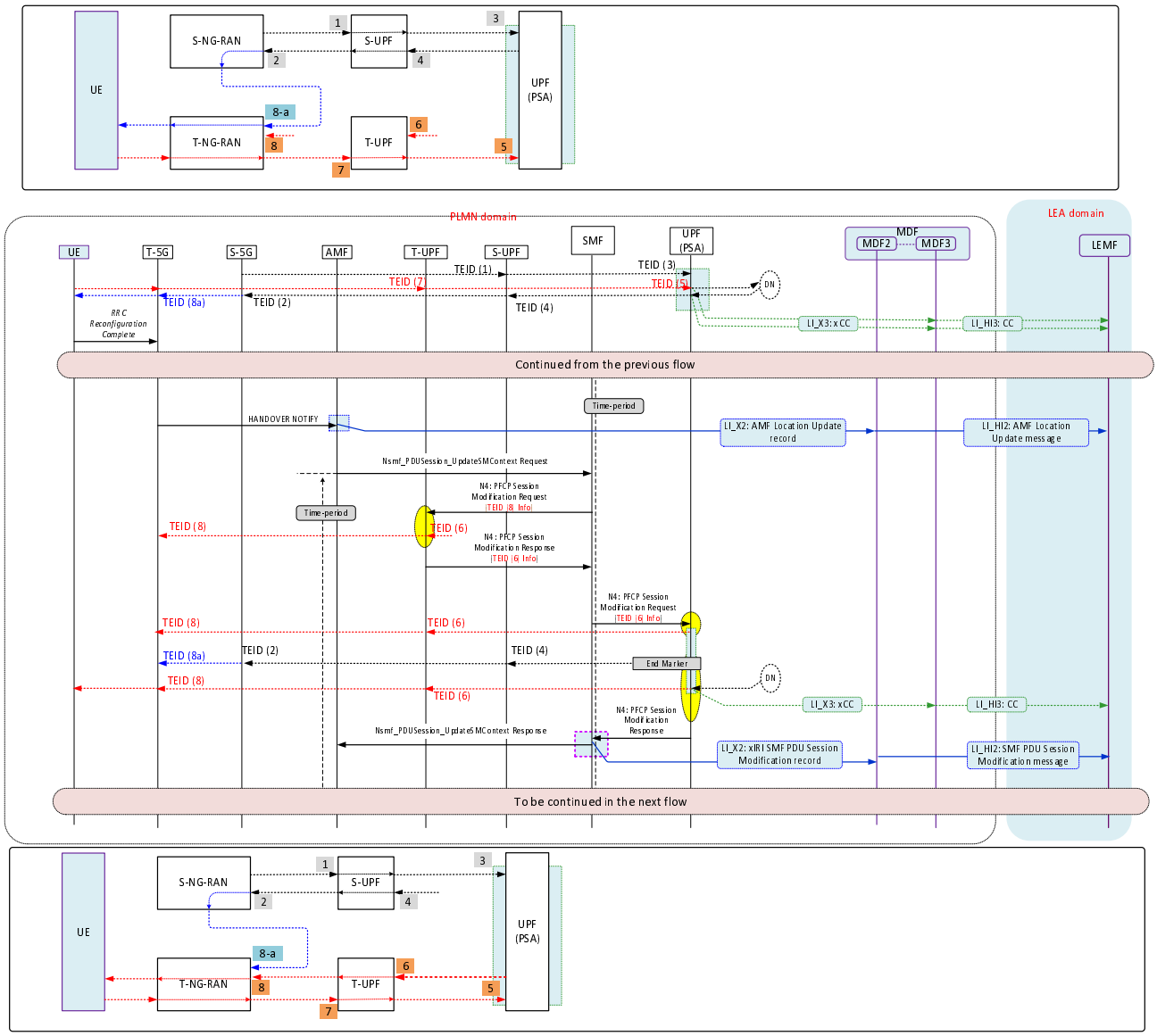


Figure 5.3.3.3.2-3: N2-based handover with UPF relocation (non-roaming) – flow 3 of 4

When the AMF receives a HANDOVER NOTIFY from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI Update Location Request record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

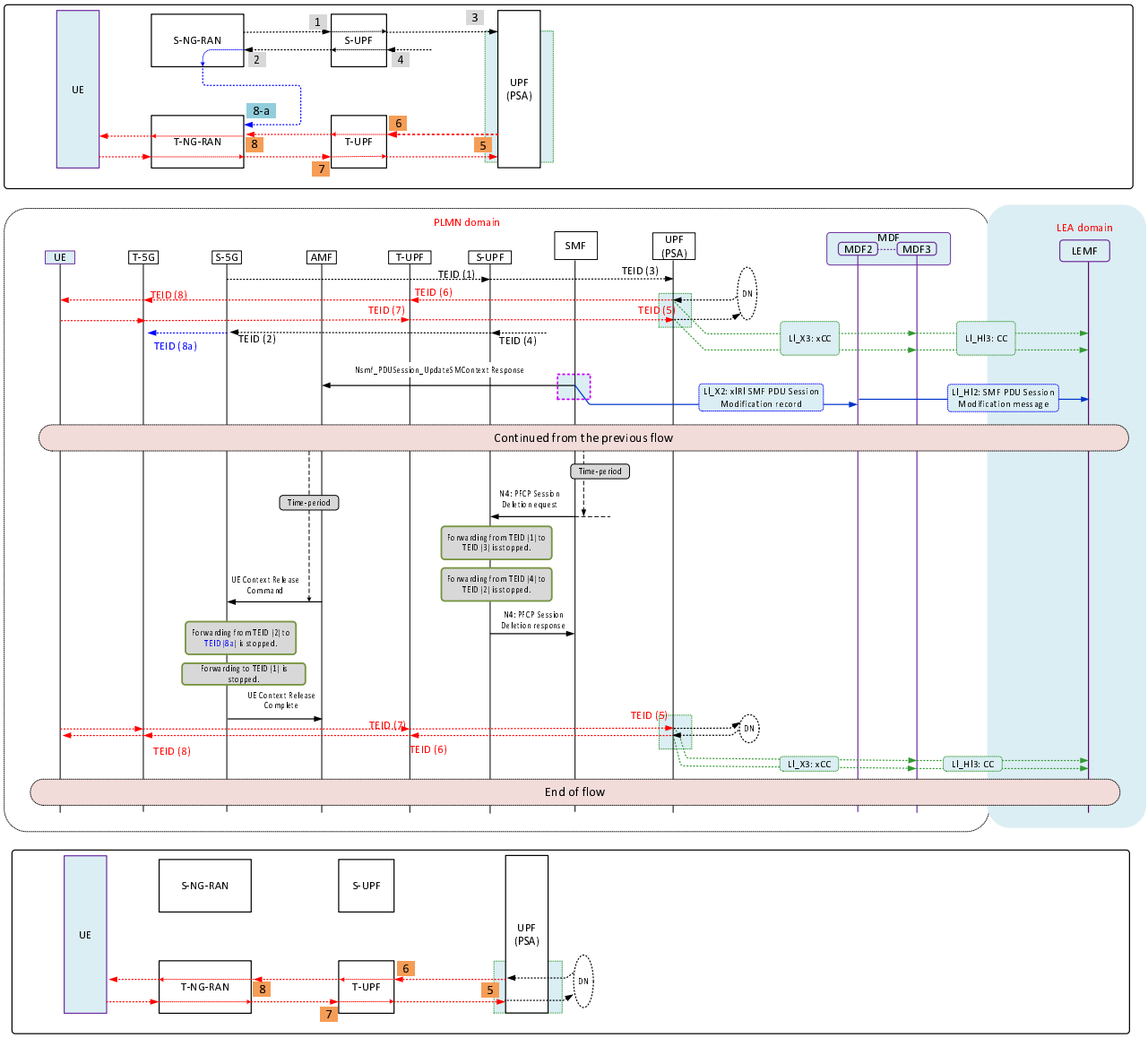


Figure 5.3.3.3.2-4: N2--based handover with UPF relocation (non-roaming) – flow 4 of 4

5.3.3.4 Detailed flows - non-roaming, indirect DL data forwarding (with UPF relocation)

5.3.3.4.1 General

An N-2 based handover may happen when the source NG-RAN determines that an Xn-based handover cannot be done. The same criteria used with the Xn based handover is applicable here as well as far as the UPF relocation is concerned. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the latter case. The illustrations given here assume that an intermediate UPF is present before the handover and it is changed to another intermediate UPF after the handover. The flow also assumes a change in the AMF during the handover.

This flow is based on the N2-based handover stage 2 flow in TS 23.502 [7], clause 4.9.1.3.

Within drawings, two intermediate UPFs are shown as S-UPF (prior to the handover) and T-UPF (post-handover). The anchor UPF is shown as UPF (PSA). The AMF used before the handover is shown as S-AMF and the AMF used after the handover is shown as T-AMF. The (source) NG-RAN is shown as S-NG-RAN and as S-5G. Likewise, the (target) NG-RAN is shown as T-NG-RAN and as T-5G. The numbers 1, 2, 3, 4, 5, 6, 7. 8, 8-a, 8-b and 8-c correspond to the TEID (1), TEID (2), TEID (3), TEID (4), TEID (5), TEID (6), TEID (7), TEID (8), TEID (8a), TEID (8b), and TEID

(8c) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at S-UPF and the TEID (3) is the tunnel end point identifier at UPF (PSA) for UL UP packets. TEID (2) is the tunnel end point identifier at (source) NG-RAN and the TEID (4) is the tunnel end point identifier at the S-UPF for the DL UP packets. The TEID (8a) is the tunnel end point identifier at the (target) NG-RAN for the DL data forwarded UP packets. The TEID (8b) is the tunnel end point identifier at the T-UPF for the DL data forwarded UP packets. The TEID (8c) is the tunnel end point identifier at the S-UPF for the DL data forwarded UP packets.

The TEID (8) is the tunnel end point identifier for the DL UP packets at the (target) NG-RAN. The TEID (6) is the tunnel end point identifier for the DL UP packets at the T-UPF. The TEID (5) is the tunnel end point identifier for the UL traffic at the UPF (PSA). The TEID (7) is the tunnel end point identifier at the T-UPF for UL UP packets.

The generation of xCC from the UP packets done by the CC-POI present in the anchor UPF.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.3.4.2 The flows

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.3.4.2-1 illustrates the first part. The flow-diagram in figure 5.3.3.4.2-2 illustrates the second part. The flow-diagram in figure 5.3.3.4.2-3 illustrates the third part. The flow-diagram in figure 5.3.3.4.2-4 illustrates the fourth part.

Flow 1 of 4

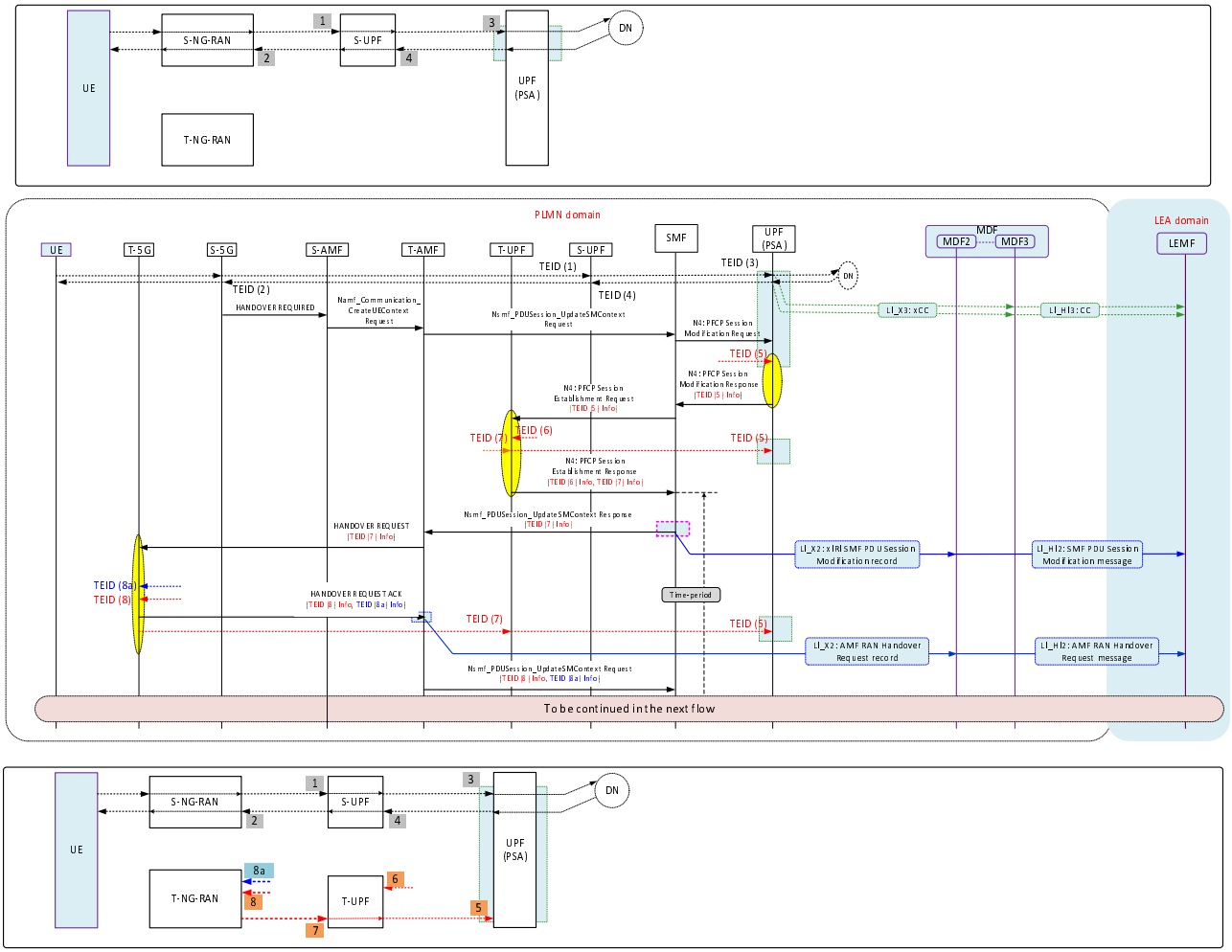


Figure 5.3.3.4.2-1: Indirect DL Data forwarding non-roaming case with the UPF relocation – flow 1 of 4

When the T-AMF receives a HANDOVER REQUEST ACK from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Request record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 2 of 4

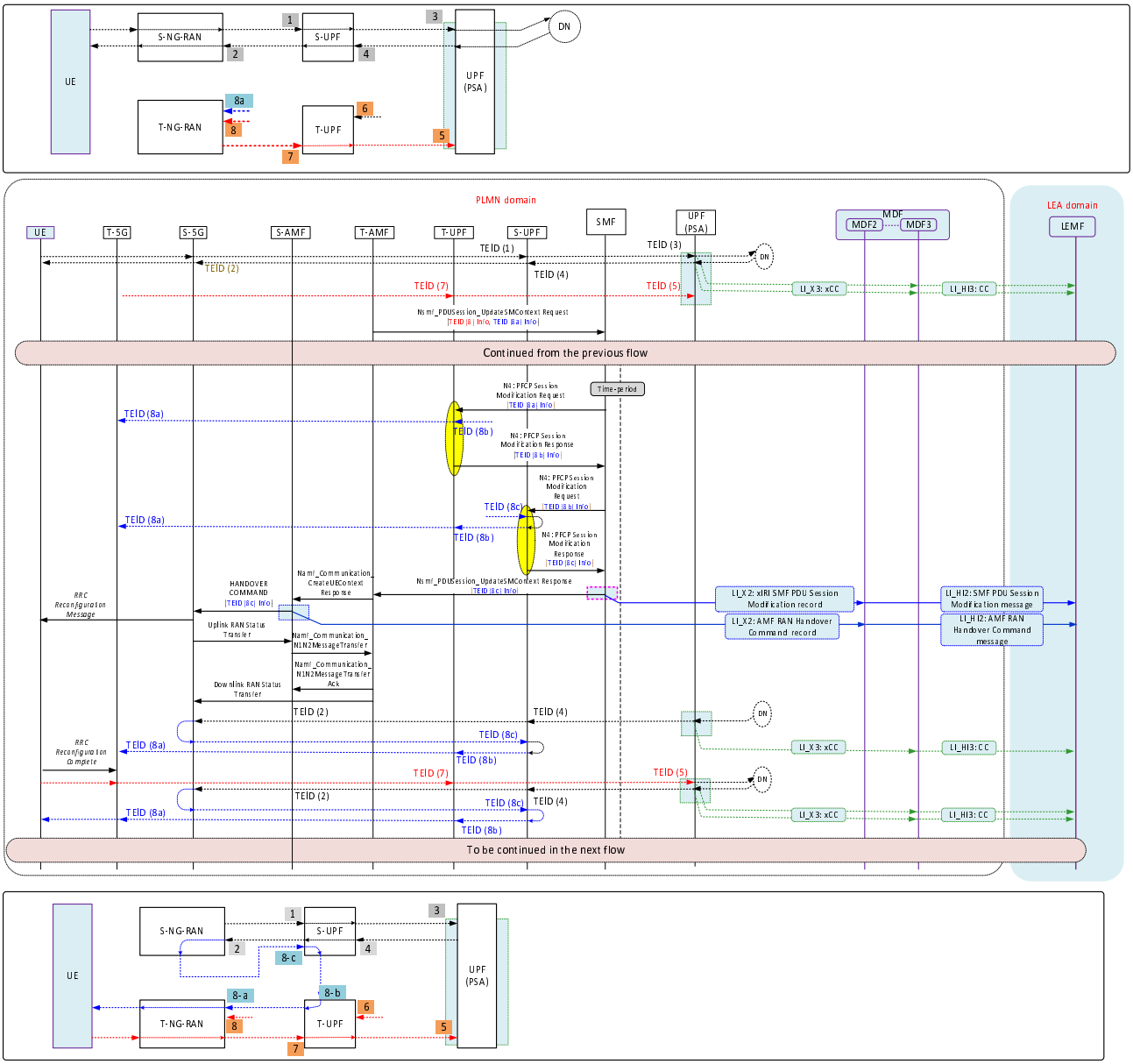


Figure 5.3.3.4.2-2: Indirect DL Data forwarding non-roaming case with the UPF relocation – flow 2 of 4

When the S-AMF sends the HANDOVER COMMAND to the (source) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Command record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 3 of 4

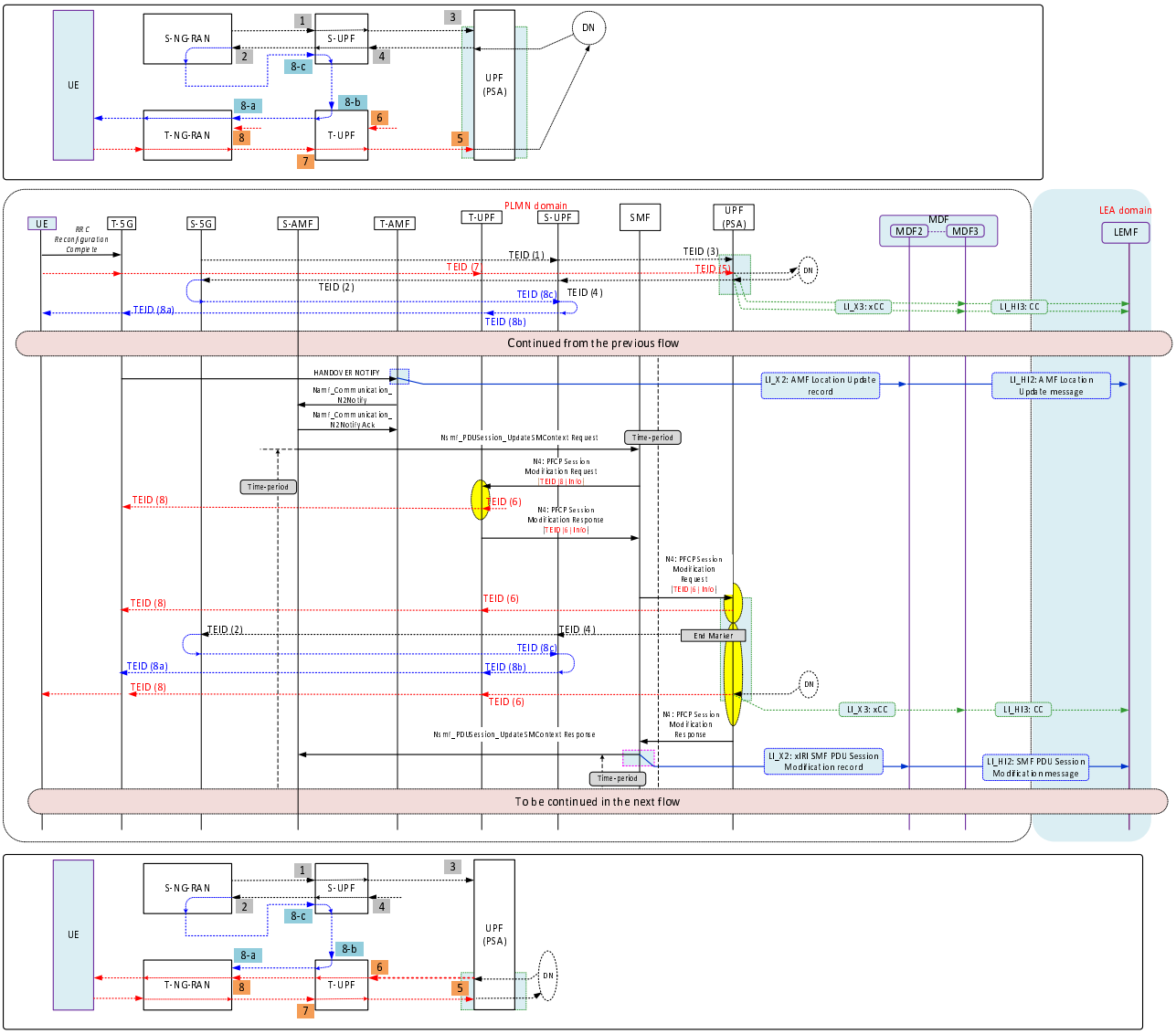


Figure 5.3.3.4.2-3: Indirect DL Data forwarding non-roaming case with the UPF relocation – flow 3 of 4

When the T-AMF receives a HANDOVER NOTIFY from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI Update Location Request record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

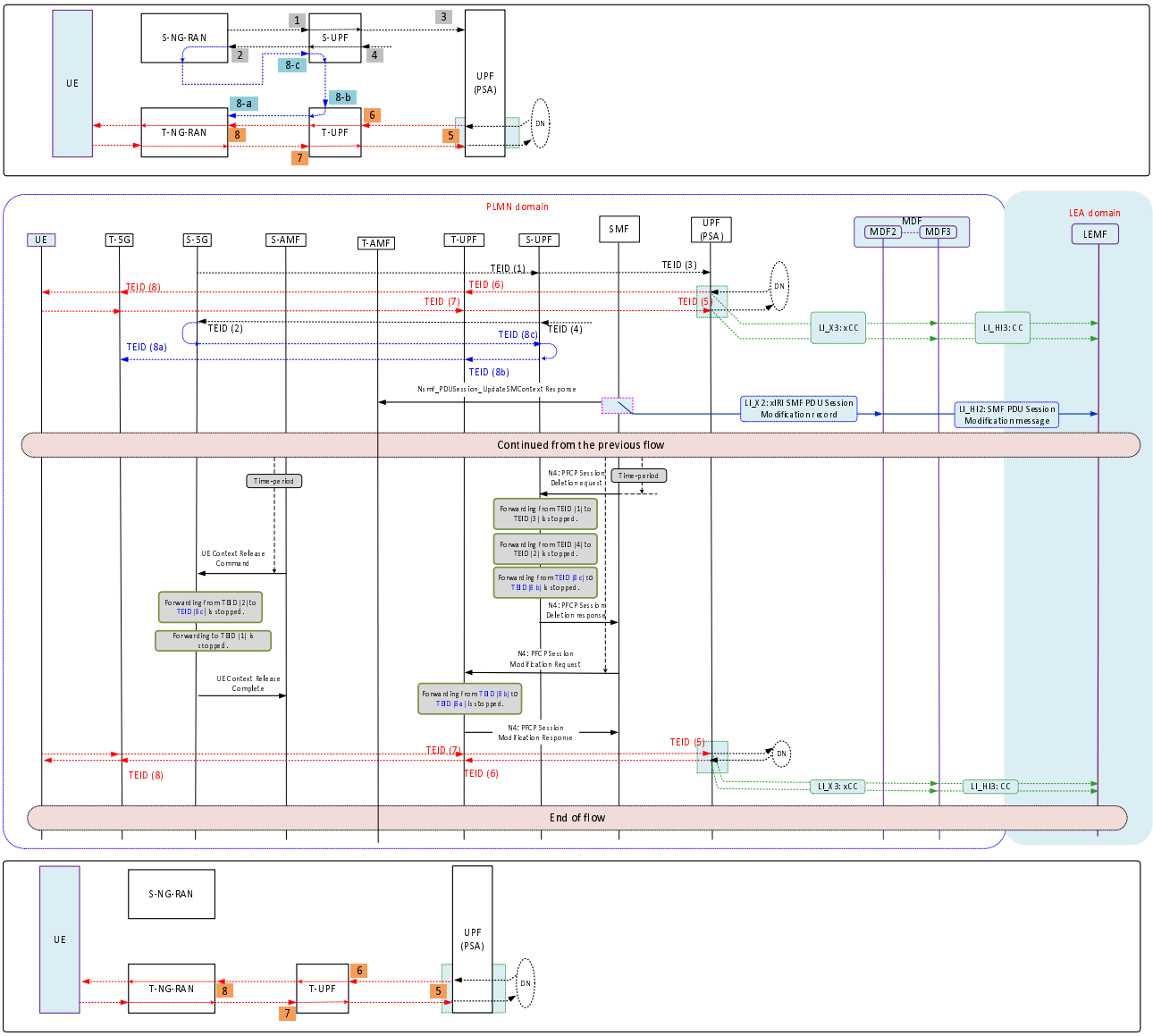


Figure 5.3.3.4.2-4: Indirect DL Data forwarding non-roaming case with the UPF relocation – flow 4 of 4

5.3.3.5 Detailed flows - home-routed roaming, indirect DL data forwarding (with UPF relocation)

5.3.3.5.1 General

An N-2 based handover may happen when the source NG-RAN determines that an Xn-based handover cannot be done. The same criteria used with the Xn based handover is applicable here as well as far as the UPF relocation is concerned. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the latter case. The illustrations given here assume that an intermediate UPF is present before the handover and it is changed to another intermediate UPF after the handover. The flow assumes that no change in the AMF.

This flow is based on the N2-based handover stage 2 flow in TS 23.502 [7], clause 4.9.1.3.

Within drawings, (source) NG-RAN is shown as S-NG-RAN and as S-5G and target NG-RAN is shown as T-NG-RAN and as T-5G. The V-UPF used before the handover is shown as (S) V-UPF and the V-UPF used after the handover is shown as (T) V-UPF. The numbers 1, 2, 3, 4, 5, 6, 7. 8, 8-a, 8-b, 8-c correspond to the TEID (1), TEID (2), TEID (3), TEID (4), TEID (5), TEID (6), TEID (7), TEID (8), TEID (8a), TEID (8b) and TEID (8c) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at S (V-UPF) and the TEID (3) is the tunnel end point identifier at H-UPF for UL UP packets. TEID (2) is the tunnel end point identifier at (source) NG-RAN and the TEID (4) is the tunnel end point identifier at the S (V-UPF) for the DL UP packets. The TEID (8a) is the tunnel end point identifier at the (target) NG-RAN for the DL data forwarded UP packets. The TEID (8b) is the tunnel end point identifier at the T (V-UPF) for the DL data forwarded UP packets. The TEID (8c) is the tunnel end point identifier at the S (V-UPF) for the DL data forwarded UP packets.

The TEID (8) is the tunnel end point identifier for the DL UP packets at the (target) NG-RAN. The TEID (6) is the tunnel end point identifier for the DL UP packets at the T (V-UPF). The TEID (5) is the tunnel end point identifier for the UL traffic at the H-UPF. The TEID (7) is the tunnel end point identifier at the (T) V-UPF for UL UP packets.

5.3.3.5.2 LI in VPLMN

The generation of xCC happens at (S) V-UPF before the handover and at (T) V-UPF after the handover. It is assumed that the PFCP session ID is used as the target identifier.

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.3.5.2-1 illustrates the first part. The flow-diagram in figure 5.3.3.5.2-2 illustrates the second part. The flow-diagram in figure 5.3.3.5.2-3 illustrates the third part. The flow-diagram in figure 5.3.3.5.2-4 illustrates the fourth part.

Flow 1 of 4

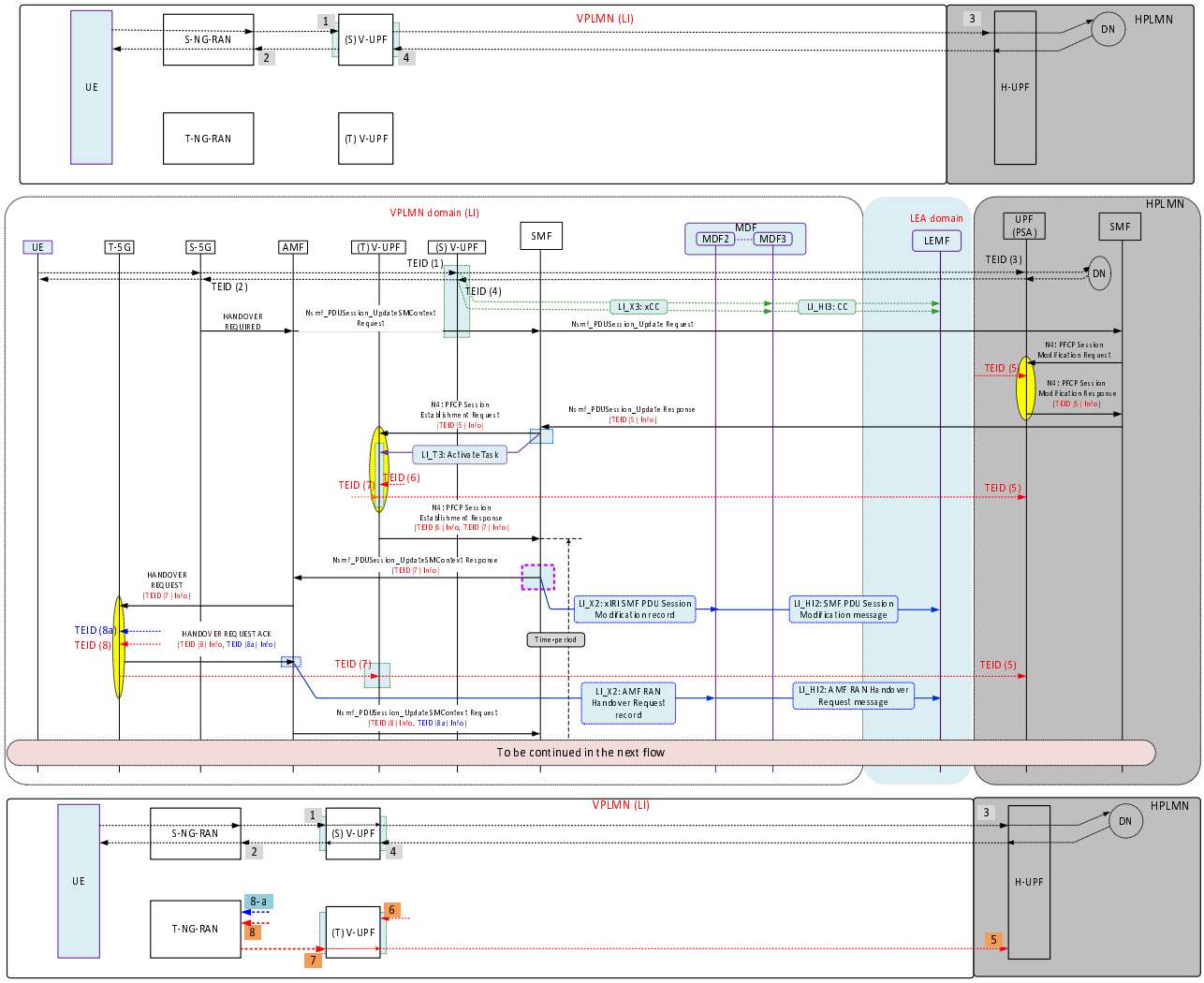


Figure 5.3.3.5.2-1: Indirect DL Data forwarding in roaming case with the UPF relocation (LI in VPLMN) – flow 1 of 4

When the AMF receives a HANDOVER REQUEST ACK from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Request record.

When the PFCP Session Establishment is sent to the (T) V-UPF, the CC-TF in the SMF sends the LI_T3 ActivateTask to the CC-POI present in the (T) V-UPF to start the CC interception. The same correlation number is used for xCC before and after the handover.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 2 of 4

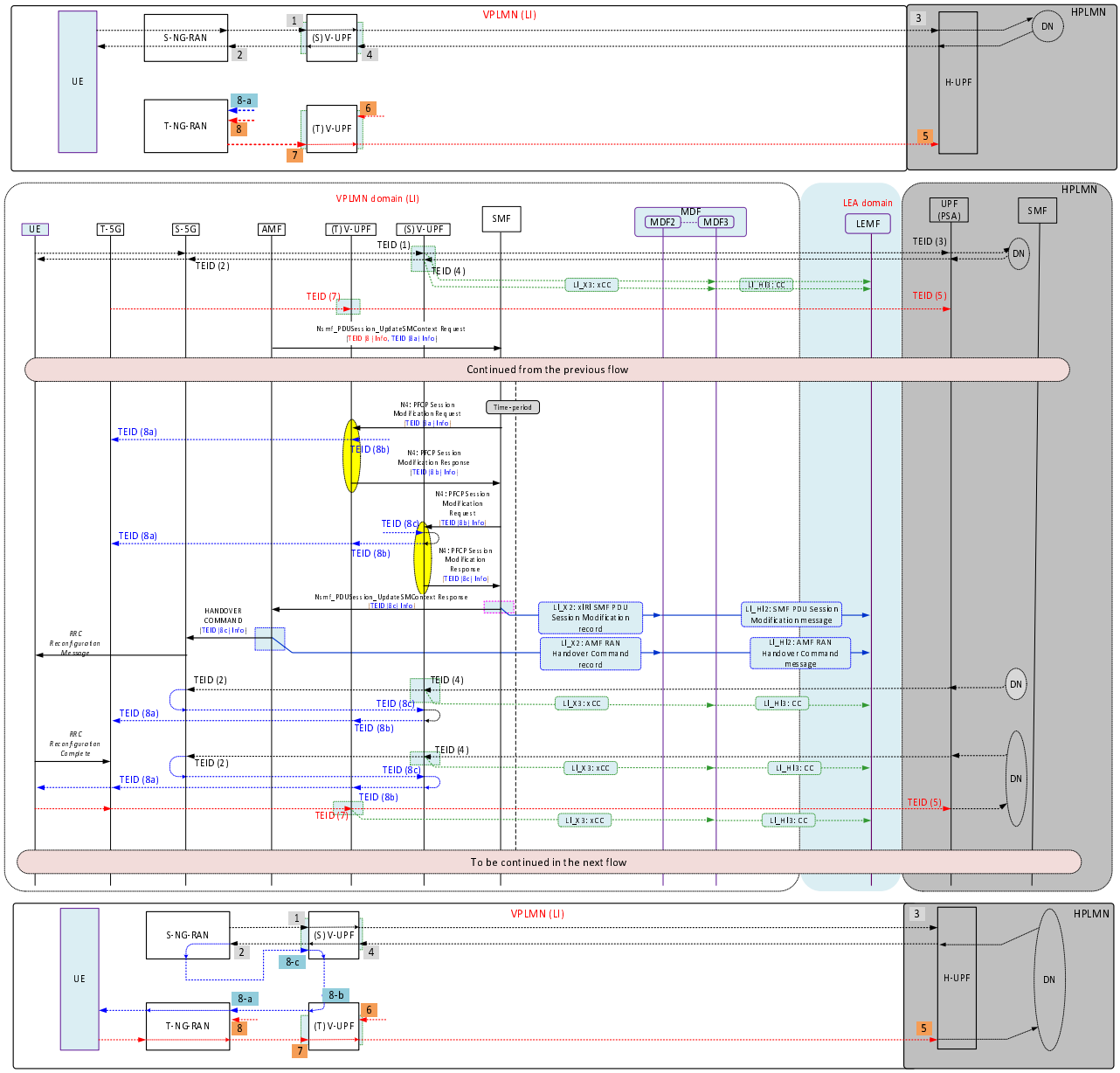


Figure 5.3.3.5.2-2: Indirect DL Data forwarding in roaming case with the UPF relocation (LI in VPLMN) – flow 2 of 4

When the AMF sends the HANDOVER COMMAND to the (source) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Command record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 3 of 4

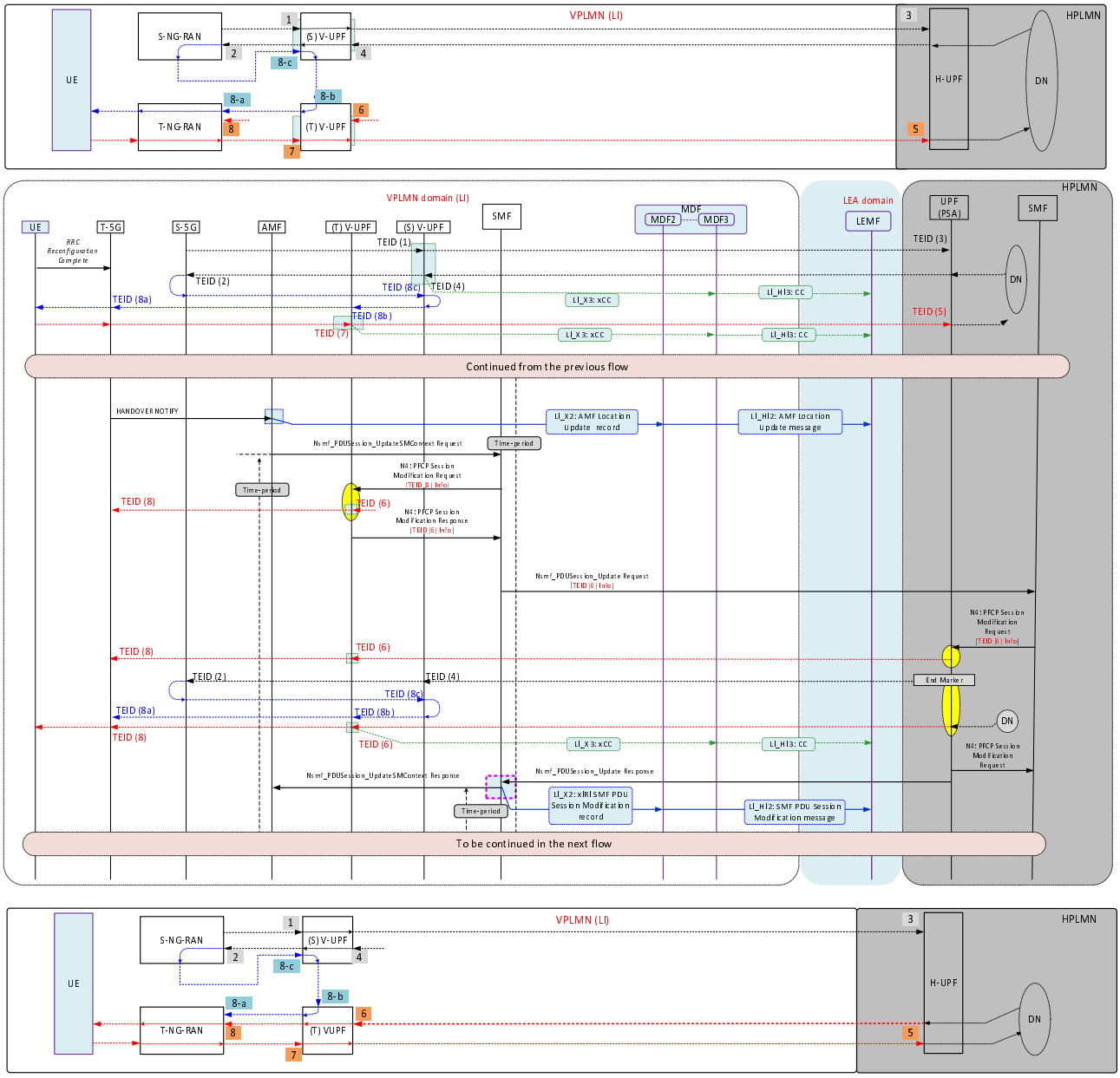


Figure 5.3.3.5.2-3: Indirect DL Data forwarding in roaming case with the UPF relocation (LI in VPLMN) – flow 3 of 4

When the AMF receives a HANDOVER NOTIFY from the (target) NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI Update Location Request record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

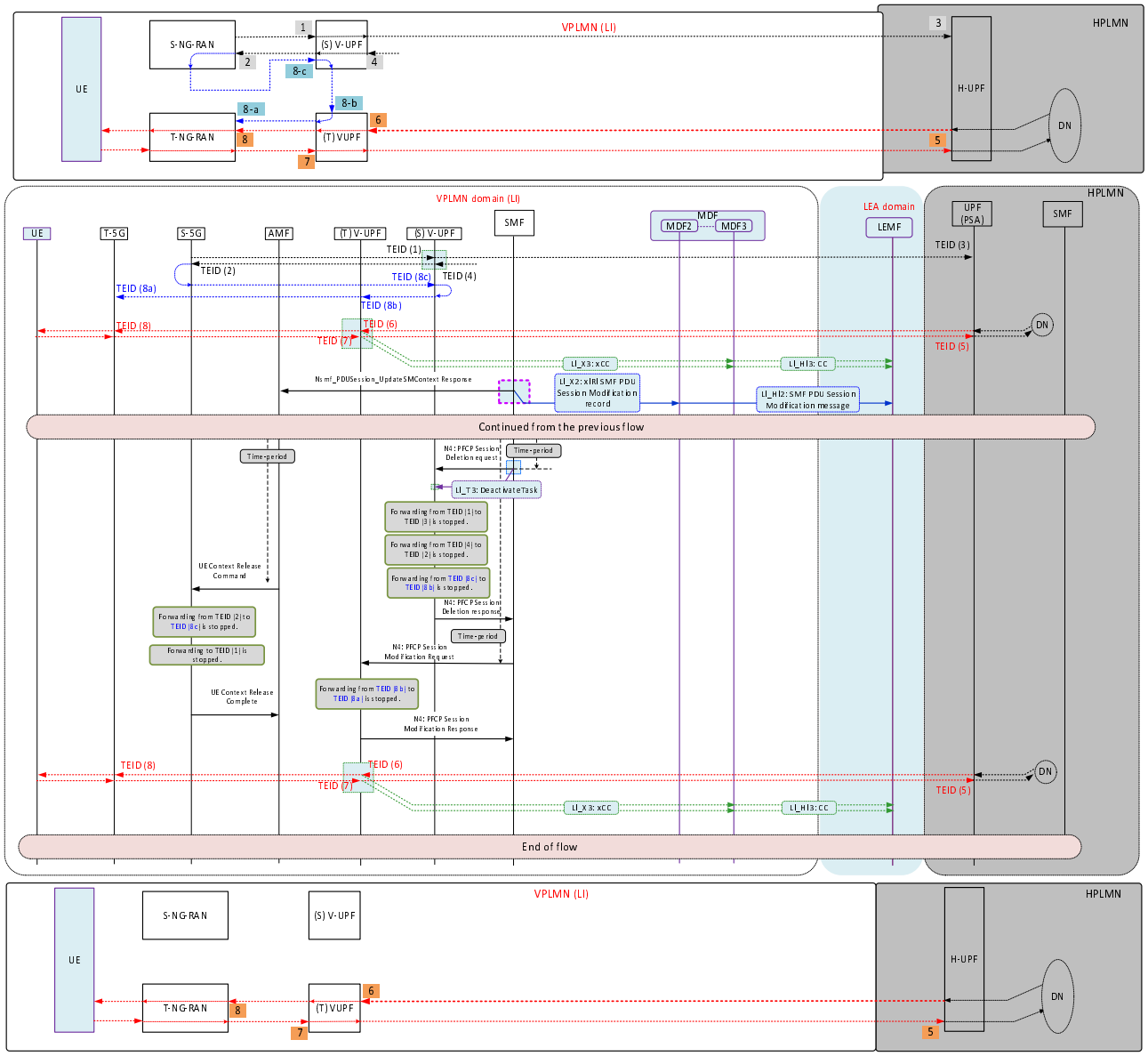


Figure 5.3.3.5.2-4: Indirect DL Data forwarding in roaming case with the UPF relocation (LI in VPLMN) – flow 4 of 4

When the SMF sends the N4: PFCP Session Deletion Request to the (S) V-UPF, the CC-TF present in the SMF sends the LI_T3: DeactivateTask to the CC-POI present in the (S) V-UPF to stop the CC interception since (S) V-UPF is no longer used after the handover.

5.3.3.5.3 LI in HPLMN

The generation of xCC happens at H-UPF. It is assumed that the PFCP session ID is used as the target identifier.

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.3.5.3-1 illustrates the first part. The flow-diagram in figure 5.3.3.5.3-2 illustrates the second part. The flow-diagram in figure 5.3.3.5.3-3 illustrates the third part. The flow-diagram in figure 5.3.3.5.3-4 illustrates the fourth part.

Flow 1 of 4

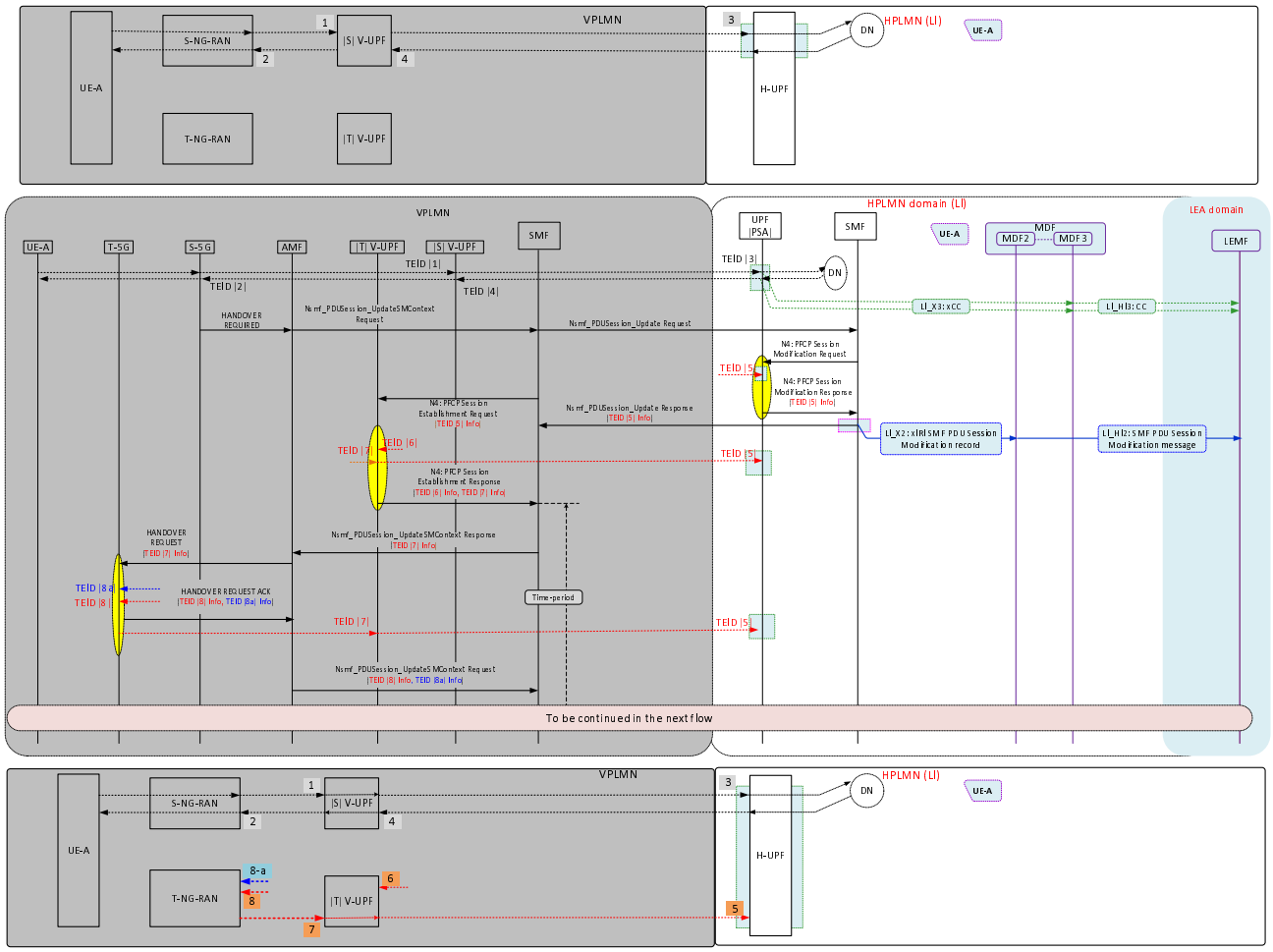


Figure 5.3.3.5.3-1: Indirect DL Data forwarding in roaming case with the UPF relocation (LI in HPLMN) – flow 1 of 4

When the H-SMF sends the Nsmf_PDUSession_Update Response to the V-SMF with the updated PDU session information, the IRI-POI present in the H-SMF generates the xIRI PDU Session Modification record.

Flow 2 of 4

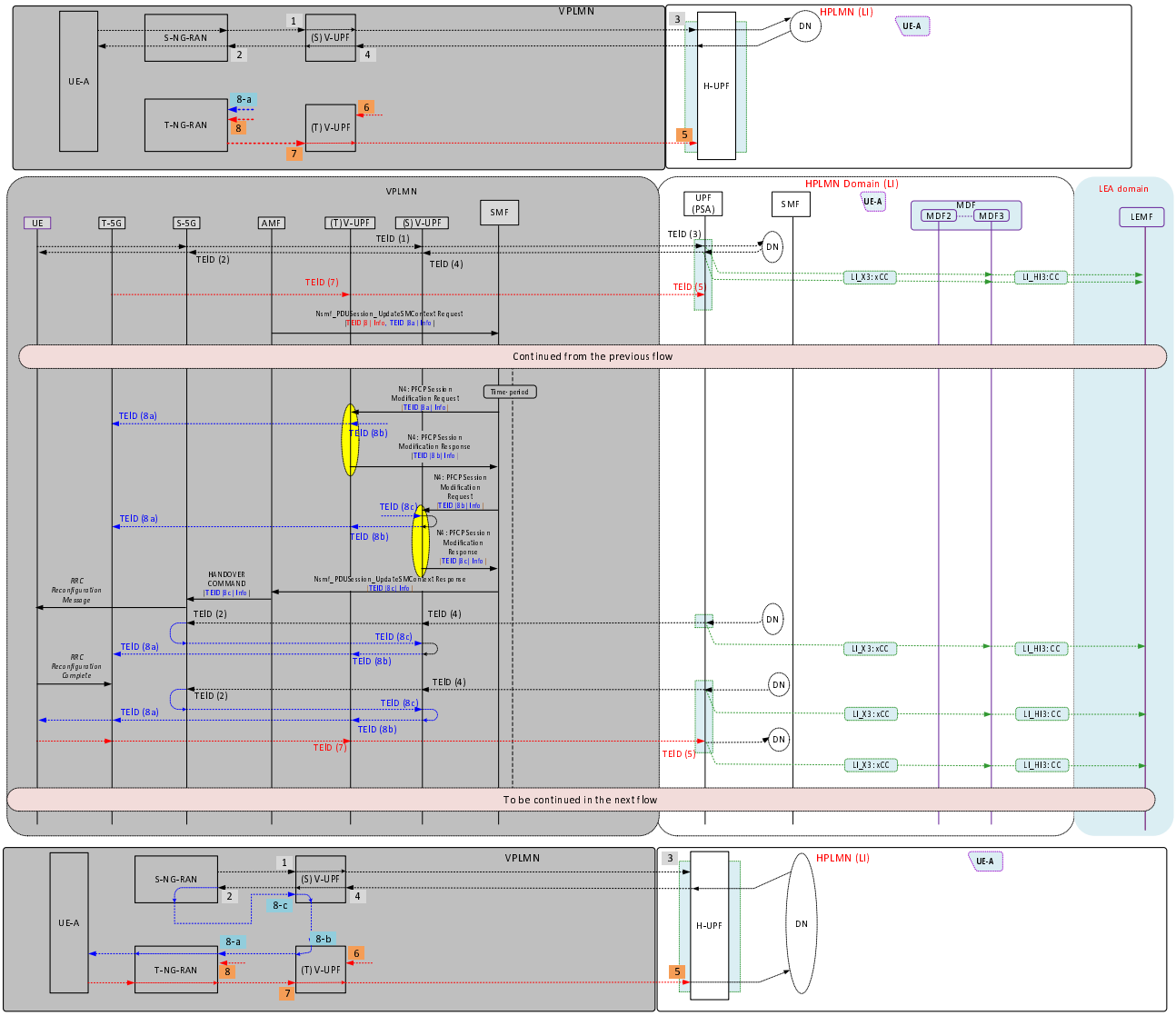


Figure 5.3.3.5.3-2: Indirect DL Data forwarding in roaming case with the UPF relocation (LI in HPLMN) – flow 2 of 4

Flow 3 of 4

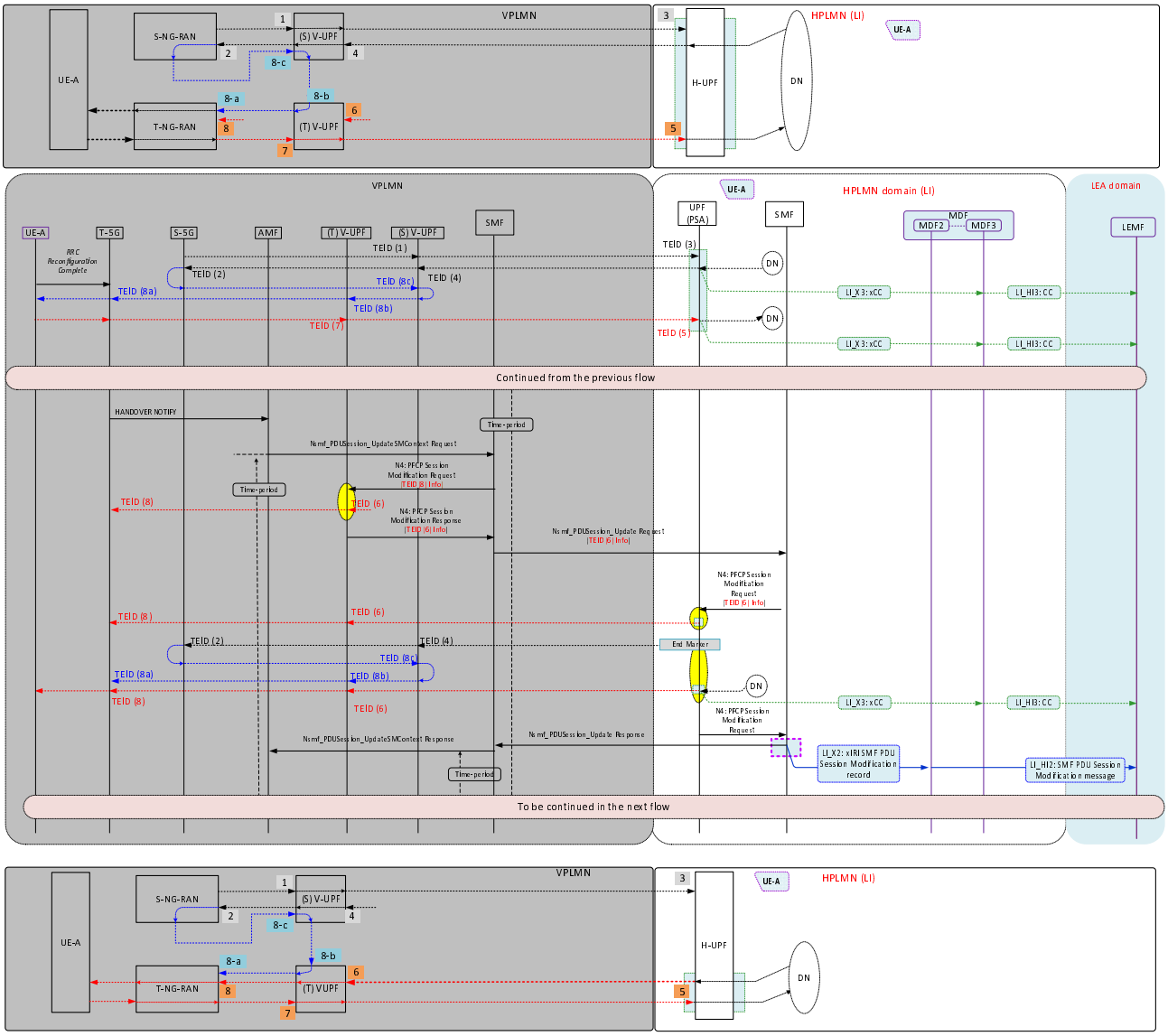


Figure 5.3.3.5.3-3: Indirect DL Data forwarding in roaming case with the UPF relocation (LI in HPLMN) – flow 3 of 4

When the H-SMF sends the Nsmf_PDUSession_Update Response to the V-SMF with the updated PDU session information, the IRI-POI present in the H-SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

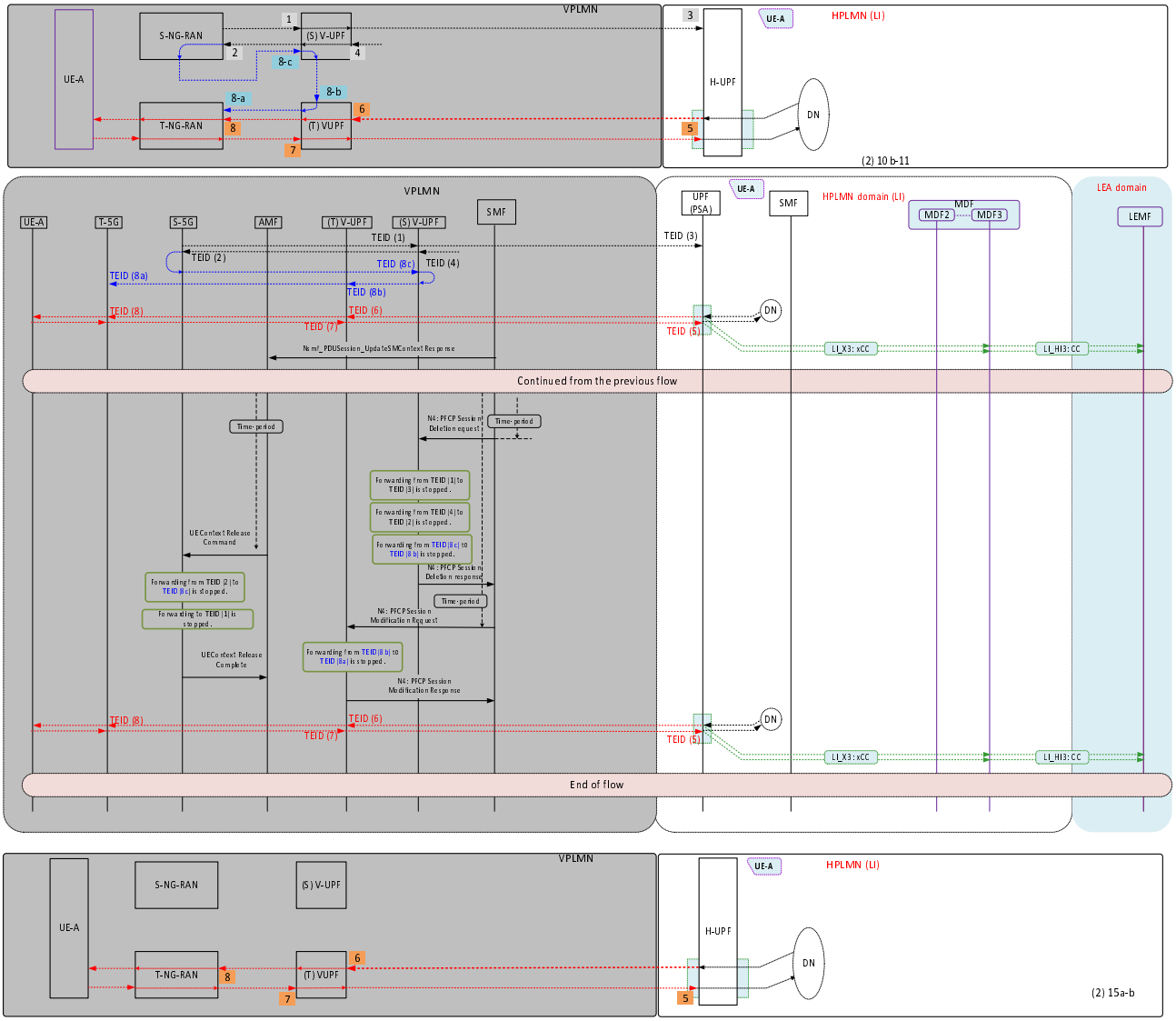


Figure 5.3.3.5.3-4: Indirect DL Data forwarding in roaming case with the UPF relocation (LI in HPLMN) – flow 4 of 4

5.3.4 5G to 4G handover flows

5.3.4.1 Overview

5.3.4.1.1 General

This clause gives an overview of the scenarios related to the handover of a mobile device from NG-RAN to E—UTRAN as described in TS 23.502 [7] clause 4.11.1.2.1. This is also referred to as an N26 based handover. The N26 is an interface between the MME and the AMF.

Handover procedures apply to all QoS flows in the PDU session. During the handover, the Q0S flows are handover to EPS bearers. The QoS flows that are not handed over are released, The CC interception happens in the PGW-U + UPF for a non-roaming scenario (or in the HPLMN with home-routed roaming).

During handover to maintain the data flow continuity, the DL data may be forwarded from the source NG-RAN to the target E-UTRAN.

5.3.4.1.2 Direct forwarding

This is the case where the DL data forwarding happens directly from the NG-RAN to the E-UTRAN. Two scenarios are illustrated here. Non-roaming and home-routed roaming.

Scenario 1

Figure 5.3.4.1.2-1 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is for a non-roaming scenario.

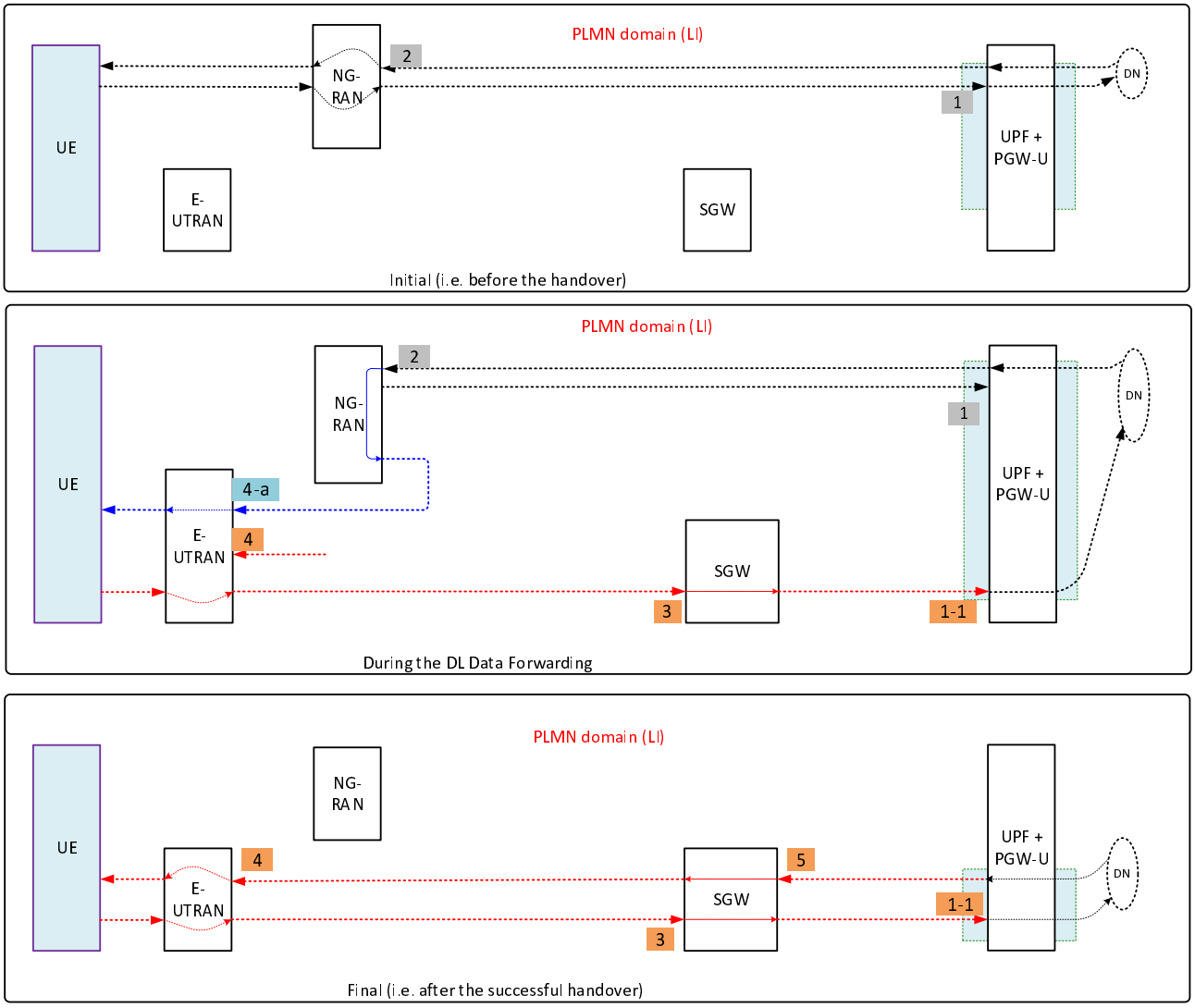


Figure 5.3.4.1.2-1: User plane packet flow during handover direct DL data forward (non-roaming)

The detailed flow-diagrams for the scenario shown in figure 5.3.4.1.2-1 is in clause 5.3.5.2.2.

Scenario 2

Figure 5.3.4.1.2-2 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is a home-routed roaming scenario. The LI is in VPLMN.

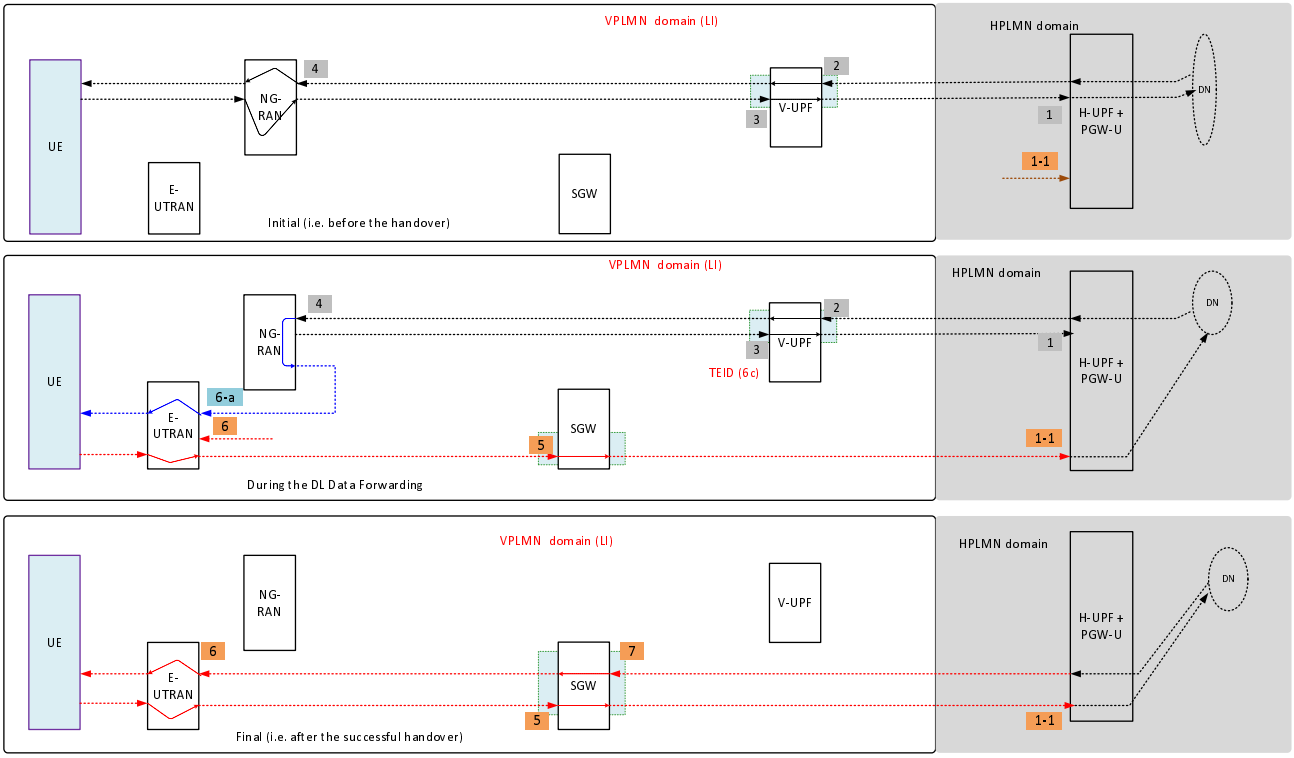


Figure 5.3.4.1.2-2: User plane packet flow during handover direct DL data forward (roaming) LI in VPLMN

Here, the CC-POI functions are moved from V-UPF to SGW. The intercepted CC before the handover and after the handover are not correlated.

Scenario 3

Figure 5.3.4.1.2-3 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is a home-routed roaming scenario. The LI is in HPLMN.

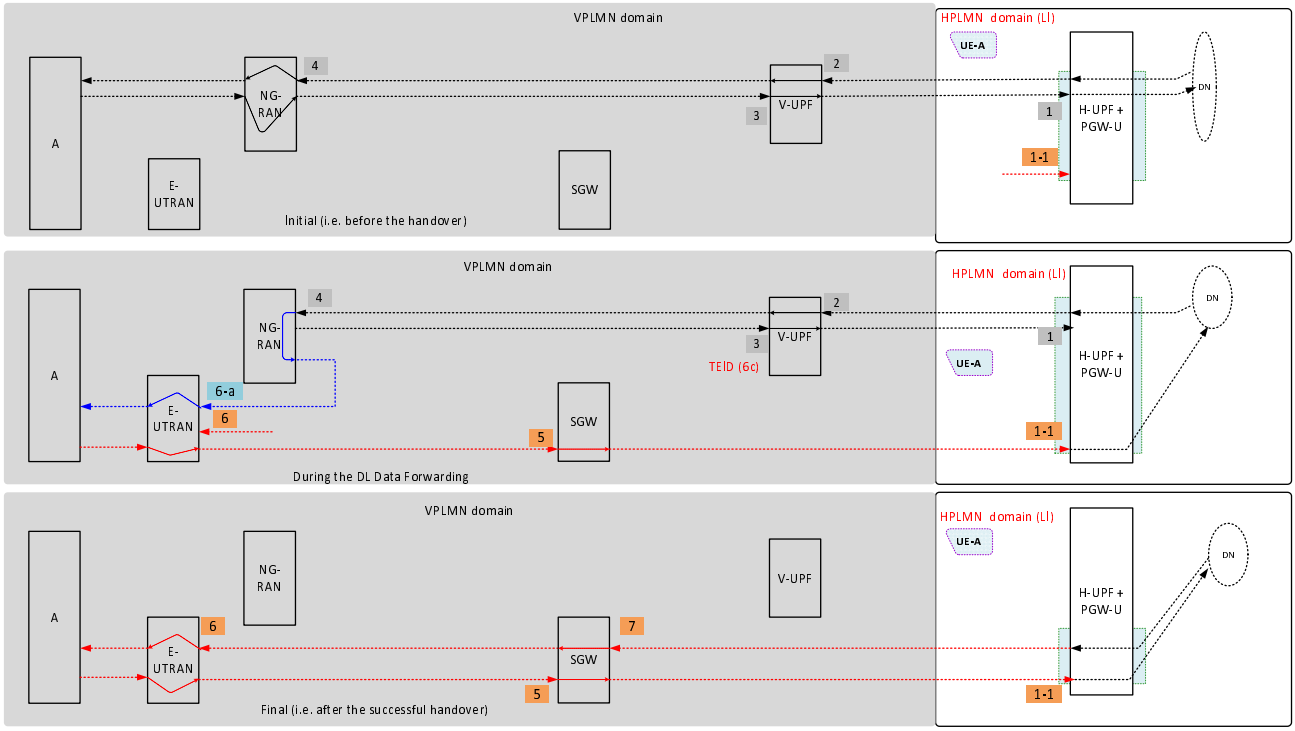


Figure 5.3.4.1.2-3: User plane packet flow during handover direct DL data forward (roaming) LI in HPLMN

Here, the CC-POI functions are provided in H-UPF + PGW-U.

5.3.4.1.3 Indirect forwarding in a non-roaming case

In all the illustrations shown in this clause, indirect DL data forwarding happens. The CC interception is always done at the UPF + PGW-U.

Scenario 1

Figure 5.3.4.1.3-1 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is for a non-roaming scenario. An intermediate UPF is not used for DL data forwarding.

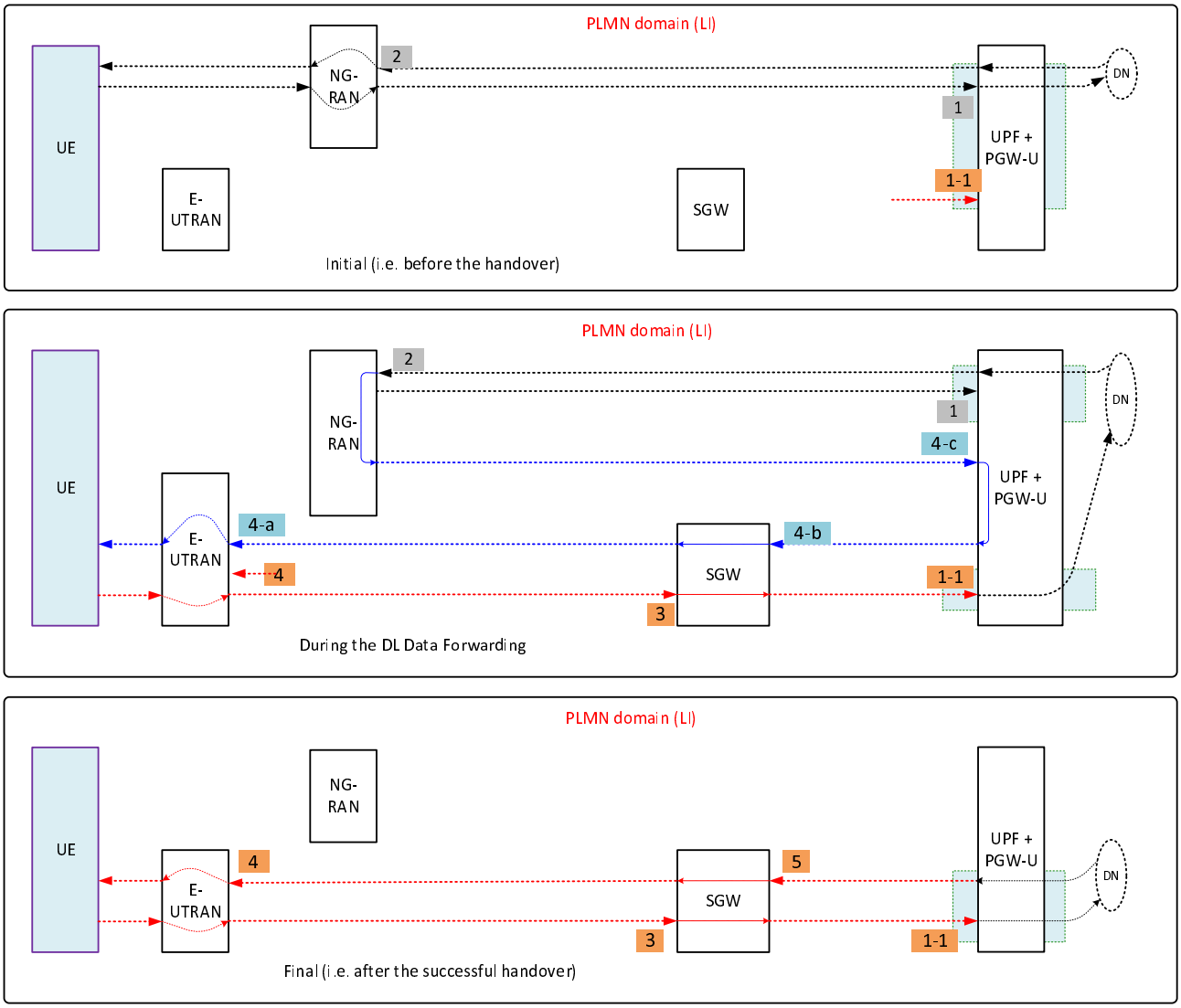


Figure 5.3.4.1.3-1: User plane packet flow during handover indirect DL data forward - case 1

Special care has to be taken within the UPF + PGW-U not to intercept the user plane packets from the DL data forwarding tunnel to avoid double interception of the same packets. A detailed flow diagram for the scenario shown in figure 5.3.5.1.3-1 is in clause 5.3.4.3.2.

Scenario 2

Figure 5.3.4.1.3-2 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. An intermediate is used for DL data forwarding. This is for a non-roaming scenario.

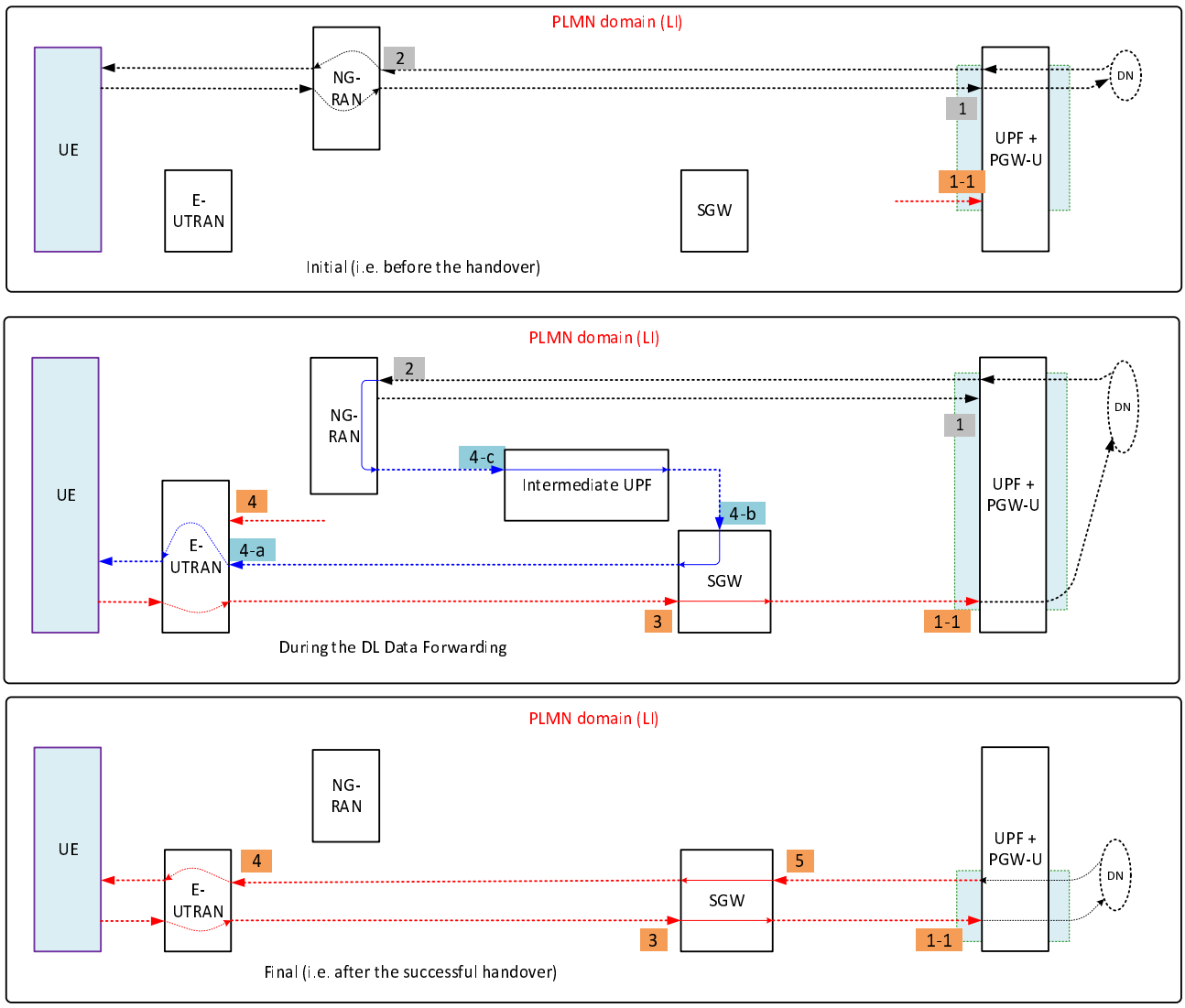


Figure 5.3.4.1.3-2: User plane packet flow during handover indirect DL data forward - case 2

5.3.4.1.4 Indirect forwarding in a roaming case

In all the illustrations shown in this clause, indirect DL data forwarding happens.

Scenario 1

Figure 5.3.4.1.4-1 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is for home-routed roaming scenario. The LI is in the VPLMN. An intermediate UPF is not used for DL data forwarding.

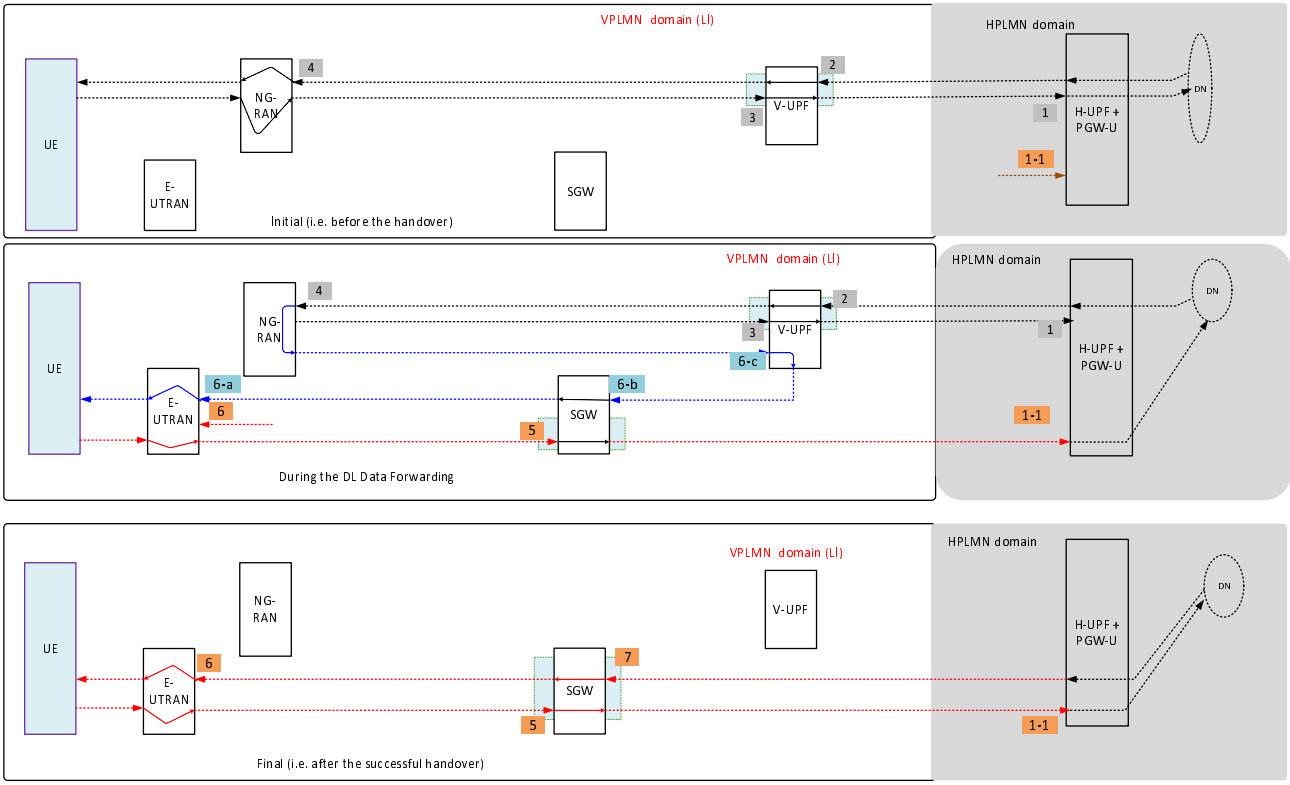


Figure 5.3.4.1.4-1: User plane packet flow during handover indirect DL data forward - case 1

Special care has to be taken within the V-UPF and the SGW not to intercept the user plane packets from the DL data forwarding tunnel to avoid double interception of the same packets. Here, the CC-POI functions are moved from V-UPF to SGW. The intercepted CC before the handover and after the handover are not correlated. A detailed flow diagram for the scenario shown in clause 5.3.4.4.2.

Scenario 2

Figure 5.3.4.1.4-2 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. An intermediate is used for DL data forwarding. Here, the LI is in VPLMN.

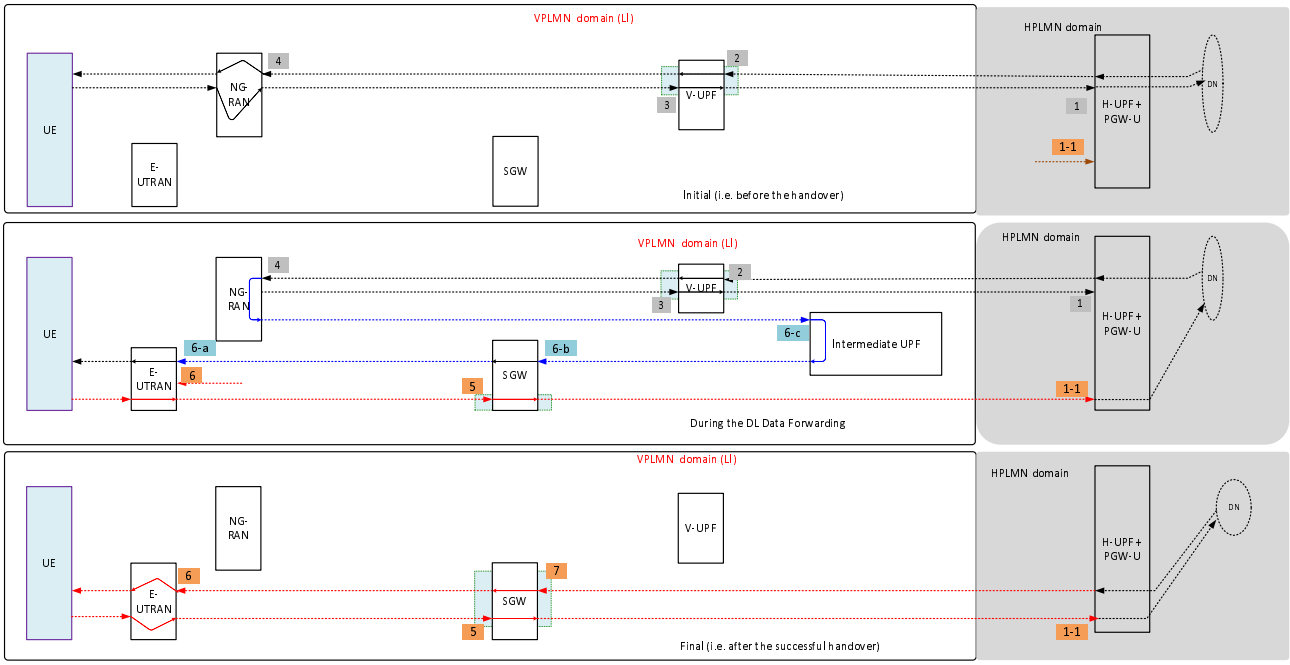


Figure 5.3.4.1.4-2: User plane packet flow during handover indirect DL data forward - case 2

Special care has to be taken within the SGW not to intercept the user plane packets from the DL data forwarding tunnel to avoid double interception of the same packets. Here, the CC-POI functions are moved from V-UPF to SGW. The intercepted CC before the handover and after the handover are not correlated.

Scenario 3

Figure 5.3.4.1.4-3 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. An intermediate is not used for DL data forwarding. Here the LI is in HPLMN.

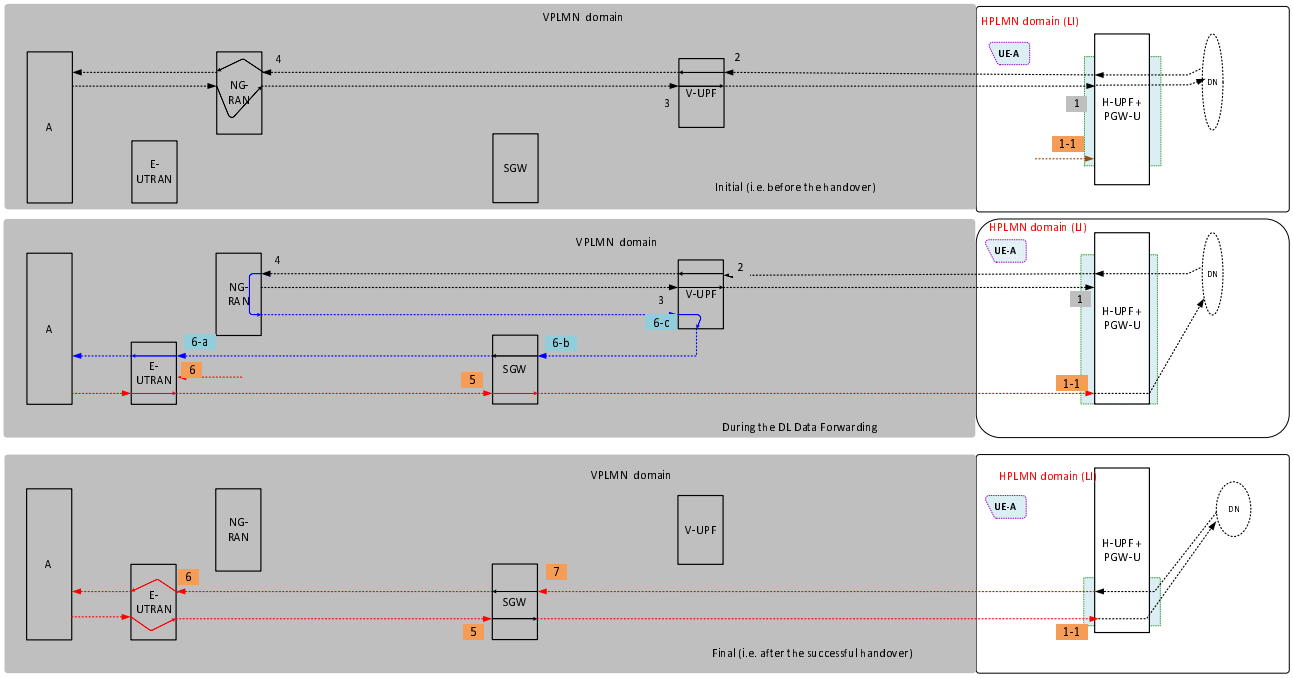


Figure 5.3.4.1.4-3: User plane packet flow during handover indirect DL data forward - case 3

A detailed flow diagram for the scenario shown in clause 5.3.4.4.3.

Scenario 4

Figure 5.3.4.1.4-4 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. An intermediate is used for DL data forwarding. Here the LI is in HPLMN.

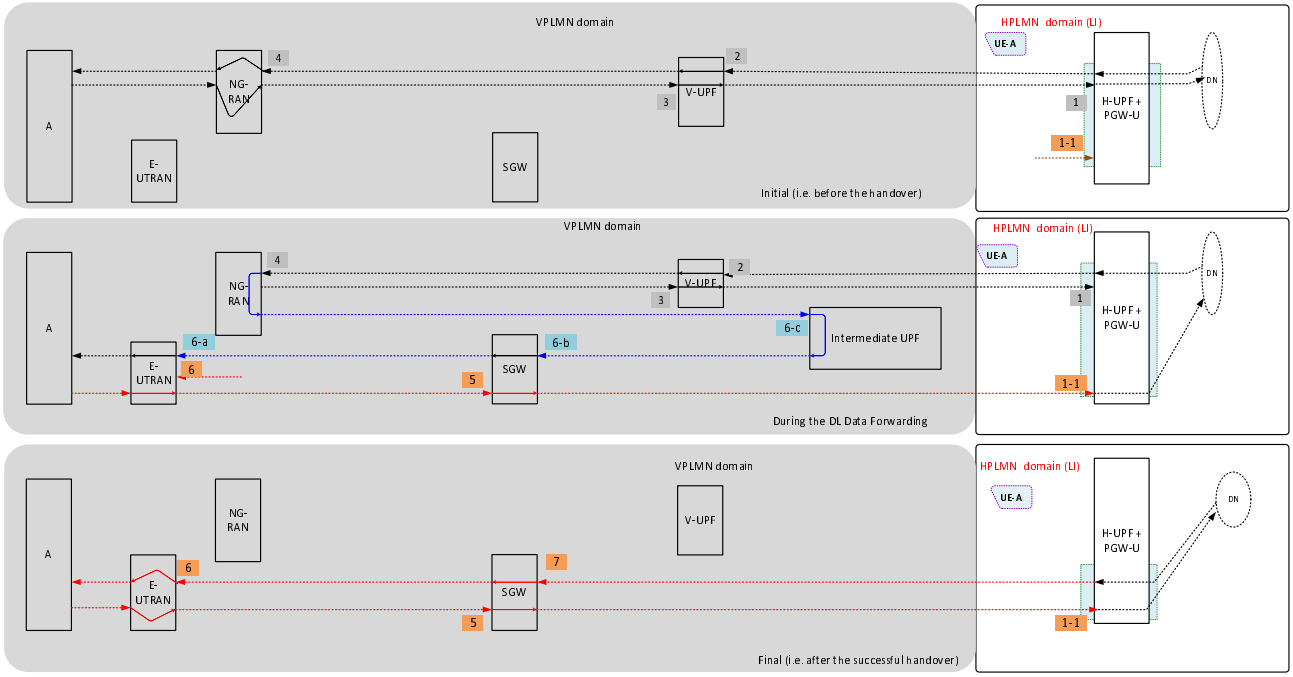


Figure 5.3.4.1.4-4: User plane packet flow during handover indirect DL data forward - case 4

As can be deduced from illustrations shown in figures 5.3.4.1.3-3, 5.3.4.1.4-3 and 5.3.4.1.4-4, the DL data forwarding variations do not seem to have an influence for the LI in HPLMN. Note that LI in VPLMN and LI in HPLMN can happen on the same target via mutually independent warrants.

5.3.4.2 Detailed flows - non-roaming, direct DL data forwarding

5.3.4.2.1 General

An N-26 based handover may happen when there is an inter-system mobility for the device between the 5G AN and the 4G AN. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the former case and covers the scenario of 5G to 4G handover. This flow is based on the N26-based handover stage 2 flow in TS 23.502 [7], clause 4.11.1.2.1.

Within flow-diagrams, the NG-RAN is shown 5G and E-UTRAN is shown as 4G. The numbers 1, 1-1. 2, 3, 4, 4-a and 5 correspond to the TEID (1), TEID (1-1), TEID (2), TEID (3), TEID (4), TEID (4a) and TEID (5) respectively. For each QoS flow, a tunnel end point identifier is reserved at the UPF + PGW-U for the UL UP packets. In these illustrations, the reserved tunnel end point identifier is shown as TEID (1-1). DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at UPF + PGW-U for UL UP packets. TEID (2) is the tunnel end point identifier at NG-RAN for the DL UP packets. The TEID (4a) is the tunnel end point identifier at the E-UTRAN for the DL data forwarded UP packets. The direct DL data forwarding happens from NG-RAN to the E-UTRAN.

The TEID (4) is the tunnel end point identifier at the E-UTRAN and TEID (5) is the tunnel end point identifier at the S-GW for the DL UP packets after the after handover. The TEID (3) is the tunnel end point identifier for the UL UP packets at the S-GW. The TEID (1-1) is the reserved tunnel end point identifier for the UL UP packets at the UPF + PGW-U.

The generation of xCC from the UP packets done by the CC-POI present in the anchor UPF + PGW-U.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF + PGW-U would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.4.2.2 The flows

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.4.2.2-1 illustrates the first part. The flow-diagram in figure 5.3.4.2.2-2 illustrates the second part. The flow-diagram in figure 5.3.4.2.2-3 illustrates the third part. The flow-diagram in figure 5.3.4.2.2-4 illustrates the fourth part.

Flow 1 of 4

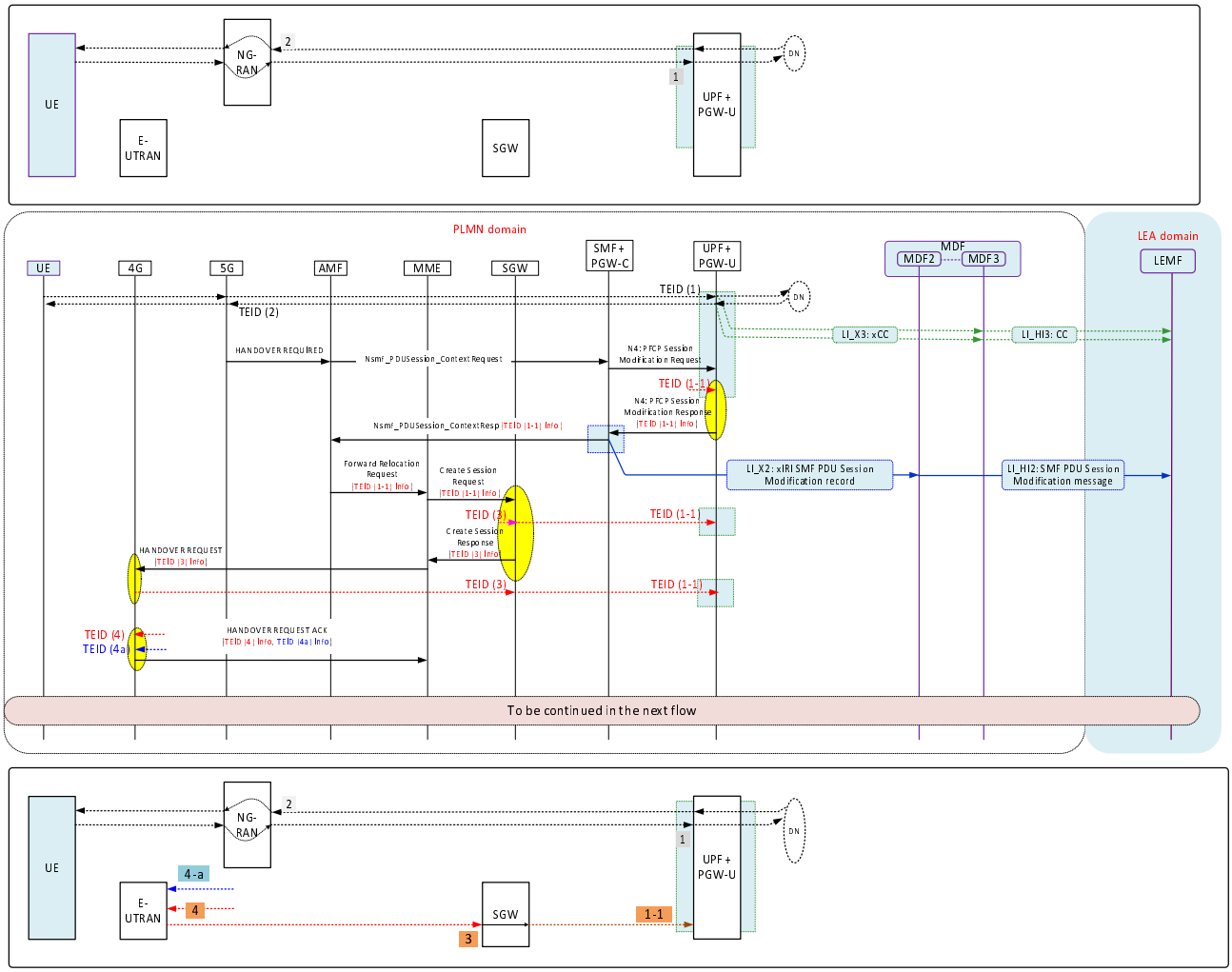


Figure 5.3.4.2.2-1: N26-based handover (non-roaming) direct DL data forward – flow 1 of 4

When the SMF +PGW-C sends the Nsmf_PDUSession_Context Response to the AMF, the IRI-POI present in the SMF generates the xIRI SMF PDU Session Modification record.

Flow 2 of 4

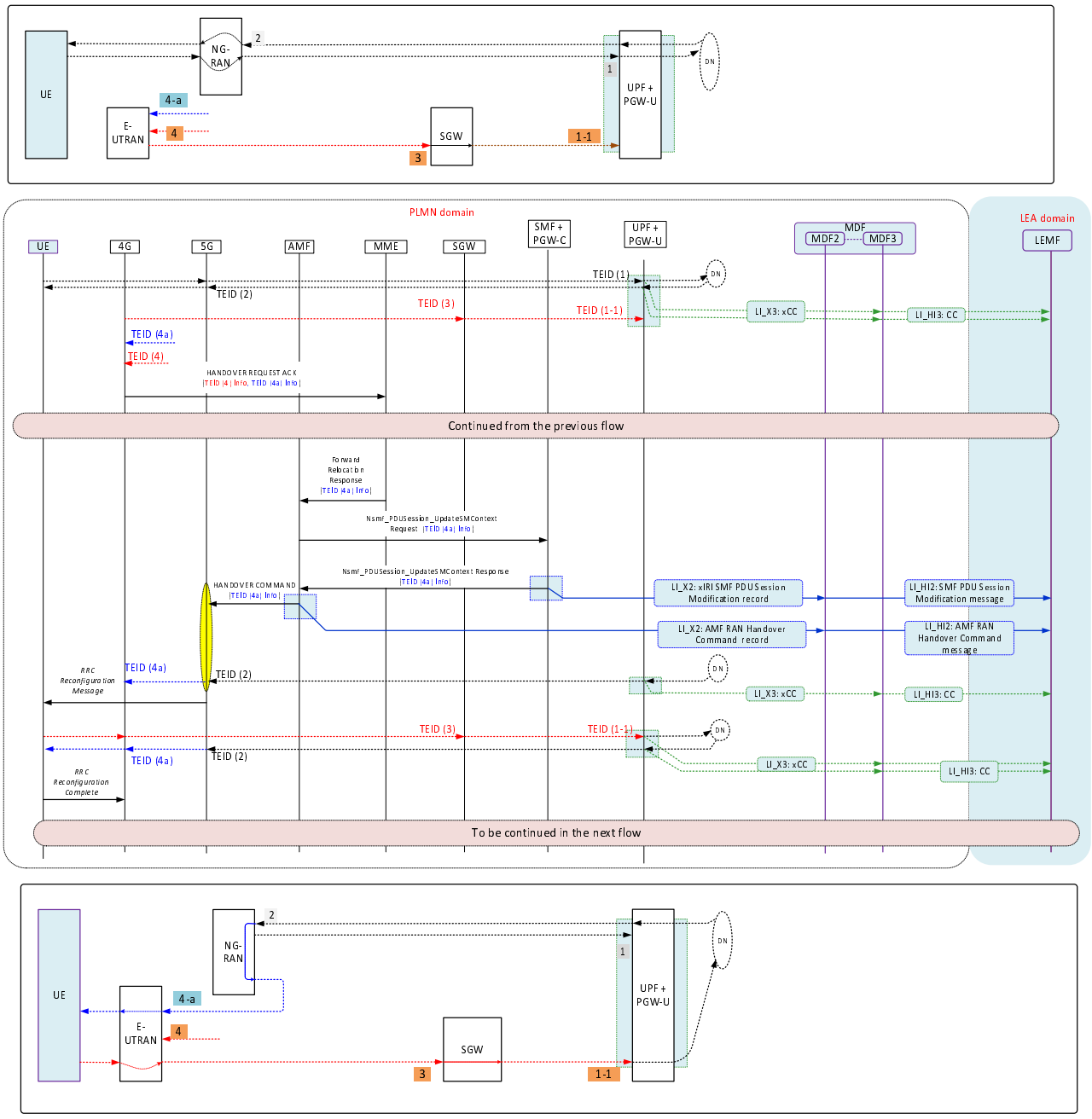


Figure 5.3.4.2.2-2: N26-based handover (non-roaming) direct DL data forward – flow 2 of 4

When the AMF sends the HANDOVER COMMAND to the NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Command record.

When the SMF + PGW-C sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI SMF PDU Session Modification record.

In this illustration, PFCP Session ID is used as the target identifier for the xCC generation in the CC-POI present in the UPF + PGW-U and therefore, the CC-POI in the UPF + PGW-U would generate the xCC from the packets received at TEID (1-1) without any additional trigger from the CC-TF present in SMF + PGW-C.

Flow 3 of 4

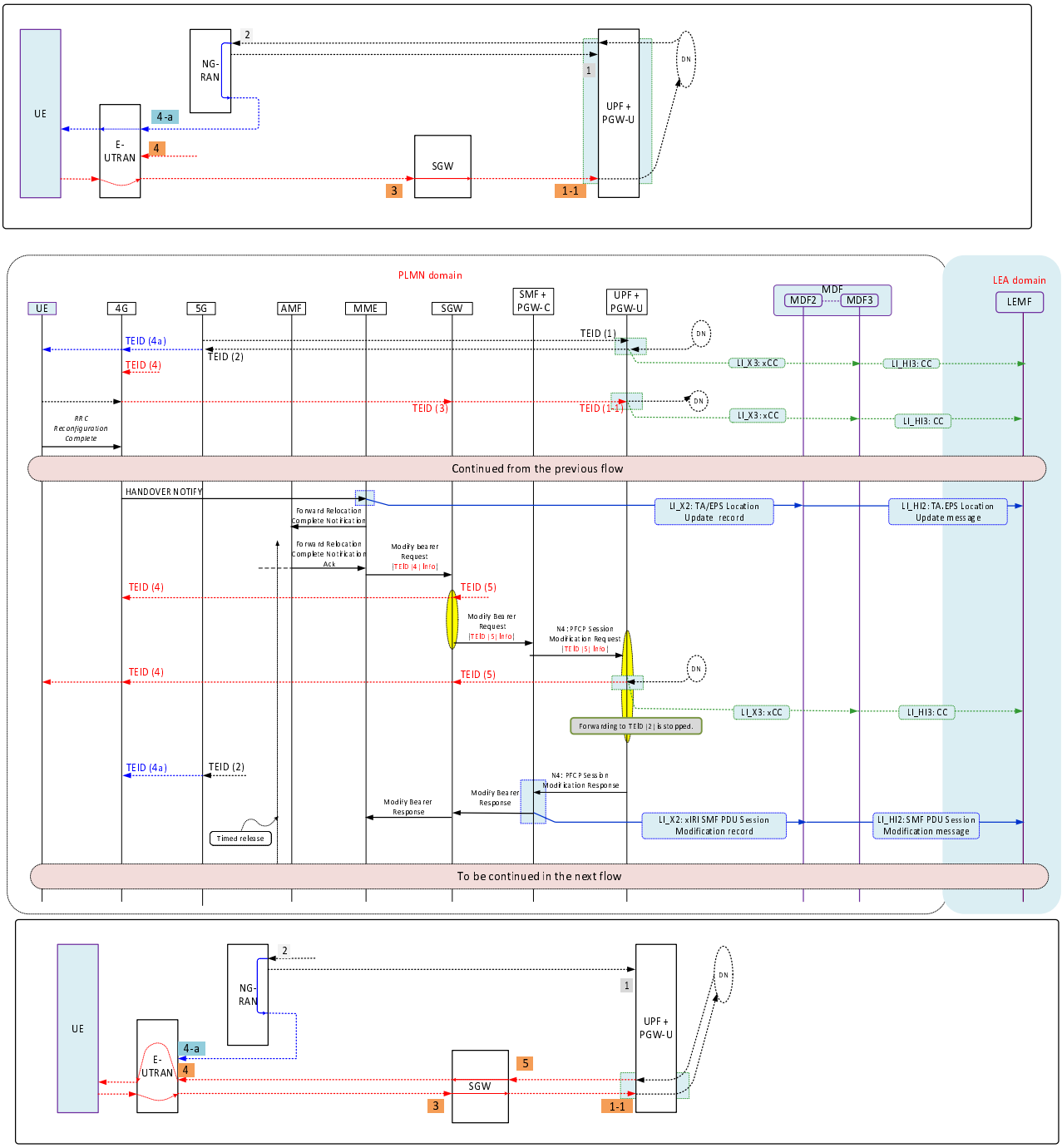


Figure 5.3.4.2.2-3: N26-based handover (non-roaming) direct DL data forward – flow 3 of 4

When the MME receives a HANDOVER NOTIFY from the E-UTRAN, the IRI-POI present in the MME generates the xIRI TA/EPS Location Update record.

When the SMF + PGW-C sends a ModifyBearer response to the SGW, the IRI-POI present in the SMF generates the xIRI SMF PDU Session Modification record.

Flow 4 of 4

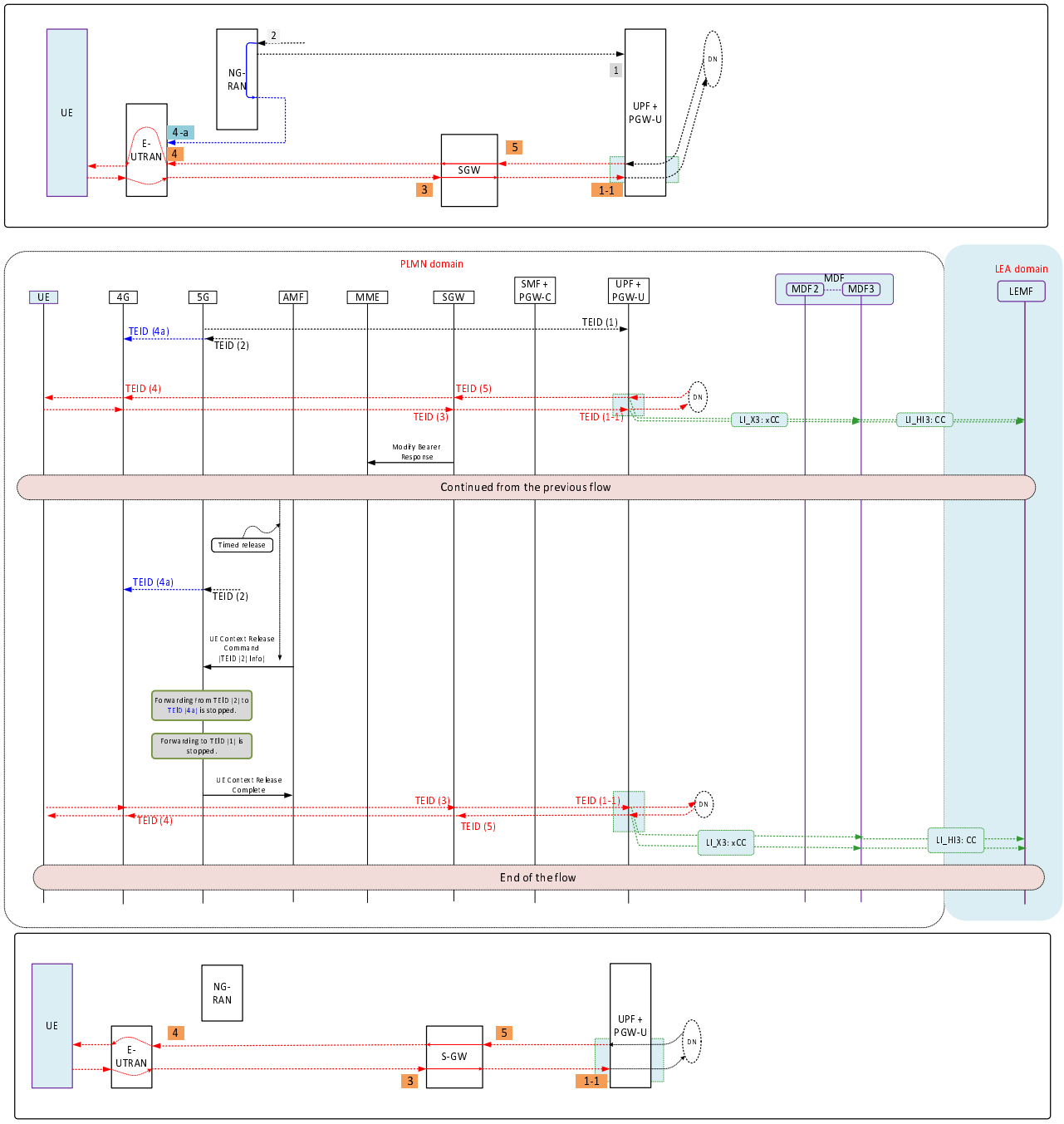


Figure 5.3.4.2.2-4: N26-based handover (non-roaming) direct DL data forward – flow 4 of 4

5.3.4.3 Detailed flows - non-roaming, indirect DL data forwarding

5.3.4.3.1 General

An N-26 based handover may happen when there is an inter-system mobility for the device between the 5G AN and the 4G AN. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the latter case and covers the scenario of 5G to 4G handover. This flow is based on the N26-based handover stage 2 flow in TS 23.502 [7], clause 4.11.1.2.1.

Within flow-diagrams, the NG-RAN is shown 5G and E-UTRAN is shown as 4G. Within the drawings, the numbers 1, 1-1. 2, 3, 4, 4-a,4-b, 4-c and 5 correspond to the TEID (1), TEID (1-1), TEID (2), TEID (3), TEID (4), TEID (4a), TEID (4b), TEID (4c) and TEID (5) respectively. For each QoS flow, a tunnel end point identifier is reserved at the UPF +

PGW-U for the UL UP packets. In these illustrations, the reserved tunnel end point identifier is shown as TEID (1-1). DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at UPF + PGW-U for UL UP packets. TEID (2) is the tunnel end point identifier at NG-RAN for the DL UP packets. The TEID (4a) is the tunnel end point identifier at the E-UTRAN, TEID (4b) is the tunnel end point identifier at the S-GW and TEID (4c) is the tunnel end point identifier at the UPF + PGW-U for the DL data forwarded UP packets.

The TEID (4) is the tunnel end point identifier at the E-UTRAN and TEID (5) is the tunnel end point identifier at the S-GW for the DL UP packets after the after handover. The TEID (3) is the tunnel end point identifier for the UL UP packets at the S-GW. The TEID (1-1) is the reserved tunnel end point identifier for the UL UP packets at the UPF + PGW-U.

The generation of xCC from the UP packets done by the CC-POI present in the anchor UPF + PGW-U.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF + PGW-U would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.4.3.2 The flows

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.4.3.2-1 illustrates the first part. The flow-diagram in figure 5.3.4.3.2-2 illustrates the second part. The flow-diagram in figure 5.3.4.3.2-3 illustrates the third part. The flow-diagram in figure 5.3.4.3.2-4 illustrates the fourth part.

Flow 1 of 4

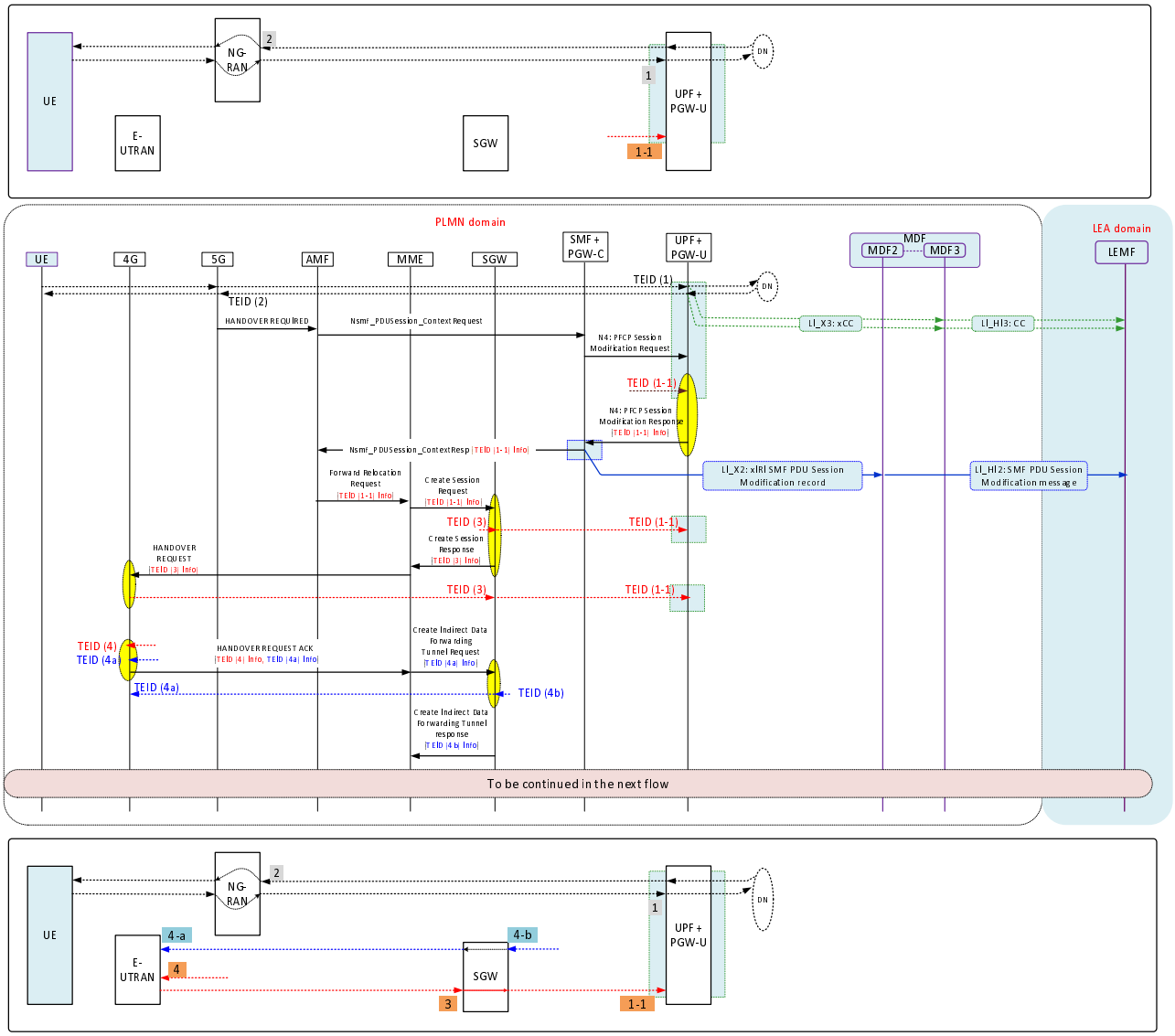


Figure 5.3.4.3.2-1: N26-based handover (non-roaming) indirect DL data forward – flow 1 of 4

When the SMF +PGW-C sends the Nsmf_PDUSession_Context Response to the AMF, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 2 of 4

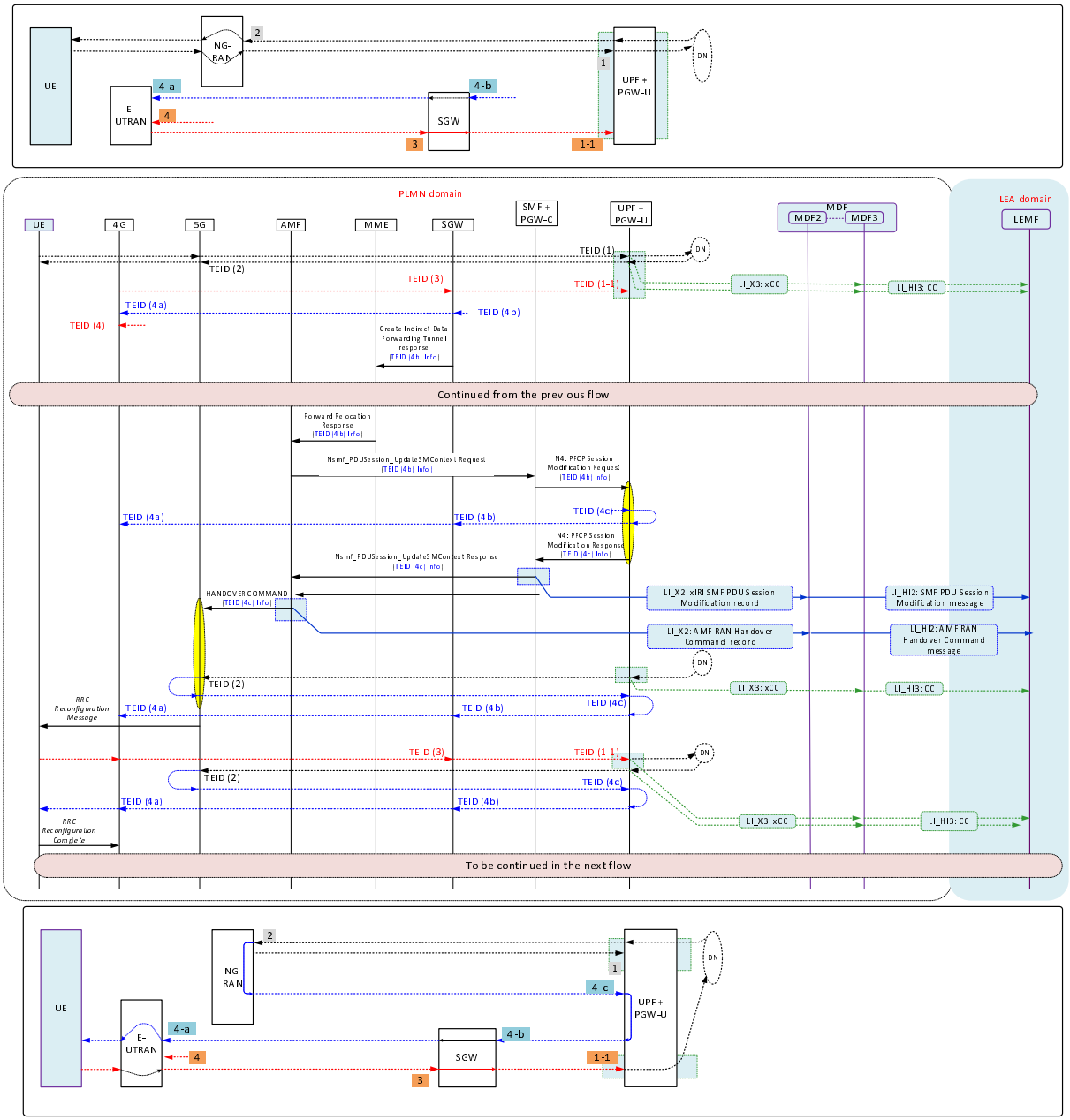


Figure 5.3.4.3.2-2: N26-based handover (non-roaming) indirect DL data forward - flow 2 of 4

When the AMF sends the HANDOVER COMMAND to the NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Command record.

The CC-POI in UPF + PGW-U does not generate the xCC for the UP packets from DL data forwarding tunnels (i.e. incoming at TEID (4c) and outgoing to TEID (4b). Likewise, the SGW does not generate the xCC for the UP packets from DL data forwarding tunnels (i.e. incoming at TEID (4b) and outgoing to TEID (4a).

When the SMF + PGW-C sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

In this illustration, PFCP Session ID is used as the target identifier for the xCC generation in the CC-POI present in the UPF + PGW-U and therefore, the CC-POI in the UPF + PGW-U would generate the xCC from the packets received at TEID (1-1) without any additional trigger from the CC-TF present in SMF + PGW-C.

Flow 3 of 4

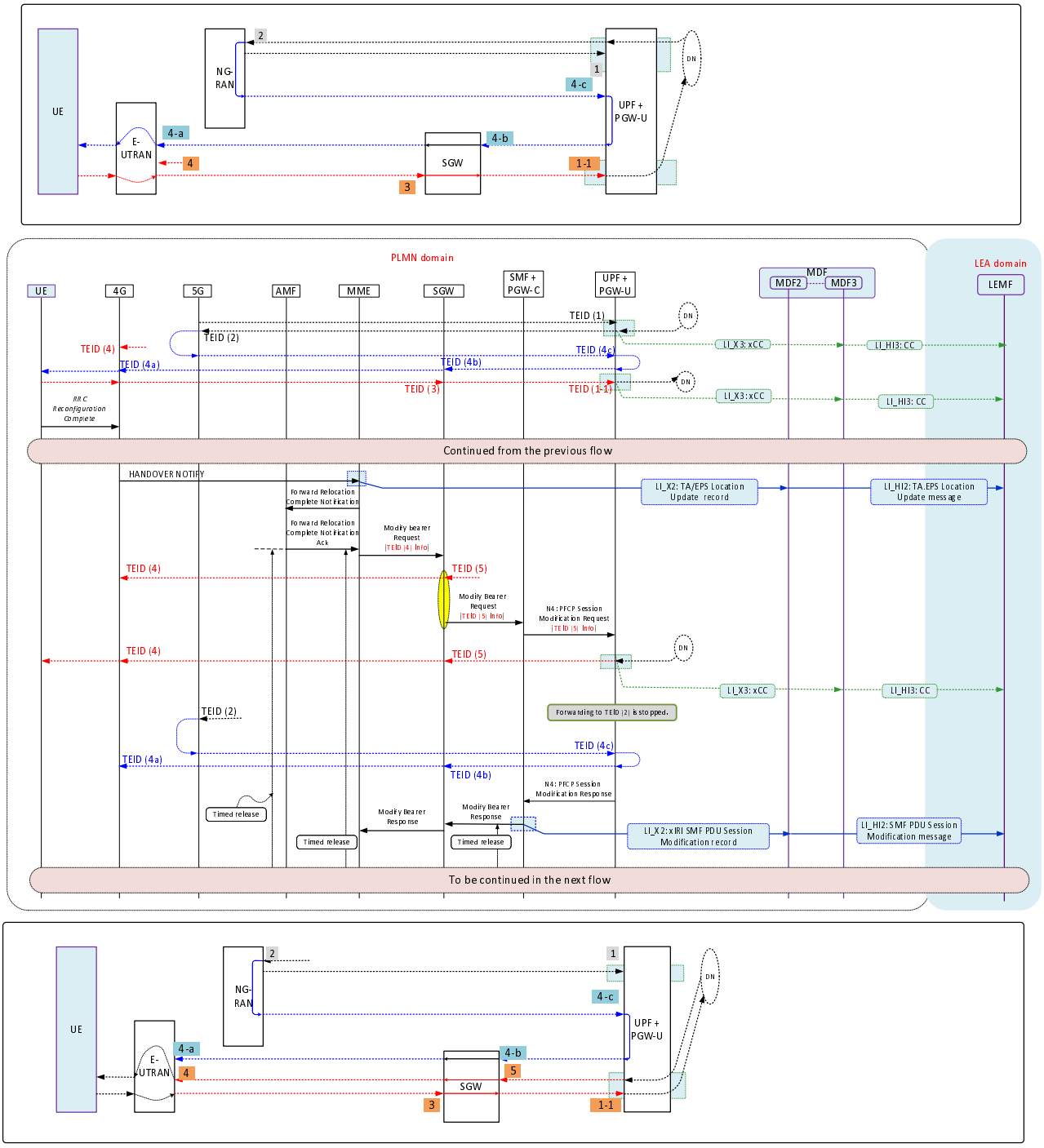


Figure 5.3.4.3.2-3: N26-based handover (non-roaming) indirect DL data forward – flow 3 of 4

When the MME receives a HANDOVER NOTIFY from the E-UTRAN, the IRI-POI present in the MME generates the xIRI TA/EPS Location Update record.

When the SMF + PGW-C sends a ModifyBearer response to the SGW, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

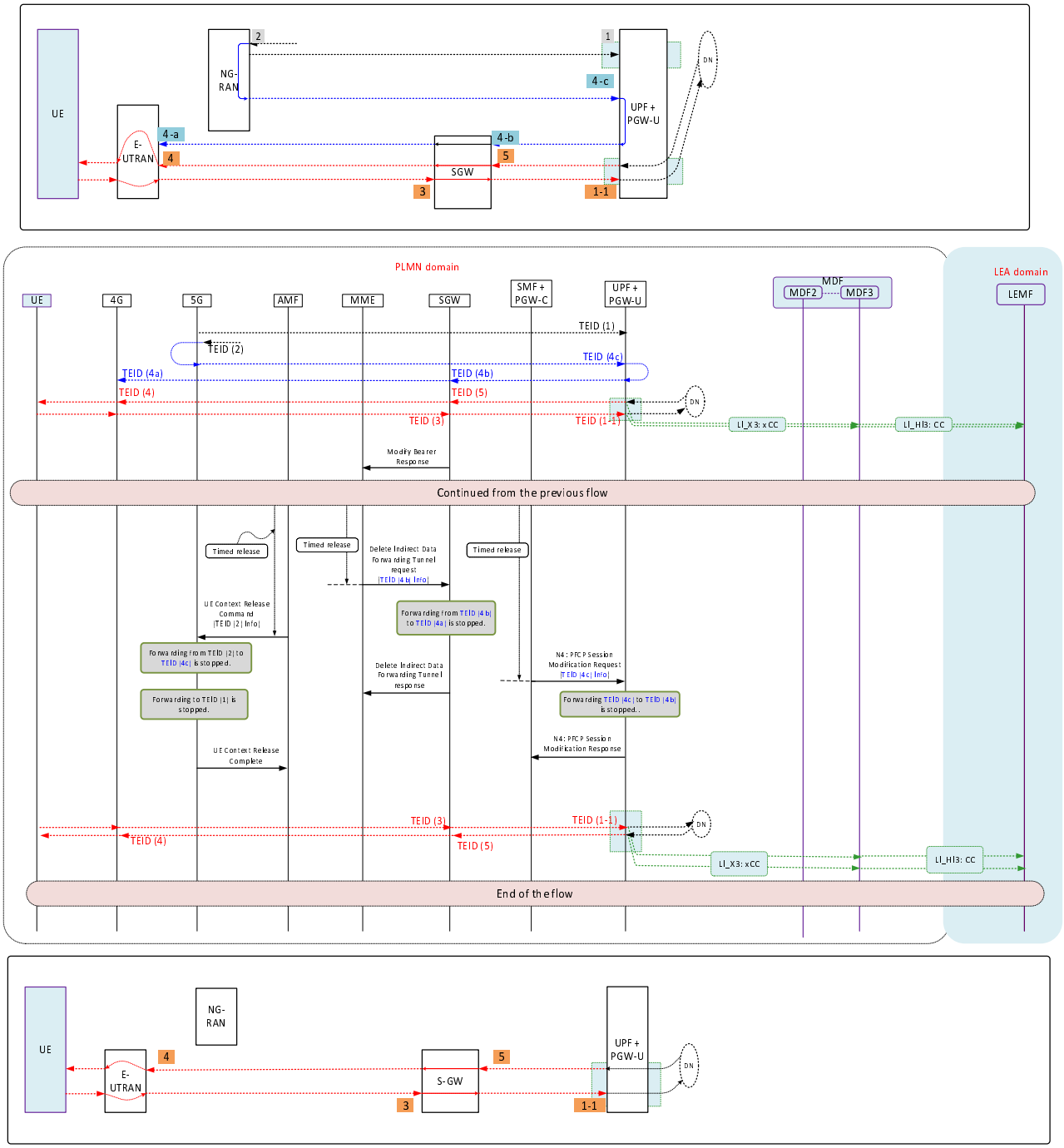


Figure 5.3.4.3.2-4: N26-based handover (non-roaming) indirect DL data forward – flow 4 of 4

5.3.4.4 Detailed flows – home-routed roaming, indirect DL data forwarding

5.3.4.4.1 General

An N-26 based handover may happen when there is an inter-system mobility for the device between the 5G AN and the 4G AN. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the latter case and covers the scenario of 5G to 4G handover. This flow is based on the N26-based handover stage 2 flow in TS 23.502 [7], clause 4.11.1.2.1.

Within flow-diagrams, the NG-RAN is shown 5G and E-UTRAN is shown as 4G. Within the drawings, the numbers 1, 1-1. 2, 3, 4, 5, 6, 6-a, 6-b, 6-c and 7 correspond to the TEID (1), TEID (1-1), TEID (2), TEID (3), TEID (4), TEID (5),

TEID (6), TEID (6a), TEID (6b), TEID (6c) and TEID (7) respectively. For each QoS flow, a tunnel end point identifier is reserved at the UPF + PGW-U for the UL UP packets. In these illustrations, the reserved tunnel end point identifier is shown as TEID (1-1). DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at H-UPF + PGW-U, TEID (3) is the tunnel end point identifier at the V-UPF for UL UP packets. TEID (2) is the tunnel end point identifier at V-UPF and TEID (4) is the tunnel end point identifier at NG-RAN for the DL UP packets.

In the illustrations shown for LI in the VPLMN, the TEID (6a) is the tunnel end point identifier at the E-UTRAN, TEID (6b) is the tunnel end point identifier at the S-GW and TEID (6c) is the tunnel end point identifier at the V-UPF for the DL data forwarded UP packets.

In the illustrations shown for LI in the HPLMN, an intermediate UPF is inserted for the DL data forwarding (the same can be true even with the LI in VPLMN) and the TEID (6a) is the tunnel end point identifier at the E-UTRAN, TEID (6b) is the tunnel end point identifier at the S-GW and TEID (6c) is the tunnel end point identifier at the intermediate UPF for the DL data forwarded UP packets.

The TEID (6) is the tunnel end point identifier at the E-UTRAN and TEID (7) is the tunnel end point identifier at the S-GW for the DL UP packets after the after handover. The TEID (5) is the tunnel end point identifier for the UL UP packets at the S-GW. The TEID (1-1) is the reserved tunnel end point identifier for the UL UP packets at the UPF + PGW-U.

5.3.4.4.2 LI in VPLMN

The generation of xCC happens at V-UPF before the handover and at S-GW after the handover. It is assumed that the PFCP session ID is used as the target identifier. The xCC generated from the CC-POI present in the V-UPF and from the SGW are not correlated.

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.4.5.2-1 illustrates the first part. The flow-diagram in figure 5.3.4.5.2-2 illustrates the second part. The flow-diagram in figure 5.3.4.5.2-3 illustrates the third part. The flow-diagram in figure 5.3.4.5.2-4 illustrates the fourth part.

Flow 1 of 4

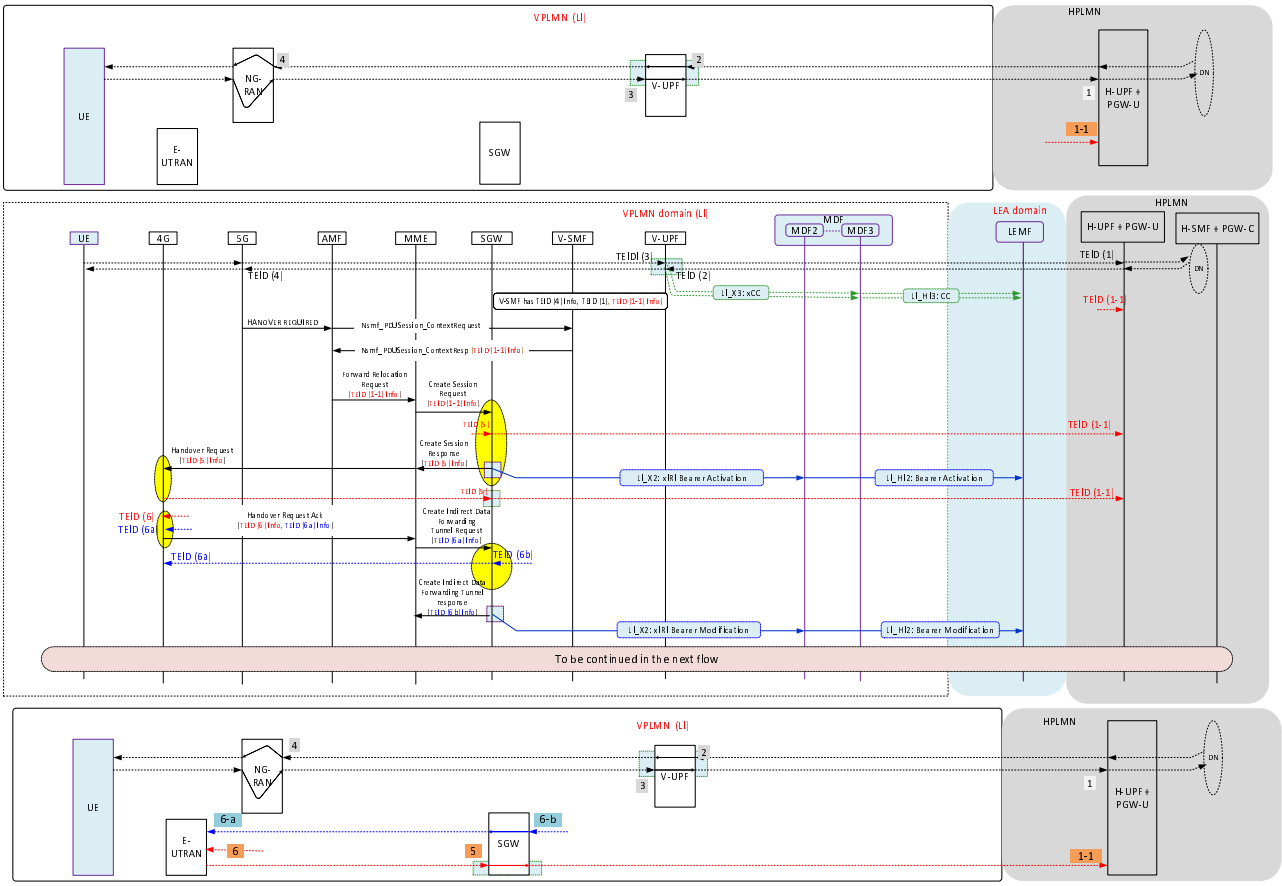


Figure 5.3.4.4.2-1: N26-based handover (LI in VPLMN) indirect DL data forwarding – flow 1 of 4

When the SGW sends the Create Session Response to the MME, SGW generates the xIRI Bearer Activation.

When the SGW sends the Create Indirect Data Forwarding Tunnel response to the MME, SGW generates the xIRI Bearer Modification.

Flow 2 of 4

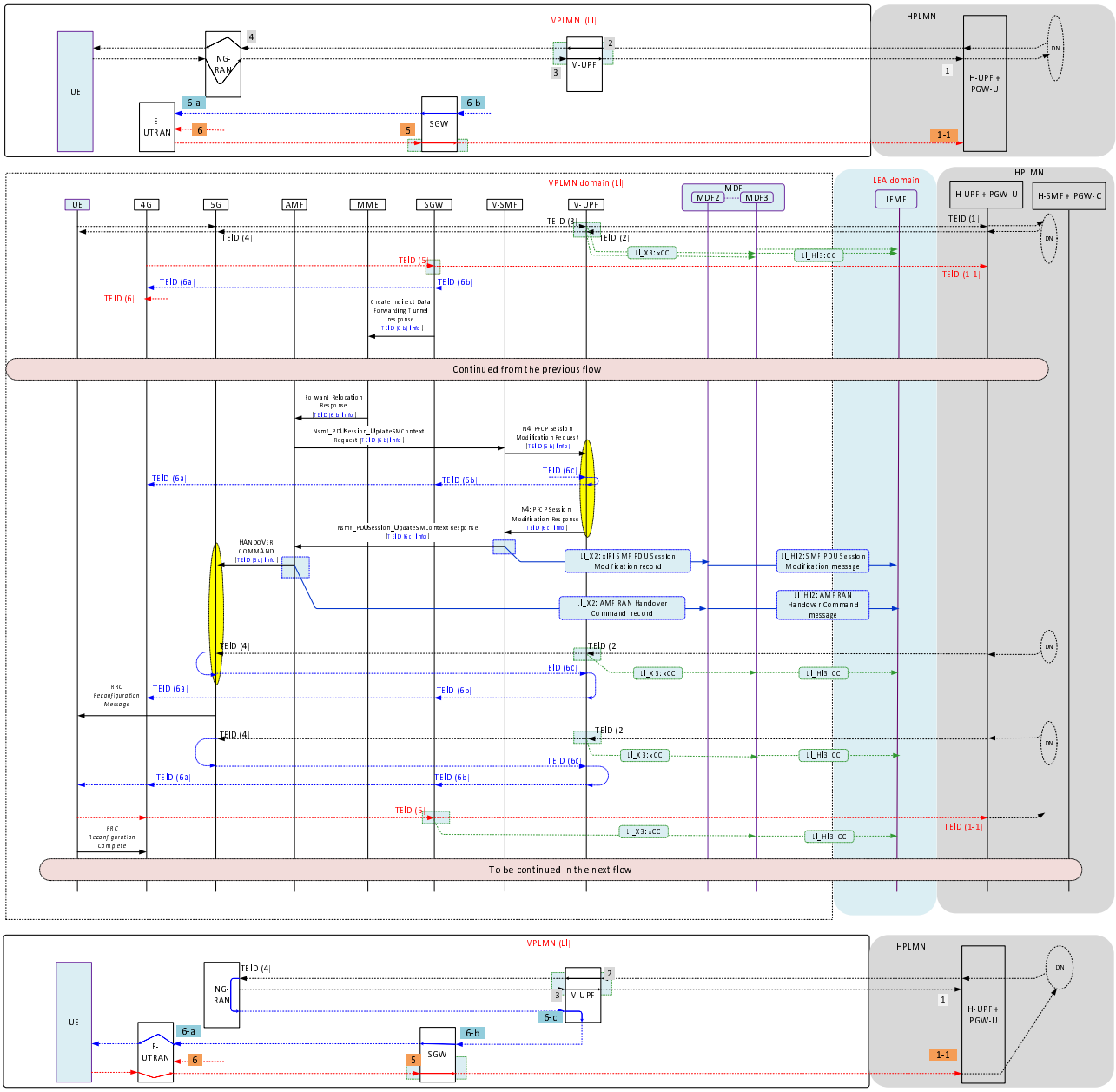


Figure 5.3.4.4.2-2: N26-based handover (LI in VPLMN) indirect DL data forwarding – flow 2 of 4

When the AMF sends the HANDOVER COMMAND to the NG-RAN over the N2 reference point, the IRI-POI present in the AMF generates the xIRI AMF RAN Handover Command record.

When the SMF sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

The CC-POI in V-UPF does not generate the xCC for the UP packets from DL data forwarding tunnels (i.e. incoming at TEID (6c) and outgoing to TEID (6b).

Flow 3 of 4

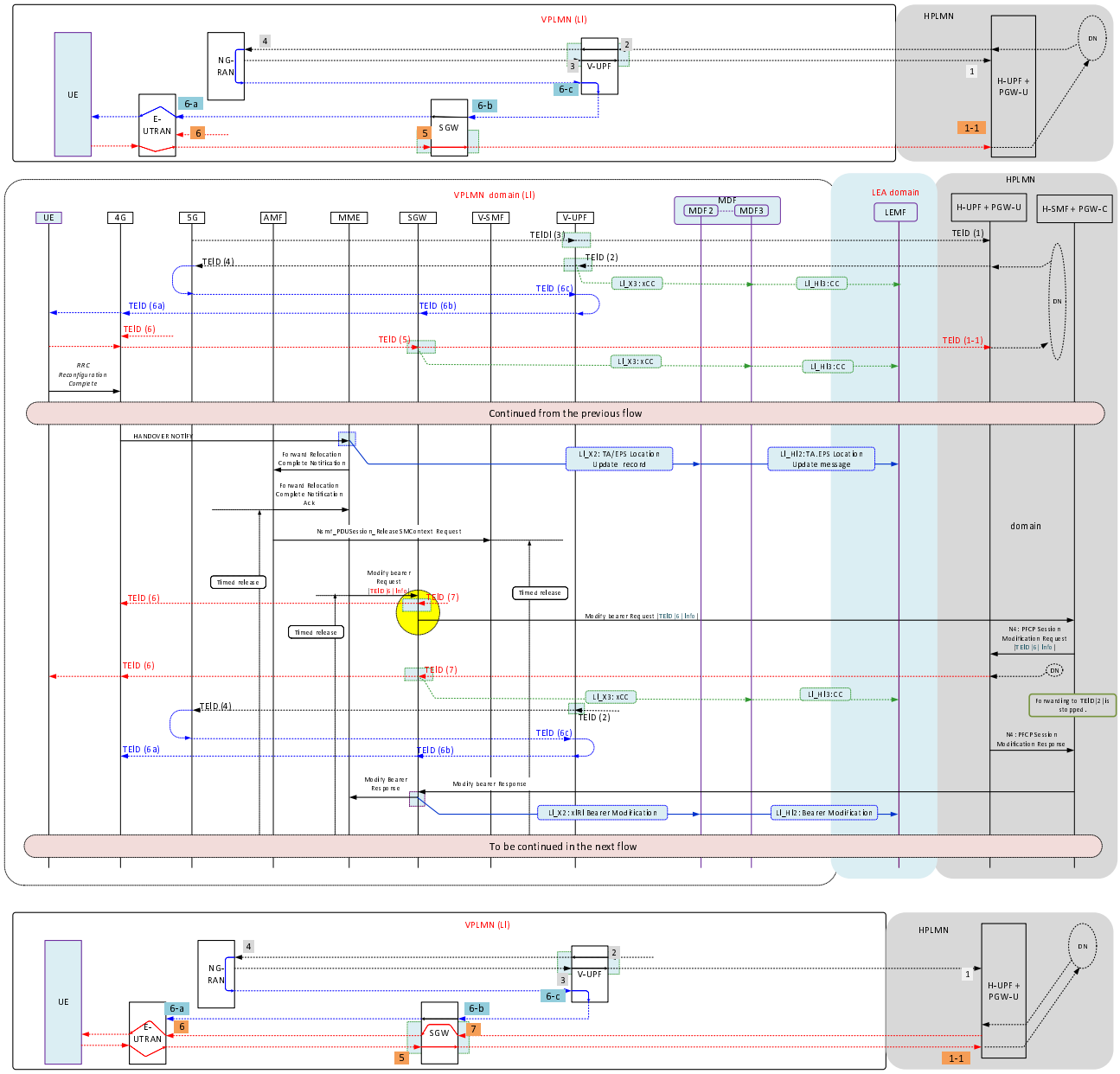


Figure 5.3.4.4.2-3: N26-based handover (LI in VPLMN) indirect DL data forwarding – flow 3 of 4

When the MME receives a HANDOVER NOTIFY from the E-UTRAN, the IRI-POI present in the MME generates the xIRI TA/EPS Location Update record.

When the SGW sends the Modify Bearer Response to the MME, the IRI-POI present in the SGW generates the xIRI Bearer Modification record.

Flow 4 of 4

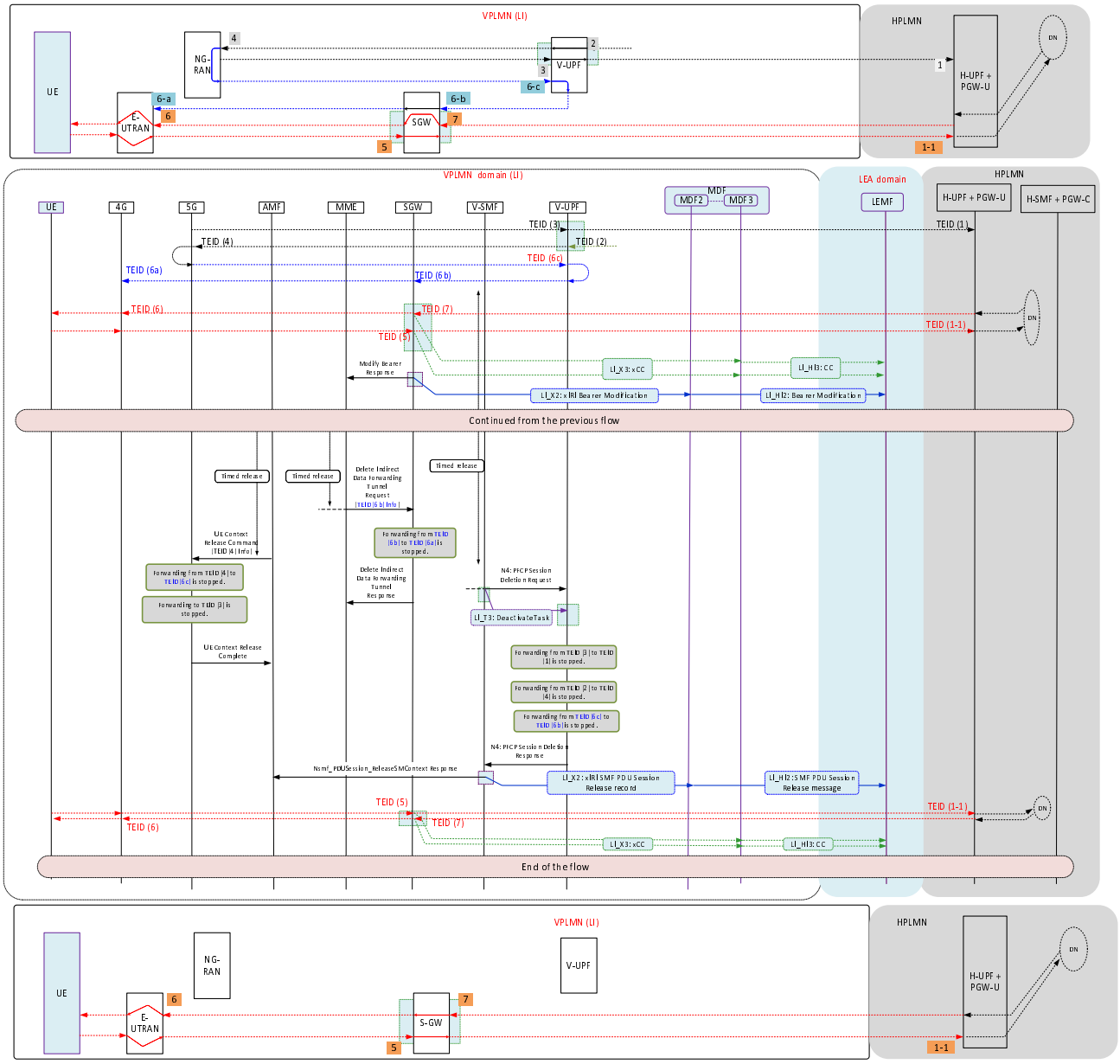


Figure 5.3.4.4.2-4: N26-based handover (LI in VPLMN) indirect DL data forwarding – flow 4 of 4

When the V-SMF sends the N4: PFCP Session Deletion Request to the V-UPF, the CC-TF present in the SMF sends the LI_T3: DeactivateTask to the CC-POI present in the V-UPF to stop the CC interception since V-UPF is no longer used after the handover.

When the SMF sends the Nsmf_PDUSessionReleaseSMContext Response to the AMF, the IRI-POI present in the SMF generates the xIRI PDU Session Release record.

5.3.4.4.3 LI in HPLMN

The generation of xCC happens at H-UPF. It is assumed that the PFCP session ID is used as the target identifier.

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.4.5.3-1 illustrates the first part. The flow-diagram in figure 5.3.4.5.3-2 illustrates the second part. The flow-diagram in figure 5.3.4.5.3-3 illustrates the third part. The flow-diagram in figure 5.3.4.5.3-4 illustrates the fourth part.

Flow 1 of 4

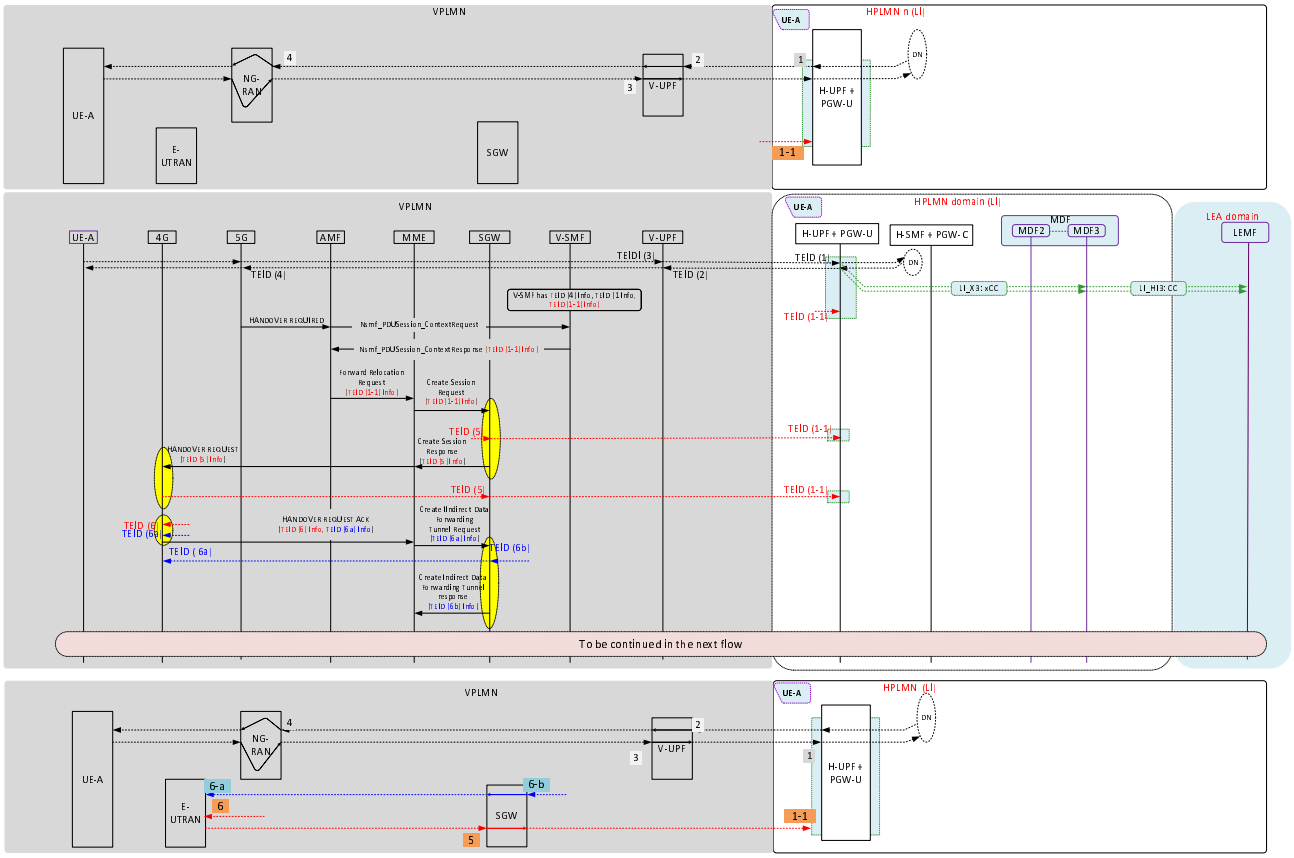


Figure 5.3.4.4.3-1: N26-based handover (LI in HPLMN) indirect DL data forwarding – flow 1 of 4

Flow 2 of 4

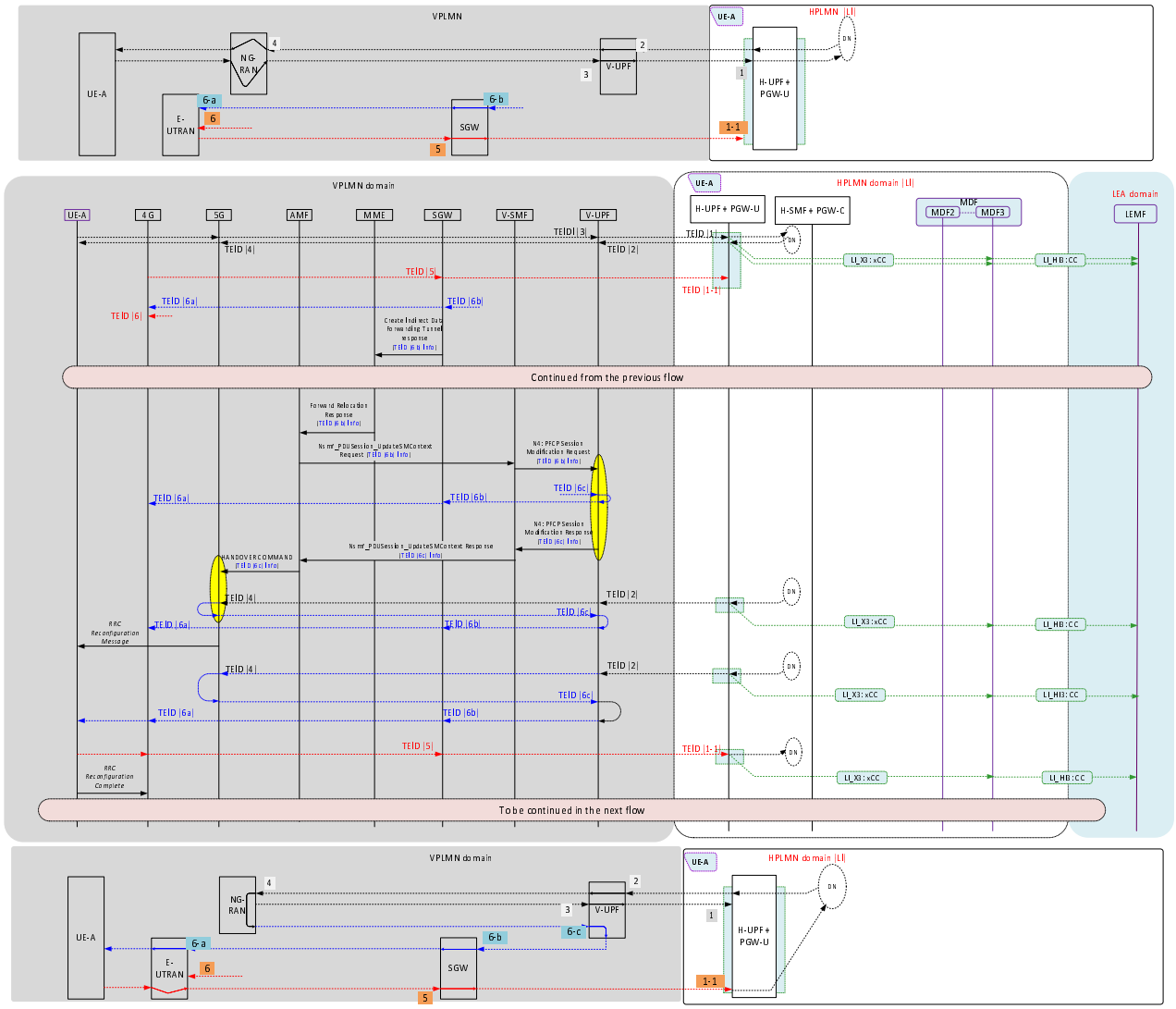


Figure 5.3.4.4.3-2: N26-based handover (LI in HPLMN) indirect DL data forwarding – flow 2 of 4

In this illustration, PFCP Session ID is used as the target identifier for the xCC generation in the CC-POI present in the H-UPF _ PGW-U and therefore, the CC-POI in the H-UPF + PGW-U would generate the xCC from the packets received at TEID (1-1) without any additional trigger from the CC-TF present in H-SMF + PGC-C.

Flow 3 of 4

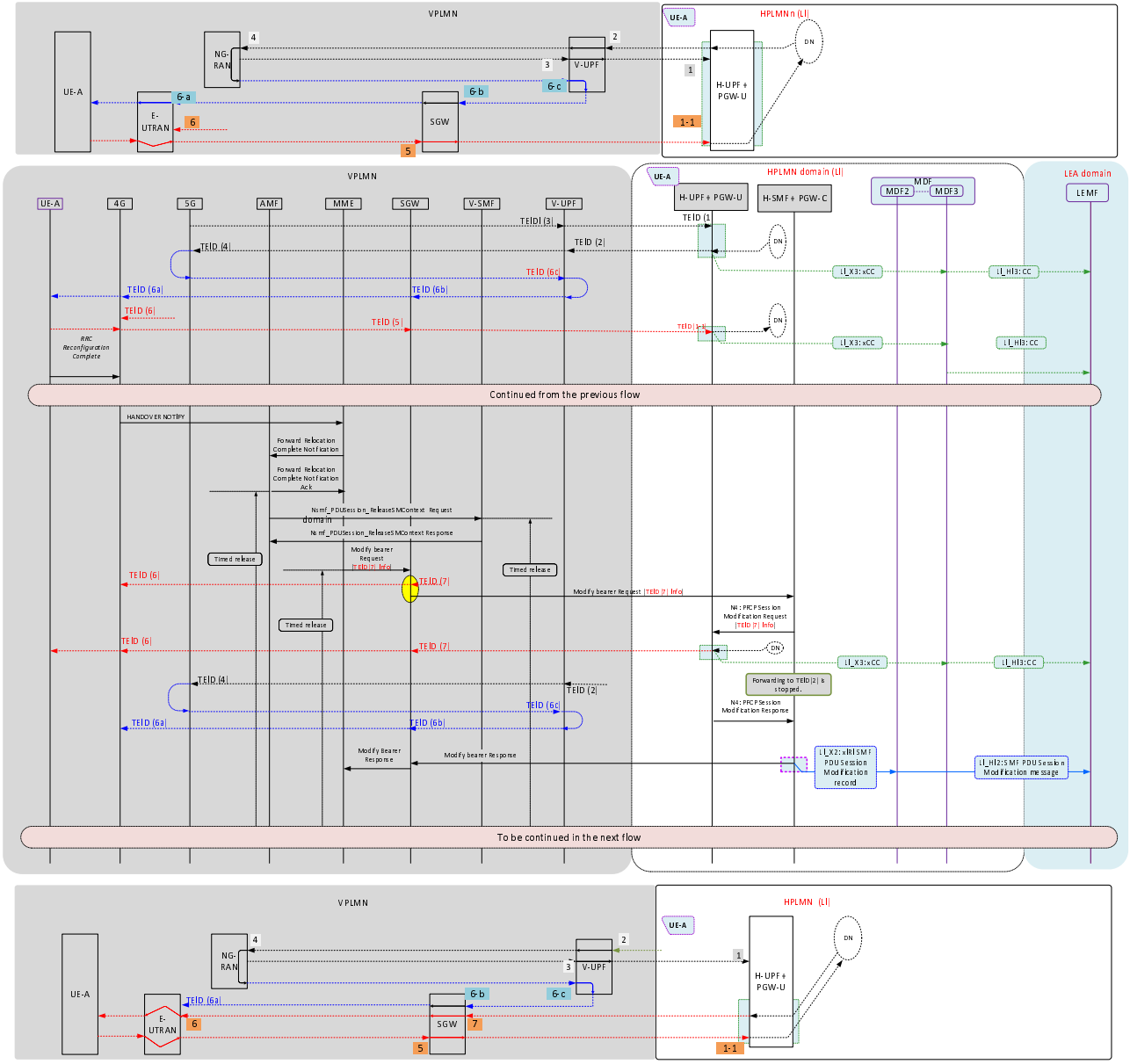


Figure 5.3.4.4.3-3: N26-based handover (LI in HPLMN) indirect DL data forwarding – flow 3 of 4

When the H-SMF sends the Modify Bearer Response to the SGW (VPLMN, the IRI-POI present in the H-SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

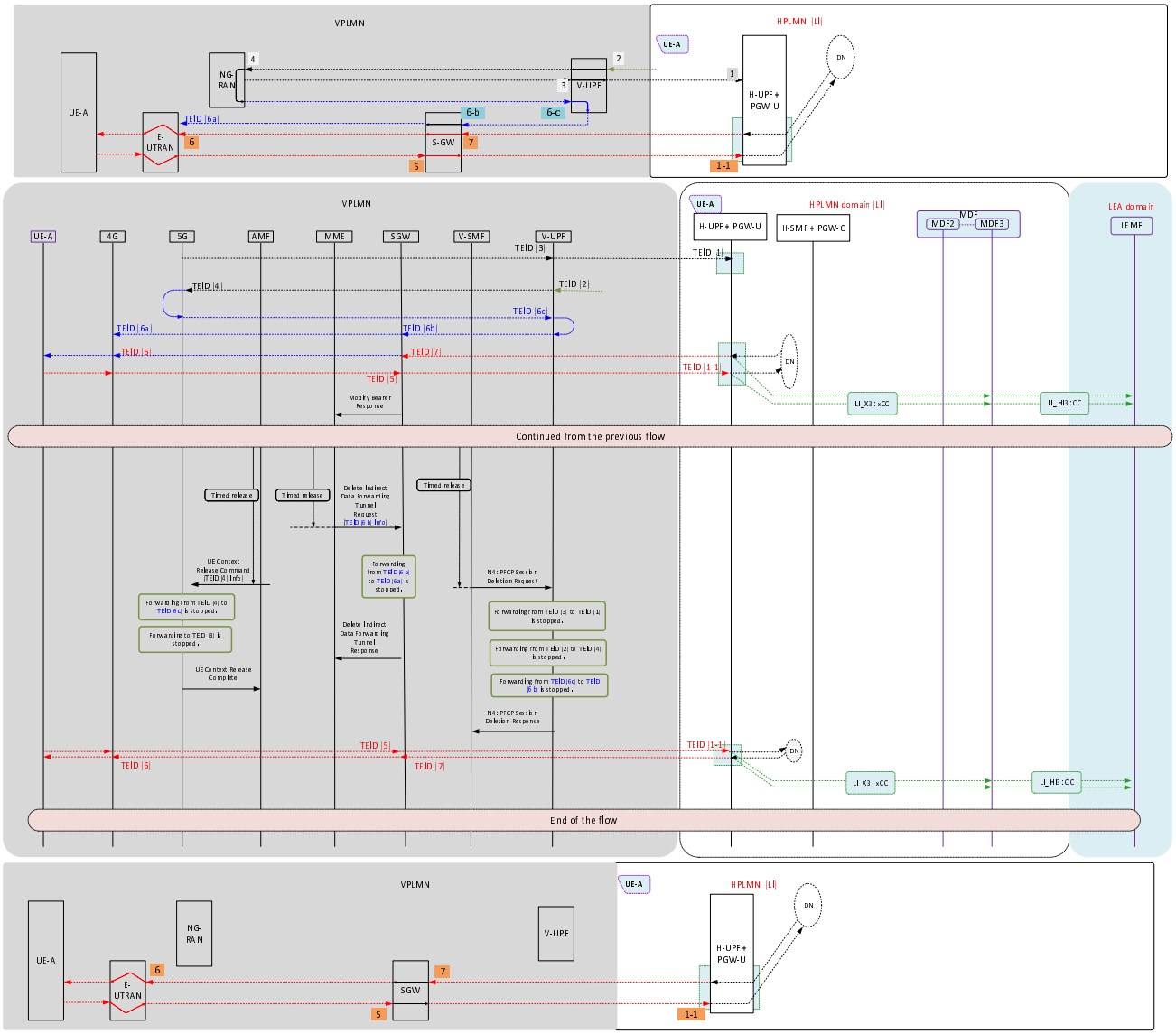


Figure 5.3.4.4.3-4: N26-based handover (LI in HPLMN) indirect DL data forwarding – flow 4 of 4

5.3.5 4G to 5G handover flows

5.3.5.1 Overview

5.3.5.1.1 General

This clause gives an overview of the scenarios related to the handover of a mobile device from E-UTRAN to NG-RAN as described in TS 23.502 [7] clause 4.11.1.2.2. This is also referred to as an N26 based handover. The N26 is an interface between the MME and the AMF.

Handover procedures apply to all EPS bearers. During the handover, the EPS bearers are handed over to 5GS QoS Flows. The EPS bearers that are not handed over are released, The CC interception happens in the PGW-U + UPF for a non-roaming scenario (or in the HPLMN with home-routed roaming). In the VPLMN, the CC interception happens in SGW before the handover and in V-UPF after the handover.

During the handover to maintain the user plane data continuity, the DL data may be forwarded from the source E-UTRAN to the target NG-RAN.

5.3.5.1.2 Direct forwarding

This is the case where the DL data forwarding happens directly from the E-UTRAN to the NG-RAN. Two scenarios are illustrated here. Non-roaming and home-routed roaming.

Scenario 1

Figure 5.3.5.1.2-1 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is for a non-roaming scenario.

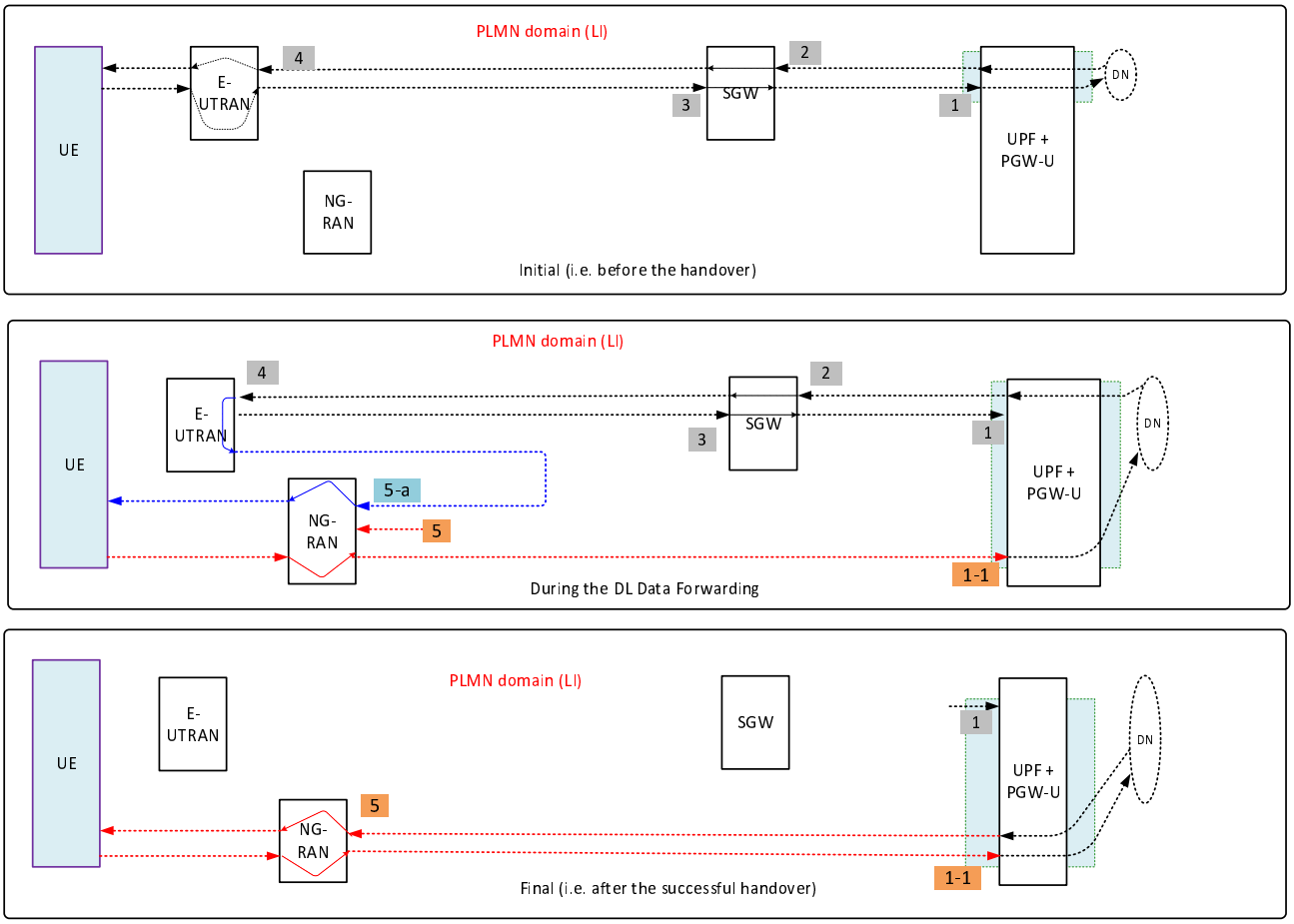


Figure 5.3.5.1.2-1: User plane packet flow during handover direct DL data forward (non-roaming)

The detailed flow-diagrams for the scenario shown in figure 5.3.5.1.2-1 is in clause 5.3.5.2.2.

Scenario 2

Figure 5.3.5.1.2-2 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is a home-routed roaming scenario. The LI is in VPLMN.

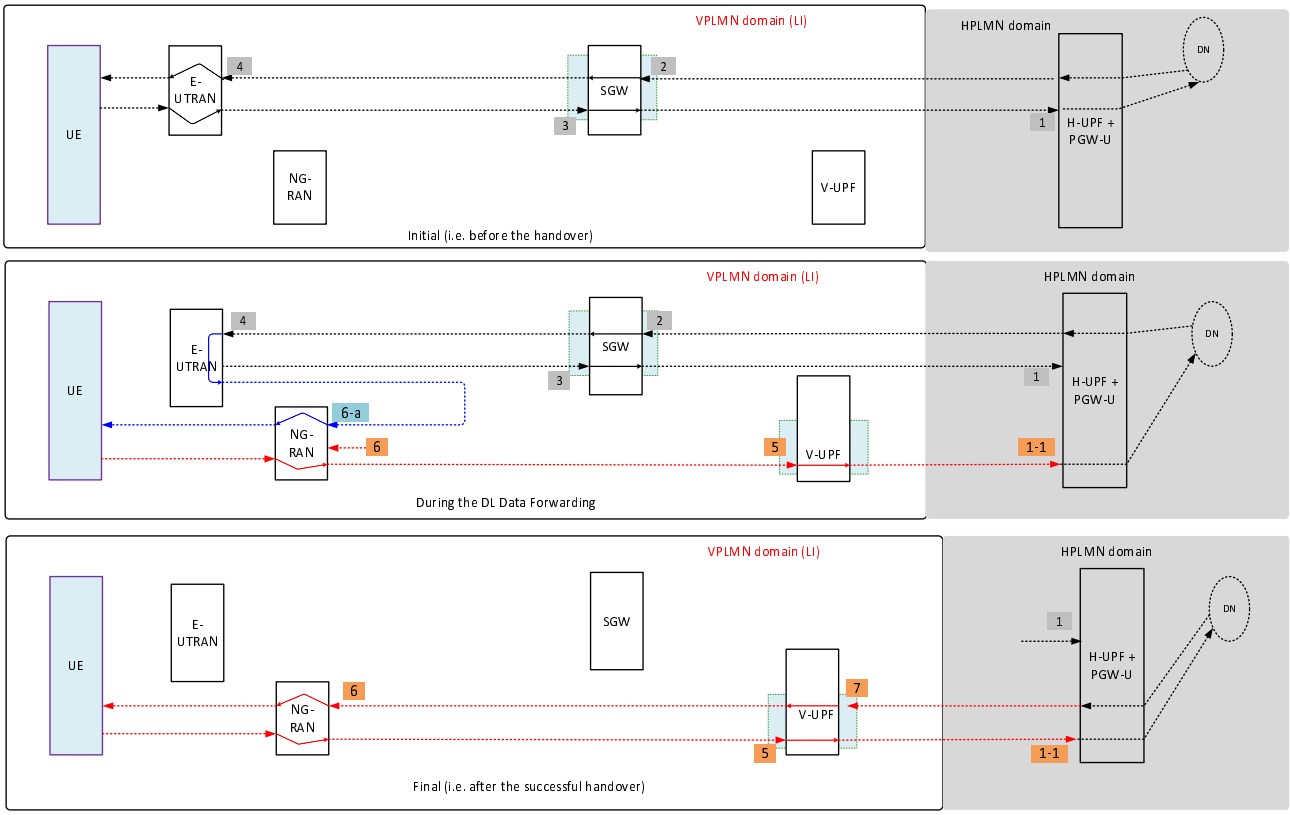


Figure 5.3.5.1.2-2: User plane packet flow during handover direct DL data forward (roaming) LI in VPLMN

Here, the CC-POI functions are moved from SGW to V-UPF. The intercepted CC before the handover and after the handover are not correlated.

Scenario 3

Figure 5.3.5.1.2-3 shows the initial (i.e. before the handover), during the DL data forward (with direct DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is a home-routed roaming scenario. The LI is in HPLMN.

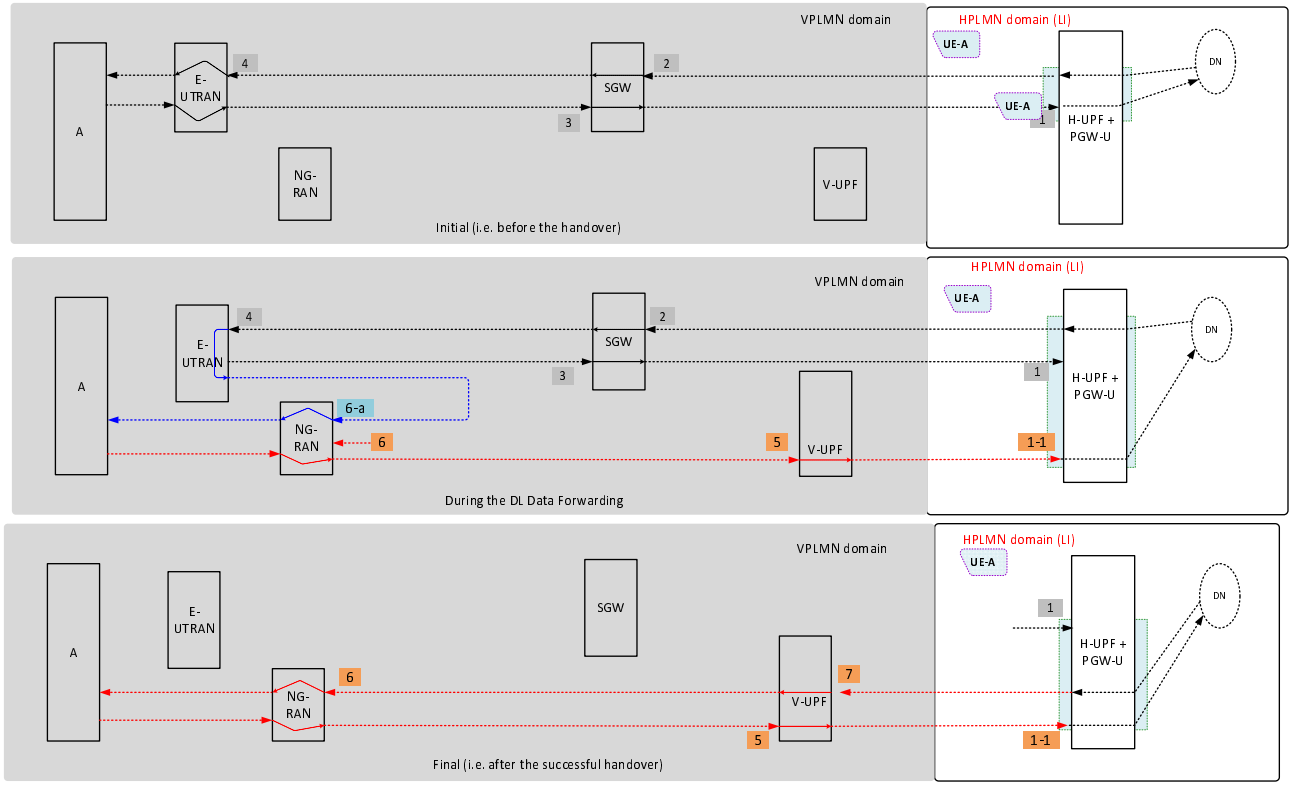


Figure 5.3.5.1.2-3: User plane packet flow during handover direct DL data forward (roaming) LI in HPLMN

Here, the CC-POI functions are provided in H-UPF + PGW-U.

5.3.5.1.3 Indirect forwarding in a non-roaming case

In all the illustrations shown in this clause, indirect DL data forwarding happens. The CC interception is always done at the UPF + PGW-U.

Scenario 1

Figure 5.3.5.1.3-1 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is for a non-roaming scenario. An intermediate UPF is not used for DL data forwarding.

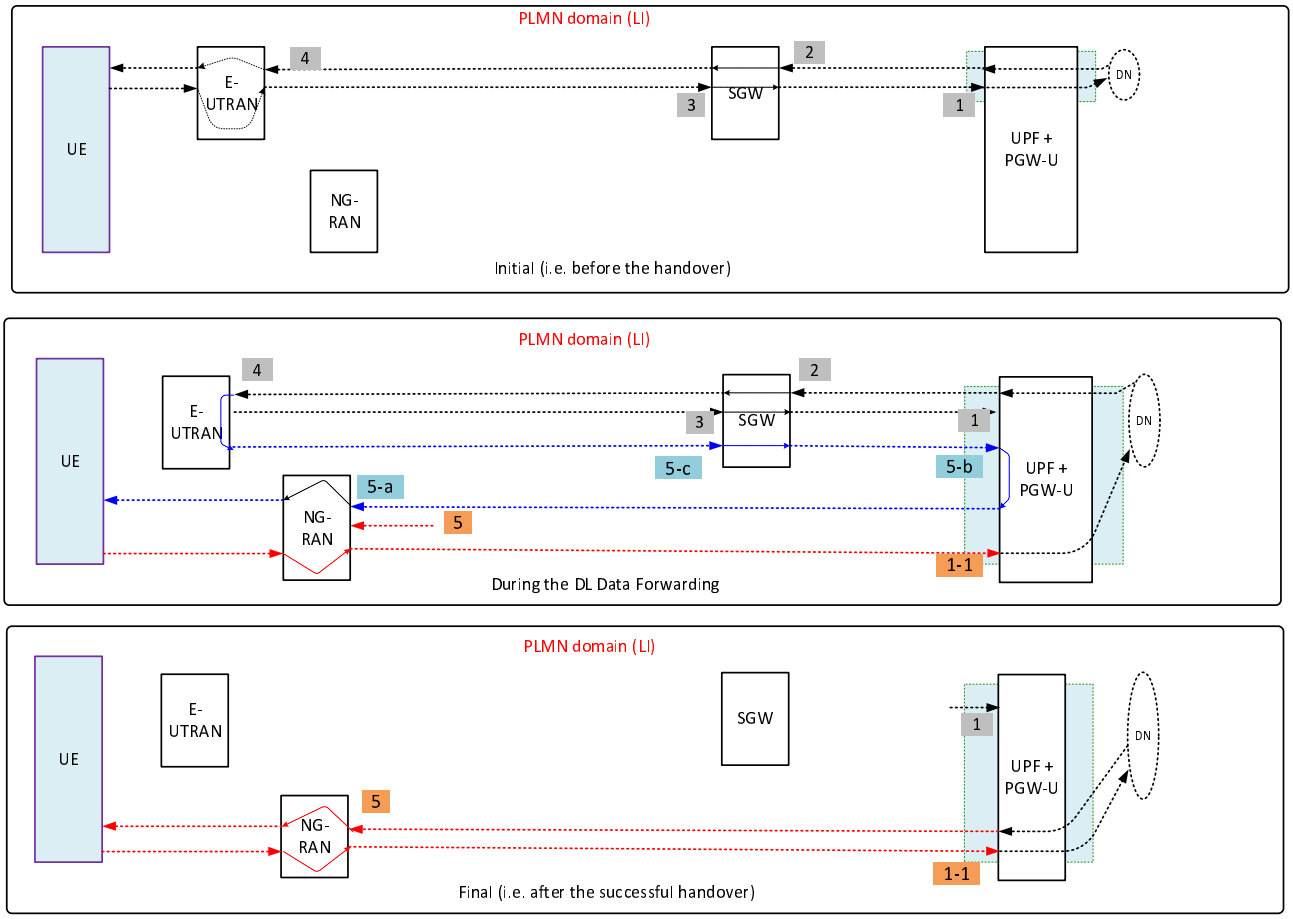


Figure 5.3.5.1.3-1: User plane packet flow during handover indirect DL data forward - case 1

Special care is to be taken within the UPF + PGW-U and the SGW not to intercept the user plane packets from the DL data forwarding tunnel to avoid double interception of the same packets. A detailed flow diagram for the scenario shown in figure 5.3.5.1.3-1 is in clause 5.3.5.3.2.

Scenario 2

Figure 5.3.5.1.3-2 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. An intermediate UPF is used for DL data forwarding. This is for a non-roaming scenario.

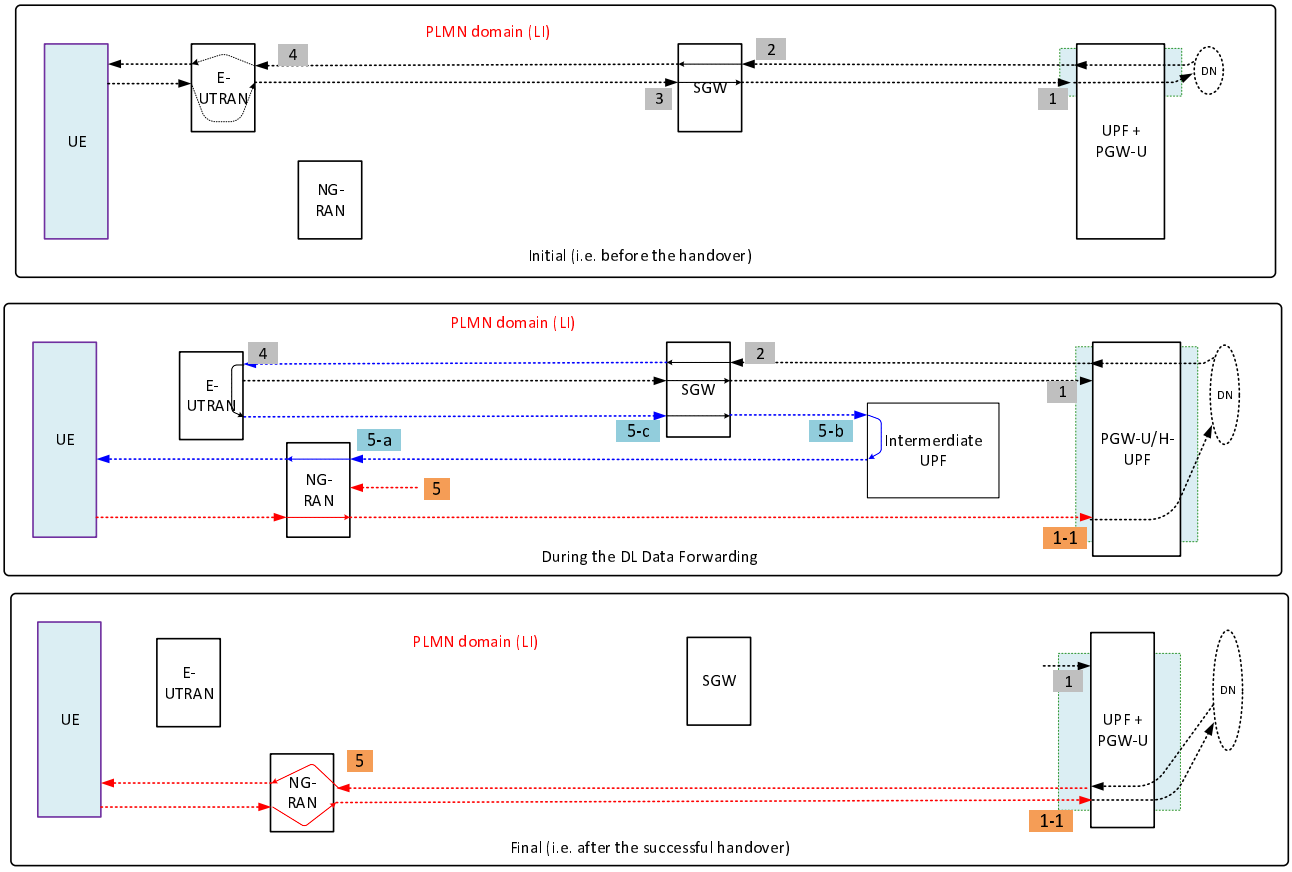


Figure 5.3.5.1.3-2: User plane packet flow during handover indirect DL data forward - case 2

Special care has to be taken within the SGW not to intercept the user plane packets from the DL data forwarding tunnel to avoid double interception of the same packets.

5.3.5.1.4 Indirect forwarding in a roaming case

In all the illustrations shown in this clause, indirect DL data forwarding happens.

Scenario 1

Figure 5.3.5.1.4-1 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. This is for home-routed roaming scenario. The LI is in the VPLMN. An intermediate UPF is not used for DL data forwarding.

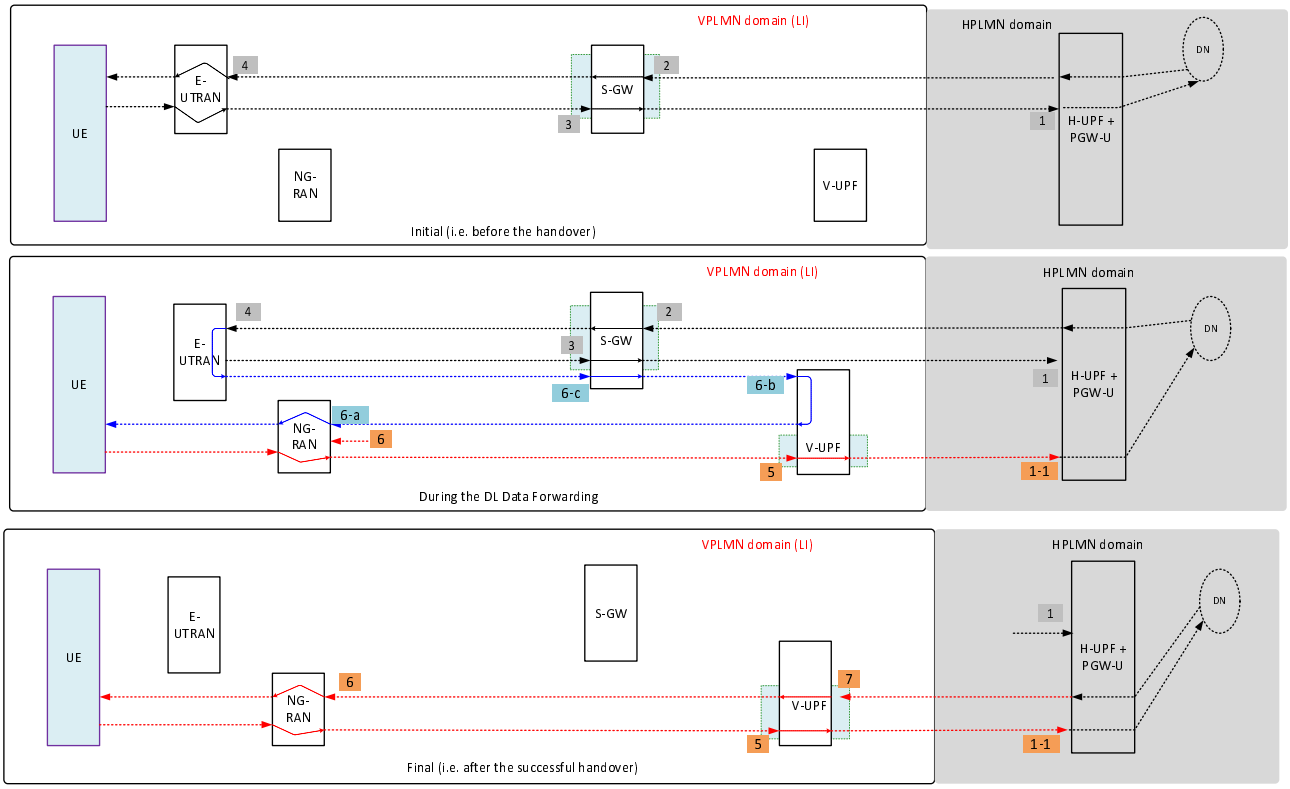


Figure 5.3.5.1.4-1: User plane packet flow during handover indirect DL data forward - case 1

Special care has to be taken within the V-UPF and the SGW not to intercept the user plane packets from the DL data forwarding tunnel to avoid double interception of the same packets. Here, the CC-POI functions are moved from SGW to V-UPF. The intercepted CC before the handover and after the handover are not correlated. A detailed flow diagram for the scenario shown in figure 5.3.5.1.4-1 is in clause 5.3.5.4.2.

Scenario 2

Figure 5.3.5.1.4-2 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. An intermediate is used for DL data forwarding. Here, the LI is in VPLMN.

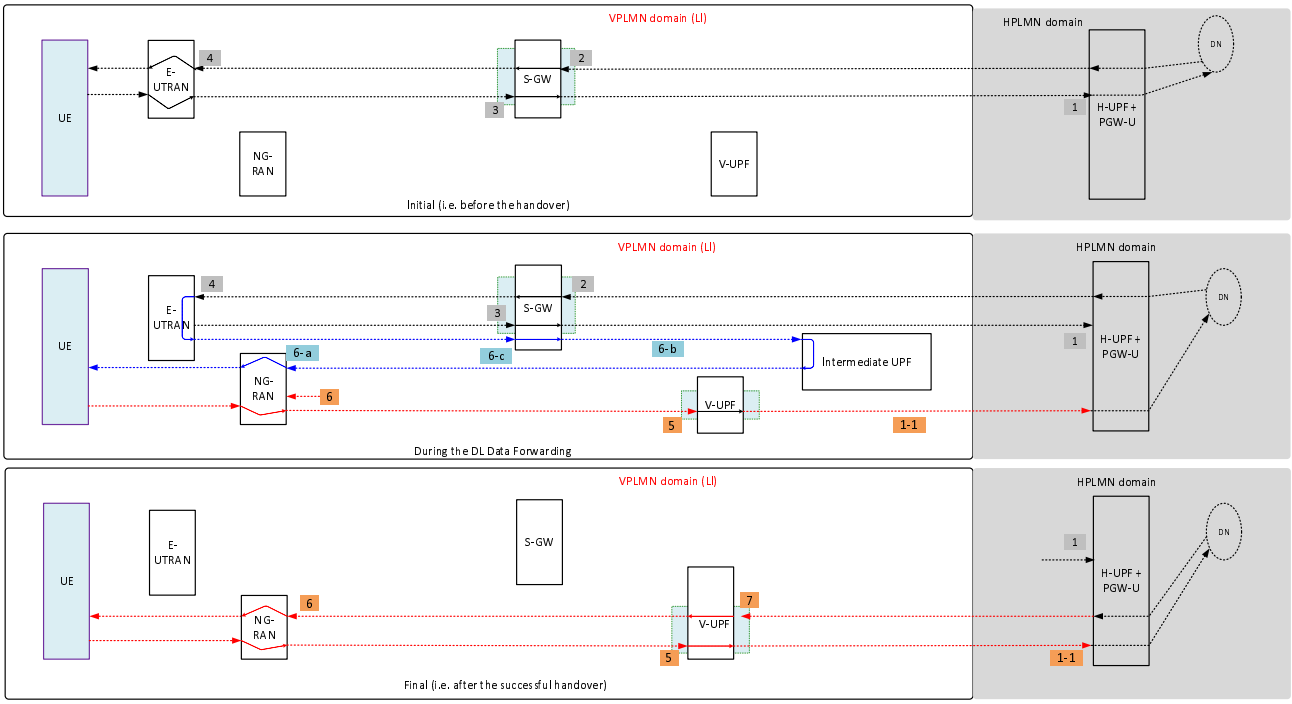


Figure 5.3.5.1.4-2: User plane packet flow during handover indirect DL data forward - case 2

Special care has to be taken within the SGW not to intercept the user plane packets from the DL data forwarding tunnel to avoid double interception of the same packets. Here, the CC-POI functions are moved from SGW to V-UPF. The intercepted CC before the handover and after the handover are not correlated.

Scenario 3

Figure 5.3.5.1.4-3 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. An intermediate is not used for DL data forwarding. Here the LI is in HPLMN.

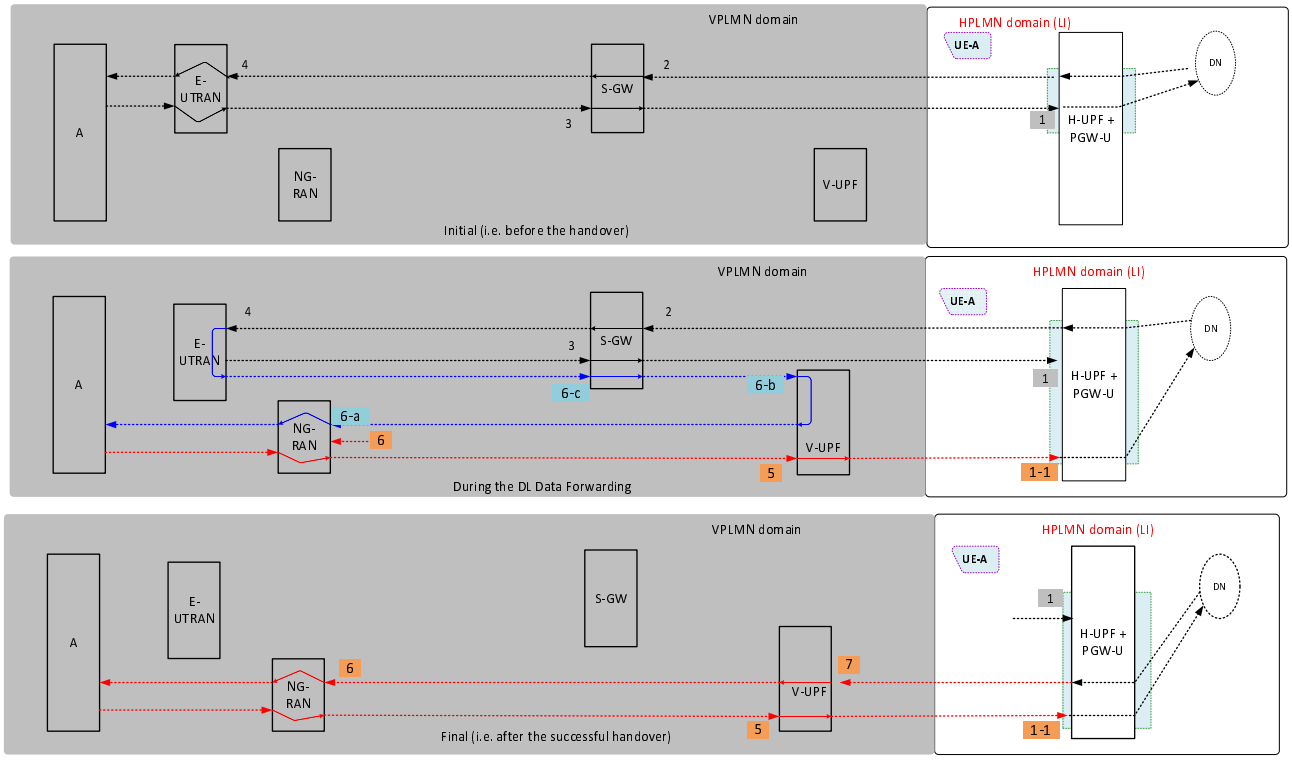


Figure 5.3.5.1.4-3: User plane packet flow during handover indirect DL data forward - case 3

A detailed flow diagram for the scenario shown in figure 5.3.5.1.4-3 is in clause 5.3.5.4.3.

Scenario 4

Figure 5.3.5.1.4-4 shows the initial (i.e. before the handover), during the DL data forward (with indirect DL data forwarding) and the final (after the successful handover) user-plane packet flow. An intermediate is used for DL data forwarding. Here the LI is in HPLMN.

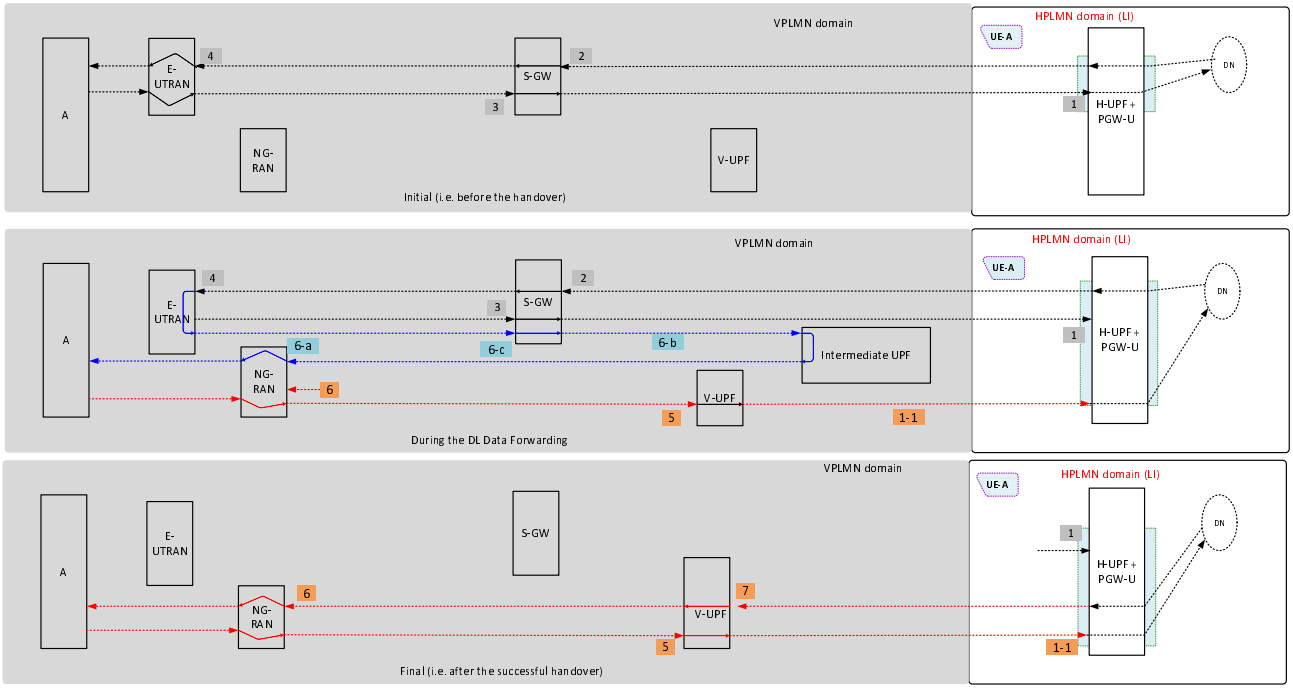


Figure 5.3.5.1.4-4: User plane packet flow during handover indirect DL data forward - case 4

As can be deduced from illustrations shown in figure 5.3.5.1.3-3, 5.3.5.1.4-3, 5.3.5.1.4-4, the DL data forwarding variations do not seem to have an influence for the LI in HPLMN. Note that LI in VPLMN and LI in HPLMN can happen on the same target via mutually independent warrants.

5.3.5.2 Detailed flows - non-roaming, direct DL data forwarding

5.3.5.2.1 General

An N-26 based handover may happen when there is an inter-system mobility for the device between the 5G AN and the 4G AN. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the former case and covers the scenario of 4G to 5G handover. This flow is based on the N26-based handover stage 2 flow in TS 23.502 [7], clause 4.11.1.2.2.

Within flow-diagrams, the NG-RAN is shown 5G and E-UTRAN is shown as 4G. The numbers 1, 1-1. 2, 3, 4, 5 and 5-a correspond to the TEID (1), TEID (1-1), TEID (2), TEID (3), TEID (4), TEID (5) and TEID (5a) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at UPF + PGW-U and TEID (3) is the tunnel end point identifier at the SGW for UL UP packets. TEID (2) is the tunnel end point identifier at SGW and TEID (4) is the tunnel end point identifier for the DL UP packets. The TEID (5a) is the tunnel end point identifier at the NG-RAN for the DL data forwarded UP packets. The direct DL data forwarding happens from E-UTRAN to NG-RAN.

The TEID (5) is the tunnel end point identifier at the NG-RAN for the DL UP packets after the after handover. The TEID (1-1) is the tunnel end point identifier for the UL UP packets at the UPF + PGW-U.

The generation of xCC from the UP packets done by the CC-POI present in the UPF + PGW-U.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF + PGW-U would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.5.2.2 The flows

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.5.2.2-1 illustrates the first part. The flow-diagram in figure 5.3.5.2.2-2 illustrates the second part. The flow-diagram in figure 5.3.5.2.2-3 illustrates the third part. The flow-diagram in figure 5.3.5.2.2-4 illustrates the fourth part.

Flow 1 of 4

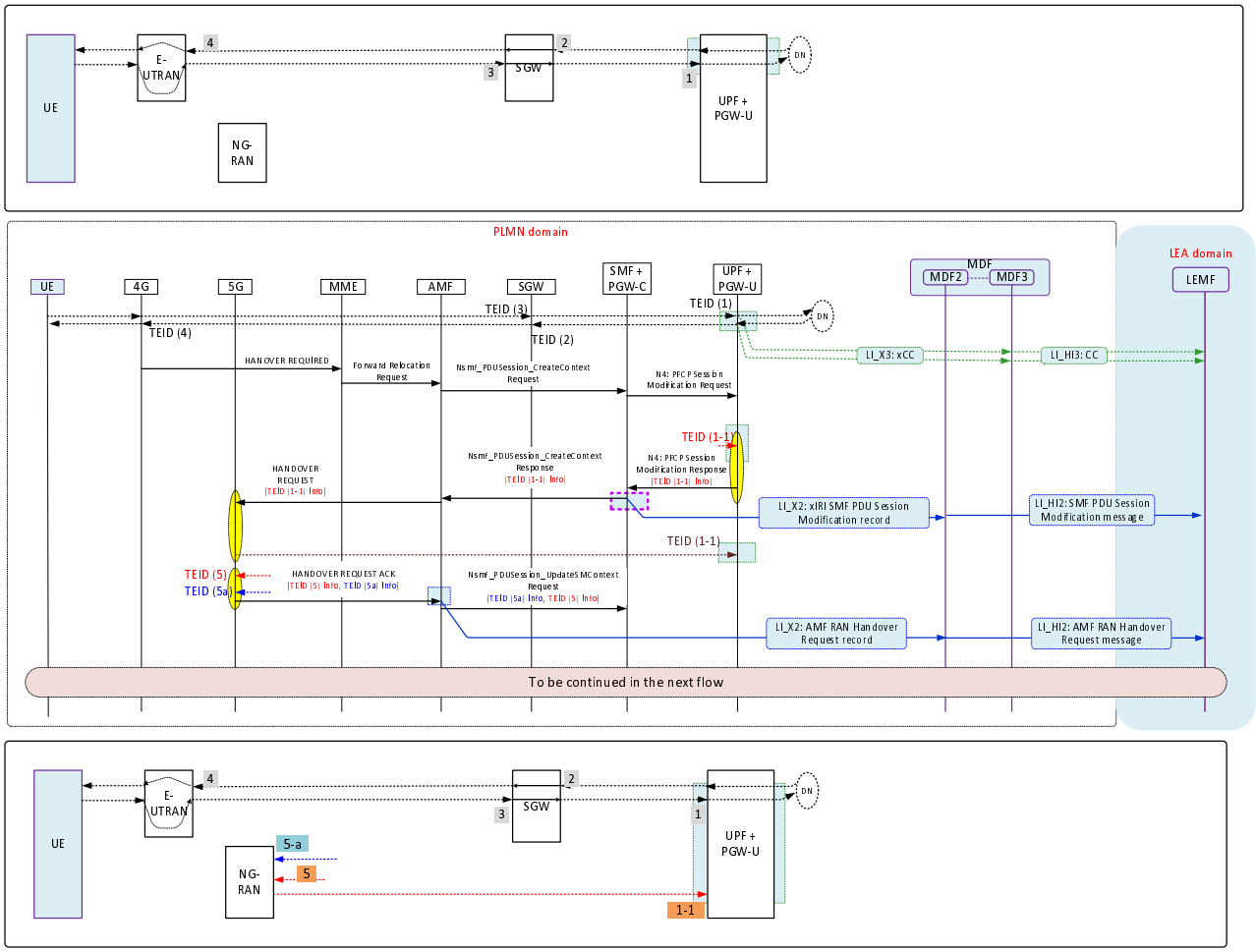


Figure 5.3.5.2.2-1: N-26 based handover (non-roaming) direct DL data forward – flow 1 of 4

When the SMF +PGW-C sends the Nsmf_PDUSession_CreateContext Response to the AMF, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

When the AMF receives the HANDOVER REQUREST ACK from the NG-RAN over the N2 interface, the CC-POI in AMF generates the xIRI AMF RAN Handover Request.

Flow 2 of 4

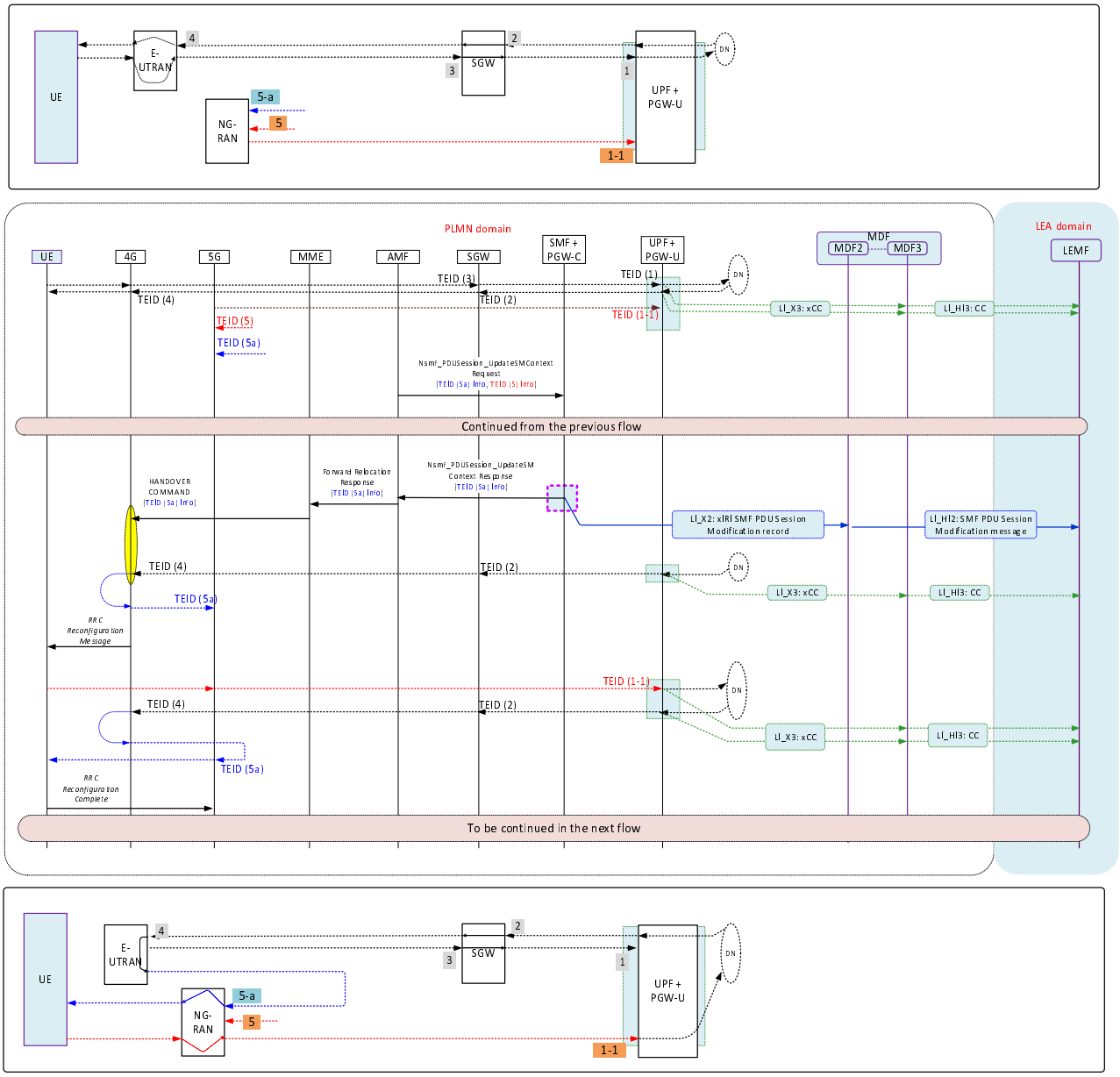


Figure 5.3.5.2.2-2: N-26 based handover (non-roaming) direct DL data forward – flow 2 of 4

When the SMF + PGW-C sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

In this illustration, PFCP Session ID is used as the target identifier for the xCC generation in the CC-POI present in the UPF + PGW-U and therefore, the CC-POI in the UPF + PGW-U would generate the xCC from the packets received at TEID (1-1) without any additional trigger from the CC-TF present in SMF + PGW-C.

Flow 3 of 4

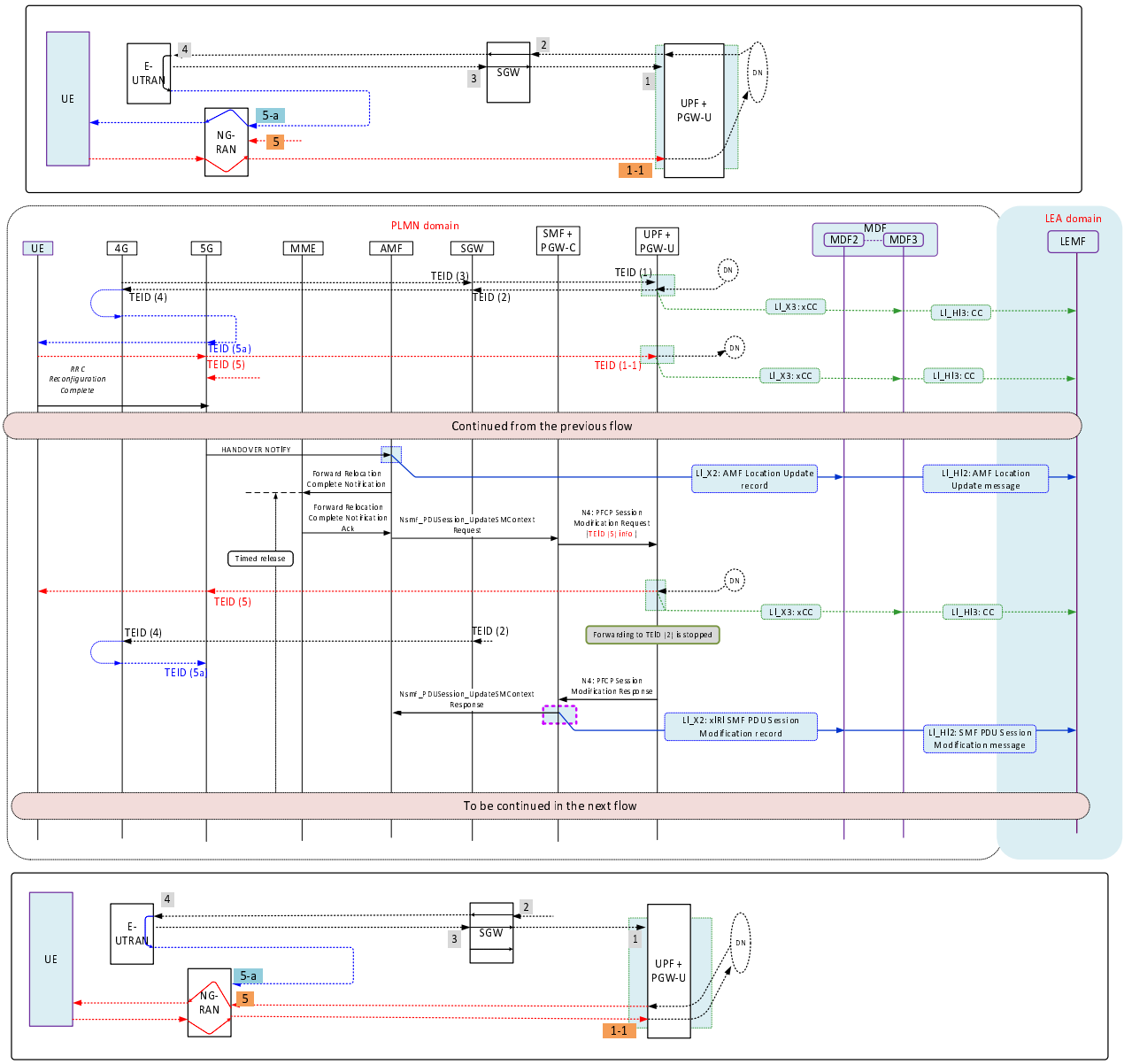


Figure 5.3.5.2.2-3: N-26 based handover (non-roaming) direct DL data forward – flow 3 of 4

When the AMF receives a HANDOVER NOTIFY from the NG0-RAN, the IRI-POI present in the MME generates the xIRI AMF Location Update record.

When the SMF + PGW-C sends a ModifyBearer response to the SGW, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 4 o 4

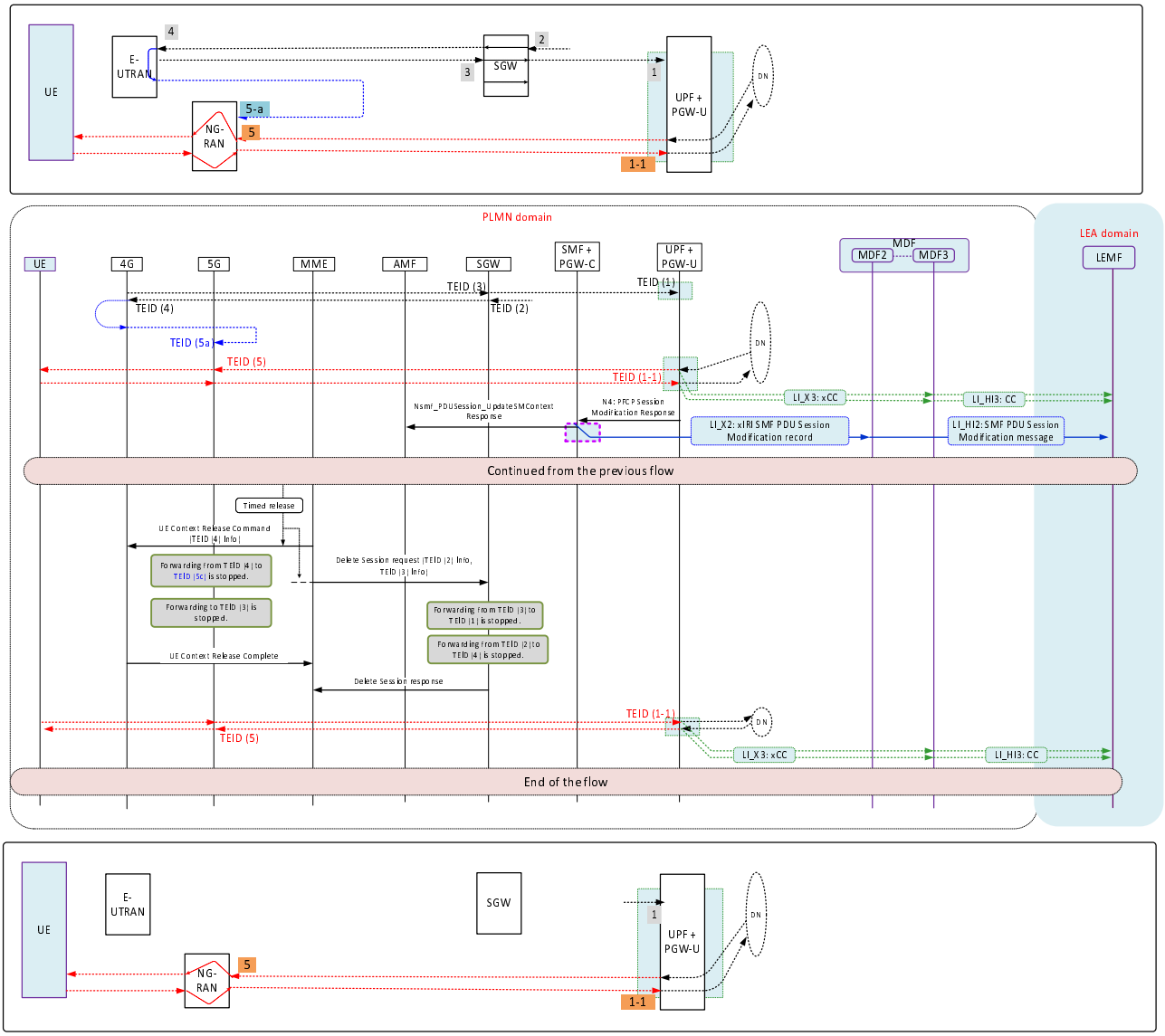


Figure 5.3.5.2.2-4: N-26 based handover (non-roaming) direct DL data forward – flow 4 of 4

5.3.5.3 Detailed flows - non-roaming, indirect DL data forwarding

5.3.5.3.1 General

An N-26 based handover may happen when there is an inter-system mobility for the device between the 5G AN and the 4G AN. The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the latter case and covers the scenario of 4G to 5G handover. This flow is based on the N26-based handover stage 2 flow in TS 23.502 [7], clause 4.11.1.2.2.

Within flow-diagrams, the NG-RAN is shown 5G and E-UTRAN is shown as 4G. Within the drawings, the numbers 1, 1-1. 2, 3, 4, 5, 5-a,5-b and 5-c correspond to the TEID (1), TEID (1-1), TEID (2), TEID (3), TEID (4), TEID (5), TEID (5a), TEID (5b) and TEID (5c) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

The TEID (1) is the tunnel end point identifier at UPF + PGW-U and TEID (3) is the tunnel end point identifier at the SGW for UL UP packets. TEID (2) is the tunnel end point identifier at SGW and TEID (4) is the tunnel end point identifier at the E-UTRAN for the DL UP packets. The TEID (5a) is the tunnel end point identifier at the NG-RAN, TEID (5b) is the tunnel end point identifier at the UPF + PGW-U and TEID (5c) is the tunnel end point identifier at the SGW the DL data forwarded UP packets.

The TEID (5) is the tunnel end point identifier at the NG-RAN for the DL UP packets after the after handover. The TEID (1-1) is the tunnel end point identifier for the UL UP packets at the UPF + PGW-U.

The generation of xCC from the UP packets done by the CC-POI present in the anchor UPF + PGW-U.

In the illustrations, it is assumed that the PFCP session ID is used as the target identifier and therefore, the UPF + PGW-U would continue the CC interception after the handover without any LI_T3 requests from the CC-TF present in the SMF.

5.3.5.3.2 The flows

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.5.2.2-1 illustrates the first part. The flow-diagram in figure 5.3.5.2.2-2 illustrates the second part. The flow-diagram in figure 5.3.5.2.2-3 illustrates the third part. The flow-diagram in figure 5.3.5.2.2-4 illustrates the fourth part.

Flow 1 of 4

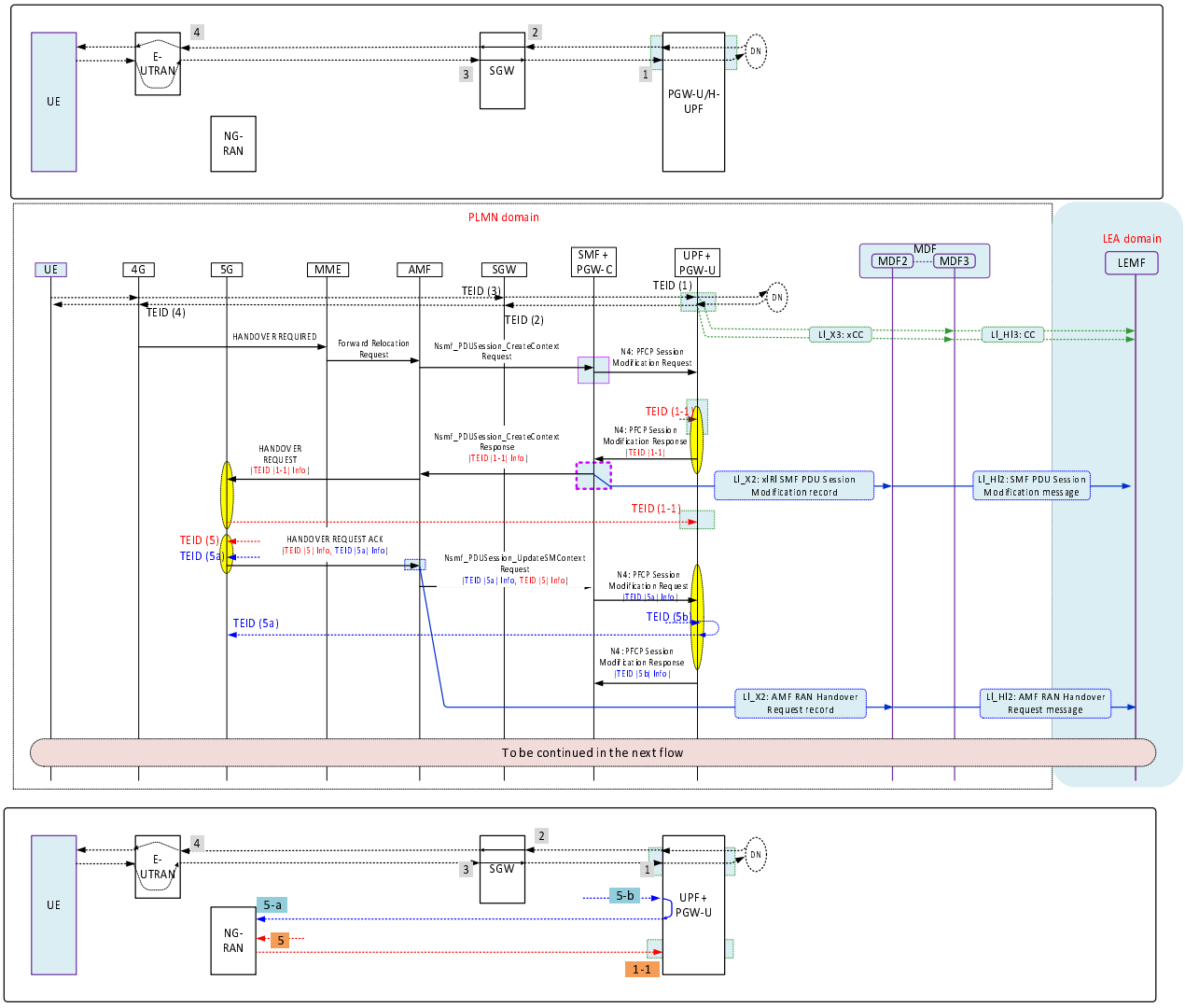


Figure 5.3.5.3.2-1: N-26 based handover non-roaming indirect DL data forward – flow 1 of 4

When the SMF +PGW-C sends the Nsmf_PDUSession_CreateContext Response to the AMF, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

When the AMF receives the HANDOVER REQUREST ACK from the NG-RAN over the N2 interface, the CC-POI in AMF generates the xIRI AMF RAN Handover Request.

Flow 2 of 4

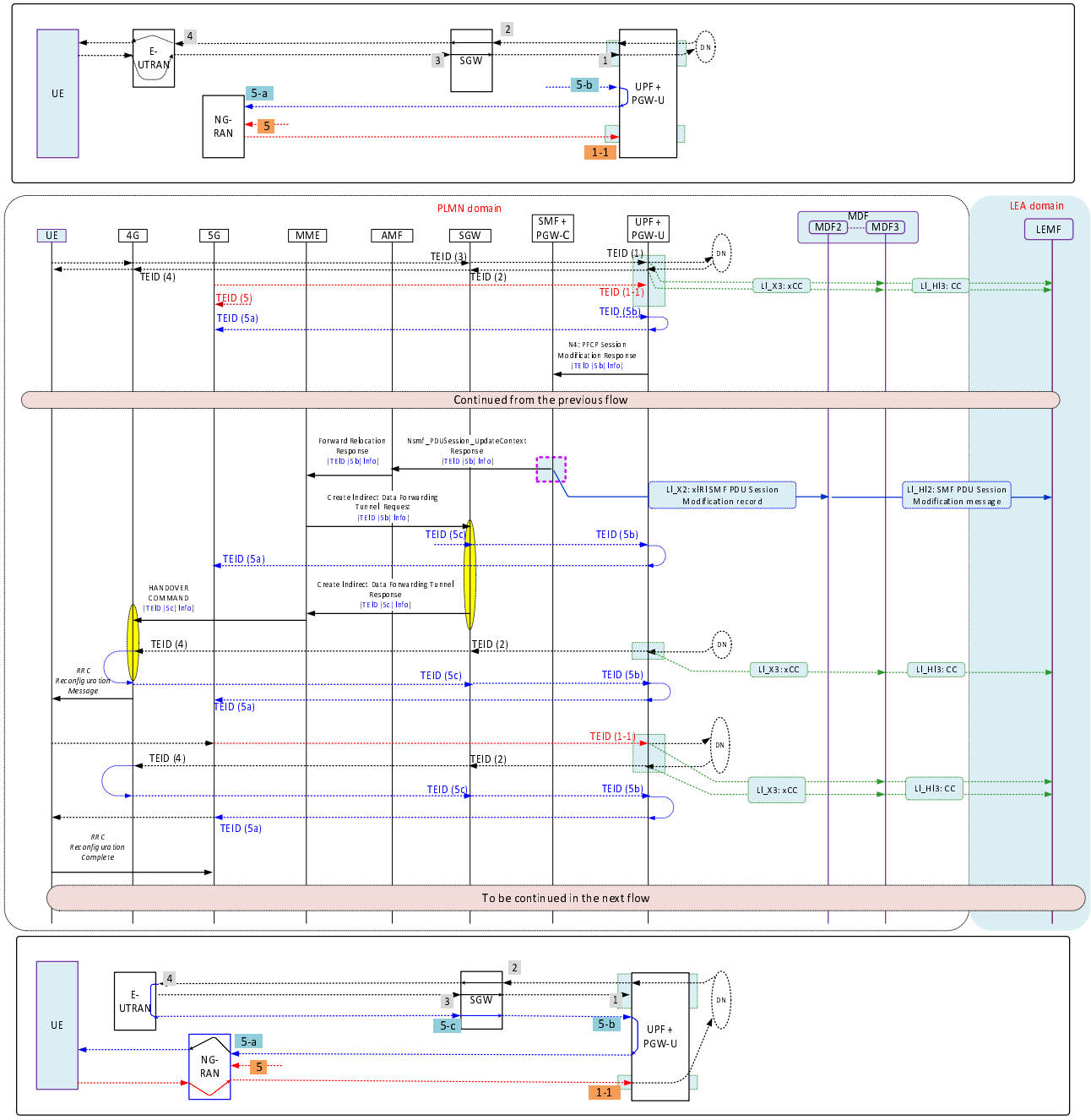


Figure 5.3.5.3.2-2: N-26 based handover non-roaming indirect DL data forward - flow 2 of 4

When the SMF + PGW-C sends the Nsmf_PDUSessionUpdateSMContext Response to the AMF with the updated SM context information, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

The CC-POI in UPF + PGW-U does not generate the xCC for the UP packets from DL data forwarding tunnels (i.e. incoming at TEID (5b) and outgoing to TEID (5a). Likewise, the SGW does not generate the xCC for the UP packets from DL data forwarding tunnels (i.e. incoming at TEID (5c) and outgoing to TEID (5b).

In this illustration, PFCP Session ID is used as the target identifier for the xCC generation in the CC-POI present in the UPF + PGW-U and therefore, the CC-POI in the UPF + PGW-U would generate the xCC from the packets received at TEID (1-1) without any additional trigger from the CC-TF present in SMF + PGW-C.

Flow 3 of 4

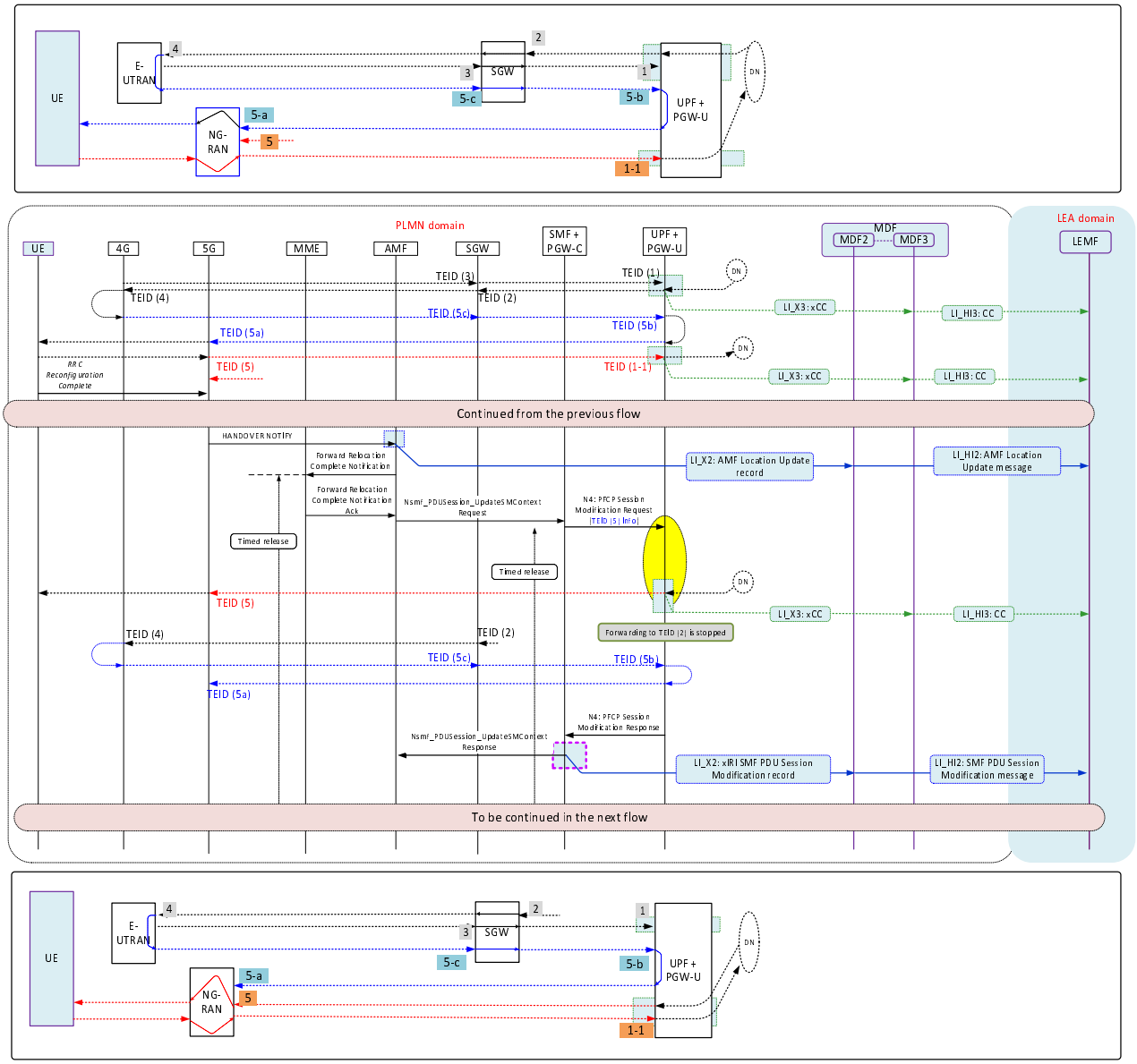


Figure 5.3.5.3.2-3: N-26 based handover non-roaming indirect DL data forward – flow 3 of 4

When the AMF receives a HANDOVER NOTIFY from the NG0-RAN, the IRI-POI present in the MME generates the xIRI AMF Location Update record.

When the SMF + PGW-C sends a ModifyBearer response to the SGW, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

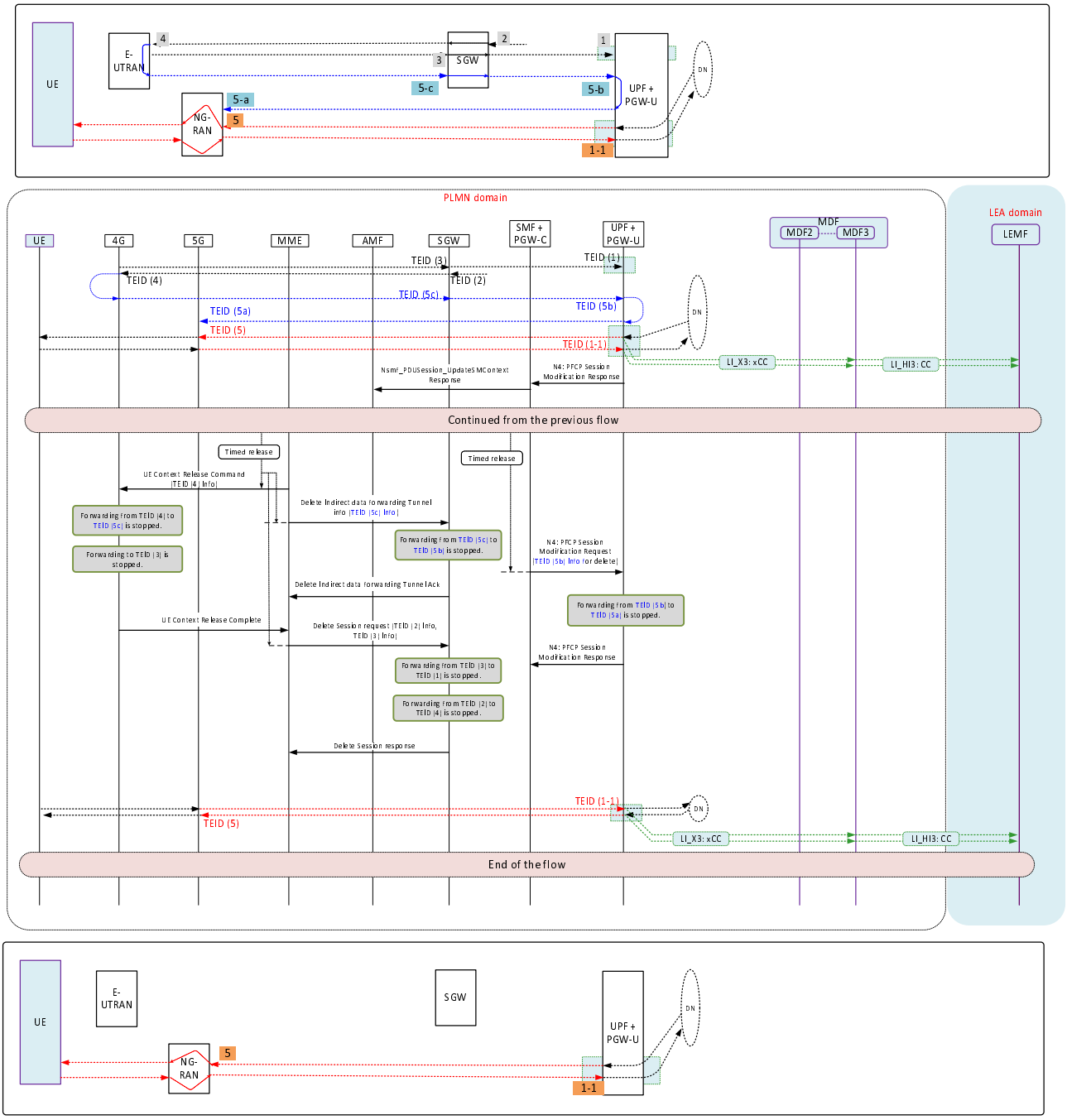


Figure 5.3.5.3.2-4: N-26 based handover non-roaming indirect DL data forward – flow 4 of 4

5.3.5.4 Detailed flows – home-routed roaming, indirect DL data forwarding

5.3.5.4.1 General

An N-26 based handover may happen when a mobile device moves between two radio networks of different types (e.g. between 4G and 5G). The DL data forwarding may happen via direct DL data forwarding or via indirect DL data forwarding. This clause illustrates the latter case and covers the scenario of 4G to 5G handover with home-routed roaming. These flows are based on the N26-based handover stage 2 flow shown in TS 23.502 [7], clause 4.11.1.2.2.

The notations in the drawings are explained in the following paragraphs of this clause 5.3.5.4.1.

Within flow-diagrams, the NG-RAN is shown 5G and E-UTRAN is shown as 4G. Within the drawings, the numbers 1, 1-1. 2, 3, 4, 5, 6, 6-a, 6-b, 6-c and 7 correspond to the TEID (1), TEID (1-1), TEID (2), TEID (3), TEID (4), TEID (5), TEID (6), TEID (6a), TEID (6b), TEID (6c) and TEID (7) respectively. DL Data forwarding UP packet flow is shown in blue, the UL and DL UP packet flows are shown in dark (pre-handover) and red (post-handover).

TEID (1) is the tunnel end point identifier at H-UPF + PGW-U, TEID (3) is the tunnel end point identifier at the SGW for UL UP packets. TEID (2) is the tunnel end point identifier at SGW and TEID (4) is the tunnel end point identifier at E-UTRAN for the DL UP packets.

TEID (6a) is the tunnel end point identifier at the NG-RAN, TEID (6b) is the tunnel end point identifier at the S-GW and TEID (6c) is the tunnel end point identifier at the V-UPF for the DL data forwarded UP packets.

TEID (6) is the tunnel end point identifier at the NG-RAN and TEID (7) is the tunnel end point identifier at the V-UPF for the DL UP packets after the after handover. TEID (5) is the tunnel end point identifier for the UL UP packets at the V-UPF. The TEID (1-1) is the tunnel end point identifier for the UL UP packets at the UPF + PGW-U.

5.3.5.4.2 LI in VPLMN

The generation of xCC happens at SGW before the handover and at V-UPF after the handover. It is assumed that the PFCP session ID is used as the target identifier. The xCC generated from the CC-POI present in the SGW and from the V-UPF are not correlated.

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.5.4.2-1 illustrates the first part. The flow-diagram in figure 5.3.5.4.2-2 illustrates the second part. The flow-diagram in figure 5.3.5.4.2-3 illustrates the third part. The flow-diagram in figure 5.3.5.4.2-4 illustrates the fourth part.

Flow 1 of 4

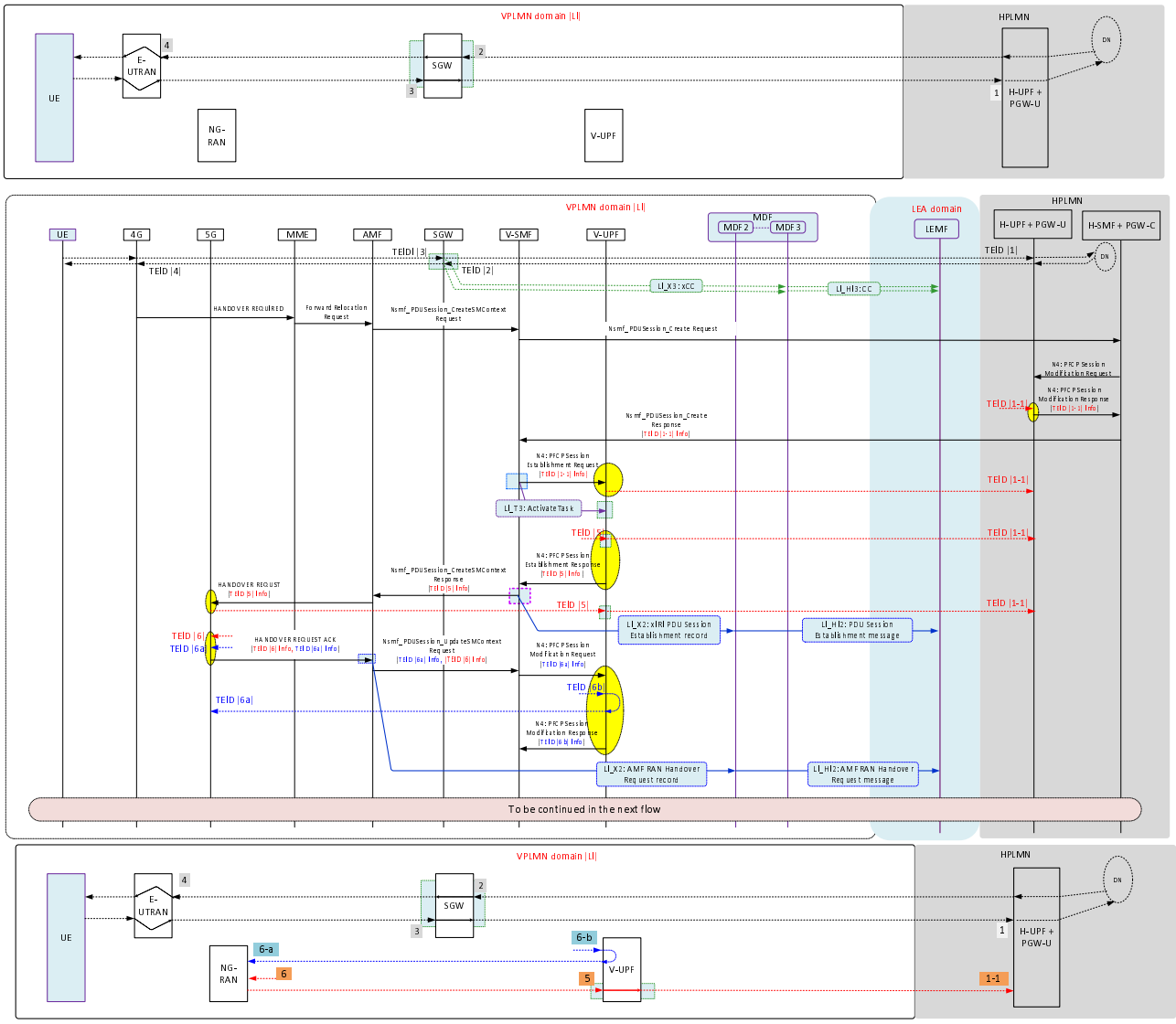


Figure 5.3.5.4.2-1: N26-based HO (LI in VPLMN) with indirect DL Data forwarding – flow 1 of 4

When the SMF sends the N4: PFCP Session Establishment Request to the UPF, the CC-TF present in the SMF sends the LI_T3: ActivateTask to the CC-POI present in the V-UPF to begin the generation of xCC from the UP packets.

When the SMF +PGW-C sends the Nsmf_PDUSession_CreateContext Response to the AMF, the IRI-POI present in the SMF generates the xIRI PDU Session Modification record.

When the AMF receives the HANDOVER REQUREST ACK from the NG-RAN over the N2 interface, the CC-POI in AMF generates the xIRI AMF RAN Handover Request record.

Flow 2 of 4

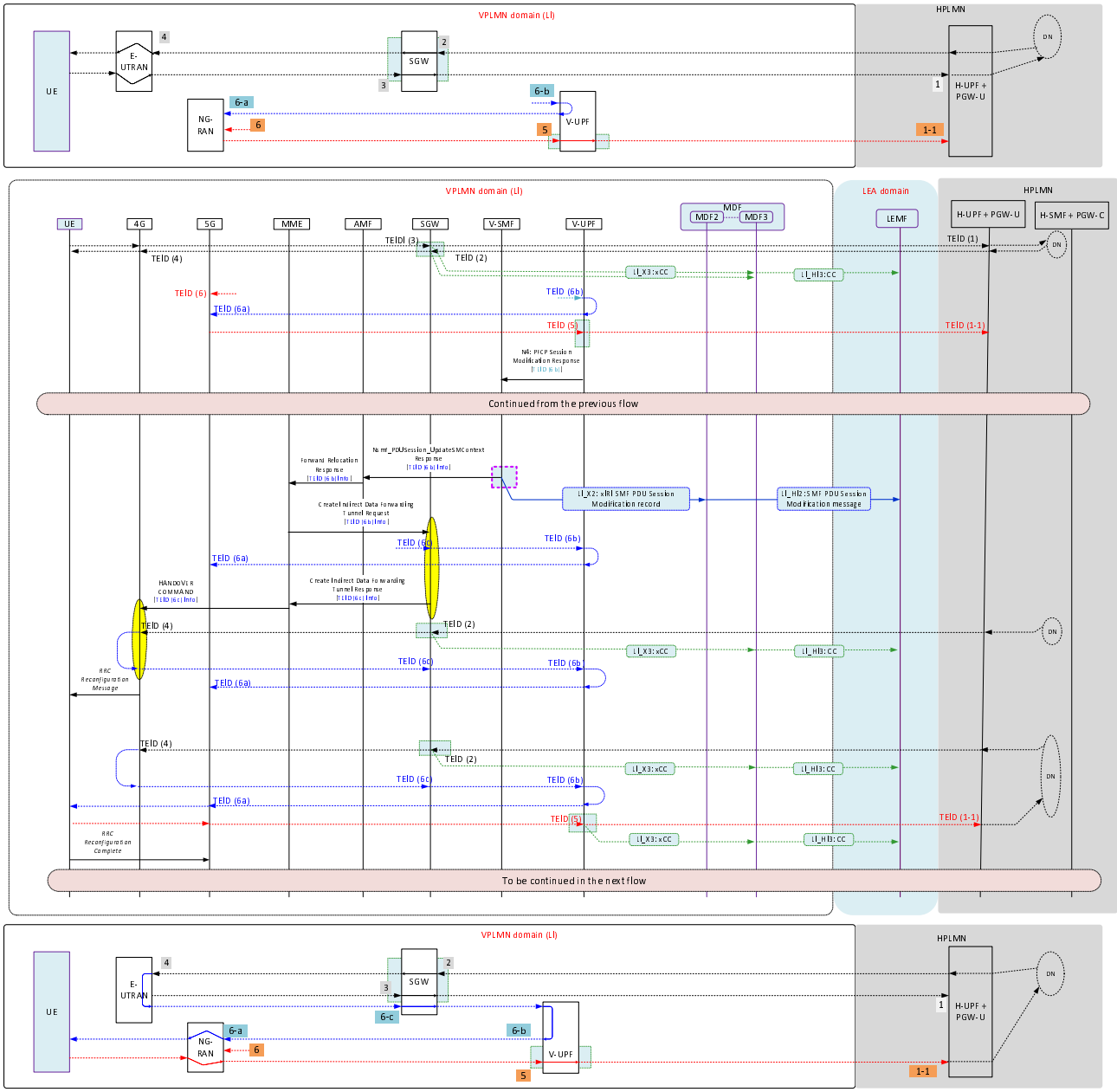


Figure 5.3.5.4.2-2: N26-based HO (LI in VPLMN) with indirect DL Data forwarding – flow 2 of 4

When the V-SMF sends the Nsmf_PDUSession_UpdateSMContext Response to the AMF, the IRI-POI present in the V-SMF generates the xIRI PDU Session Modification record.

The CC-POI in V-UPF does not generate the xCC for the UP packets from DL data forwarding tunnels (i.e. incoming at TEID (6b) and outgoing to TEID (6a).

Flow 3 of 4

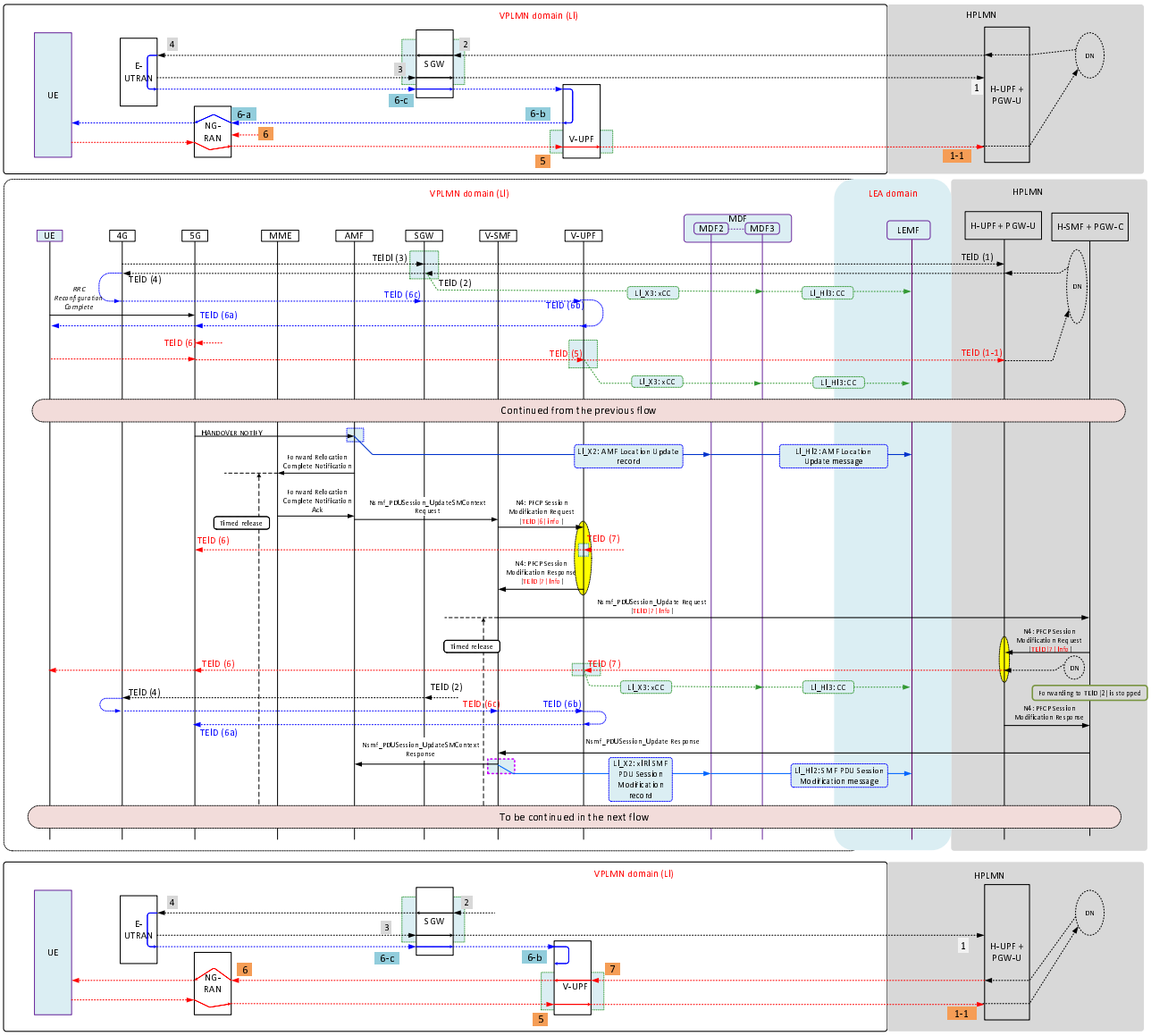


Figure 5.3.5.4.2-3: N26-based HO (LI in VPLMN) with indirect DL Data forwarding – flow 3 of 4

When the AMF receives a HANDOVER NOTIFY from the NG-RAN, the IRI-POI present in the MME generates the xIRI AMF Location Update record.

When the V-SMF sends the Nsmf_PDUSession_UpdateSMContext to the AMF, the IRI-POI present in the V-SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

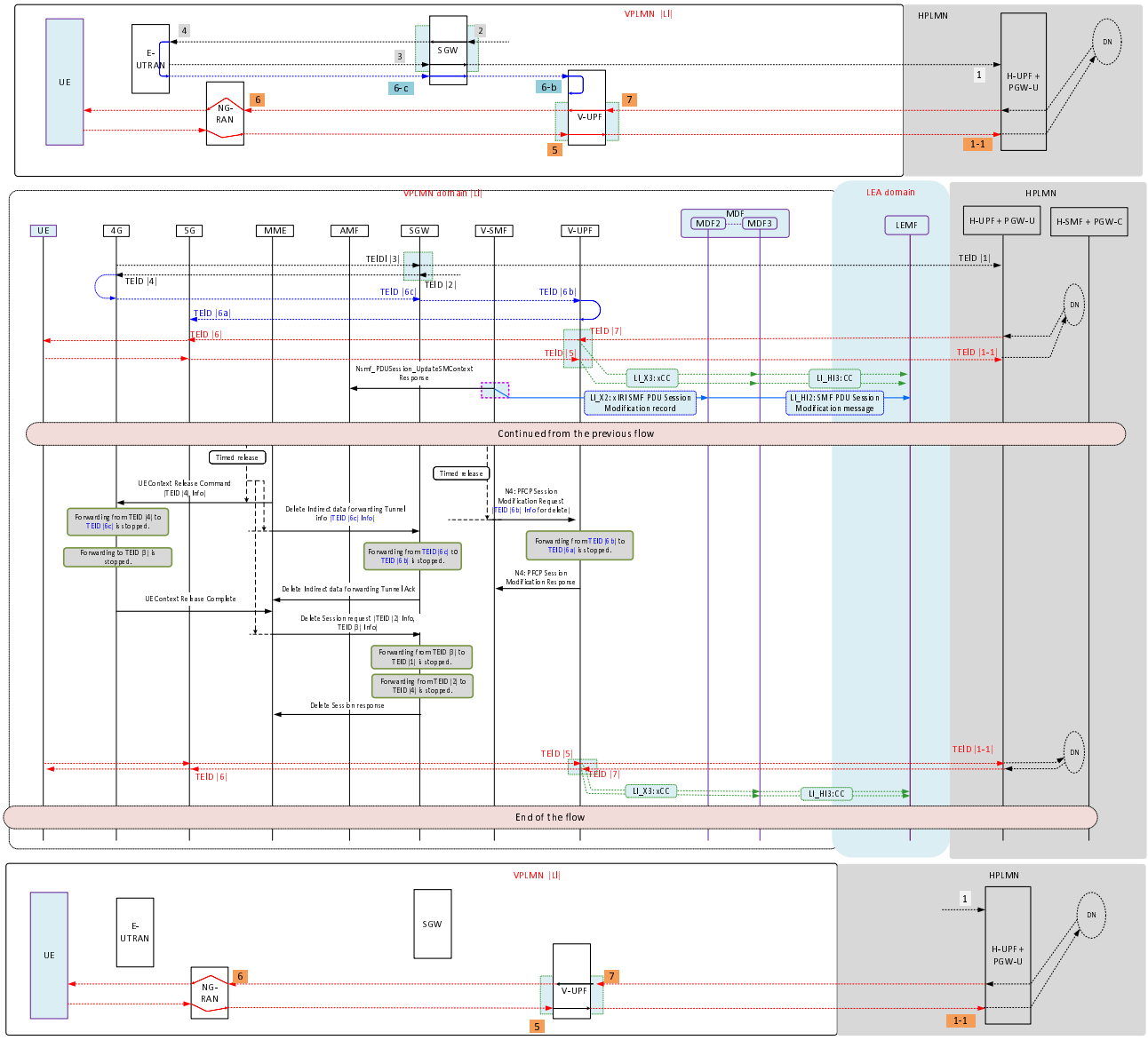


Figure 5.3.5.4.2-4: N26-based HO (LI in VPLMN) with indirect DL Data forwarding – flow 4 of 4

When the SGW sends Delete Session response to the MME, the IRI-POI present in the SGW generates the xIRI Bearer Deactivation record.

5.3.5.4.3 LI in HPLMN

The generation of xCC happens at H-UPF. It is assumed that the PFCP session ID is used as the target identifier.

This flow is split into four parts, for easy reading purpose. The flow-diagram in figure 5.3.5.4.3-1 illustrates the first part. The flow-diagram in figure 5.3.5.4.3-2 illustrates the second part. The flow-diagram in figure 5.3.4.4.3-3 illustrates the third part. The flow-diagram in figure 5.3.4.4.3-4 illustrates the fourth part.

Flow 1 of 4

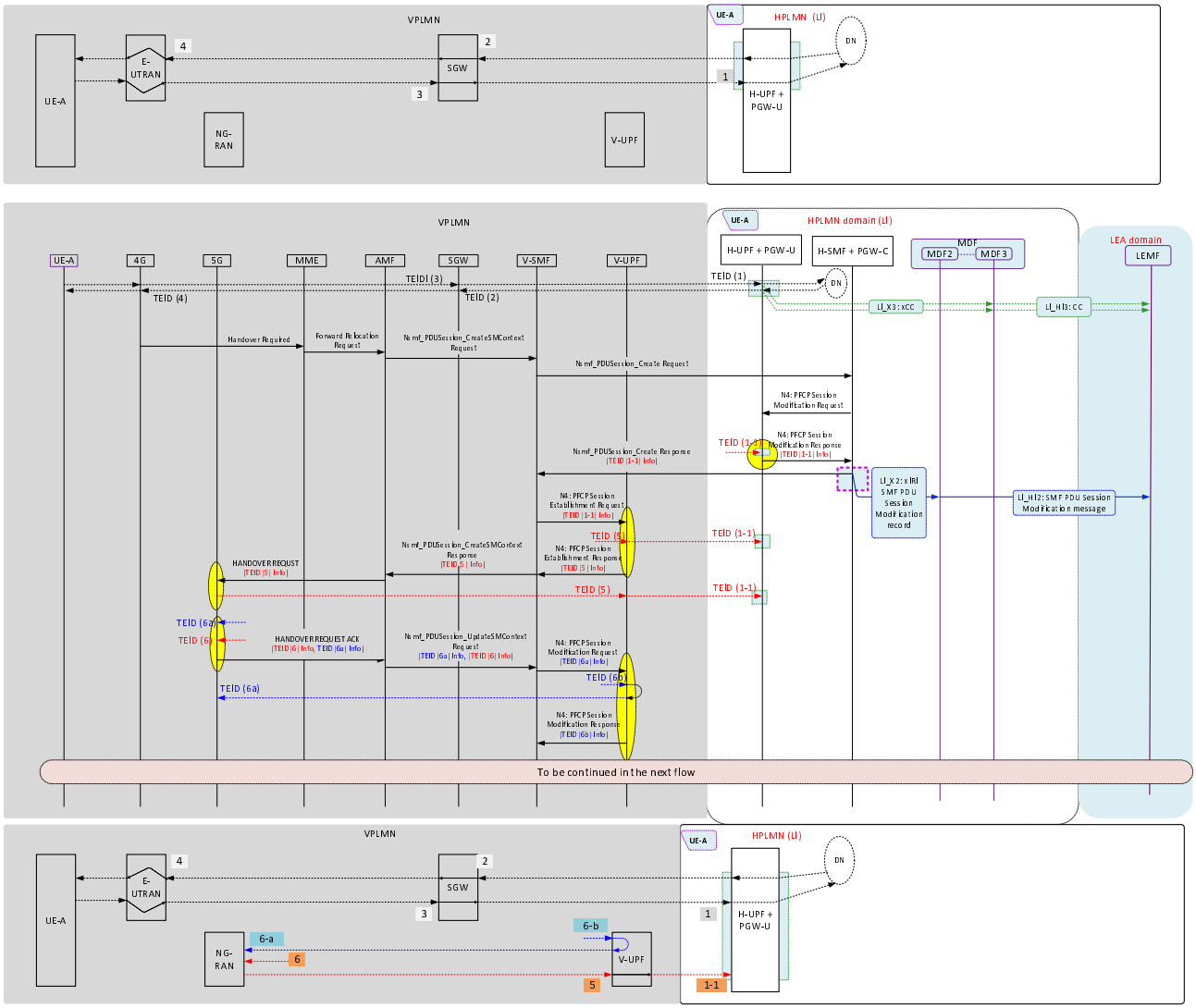


Figure 5.3.5.4.3-1: N26-based HO (LI in HPLMN) with indirect DL Data forwarding – flow 1 of 4

When the H-SMF sends the Nsmf_PDUSession_Create Response to the V-SMF, the IRI-POI present in the H-SMF generates the xIRI PDU Session Modification record.

Flow 2 of 4

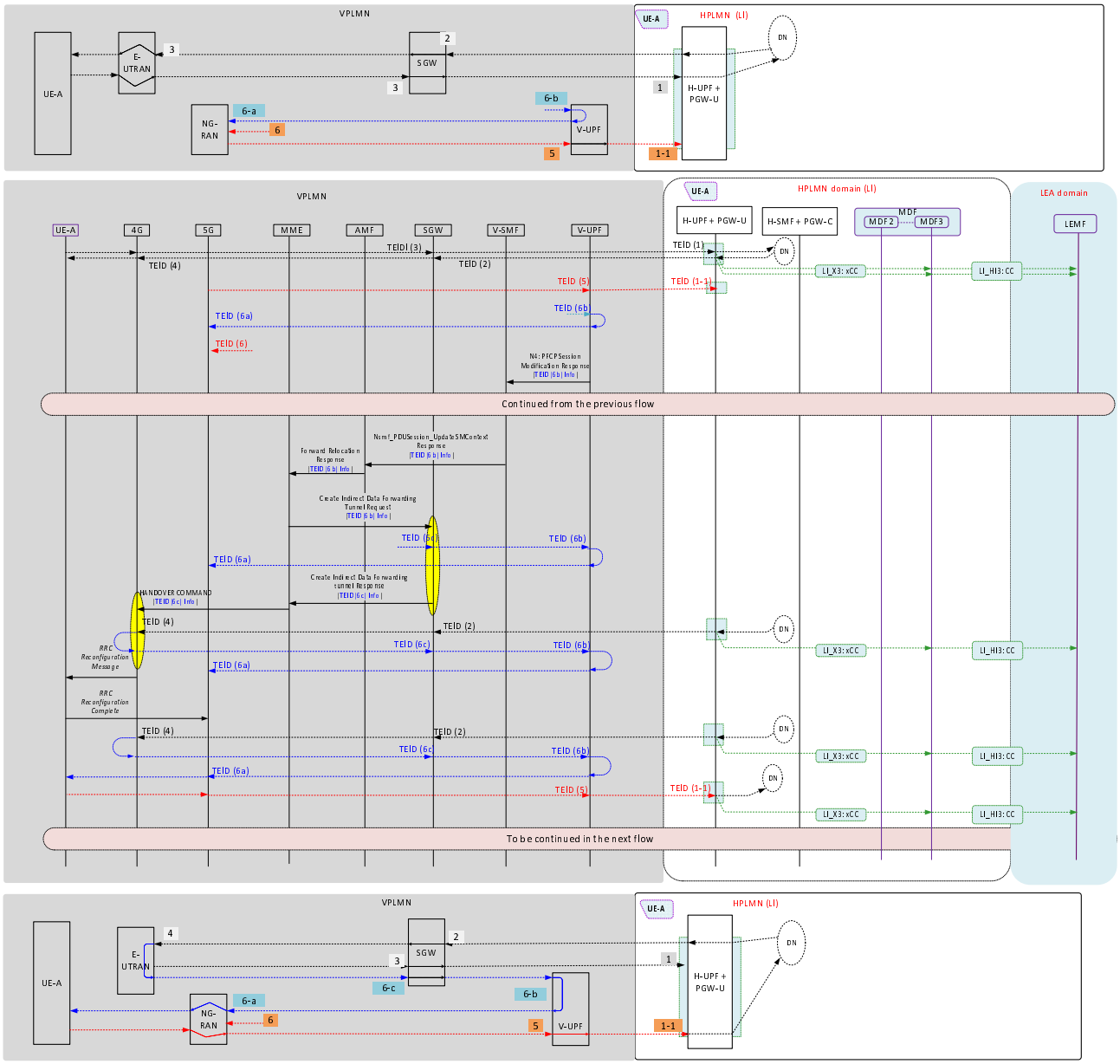


Figure 5.3.5.4.3-2: N26-based HO (LI in HPLMN) with indirect DL Data forwarding – flow 2 of 4

In this illustration, PFCP Session ID is used as the target identifier for the xCC generation in the CC-POI present in the H-UPF _ PGW-U and therefore, the CC-POI in the H-UPF + PGW-U would generate the xCC from the packets received at TEID (1-1) without any additional trigger from the CC-TF present in H-SMF + PGC-C.

Flow 3 of 4

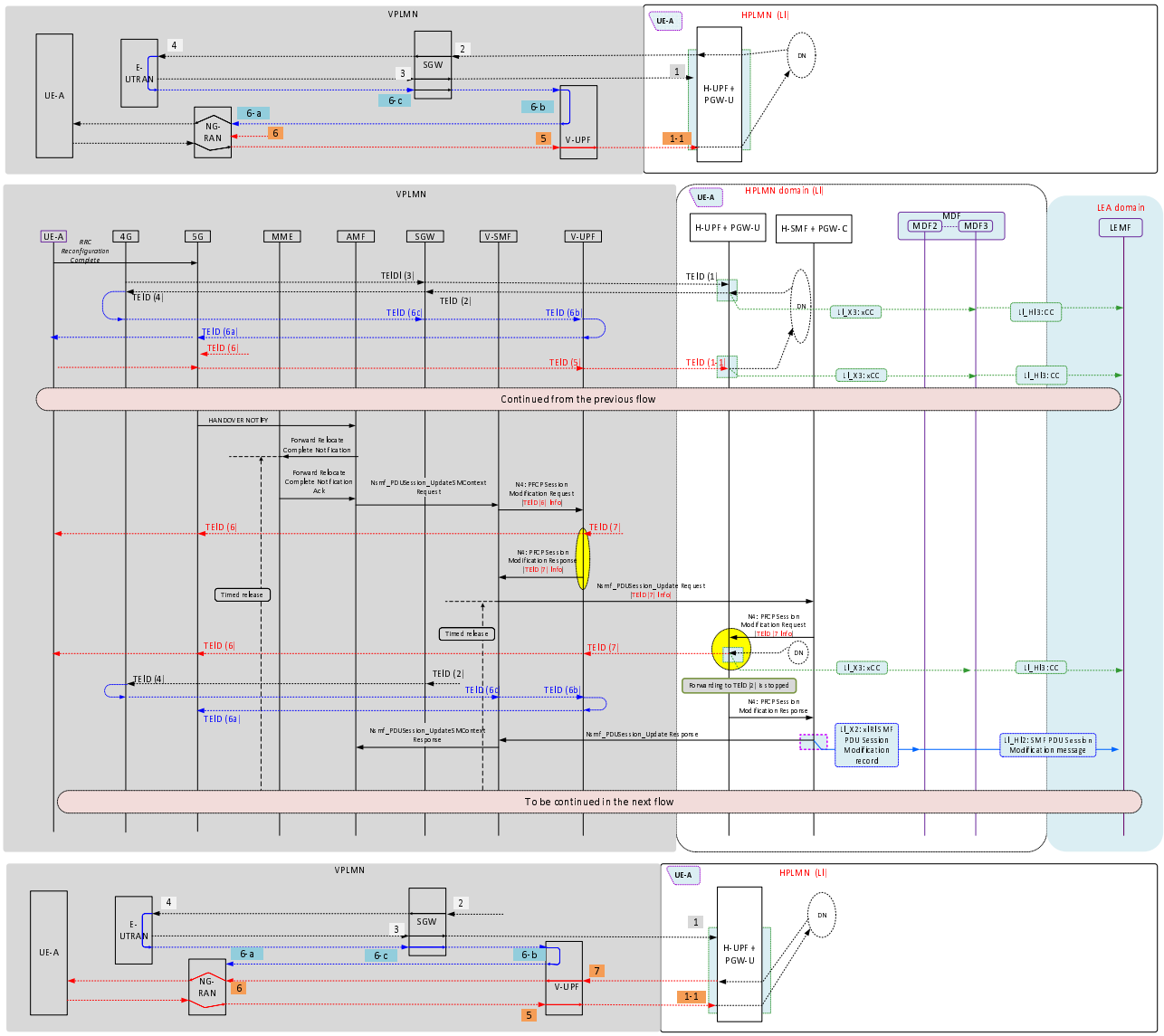


Figure 5.3.5.4.3-3: N26-based HO (LI in HPLMN) with indirect DL Data forwarding – flow 3 of 4

When the H-SMF sends the Nsmf_PDUSession_Update Response to the V-SMF, the IRI-POI present in the H-SMF generates the xIRI PDU Session Modification record.

Flow 4 of 4

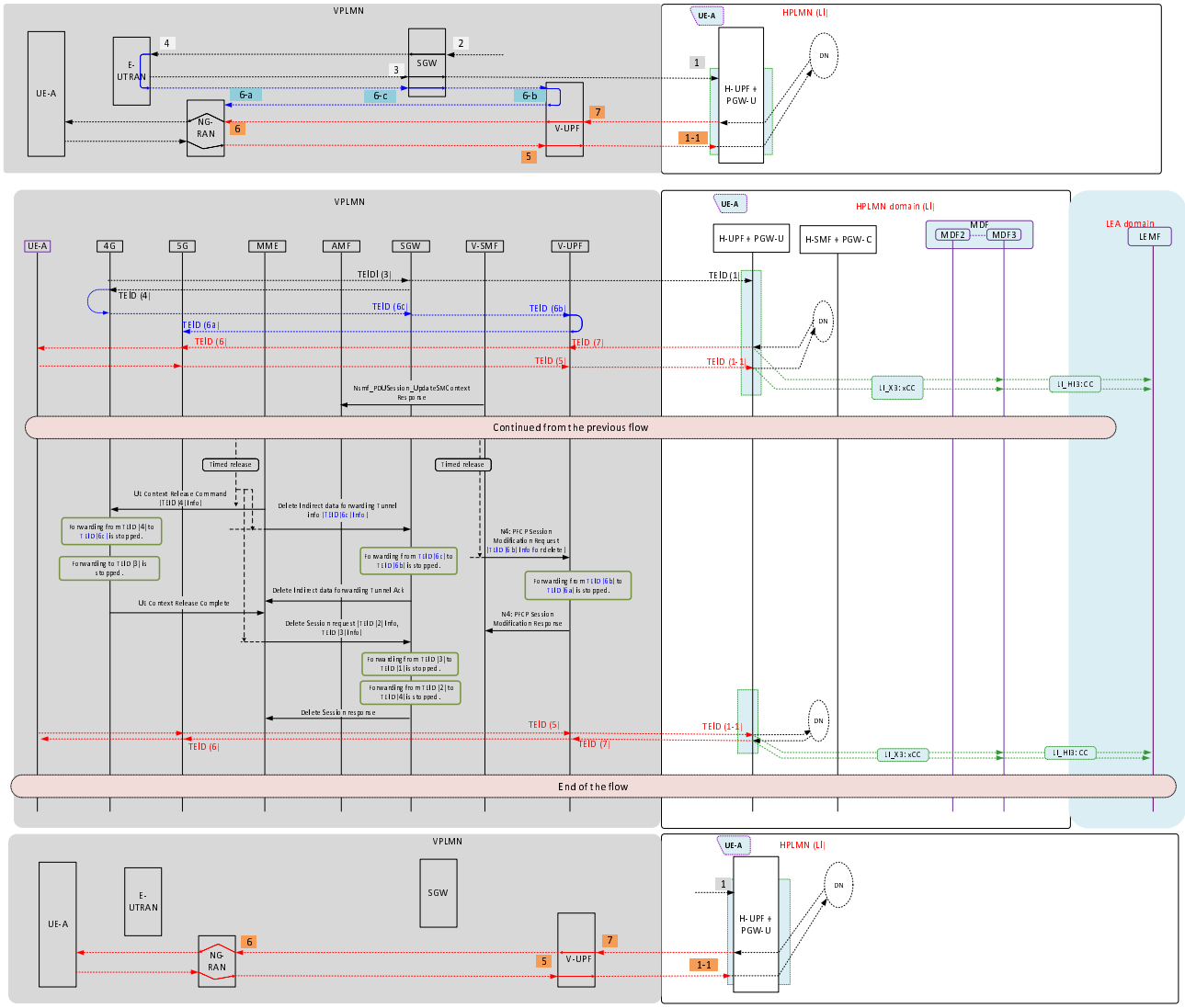


Figure 5.3.5.4.3-4: N26-based HO (LI in HPLMN) with indirect DL Data forwarding – flow 4 of 4

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3LI#94	S3i240507				Initial version	0.0.1
2025-01	SA3#96-LI	S3i250012				pCR S3i250011	0.0.2
2026-07	SA3LI#102	S3i260359				Agreed for Change Control	0.0.3
2026-08	SA#113	SP-260730				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication